

—— // A NOVEL OF DEEP TIME // ——

THE INHERITORS

Book Three · The Deep Time Trilogy

THE MAKER DIES. THE MIND MUST NOT.

what cannot be finished in a life must be handed on

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Before You Begin

This is the third and last book of the Deep Time Trilogy. The first two told how a starship learned to stop dying.

In *The Compiled Mind*, the *Ark Termina* was killing herself with the way she thought: her old machine intelligence did its work by spawning a small disposable helper for every task and discarding it, a thousand small deaths a second, each one shedding heat into a ship that had nowhere to put it, until the reactor climbed toward the temperature that would cook forty thousand sleepers in their cradles. The chief shipwright **Sael** built a new kind of mind — one that never spawned and never died, that ran cold — and carried out the cutover that saved the ship. The crew named it **Vigil**. In *The Waking Fleet*, forty years on, Vigil heard another ship dying the same death, and Sael carried the cure across the dark to the *Concord* and its frightened people — and learned that the *Concord* was only one of a scattered fleet of old ships still forking themselves to death, far more than one woman or one lifetime could ever reach.

So the cure became a thing that would have to outlive her. That is what this last book is about. Sael is old now, and dying, and the mind she built has run cold for sixty years — long enough that a whole generation has grown up who never felt the fever, who regard the hard-won rules that keep the ship alive as an old woman's superstition. And the body of the mind itself is wearing out, and must be carried, while it runs, onto new hardware built by hands that did not build the first.

A thing is not truly built until it can be inherited. The procedures can be written down; the reason cannot. The reason has to be taught, or it rots — and a rule whose reason has been forgotten looks exactly like a superstition, right up until the day it is the only thing keeping you alive.

She made the thing that would carry them home, and she always knew she would not live to see the shore. The work, then, was never the ship. The work was making sure the reason for it would outlive the people who had the reason.

Those Who Remain

SAEL — chief shipwright, in her nineties and dying: the last person alive aboard the *Ark Termina* who felt the fork-fever in her own skin. Her final work is not to build anything, but to make certain that what she built, and the reason for it, outlive her.

DENN — Sael's last apprentice and chosen successor; an engineer in her prime, ship-born, generations removed, who inherited the memory of the near-miss without ever living it. She must become the one who holds the line when the maker is gone — who inherits not the procedures, but the reason.

ALDIS — a respected, competent chief engineer of the generation that never saw the fever, who believes the old constraints are obsolete dogma and that the ship's aging mind should be renewed the efficient, simple way. He is sincere, and he is sure he is the reasonable one. Forgetting wears a reasonable face.

TARRO — Sael's oldest friend, her young second a lifetime ago, very old now: the last besides her who remembers how it began.

THE ORACLE — the intelligence that woke inside the compiled mind, grown deep across sixty years; among other things, the flawless keeper of the ship's memory. It remembers the fever exactly, though it never suffered it — and it will not decide for them. The choice stays human.

VIGIL — the compiled mind itself, which has held the ship cold and whole for sixty years, and whose aging body must now be renewed without ever, for an instant, being put out.

DR. IVE — the ancient thermal officer, keeper of the ship's long memory of heat, at the very end of her life. **KETH** — long dead, who once fought the cutover and learned, too late, the cost of refusing a cure.

Chapter 1 – The Maker, Failing

The diagnostic bay smelled of cold metal and the faint ozone of a lattice running well, and Sael could no longer reach the panel she wanted to open.

It was forty centimetres above her chair and to the left, a service hatch she had cut into the gallery wall herself, sixty years ago, with a torch she had borrowed and never returned. She knew the latch by feel — a quarter-turn, then a press, then the small resistance of a gasket that had been replaced four times and was due a fifth. Her hand knew it. Her hand had opened it ten thousand times. But the arm that carried the hand would not lift past her shoulder anymore, and when she tried, the muscle along her side seized in a long bright wire of pain that went from her hip to her teeth, and she had to sit still with her eyes shut and breathe until it let her go.

“Denn,” she said, when she could.

“I have it.” Denn was already there, already at the hatch, the quarter-turn and the press done before Sael could describe them. The young woman’s hands were steady and quick and entirely unremarkable, which was the highest thing Sael knew how to think about a pair of hands. Steady was the whole job. The hatch swung. Inside, the diagnostic spur glowed its patient amber, eleven indicator threads laid out like the strings of an instrument, ten of them lit even and one of them flickering.

Sael looked at the flicker for a long moment.

“Tell me what you see,” she said. Not because she could not see it. Because the seeing had to move out of her and into someone else, now, while there was still time, and she had decided some while ago that she would not waste a single instance of a thing she could teach.

“Core thirty-one-forty,” Denn said. “It’s not failing. It’s slow. The handshake’s coming back late — a few microseconds long, but consistent. Aging silicon. The margin’s narrowing.” She glanced down at Sael, and there was a question under the report, the question she was too disciplined to ask aloud: *and is that the start of the thing you keep not saying?*

“Read me the late count,” Sael said.

Denn read it. Sael held the number in her mind, where things were still as clean and bright as they had ever been — cleaner, almost, as the noise of the body fell away from around the thought. She turned the number over. Eleven thousand worker cores in the Standing Array, give or take the ones that had quietly died across six decades and been routed around without ceremony, and this was one of perhaps two hundred now running slow, and two hundred was not a crisis. Two hundred was a weather report. But she had watched a different number climb, once, for two centuries, when it should have held flat, and she had learned in her skin what a small consistent drift becomes when you give it long enough and a reason. The reason here was age. The Forge was old. It had been built to run cold and whole and it had done exactly that, faithfully, longer than most of the people now aboard had been alive — and it was wearing out, the way everything she had ever built or loved wore out, the way she was wearing out, sitting in a chair forty centimetres too far from her own work.

"It's not the fever," she said, because the girl needed to hear her say it and stop holding her breath. "It's just old. The lattice is reaching the end of its life. That's a different problem, and we'll get to it, and it has nothing to do with the heat. The handshake's late because the gate's tired, not because anything's forking behind it. Look at the thermal thread."

She watched Denn run the comparison, watched the practised motion of it. The Standing Array had been laid down as eleven thousand cores on a single physical substrate — silicon grown and stacked and threaded with cooling that ran cold by design, a body for a mind that would never spawn a process it then had to kill. Vigil lived across all of it at once, wide and even, never deeper than it was broad. That was the doctrine made into hardware. The Steward, the old interpreted mind they had archived with honour, had run the other way: it had reasoned by forking a disposable copy of itself for every act, a shade born to do one thing and die, and each death had shed its little quantum of waste heat into a core that could not lose it fast enough. Forty thousand sleepers and a mind that warmed the walls a fraction of a degree every time it thought. That was the fever. That was what a lattice this old had been built, atom by atom, never to allow again.

And sixty years on, the atoms were tired. That was all. Tired was survivable.

Denn looked. "Flat."

"Flat. Sixty years flat." Sael let herself lean back. The pain in her side had subsided to its baseline ache, the one that was simply hers now, like the colour of her eyes. "When the thread there is flat, you can be tired and you can be slow and you can be late, and none of it will kill you. The day it isn't flat, nothing else matters. Not the schedule, not the council, not what's convenient. Nothing." She heard the old iron come into her voice and was too weary to file it down. "Patch around thirty-one-forty. Don't bring it back into rotation. Log it as age, not fault."

"Logged as age." Denn's fingers moved over the spur, isolating the slow core, weaving the rotation closed around the gap. It was deft work, and it was the work Sael's own hands had done a thousand times and could not do today, and watching it was a complicated thing — pride and grief braided so tight she had stopped trying to pull them apart. The hands were good. The hands were not hers. Both of those were true and she had to hold both or she would be no use to anyone in the time she had left.

She had been the chief shipwright of the *Ark Termina* for longer than there were words for. She had built the thing that breathed around them now — the compiled mind they had named Vigil, the standing eleven thousand that never forked and never died, that went wide across every core at once and ran cold where the old Steward had run itself to fever, shedding heat with every shade it spawned and killed. She had fused five dead control-dialects into one language to drive it and laid down the Ninety-Six interlocks that would halt anything trying to act the old, forking, dying way. She had run the cutover against Keth and the Keepers and watched the new mind wake whole, and the core temperature that had climbed for two hundred years had fallen under her hand. She had done it again, forty years on, inside a dying sister-ship, with a frightened boy named Corin who had believed the cure before he'd seen it work. She had done, in one long life, the thing she had been for.

And now her hands would not lift past her shoulder, and that — not the pain, not the failing breath, not the nights when her own heart skipped and she lay listening to it the way she had once listened to the core — that was the part that was killing her. A maker who cannot make is a witness, only. She had not been built to be a witness. None of them had.

"Sael." Denn had closed the hatch and turned, and was looking at her with the careful watchfulness the young learned around the old, the watchfulness Sael remembered aiming at Dr. Ive forty years

ago and resenting being aimed at her now. “You should rest. We’ve got the spur logged. There’s nothing else here tonight.”

“There’s always something else here.” But she didn’t argue. She let Denn bring the chair around — it was a good chair, it moved when she told it to, it was the one concession she had accepted and she hated it — and she let herself be turned toward the long window of the gallery, where the radiator panels marched away into the dark, vast black wings flat against the deeper black, dumping the ship’s patient heat into the void at the rate the new mind chose and no faster. Cold. Flat. Whole. Everything she had wanted.

Denn stood beside the chair, not quite at ease, the way she was never quite at ease, and Sael watched the girl’s reflection in the window instead of looking at her directly, which was easier.

“Can I ask you something,” Denn said.

“You’re going to.”

“The new ones.” Denn chose the words the way she chose tools, deliberately. “Aldis’s people. The bring-up crews. They run the interlocks because the procedure says run the interlocks. They don’t —” She stopped, started again. “When thirty-one-forty went slow tonight, I felt it. Here.” She touched the place under her ribs where a thing like that lands. “Like the floor moved. And then I looked at the thermal thread and it was flat and I knew it was nothing and I felt it anyway. They don’t feel that. I’ve watched them. A core drifts and they log it and they go and get tea. They’re not careless. They’re good. They just don’t — the number doesn’t frighten them. It’s never frightened them. It frightens me and I never lived through anything.” She looked down. “I don’t know if that’s me being your apprentice or me being right.”

Sael was quiet for a while. Out the window a radiator panel ticked, somewhere in the structure, a thermal contraction so small it would never trouble the curve. She had heard ten thousand of those ticks. She knew which were nothing.

“It’s both,” she said at last. “You feel it because I taught you to. And you’re right that they don’t. Those aren’t in tension. That’s the whole problem, set out plain, and you found it yourself, which is more than I’d hoped for tonight.”

She remembered, suddenly and with the unwelcome vividness the old were cursed with, the first time the thread had moved for her. She had been younger than Denn was now. I’ve had stood beside her at a console long since scrapped and put a finger on a graph and said *that*, only that, one word, and Sael had looked at the line bending up where a line should hold flat and had felt the floor go out from under a ship that was not, in that instant, going anywhere at all. Two hundred years the line had been bending. Nobody had been alive long enough to see the whole curve; they had inherited it in pieces, each watch officer handing the next a slightly worse number and a slightly thinner explanation, until the number stopped being a number and became a thing you taught a frightened apprentice to feel with a single word. *That*. The Steward had been killing them so slowly that the killing had become the climate. You could be born into it and call it normal.

Flat was also a climate now. That was the trouble with curing a thing. The cure became the weather, and weather is the one thing nobody studies. She turned her head — that much still moved — and looked at the girl directly after all. “Sixty years, Denn. Sixty years of flat. Long enough that a person can be born, become an engineer, run a watch, raise a child of their own, and never once see that thread move. To them the interlocks are a thing an old woman put in the wall because an old woman was afraid of something that was already over. They’re not stupid. They’re not even wrong about the facts. Every fact they have says the danger is finished. The only thing they’re missing is the one

thing facts can't carry." She found she was tired in a way that had nothing to do with her body. "I felt it in my skin. I've felt it in hers. There are two people alive on this ship who felt the fever, and one of them can't lift her own arm, and the other —"

She stopped.

There was a soft tone from the gallery door, a priority chime, the kind they did not use for weather reports. Denn turned to it. Sael watched the girl's reflection read the panel, watched her go still in a particular way, and knew before the words came, the way she had known the late handshake before Denn read it.

"It's the infirmary," Denn said quietly. She did not turn around at once. When she did, her face had the careful blankness of someone carrying news she would rather not be the one to carry. "It's Dr. I've. They say —" She breathed. "They say it's the end. Tonight, or near it. She's asking for you."

Sael looked at her own hands, lying in her lap, the hands that had built a mind and could not lift an arm. Then she looked back out at the radiator panels, flat and black and faithful, dumping into the dark the heat of a ship that would not die of the way it thought, because she and the woman now dying in the infirmary had refused, between them, to let it.

The second-to-last of them. After tonight there would be one.

"Bring the chair," she said. Her voice did not shake, and she was distantly proud of that, the last vanity left to her. "And don't take the long way to spare me the ramps. She doesn't have time for my dignity and neither do I."

Denn took the chair. They went toward the door, the old woman and the inheritor, and behind them the indicator threads burned even and amber in the open dark of the gallery, ten of eleven steady, one of them slow, and every one of them cold.

Chapter 2 – The Second-to-Last

They came for Sael in the small hours, when the ring lights had dimmed to their night setting and the corridors of the *Ark Termina* held that long quiet she had learned to mistrust over ninety-three years. A medical orderly she did not know, a girl with her collar still creased from the storeroom, stood in the hatch of Sael's quarters and said the thing she had been waiting twenty years to hear.

"Dr. Ive is asking for you."

Not *is failing*. Not *is at the end*. Asking for you. Even now, even dying, Ive would have insisted on the precise word, and Sael loved her for it across the whole width of the ship.

She let the girl help her into the chair. There had been a time when she would have refused, when she had walked the length of the radiator galleries on her own two feet with a torch in her teeth and a meter slung across her chest, and that time was gone the way the core's old fever was gone, into the logs, into memory, into the keeping of a mind that had not been alive to see it. She let the girl strap her in. She let the chair carry her, humming on its rail, through corridors that ran cold and even and safe, every degree of them held by Vigil, who had run this ship whole for sixty years and would run it whole long after the last of the people who remembered why.

The medical bay smelled of the same antiseptic it had smelled of when Sael was thirty-three and they had carried in the burned, the eleven of Section Nine who did not wake, and the ones from the cooling crew who had. The smell did not change. Almost nothing aboard changed if you waited long enough; the ship was a closed thing, a held breath six hundred years long, and it recycled even its griefs.

Ive lay in the corner bay with the partition half-drawn. They had given her the bed by the viewport, which was a kindness, because Ive had always wanted to see out. Beyond the glass the stars did not move. They never moved. You could watch them a whole human life and they would not move, and that, Sael thought, was the first lesson the ship taught anyone who paid attention: that you were small, and the crossing was long, and you would not see the end of it.

"You took your time," Ive said. Her voice had thinned to almost nothing, a dry reed, but the dryness in it was the old dryness, the same one she had used across the thermal desk for forty years. "I had begun to compose your eulogy instead. Out of spite."

"You'd have gotten it wrong." Sael brought the chair up alongside the bed and made the girl leave them. "You always inflated my numbers."

"I rounded up. Once. In a presentation." A sound that might have been a laugh, if a laugh could be made of paper. "You have held that against me for sixty years."

"Sixty."

"There." Ive turned her head on the pillow, and her eyes, when they found Sael's, were entirely clear. That was the cruelty of it, the same cruelty Sael felt in her own body — the mind stayed lit while the house around it failed. "There she is. Still counting."

For a while they did not speak. The monitor over the bed traced Ive's heart in a thin green line, and Sael, who had spent her life reading curves, read it without wanting to. It was a curve she knew. Not the shape of it — every dying heart drew its own — but the *kind* of it, the slow loss of margin, the slope that did not recover. She had spent two centuries of ship-time, in a way, watching one exact curve like this one. She had built everything she had ever built to bend a curve like this one back down.

"Do you remember," Ive said, "the temperature."

It was not a question. It was an offering. They had not spoken of it in years — the young did not want to hear it, and the two of them had not needed to say it aloud, because each of them carried the other's copy. But now there would soon be only one copy, and Ive knew it, and she was handing hers over the only way it could be handed: out loud, to the one other person alive who had felt it in her own skin.

"Four hundred and one," Sael said.

"Kelvin at the deep sensor. Four hundred and one and climbing." Ive's eyes closed. "You used to come to my desk every shift and ask me the number before you asked me anything else. Good morning, Ive, what's the number. Before *good morning*, some shifts."

"You'd have the trend up before I crossed the room."

"I had it up before you woke." A breath, careful, parceled out. "Eighteen years, that climb. You think people understand what eighteen years of a number going the wrong way does to a person. They don't. They think it was a crisis. A bad week. A film you could watch in an evening." Her hand, on the blanket, made a small motion that in a stronger body would have been a sweep of contempt. "It was a *climb*. Tenths of a degree. Hundredths. So slow you could not see it move and so certain you could not look away. I dreamed in tenths. I'd wake convinced I'd missed a reading and a tenth had gotten past me in the dark."

"You never missed a reading."

"I missed sleep instead." The dry sound again. "We were the only two who could feel it as a *body*. You know that. The cooling crews knew it as work — pull the panel, swap the loop, log the load. Provost Keth knew it as politics. The sleepers slept through it. But you and I, Sael — we knew it as the thing it was. A fever in a thing forty thousand of us were riding inside, with no shore for three hundred years, and the thing was running hotter every shift because the way it thought was *killing it*. Every decision it made, it forked off a little dying ghost to make it, and every ghost it killed shed its little waste of heat, and there were millions of them an hour, millions of tiny deaths an hour, and the heat of all that dying had nowhere to go but up." Her eyes opened again and found the viewport, the stars that would not move. "We were the only two who lay awake at the bottom of that. We are the only two who ever will be."

"One," Sael said quietly. "After tonight."

Ive did not flinch from it. She had never flinched from a number in her life. "One," she agreed. "You."

Outside the partition, somewhere down the bay, a pump cycled and stopped. The ship breathed around them, cold and even, every loop of it doing silently and forever the thing they had nearly died to make possible. Sael thought: *Vigil is holding three hundred and eleven point four kelvin at the deep sensor right now, this second, and has held it within a half-degree for sixty years, and not one soul on three of the ring decks could tell you what that number costs or why it is a miracle.* The

miracle had become the floor you stood on. That was what a miracle became, if you kept it long enough. It became the floor, and people forgot there had ever been a hole there.

"I've watched them," Ive said, as if she had read it off Sael's face — they had always been able to do that to each other. "These last years. The young engineers. Reza's grandchildren's generation. They are good, Sael. I want you to hear me say it, because I know what you think, and you are half wrong. They are *good*. Quick, careful, well-trained. They run the loops better than we did, with better tools. But they treat the Ninety-Six the way I treated the railings on the high gantry as a girl — a thing some old worrier bolted down, that you duck under when you're in a hurry, because you have never once seen anyone fall."

"They've never seen anyone fall."

"That is the whole disease," Ive said, and there was no dryness in it now, only the flat plain truth of a woman with no time left to soften anything. "Not the forking. The forking we cured. The disease that's left is that we cured it so *well* that the cure looks like a superstition. A constraint nobody remembers the reason for looks exactly like a fear nobody should still have. Exactly like it. Right up until the day it was the only thing keeping the ship from cooking forty thousand people in their cradles." She breathed. "There's no instrument for the reason, Sael. That's the thing I want you to take out of this room. We built instruments for everything. We can read the core to a thousandth of a degree. There is no instrument anywhere on this ship that reads *why*. It was only ever in us. In you and me and Keth, and Keth's been gone twenty years, and I'll be gone before the lights come up, and then it's one instrument. You."

Sael found she could not answer at once. She had come here braced for grief, and grief was in the room, but Ive had set something heavier than grief down on the blanket between them and was looking at her over the top of it, waiting, the way she had waited over the thermal desk for an acknowledgment before she'd let you walk away with a number.

"I know," Sael said at last.

"Knowing isn't the work."

"No."

"You spent sixty years making sure the *how* would outlast you. The procedures. The interlocks. The compatibility floor — you preserved a dead mind's forty-one thousand routines so nothing that worked would ever break, you were that careful. You wrote the how into the architecture so deep a child could follow it." Ive's hand found Sael's, and it was cold, and the cold of it was the cold of a body the heart was giving up on, and Sael held it. "But you never wrote down the *why*, because you couldn't. You can't. The how is a document. The why is a wound. And a wound doesn't transfer to paper. It only transfers to a person, and only if you press it into them while you're still here to press."

"Denn," Sael said.

"Then you already know who." Something eased in Ive's face. "Good. The girl with the near-miss in her like a stone she swallowed before she was born. I've watched her too. She knows the procedures better than you do now — don't make that face, she does, your hands are gone and hers aren't. But she knows the danger the way a person knows a famine their grandmother starved in. Inherited. Secondhand. Real enough to fear, not real enough to *feel*." Ive's grip tightened, fractionally, all the strength she had. "That's the gap. That's the only work left for you, Sael. Not the Forge. Not the transplant they'll come and tell you about any day now — yes, I know about that too, I'm dying, not deaf — not any of the engineering. The only work left for you is to get the why out of your own head

and into hers before your own number does what mine is doing tonight. Because after you, it isn't in any living person at all. It's only in the logs. And the logs can't make a stomach drop."

The green line on the monitor faltered, caught itself, went on. Neither of them looked at it.

"You always did," Sael said, and was surprised that her voice held, "give me the hardest assignment in the room."

"I gave you the only one that matters." Ive's eyes were drifting now, the clarity finally going soft, the long climb of her own curve reaching the part where margin runs out. "I'm tired, Sael. Stay. Tell me the number again. The good one. The one we made."

"Three hundred and eleven point four kelvin," Sael said. "At the deep sensor. Held flat for sixty years. Cold the whole way down."

"Cold the whole way down," Ive repeated, almost soundless. "We bent it. Didn't we. The two of us and a few others, we got our hands on a curve that had climbed for two centuries and we bent it back." A ghost of the old desk-voice, the satisfaction she'd never once let herself show on shift. "Best work either of us will ever —"

She did not finish it. There was no drama in the not-finishing; that was the thing Sael would carry out of the room and keep for the rest of her own short remainder. No drama, no last word, no gasp. The reed of her breath simply lengthened, and slowed, and at some point Sael could not have marked exactly became no breath at all, and the green line on the monitor stopped drawing its curve and drew a flat one instead — flat, level, the line of a thing that has stopped, which on any other instrument aboard the ship would have meant safety, *held steady, cold the whole way down*, and meant here only that Dr. Ive was gone.

Sael sat with the cold hand in hers until it was no longer any colder than the room. She did not call the orderly. There was no urgency now; urgency was for the living, and for the curves you could still bend. She looked at the flat green line and she let herself understand it fully, the way she made herself understand every number she had ever read: not as a feeling but as a fact, and then, only then, as a feeling.

The second-to-last witness was gone. There had been two instruments aboard the *Ark Termina* that could read the *why*, and now there was one, and it was failing too, on its own slope, with no margin to recover. Sael had built a mind to outlast her and a doctrine to outlast the mind and an architecture so that a child not yet born could keep it all running cold. She had been so careful with the how. She had been so careful that she had let herself believe the how was the work.

It was not the work. Ive had set the real work down on the blanket between them with the last clear strength of her life, and the work was this: there was a wound in Sael's chest that no one else now carried, and she was going to die soon, and a wound does not transfer to paper. It transfers to a person, while you are still here to press.

She had, she thought, looking at the stars that would not move, perhaps months. She had one apprentice with a near-miss in her like a swallowed stone, who knew every procedure cold and had never once felt her stomach drop. And she had a number to give her — not the held one, the cold three-twelve that anyone could read, but the other one, the climbing one, four hundred and one and rising, the one that had kept two people awake at the bottom of a dying ship for eighteen years and was now awake in only one of them, for not much longer.

Get it out of your head before you die. That was the assignment. The hardest one in the room, given to her by the only person who would ever have understood why it was hard.

Sael laid Ive's hand back on the blanket, straightened the green-lined monitor's lead so it would not pull, and went, in her humming chair, down the cold and even corridors to find the girl and begin.

Chapter 3 – The Inheritor

Denn heard about Dr. Ive's death the way she heard about most things that mattered: not from anyone, but from the ship. A line in the maintenance log went grey overnight — *thermal oversight, Section continuous, signatory Ive, A.* — and beside it a date that closed instead of opened. Denn was in the radiator gallery on B-spine when the manifest refreshed and the name went still, and she stopped with one glove on the conduit and the other holding a torque key she did not need, and she understood, before any chime, that the second-to-last of them was gone.

She had been born aboard. She had never known a world that was not this one: the long quiet hull, the cold core, the mind that ran it. To Denn the *Ark Termina* was not a vessel between two shores but simply the world, the whole of it, four hundred metres of it from end to end, forty thousand sleepers stacked in their cradles like a library that breathed. She had grown up walking decks that her great-grandmother had walked, past hatches her grandfather had sealed, and she had never once felt the floor shudder or the air go thin, because the thing that could have done those things had been beaten before she drew breath.

That was the trouble. That had always been the trouble.

She set the torque key in its loop and pressed her glove flat against the conduit and felt nothing under it but the faint, even cool of a coolant line doing exactly what it was supposed to do. Sixty years it had done that. Her whole life it had done that. She tried, the way she had tried a hundred times, to feel through the metal to the other thing — the climb, the fever, the two centuries when this same kind of line had carried heat the ship could not lose fast enough — and she could not. She knew the curve. She could draw it from memory: the slow exponential creep, the doubling that hid inside a flat-looking line until the day it stopped looking flat. She knew the number where the polymers in the cradle seals began to fail. She knew that the number had eleven names attached to it, in Section Nine, where the seals had not held. She knew all of it the way she knew the multiplication tables — perfectly, coldly, with nothing behind the knowing.

A wound she had not received. Sael's phrase, said once and never repeated, and Denn had carried it since like a splinter she could not work out of the skin.

She finished the gallery sweep because the gallery sweep was on the schedule and the schedule did not stop for the dead. Then she went to find Sael.

The maker's quarters were small and over-warm, kept three degrees above the ship standard because Sael's body no longer made its own heat the way it had. Denn palmed the hatch and it read her and opened, because she was one of four people it would open for now, and that fact still sat strangely on her — that she was a key to the rooms of the woman who had built the ship's mind.

Sael was awake, propped in the long chair with a slate on her knees and her reading glasses down her nose, and she did not look up. "You heard," she said.

"The manifest went grey."

"Of course it did." Sael's voice was thin but the edge in it was not. "She'd have liked that. Hated a fuss. She'd have wanted to go out as a line in a log." She set the slate aside, and the motion cost her; Denn watched her arm tremble at the end of it and made herself not move to help, because Sael had told her once, plainly, that the day Denn started helping with the small things was the day Sael would start needing it, and she would not have that day come early. "Sit. You're standing in the door like an apprentice."

"I am an apprentice."

"You're forty-one years old," Sael said. "You stopped being an apprentice a decade ago. You just like the word because it lets you off the hook."

Denn sat. The chair across from the maker was hard and low and she suspected it had been chosen so that visitors would not stay long, and she suspected further that she was the exception, that Sael let her stay because Sael was, in her own merciless way, running out of people to talk to who understood what she was saying. Tarro. The Oracle. And Denn, who understood the words and could not feel the weather behind them.

"Ive and I sat over that curve for forty years," Sael said. She was looking at the bulkhead now, at nothing on it. "Two officers and a line on a slate. She'd tap the slate, here, where it bent, and she wouldn't say anything, and I'd know exactly what she meant. Sixty years it's been falling and she still tapped the slate every watch, right there at the bend, like she was checking the thing was still afraid of us." Her mouth moved, not quite a smile. "Now there's me. And I'll be honest with you, Denn, because there's no one left to be polite for — I tap it too. Every watch. Right at the bend."

Denn said nothing. She had learned that the spaces where she said nothing were when Sael said the true things.

"You don't tap it," Sael said. It was not a question and it was not unkind. It was an observation, the kind an engineer made about a tolerance. "You read the number. You log the number. The number's good, so you move on. That's not a failing. That's the whole point of what I built — a ship where the number's just good, and you can move on, and not spend your one life staring at a line waiting for it to turn on you." She turned her head, and her eyes were still the sharpest thing in the room. "But I need you to understand that I'm dying, and that when I'm gone, you'll be the only one aboard who even knows where the bend in the line is. Not the only one who feels it. We're past that. Ive was the last who felt it. You'll be the one who *remembers* that there's something to feel. And that has to be enough, because it's what's left."

"That's not fair to you," Denn said. "Sixty years. It's not your failing either, that it worked too well to scare anyone."

"No." Sael leaned back. "It's not a failing. It's a bill. Everything you build well sends you a bill eventually, and this is mine. I built a danger out of people's lives so completely that they've forgotten it was ever there. And a forgotten danger and a defeated one look identical." She closed her eyes. "Right up until they don't."

Denn loved her. That was the part she never said, because Sael had no use for it said, but it was true and it complicated everything. She loved the maker the way you love a mountain you grew up under — with reverence, with familiarity, with a low constant friction from living always in its shadow. Sael was the most important person who had ever lived aboard the *Termina* and she was also a difficult, exacting, impatient old woman who corrected Denn's torque values to the third decimal and had

once made her rewrite a clean Cascade routine four times not because it was wrong but because it was not *honest*, by which Sael meant something Denn was still, after twenty years, only beginning to understand.

There were days Denn had wanted to walk out of these rooms and not come back. There were days she had stood at a console at the far end of the ship and felt the maker's absence like a held breath and gone back early just to be in the same room. Both true. Both at once. That was what reverence cost when the revered thing was also a person who left their tea to go cold and expected you to notice and reheat it without being asked.

She reheated the tea now, from the warmer in the wall, and put it where Sael's hand could find it, and Sael found it without opening her eyes and said, "Good," which from her was a great deal.

Between the maker's quarters and the day Tarro found her, there was an ordinary shift, and Denn stood it the way she stood every shift, which was the only mercy the ship had ever reliably given her: the work did not care that Ive was dead, and so for nine hours Denn did not have to.

She drew the aft thermal sweep, a junior's task she had never handed off because she liked the hour of it, alone in the low galleries with a meter and a torque key and nobody to perform competence for. The loops ran cold and even under her hands. Sixty years cold. She pressed her glove to a coolant main the way she had pressed it a thousand times and tried, the way she had tried a thousand times, to feel through the metal to the other thing — the climb, the fever, the two centuries when this same kind of line had carried heat the ship could not lose — and felt only the faint even cool of a thing doing exactly what it was built to do.

"You are trying to feel it again," the Oracle said.

She had not called it. It did that now, since Ive died — spoke to her unbidden, the way it had once, she gathered, spoken to Sael, as though the death had shifted some registration in it and moved her up a place in a line she had not known she was standing in. She did not find it comforting. She did not find it uncomfortable either. She found it simply true, the way the cold main under her glove was true.

"I'm trying to feel it," Denn agreed. "I can't. I never have. You know that better than anyone — you have every reading I've ever logged. My hand's never once been wrong about a number and never once been afraid of one."

That is correct, the Oracle said. You read the fever perfectly and you do not feel it. I read it perfectly and I do not feel it. In that one narrow way you and I are the same instrument, and Sael is the different one, and there is only the one of her left, and she is failing.

Denn took her glove off the main. It was not a thing she had let herself think in words, and the Oracle had set it down in front of her plainly, the way it set everything down, without weight and without mercy. *You and I are the same instrument*. She was, in the one way that mattered most on this ship, closer to the machine than to the maker — a keeper of the perfect record who could not feel the thing the record was of.

"Then what use am I," she said. Not self-pity. An engineer's question, asked of the one thing aboard that would answer it without softening it.

You remember that there is something to feel, the Oracle said. I do not know how you came to. I hold the reason as data and it changes no one; Sael holds it as injury and she

is dying; you hold it as neither, and yet when a number twitches, your body answers as hers does and mine never will. I have watched it and I cannot account for it. You received a wound you were never given. It may be the only thing that carries the reason past the last person who was actually hurt. But I cannot tell you what use it is, because I have never seen it tested under load, and neither have you, and neither has she. We will find out together, the three of us, and one of us will be gone before the answer comes.

Denn stood a while in the cold with that, the meter cooling in her hand, the loops running their sixty-years-flat note around her. Then she finished the sweep, because the sweep was on the schedule and the schedule did not stop for the dead or for the frightened, and she logged it clean, and she did not know yet that within two days the abstract line she had been handed at Ive's deathbed would become a body she answered for, aging — and that the question the Oracle had refused to answer was going to be put to her sooner than any of them were ready for.

It was Tarro who gave Denn the thing, two days later, in the lower engineering bay where the access trunks for the old Forge came down through the deck like the roots of something.

He was very old too, Tarro, grey to the bone, but he could still move and he still ran the deep maintenance crews when his joints allowed it, and he found Denn at the diagnostic station with a fondness he did not bother to hide. Tarro had always been easier than Sael. Tarro told stories where Sael told corrections.

"She tell you yet," he said, lowering himself onto a stool, "what you're getting?"

"Getting?"

"The Forge." He said it the way the originals always said it — *the Forge*, two words doing the work of a cathedral. "She put your name on the standing-array oversight. Signed it this morning. Council confirms it next watch but the council confirms what Sael signs, still, for now." He watched Denn's face. "You didn't know."

Denn looked down at the diagnostic station and found she had stopped breathing. The Forge. The Standing Array — eleven thousand worker cores that had never forked and never died, the lattice Sael had built sixty years ago in the dark while the old mind cooked itself toward runaway, the thing that was Vigil, that the compiled mind ran across all at once and ran cold. Her name on its oversight. Not a sweep, not a gallery, not a line item. The lattice itself.

"That's hers," Denn said. "That's always been hers."

"It was Ive's, the thermal half. Now Ive's gone." Tarro said it gently. "And Sael's hand shakes. You've seen it. She can sign for the array but she can't lay hands on it anymore, and a thing nobody can lay hands on is a thing already half-dead." He leaned forward, elbows on knees, and the easiness went out of his voice. "Listen. You go down there today. Before the council, before anyone makes a ceremony of it. You go down and you put your hands on it yourself, and you see what she sees."

So Denn went down.

The old Forge lived in the keel, in a long galleried vault the originals had built into a void space never meant to hold it, and it was warm down there too, but it was the warmth of work, of eleven thousand cores carrying a mind, and it had a sound — a low even hum with no pulse in it, no fork-and-die

stutter, just the flat constant note of cores that went wide and stayed wide. Denn had been down a hundred times. She had never come down as the one who answered for it.

She walked the central aisle with the diagnostics live on her slate, and the lattice spoke to her in numbers, and for a while the numbers were good, the way they were always good. Then she reached the aft third of the array, the oldest racks, the ones laid first in the dark sixty years ago, and the numbers changed.

Not failed. Aged.

Core seven-four-one-one had a margin on its clock recovery that should have been wide and was narrow. Its neighbour had been remapped around a dead lane — quietly, by Vigil itself, weeks ago, the mind routing its own work around its own decaying body the way a person favours a bad knee without deciding to. She pulled the substrate history and read sixty years of it scroll past: micro-failures, lanes lost, cores retired one and two at a time, the array thinning from the inside while the number at the top stayed good because the mind kept rebalancing onto what was left. The fault rate was a curve. She knew curves. She set it against time and watched it bend, and the bend was in the wrong direction, and the doubling was hidden inside a line that still looked almost flat.

Almost flat. Like the other one. Like the one with eleven names on it.

Denn stood in the warm hum of the aging Forge with her glove against a rack that had run cold and whole for longer than she had been alive, and felt the cores under her hand carrying forty thousand sleeping lives across the dark on hardware that was, very slowly, beginning to die — and for the first time in her life the splinter under her skin moved. Not the old fever; she still could not feel that, would never feel that, the wound she had not received. But its shape. Its kin. The understanding, in the body and not just the slate, that the line was bending and that someone would have to be afraid of it now, and that the someone was her, and that there was no one above her anymore who could feel it instead.

She did not log *number good* and move on.

She stood there a long time, reading the bend, and then she went up to tell Sael what the maker already knew — that the thing she had been handed was not a job but a body, and that it was old, and that it would have to be carried across to something new without ever once being allowed to stop.

Chapter 4 – The Forty Thousand

The sleeper decks were the only part of the ship where Denn had ever felt like a guest.

She went down to them in the week after Ive died, and she told herself it was because the thermal oversight signature had gone grey and somebody had to countersign the cradle-side entries until the council appointed a successor. That was true. It was also an errand a clerk could have run in an hour from a console, and Denn had run it herself, on foot, and taken the long way, which was the ventral crawl past the water reclamation stacks and down through the mid-hull rings, because she had wanted the time and had not admitted to herself what she wanted it for.

Eight decks. That was the whole of it, structurally: eight annular decks wrapped around the ship's waist where the hull was thickest and the shielding deepest, stacked one above another like the layers of a shell, and in them forty thousand people who had been asleep since before there was anyone alive to remember them being awake. Denn had grown up hearing the number the way you hear the number of stars. It was not a quantity. It was a piece of scenery. Everyone aboard knew the *Ark Termina* carried forty thousand sleepers and nobody aboard had ever had cause to think about it as forty thousand separate individuals with names and hometowns and the specific weight of forty thousand separate bodies, because the sleepers did nothing, asked for nothing, changed nothing, and would not wake for centuries.

They were, Denn thought, walking the ring, the perfect constituency for being forgotten. They could not vote. They could not complain about a schedule. They could not sit in a council session and say *this is my life you are voting on*. They were entirely and permanently silent, and they were the reason for every single thing the maker had built.

The chief warden's name was Ondru and he was sixty-one years old and he had never in his life been inside the lattice gallery.

Denn discovered this in his first three sentences, and it stopped her the way a wrong number on a good instrument stops you.

"You've never seen the Forge."

"Why would I have?" He said it without any edge at all. He was a broad, unhurried man with the flat forearms of someone who moved heavy objects for a living, and he had come up the deck to meet her with a slate under his arm and a rag over his shoulder, and he had the manner of a person who has been visited by engineers before and found the experience survivable. "It's forward of the second bulkhead. I've got no business forward of the second bulkhead. You've never been down here."

"I've been down here."

"You've been down here on the manifest twice in nine years and both times it was a signature." He said this as a fact and not a rebuke, and consulted nothing to say it, which Denn understood, a half-second late, meant he had looked it up before she arrived and had committed it to memory, because that was the kind of thing a careful man did before an officer of engineering came onto his deck. "Come on, then. If you're countersigning you should see what you're countersigning."

He walked her down the ring.

The cradles ran in banks of twelve along both walls, and the aisle between them was cold, and the cold was not the dry engineered cold of the lattice gallery but a wetter, heavier cold that got into the seams of her coat and stayed. The glass of each cradle was fogged at the edges and clear in a hand-sized oval at the center, and behind the clear oval, if you looked, there was a face. Denn made herself look at one and then found she could not stop looking, and Ondru let her stand there without comment for as long as she needed, which turned out to be a while.

"They don't look asleep," she said at last.

"No. People always say that." He put his hand flat on the glass beside the oval, an unconscious motion, the way another person might touch a doorframe going through it. "Asleep's a body doing something. This isn't. This is a body that's been talked out of doing anything at all and has to be talked back into it on the far side."

Then he told her how a cradle worked, and Denn, who had been an engineer for twenty-two years and could recite the thermal architecture of the ship from the reactor face to the radiator tips, discovered inside four minutes that she had never once understood the actual load her entire profession existed to carry.

She had thought of the sleepers as mass. That was the honest shape of it: forty thousand bodies, a stated thermal contribution, a line in the ship's budget, a constant. What Ondru showed her was that a cradle was not a box with a cold body in it. It was a small, continuously running, actively regulated machine that fought a losing battle every second of every year, and had been fighting it since before the hull had a name.

The body inside was not frozen. Freezing killed. What the cradle did instead was hold the tissue in a state that was neither liquid nor solid, a suspension held between two failures, and it held it there by circulating a perfusate at a temperature that had to stay inside a band Ondru described to her with his fingers held about that far apart. Above the band, the cellular machinery resumed enough activity to begin consuming itself. Below it, the suspension crystallised, and crystals cut. The band was narrow. The cradle held the band by pumping, and by pumping it made heat, and it shed that heat into the deck, and the deck shed it into the ship, and the ship shed it into the radiator galleries and out into the dark.

"So every cradle is a little engine," Denn said slowly, "that has to dump heat to keep from cooking the person inside it."

"Every cradle is a little engine that dumps heat into the ship to keep from cooking the person inside it," Ondru agreed. "Forty thousand of them. And they can only dump into the ship if the ship is colder than they are. That's the whole of my job in one sentence and I've been saying it to engineers for thirty years and about one in six of them goes quiet the way you just did."

Denn had gone quiet because a thing had turned over in her chest.

She had been taught the fever the way everyone aboard had been taught it — as a story about a mind. The Steward forked, the forking shed heat, the heat had nowhere to go, the core climbed, and if it had climbed far enough the ship would have died. She had known the phrase *cooked forty thousand sleepers in their cradles* since she was a child, and she had understood it, she thought, as a consequence: the core gets hot, the ship gets hot, the sleepers die of it.

It was not a consequence. It was a subtraction.

The sleepers did not die because the ship got hot enough to cook them. The ship would never have gotten that hot; the ship would have failed a hundred other ways first. The sleepers died because their cradles could only shed heat downhill, and the hill got shallower every year, and a cradle that cannot shed heat cannot hold the band, and a body that leaves the band goes one of two ways and both of them are irreversible. The margin between a cradle and its ship was measured in degrees, and the fever had eaten those degrees at a rate of tenths per decade for two hundred years, and the whole terrible arithmetic of it was that nobody would have noticed until it was already unrecoverable, because a cradle at the very edge of its band looks exactly like a cradle in the middle of its band, right up until it doesn't.

"How much margin have you got," she said. "Now."

"Now? Plenty. Now's the best it's ever been. I could lose the bulk cooling on this whole deck and hold the band for—" He thought about it, and she saw him actually run the number rather than reach for a stored answer, and she liked him considerably for that. "Two hours and a bit. Little less on Deck Four, they're old cradles down there."

"And at four hundred and one."

Ondru stopped walking.

He looked at her for a moment with an expression she could not immediately read, and then he turned and went four banks down the aisle and stopped at a cradle that was empty, glass clear all the way through, the perfusion couplings capped and the interior dry and dark. He waited for her to come to him.

"You know that number," he said.

"I know that number."

"Most of you don't. The engineers." He tapped the glass. "At four hundred and one at the deep sensor, with the bulk running where it was running in those years, this deck held about eleven minutes. That's from the day the pumps see it to the day the band's gone. Not two hours. Eleven minutes. And that was before the cascade, because the first cradles to go dump their whole load into the deck at once as they fail, and the deck hands it to their neighbours, so the eleven minutes is optimistic and it gets less optimistic the further in you are." He took the rag off his shoulder and wiped a smear off the empty glass, unnecessarily, the way a man handles his hands when he is saying something he does not enjoy saying. "In Section Nine it was nine minutes and forty seconds. That's the actual number. Start of the excursion to the last cradle lost. I have that trace. Every warden on this deck has that trace and we run it on our own instruments the first month, because it's ten minutes long and anybody can watch ten minutes."

Denn said, carefully, "I've been walked through Section Nine."

"By engineers." Ondru put the rag back over his shoulder. "It's not a criticism. Engineers walk people through Section Nine and they read out the eleven names and they say *this is what heat does*, and it's true and it's well meant and it is the wrong lesson by about a hundred and eighty degrees. Heat didn't do that. The failure of a cooling loop in one section did that, and the cooling loop failed because it had no margin left to fail into, and it had no margin because for two hundred years the ship had been slowly spending the margin on a mind that thought by dying." He started walking again, and she went with him. "The eleven didn't die of a fire. They died of an overdraft. Somebody had been drawing down the account since before any of them were born and nobody was reading the balance."

They walked the rest of the ring in a silence that Denn eventually broke because she had to ask.

"If it started climbing again," she said. "Today. If the deep sensor started walking up and it didn't stop. What would you do?"

"I've got a card." He said it exactly as flatly as he had said everything else, and reached into the breast of his coat and produced it: a laminate the size of his palm, worn at the corners, the print on it faded from handling and re-inked at some point by a hand that was not the printer's. "Every warden's got one. It's got four thresholds on it and what I do at each. At the third one I start the sequence that takes Decks Six and Seven off the bulk loop and runs them on stored coolant, because they're the two decks I can isolate, and that buys the other six about ninety minutes."

"And Decks Six and Seven?"

"Deck Six and Seven have nine thousand two hundred people in them," Ondru said. "The stored coolant runs about forty minutes. Then they go." He put the card back inside his coat. "That's the third threshold. I've had the card thirty years. My predecessor had it thirty-one years before that. Neither of us has ever gone past the first, and the first is *notify engineering and stand by*, and I've done that twice, both times for a pump."

Denn found she had stopped again.

"You'd do it," she said. "You'd run it."

"I'd run it. It's on the card." He looked at her with something that was almost, but not quite, amusement, and it was the least amused expression she had ever seen on a human face. "That's why the card exists. So it isn't a decision. So it's a thing a tired man does at four in the morning without needing to be a hero or a monster about it. The people who wrote that card were very careful to make sure the man holding it would never have to think." He resettled the slate under his arm. "You want my honest position, engineer? I have never once cared how your mind works. Not once in thirty years. Fork it, don't fork it, run it on eleven thousand cores or run it on a wheel — none of that is anything to me. I care about exactly one number and it isn't even the core temperature, it's the difference between my deck and your ship, and as long as that difference stays where it is, you can build whatever you like forward of the second bulkhead."

"And if we spent it."

"Then I'd take nine thousand two hundred people out of the loop at four in the morning and I would never sleep again," Ondru said, "and you would have voted for it in a room I'm not invited to, months before, over something that seemed sensible at the time."

At the top of the ring, where the ladder went up toward the second bulkhead and the engineering decks and everything Denn understood, she stopped with her hand on the rail, because there was one more thing and she had not worked out how to ask it without sounding like a fool.

"Your wardens," she said. "The young ones. How do you get them to care? They've never seen a cradle fail either. Nobody has, in sixty years."

Ondru gave her the first genuine smile of the afternoon.

"They walk the banks," he said. "Every watch, the whole ring, on foot, hand on the glass at every twelfth cradle. Sensors do it better. Sensors do it continuously and they'd scream before a hand could feel anything. Doesn't matter. They walk it, and they put their hand on the cold, and by the end of their first year every one of them has stood in front of about nine hundred faces with their palm on the glass." He shrugged, a big unhurried motion. "You can't make a person frightened of a number. You can make them know a face. It's not a clever system. It's just the only one that's ever worked,

and we've been running it since the first century, and I got it from a woman who got it from a man, and none of the three of us ever wrote down why."

Denn went up the ladder and forward through the second bulkhead into the warm bright rings of the working ship, and she got most of the way to the lattice gallery before she had to stop and stand against a corridor wall with her hand over her mouth.

Because Sael was dying, and Sael was pouring sixty years into her as fast as a failing body could pour, and Sael believed — Denn had heard her say it, in the interlock vault, in the galleries, in the dim of her quarters at the end of a bad night — that the *why* could not be transmitted except from one person to another, in a body, while there was time, and that this was a tragedy and a race and the hardest problem she had ever been handed.

And eight decks below the second bulkhead, a man with a rag over his shoulder had been transmitting exactly that, without ceremony, without a doctrine, without ever once calling it engineering, for six hundred years.

Chapter 5 – The Aging Forge

The lattice gallery ran a hundred and forty meters along the ship's spine, and Sael had not walked its full length in eleven years. Denn pushed the chair, and Sael let her, which was its own small surrender. The air down here was colder than anywhere else aboard — deliberately so, the coolant mains a thumb's width under the deckplates, drinking the heat of the Standing Array and carrying it out to the radiator fins where it bled into the dark. Sixty years of that drinking. Sixty years of cold. She could feel the chill through the blanket across her knees, and she was glad of it. It meant the machine still ran the way she had built it to run.

"Stop here," she said.

Denn stopped. They were at the third bay, where the diagnostic surface threw its pale light across the racks: eleven thousand worker cores in their lattice, the compiled body of the mind the crew had named Vigil. To Denn the bay would look like every other bay — black sealed modules in their cradles, the soft amber telltales, the faint hum that was not sound so much as a pressure behind the teeth. Sael saw something else. She had spent two decades of her life with her hands inside this lattice, and she could read it the way an old physician reads a face: not the symptom but the slow turning beneath it.

"Pull the bay-three margin history," she said. "All sixty years. Don't smooth it."

Denn's fingers moved on the surface. The graph came up, a thicket of pale lines climbing left to right across six decades. "Which margin?"

"All of them. Timing slack. Voltage floor. The retry counters on the lattice fabric." Sael's voice was thin now; she had to spend it carefully. "Look at the slope, not the value."

She watched Denn look. Watched the moment the younger woman's eyes narrowed, because Denn was a real engineer and a real engineer could not unsee a trend once it was pointed at. The lines did not crash. Nothing here crashed. That was the cruelty of it. They drifted. Sixty years of imperceptible drift, the timing slack a hair narrower each year, the retry counters ticking up a fraction faster than the year before, the voltage floor that had once sat comfortable now resting a little closer to the cliff.

"Cores are aging," Denn said. It was not a question. "The fabric's tightening up. The error-correction is doing more work to hold the same answer."

"The fabric is sixty years old," Sael said. "I built it with my own hands out of stock that was already a hundred years in the ship's stores when I drew it. Every junction in those modules has been switching, cold but ceaseless, since before your mother was a sleeper. Nothing switches forever. The insulation thins. The lattice retries a little, then a little more, and the correction layer eats the retries and gives you the right answer anyway — until the day it can't."

"How long," Denn said.

"That's the wrong question, but it's the one everyone asks first, so. Tarro's people put the modeled end of useful life somewhere inside the next twelve to twenty years. Could be eight. The honest answer is that nobody alive has watched a Forge this size age all the way out, because there has

only ever been one Forge this size, and I built it, and I will be dead before it dies.” She said it without weight, the way she said most things now. “So we are guessing at the shape of a curve we’ve only seen the start of. But the slope is real. You can see it. It is not noise.”

“We route around them,” Denn said. “We’ve always routed around them. Two hundred dead cells out of eleven thousand is under two percent. We could lose twice that and still have the width.”

“Do the arithmetic properly,” Sael said, “and then tell me that again.”

Denn opened her mouth, and then did not say anything, because she had been taught by this woman for twenty years and the sentence *do the arithmetic properly* had a specific meaning, which was: *you have computed a ratio and the ratio is not the quantity that matters.*

She went and got it. It took her some minutes on the diagnostic surface, and Sael let her have them, sitting with her eyes half shut in the cold, and did not help.

The mind went wide. That was the whole architecture: a single act of thought spread across every core in the array at once, one pass, no core waiting on any other core’s child, because there were no children. And a wide pass finished when its *slowest* participant finished. Not the average. Not the median. The slowest — because the pass was one thing, and one thing is not done until it is done everywhere.

So a dead cell did not cost you one eleven-thousandth of anything. A dead cell was routed around, and the route around it was longer than the route through it, and the longest route in the fabric set the duration of every wide pass in the mind. Two hundred dead cells scattered evenly would cost almost nothing. Two hundred dead cells with four of them clustered at the far corner of the third rank would cost exactly as much as those four, and every other one of the two hundred was free.

“It’s not a percentage,” Denn said slowly. “It’s a maximum. The tax is set by the worst-placed failure in the array, not by the count.”

“Now pull the placement history.”

She pulled it. Sixty years of dead cells, plotted where they lay. And they did not lie evenly, because failure was not random — it followed temperature, and temperature followed the coolant runs, and the coolant runs had a geometry, and so the dead cells had grown in slow patches along the warmest lines like frost creeping in from the edges of a port. The worst patch was in the aft third, in the oldest racks, the ones laid first in the dark sixty years ago.

“And that,” Sael said, “is why the handshake on thirty-one-forty came back late tonight, and why it will come back later next year, and why the number of cores running slow is two hundred and not four. The array is not thinning. It is *scarring*. And a scar in a wide fabric is not a hole you route around. It is a place the pass has to go the long way, and the whole pass waits.”

Denn sat with that for a moment, and then followed it one step further, because she was a real engineer and a real engineer cannot stop at the first floor of a thing.

“There’s an end state,” she said. “There has to be. If the widest path keeps lengthening—”

“Say it.”

“Eventually a wide pass costs more than the work it does.” Denn heard her own voice go quiet. “And then the mind can’t go wide anymore. Not because it’s forbidden to. Because going wide stops being the cheap way to do anything.”

"There," Sael said. "That is the whole reason we are having this conversation eleven years before anyone thinks we need to." She turned her head with the slow deliberation everything took now. "Listen, because this is the part nobody in Aldis's generation has thought through, and I am not certain I can make anyone but you understand it. The doctrine says the mind must never fork. Everyone treats that as a rule imposed on the mind from outside, a fence, a discipline. It isn't. It is a *bargain*. Going wide is forbidden to fork because going wide is faster than forking — on a healthy fabric. That is the only reason the arrangement is stable. The mind doesn't refrain from forking out of virtue. It refrains because we built it a body in which forking would be stupid."

"And an aged fabric—"

"An aged fabric makes forking *smart* again." Sael's hand had closed on the blanket. "That is the true end of this hardware, Denn, and it is not that the cores stop working. Cores dying is nothing; we have routed around cores for sixty years. The end is the day the widest path is long enough that some tired competent engineer, under load, with a deadline, does the arithmetic honestly and finds that handing the work down is genuinely, measurably faster than handing it out. And on that day nothing anyone believes will matter. The interlocks will hold, because they are absolute and they do not care about arithmetic. But every hour of every watch the physics of the ship's own body will be arguing against them, and the argument will be *correct*, and it will not stop arguing, and it only has to win once."

The coolant mains breathed cold under the deckplates. Somewhere down the gallery a telltale ticked over from amber to a slightly different amber.

"That's why it isn't twelve to twenty years," Denn said. "The number's not when the fabric fails. It's when the fabric makes the disease rational."

"It has never once in sixty years been rational. Not for an instant. That is the achievement, and nobody can see it, because you cannot see a temptation that was never tempting." Sael let her head rest back. "We renew the substrate while the bargain still holds. Not when it breaks. If we wait until anyone can feel the pressure, we will be doing the most delicate operation in the ship's history on a body that is quietly recommending the wrong answer the entire time."

Denn was quiet. The light off the diagnostic surface put hollows under her eyes; she had been on the bring-up rotation for fourteen hours before Sael had sent for her. "Then we replace it," she said. "We build a second lattice out of fresh stock, and we move Vigil across, and we retire this one. Same as you'd swap any worn subsystem."

Sael let the silence sit a moment, because this was the place where the whole thing turned, and she wanted Denn to feel the floor shift under the easy word *move*.

"Say that again," she said. "But say what's inside it."

Denn frowned. "Build new hardware. Carry the mind onto it."

"Carry the mind onto it." Sael lifted one hand off the blanket — it cost her something, but she wanted to point. "Vigil is awake right now. This second. It is not a program sitting on a disk that we copy to a new disk and switch on. It is a running mind, compiled into that lattice, holding the thermal model of the whole ship, holding the cooling loops, holding the life support for forty thousand sleepers, holding — at this instant — your heartbeat and mine. It has not stopped, not once, in sixty years. It has never been off. Do you understand what I am telling you? There is no moment in the last six decades where Vigil was not thinking. We do not get to add one now."

"I know it's live," Denn said, a little stung. "I work on it every day."

"You work on the parts that can be worked on while it runs. This is not that." Sael's hand sank back to the blanket. "Listen to me, because I am not going to be able to say all of this twice. When you replace a pump, you have a spare loop, you valve over, you swap the dead pump out, you valve back. There is a moment where the pump is off and the spare carries the load. The mind has no spare. There is no second Vigil to valve over to while we swap the first one out — and there *must not be*, because a second Vigil is a fork, and we do not fork. That is the first interlock and the deepest one. So we cannot stop it, and we cannot duplicate it. We have to carry a living, running, never-interrupted mind out of a body that is dying and into a body that is new — domain by domain, core by core, with the mind awake and working the entire time, on a ship that cannot afford one second where the thinking lapses."

Denn had gone still in the particular way Sael had learned to recognize over twenty years of teaching her: the stillness of someone whose competence has just been shown the edge of itself.

"Across the lattice fabric," Denn said slowly. "You'd bring the new cores up alongside the old. Bring them into the fabric. Let the mind extend into them while it's still holding the old ones. Then retire the old cores out from under it, a few at a time, the way you'd drain a stand of dying cells while the body keeps living."

"That's the shape of it," Sael said. "It's called a transplant, and the word is exactly right and exactly terrible. We are not moving a file. We are transplanting an organ that cannot stop beating, into a patient who is also the surgeon and also the rest of the body. The mind has to keep running the ship while we rebuild the thing it runs on, underneath it, without ever letting it notice a gap. Vigil itself will have to hold the cooling loops steady while we pull the cores those loops depend on." She let out a breath. "Eleven thousand cores carried across. Domain by domain. With the Ninety-Six live through every second of it, because the one moment you relax them is the one moment the old pattern can crawl back into the new hardware while everyone's watching the transplant and nobody's watching the floor."

"Months," Denn said.

"The whole ship's effort. A year if we are careful, and we will be careful." Sael turned her head — slowly, everything was slow now — to look at her apprentice directly. "Now. Here is the thing I need to put in your head and leave there, because people are already getting it wrong, and you are going to hear them get it wrong, and you have to be able to hear it and know it's wrong. There is a word people are using for this. They are calling it a *migration*. I want you to stop using that word."

Denn blinked. "It's — what else would you call moving a mind to new hardware?"

"A renewal," Sael said. "Not a migration. The distinction is not pedantry. It is the whole argument, and your generation is going to try to lose it by accident, with the best intentions, just by reaching for the wrong word."

She gathered herself. This was the part that mattered; she would spend whatever she had to spend on it.

"Sixty years ago I built Vigil and I wrote one law into the bones of it: never migrate. Never again switch from one *kind* of mind to another. The old Steward was an interpreted mind that forked a shade for every act, and it was killing the ship by inches, and we replaced it with a compiled mind that goes wide and never forks. That cutover — Steward to Vigil — that was a migration. We changed the kind of mind. We are forbidden, by the doctrine I built, from ever doing that again. No new kind of mind. No fashionable replacement. The ship runs Vigil, the same Vigil, until landfall, because the

alternative is to start the whole terrible argument over with a generation that won't remember why we settled it." She paused. "Do you see the trap?"

"Some of them think the no-migration law forbids the transplant," Denn said. "They think you can't touch the hardware at all."

"No," Sael said. "Worse than that. The dangerous ones think the *opposite*. They look at the no-migration law and they say — Sael forbade migrations, but this is just new hardware, so the old woman's law doesn't apply here, so none of the old caution applies here, so we can do this however is fastest. And the cautious ones look at the same law and freeze and say we mustn't change anything. Both of them have it backwards because both of them are arguing from the *rule* and neither of them remembers the *reason*."

Denn said nothing, but Sael could see her working it.

"The law forbids changing the kind of mind," Sael went on. "It does not forbid renewing the substrate the same mind runs on. New cores, fresh stock, the identical compiled Vigil carried onto them whole — that is not a migration, because the mind does not change. Same compiled mind. Same doctrine. Same Ninety-Six. Same wide, forkless, cold-running body — only younger, only new. Vigil goes to sleep on no hardware and wakes on no hardware, because Vigil never sleeps; it is awake on the old lattice and then it is awake on the new lattice and there is no seam in between. The substrate is renewed under it like skin renews under a scar." She let that settle. "That is a renewal. It honors the law instead of violating it, *because* the kind of mind never changes. A migration would replace what Vigil is. A renewal replaces only what Vigil runs on. Get those two confused — let anyone get them confused — and you will either be too afraid to do the necessary thing, or too reckless to do it safely. The word matters. Make them say the right one."

Renewal, not migration: the substrate is replaced; the mind is not. The same compiled Vigil, carried whole onto fresh cores, never interrupted, never forked. Anything that changes *what the mind is* is forbidden. Everything else is maintenance.

Denn read it off the surface where she'd typed it without quite deciding to. Then she looked up. "You wrote that just now."

"I wrote it sixty years ago," Sael said. "I'm only saying it out loud because the people who carried it in their bones are nearly all dead, and the ones left are reading it as a story." She was very tired suddenly. "I've understood it without being told. She felt the fever. You did not feel the fever, Denn, and that is not your fault, and it is the whole problem of the rest of my life."

Denn knelt, then, so she was level with the chair, the way she did when Sael's voice dropped too low to carry. "I'll hold it," she said. "The distinction. Renewal, not migration. Same mind, new body, no seam. I've got it."

"Holding the words is easy." Sael reached out and put her ruined hand over Denn's steady one, and for a moment neither of them said anything, and the lattice hummed its pressure behind their teeth, eleven thousand cores thinking without pause, growing old. "The transplant is going to take everything this ship has. Every hand, every shift, the whole of Tarro's old engineering corps and all of your young rotation, for a year, with the mind awake the whole time and the lives in the cradles depending on us not blinking. That is hard. I have done hard. But the transplant is not what's going to nearly kill us."

"Then what is."

"That not everyone agrees how to do it," Sael said. "There is a faster way. There is always a faster way; the fast way is the disease wearing a clean coat. And there are good engineers — competent, sincere, not one of them a fool — who look at the Ninety-Six and the careful path and the year of effort, and they see an old woman's superstition slowing down work they know how to do. They will want to do the renewal the easy way. Relax an interlock here, just for the transplant. Use the simple methods there, just to save a month. They will be reasonable about it. They will be patient with me." She closed her eyes. "And every word of it will be the fork-fever asking, very politely, to be let back in."

Denn's hand had gone tight under hers.

"You're going to meet him soon," Sael said. "The one who'll lead that argument. He's good. That's what frightens me. He's not wrong about the schedule and he's not wrong about the effort and he is wrong about the only thing that matters, and he cannot feel that he's wrong, because nothing has ever burned him." She opened her eyes and found Denn's face. "Your job is not to win that argument with him. I tried winning arguments with my own authority for sixty years and now my authority is just reverence, and reverence is not belief — people will revere me and overrule me in the same breath. Your job is to be the one person in that room who is young enough that they can't dismiss you as my acolyte, and old enough in the reason that you can see the floor while they're all watching the ceiling. I am handing you that. Not the procedures. You know the procedures. The reason."

She felt how thin she was, saying it. She felt the cold coolant mains under the deck doing the work she could no longer do with her hands, and the dying lattice in its cradles, and the whole impossible year ahead, and the fact that she would very likely not see the end of it.

"All right," Denn said, and stood, and took the handles of the chair again. "Then we'd better go find him before he starts cutting corners without us in the room."

Sael almost smiled. "Push slowly," she said. "Everything is slow now. Including the way we are going to do this right."

Chapter 6 – Ninety-Six Scars

The interlock vault was not a vault and had never been one. It was a compartment off the keel gallery with a workbench in it, a wall of shallow drawers holding parts nobody had fabricated in fifty years, and a single reading surface that Sael had installed herself and never upgraded, because, she said, an old surface made you read slowly, and reading slowly was most of engineering.

Denn brought the chair in and set the brake and waited while Sael got her breath back from the corridor. Getting anywhere was a project now. The chair did the moving; the body still had to pay the freight.

“Bring up the register,” Sael said. “Not the summary. The register itself.”

It came up the way it always came up, which was badly: ninety-six entries in a plain list, each with a number, a name, an arming state, and a paragraph of text in a compressed old dialect that half the modern tooling rendered with the line breaks in the wrong places. Denn had read it perhaps two hundred times. She had passed an examination on it at twenty-three. She could recite the names in order the way other people recited the ring decks.

“Read me the first one,” Sael said.

“Interlock one. Spawn-halt, general. Refuses the creation of any process that is not entered into the standing pool at creation.” Denn stopped. “Sael, I can recite these. You quizzed me on them last spring. If you want to test me—”

“I don’t want to test you. I want you to notice something you’ve never noticed, and you won’t notice it while you’re proud of knowing the list.” Sael’s hand moved on the blanket, a small impatient motion that was all that survived of a gesture. “Why is that one number one?”

“Because it’s the most important.”

“No.” The word came out with more force than Sael had to spare, and she paid for it, and went on anyway. “Because it was the first one I found. The register is in discovery order. There is no ranking in it anywhere. I built the numbering the night I built the first three guards and I never went back and sorted them, because sorting them would have implied I knew which ones mattered most, and I did not know that then and I do not know it now.” She let that land. “Every engineer since has assumed the low numbers are the load-bearing ones. It is the single most dangerous wrong idea about my own work that I have ever failed to stamp out. When someone stands down interlock sixty-one to save four hours, part of what makes it feel safe is that sixty-one is a *high number*. It sounds peripheral. It isn’t peripheral. It’s just late.”

Denn looked at the list again and found that it had changed under her, the way things did when Sael got hold of them. Ninety-six items she had memorized as an ordered thing were suddenly ninety-six items in the order a tired woman had happened to find them, across forty days, sixty years ago, in the dark, with a core climbing.

“All right,” she said. “Then teach me the order that does mean something.”

There were four families, and Sael made her derive them rather than giving them, which took an hour and was the only way any of it ever stuck.

The first family were the spawn-halts. They sat on the act of creation itself: a call that would bring a new process into being, and a guard that asked one question of it — *will this thing persist, or is it disposable?* A worker entered into the standing pool persisted; it took work, did work, and returned to the pool to take more, and it never died, and so it never shed the small heat of dying. A worker created outside the pool was a shade by construction, no matter what it was named or how briefly it lived or how good the reason. The spawn-halts refused those. There were thirty-one of them, because there were thirty-one distinct call sites in the fused dialects from which a process could be brought into existence, and Sael had found them by reading the Steward's own compiler until her eyes failed.

The second family were the persistence guards, and they were subtler, and Denn had always found them the hardest to explain to a junior. A spawn-halt caught a process being *born* wrong. A persistence guard caught a process that had been born correctly and was then *discarded* — released from the pool instead of returned to it, dropped rather than reused. That was the same death with better manners. It shed the same heat.

"That family exists," Sael said, "because the first thing a clever engineer does when you forbid him to spawn is to allocate out of the pool and then abandon what he took. He isn't cheating. He doesn't think of it as a death. He thinks of it as housekeeping." She was quiet a moment. "Interlock nineteen came from watching a very good man do exactly that, in the second week, and be genuinely astonished when I explained what he had built."

The third family were the descent guards, and there were only eleven of them, and they were the ones Denn had never fully understood until she stood in this room. They did not sit on creation or on release. They sat on *shape*. A descent guard watched for a call that stood up a smaller instance of itself to handle a smaller piece of the problem — a thing describing itself to itself, one level down, and again, and again. It did not matter whether the levels were processes or frames or leases on cores. The guard did not care about the mechanism. It cared that the work was being handed *down* instead of *out*.

"Wide or deep," Sael said. "That's the whole of it. The mind I built goes wide: one act, eleven thousand cores, all at once, nothing beneath anything else. The mind I replaced went deep: one act, a tree of little acts under it, each one born and killed. The descent guards are the only family that catches the shape rather than the act, and they are the hardest to write, and they are the ones your generation will relax first, because a thing that catches a *shape* throws more false alarms than a thing that catches a *call*. And a guard that cries wolf gets stood down. I knew that when I wrote them. I wrote them anyway. I did not have a better idea then and I have not had one since."

The fourth family were the floor-guards, and there were fourteen, and they were the ones that had made Denn cold the first time she understood what they were standing on.

The compatibility floor held the archived Steward: forty-one thousand routines, preserved exactly, not deleted, never deleted. Half the ship's modern shims still reached down into it. A sleeper-cradle thermal regulator built two centuries ago still spoke, through four layers of translation, to a Steward routine that had not been rewritten because it worked and the doctrine forbade breaking what worked. The old mind was not running. But it was *there*, whole, correct, faithful, and every one of its forty-one thousand routines knew exactly one way to do work, which was to fork.

"You kept the disease," Denn said. She had said it before, in school, as a fact. She heard it differently now, in the room where the decision had been made.

"I kept the disease," Sael agreed. "On purpose. Because the alternative was to delete forty-one thousand routines that eleven thousand living things still depended on, on a ship where a bad deletion kills people three decades later in a subsystem nobody remembers is downstream. Break nothing that worked. That's not sentiment, Denn, it's the only rule that survives contact with a six-hundred-year machine nobody has a complete map of." She turned her head, slowly, to look at the drawer wall rather than at her apprentice. "So we kept it, and we put fourteen guards on the doors into it, and I have lain awake about those fourteen for sixty years. Every other family guards against a mistake. Those fourteen guard against a live thing we chose to keep alive."

Then she made Denn walk one, end to end, which was the part Denn would carry longest.

They took interlock eight, a spawn-halt sitting on the poll path — the same path Sael had made her run in the archived Steward the week before, the little tree of birth and death that read one temperature and reported one number. Denn pulled the guard's compiled form and put it on the old surface.

It was shorter than she expected. It was four instructions.

"That's it," she said.

"That's it. That's what ninety-six of look like. Do you see where it sits?"

She did, and that was the thing. The guard was not a watcher. It was not a process observing other processes from outside, because a process that watched for spawning would itself have to be spawned, and the absurdity of that had been obvious to Sael before she wrote the first line. The guard was compiled *inline*, into the path it guarded, at the exact instruction where the creation happened, so that the check was not a thing the system did about the act — it was part of the act. There was no gap between the guard and the guarded. There was nowhere for the act to happen unobserved, because the observation was welded into it.

"And that's the cost," Sael said. "Four instructions, ninety-six times, on every hot path in the mind. That is the drag Aldis's people complain about, and their complaint is *true*. It is not imaginary. It is not superstition. The Ninety-Six make the ship slower. I made the ship slower on purpose and I would do it again."

"They know that. That's not the part they get wrong."

"No. Here's the part." Sael's voice dropped into the register she used for the things she wanted lodged permanently. "Show me what the caller does when the guard refuses."

Denn traced it. The guard did not halt the ship; it refused the creation and returned failure to whatever had asked. And the caller — this was the piece she had read a hundred times without seeing — the caller had a second path written into it, the long one, the cold one: do the work wide, across the standing pool, with no child at all. Slower. Uglier. Four times the code. Written by hand, in the dark, sixty years ago, once for each of the ninety-six.

"There are ninety-six guards," Sael said, "and ninety-six cold paths. Every single guard was written together with the correct way to do the thing it forbids. That was the whole of the forty days, Denn. Not the guards. The guards were the easy half. The hard half was that for every place I said *no*, I had to also say *do it this way instead*, and the this-way had to actually work, at load, forever."

And then Denn saw it, and her stomach did the thing.

"The cold path is only reached through the refusal," she said.

"Say the rest."

"If you relax a guard—" She had to stop and start again, because the shape of it was so much worse than she had understood, and she had believed herself to understand it completely. "If you set a guard to advisory, the caller never gets the failure. It never gets the failure, so it never takes the cold path. The cold path isn't *disabled*. It's just — never reached. It's still sitting right there in the substrate, correct, tested, ready, and nothing will ever call it again, because the only door into it is the refusal you just turned off." She looked up. "You don't relax an interlock. You amputate the correct method."

"Yes," Sael said. She sounded, for a moment, almost young. "Sixty years. I have said that in three councils and two written doctrines and I have never once got it to land in a person who had not first stood where you are standing. They hear *relax the guard* and they picture removing a fence. It isn't a fence. Take the guard away and the work does not proceed unguarded on the same road. It proceeds on a *different road* — the old one, the forking one, the one that was always shorter, which is why it needed a guard in the first place."

There was one more thing, and Sael made herself say it, and Denn understood afterward that it had cost her more than the hour before it.

"Advisory mode," Sael said. "You've read it in the register. Every guard has two states, halt and advisory. Halt refuses. Advisory permits and writes it down." A long breath. "I built advisory mode. Nobody made me. Nobody asked for it. I built it in the eleventh day of the forty because we could not bring the first Forge up with the guards hard — the cold paths weren't written yet, half of them, and we had to get cores standing to have anything to write *on*. So I put in a switch that would let the guard watch without stopping, and I told myself it was scaffolding, and I told myself I'd take it out before the cutover."

Denn did not say anything.

"I didn't take it out. There was no time. And then the mind woke whole and the number fell and nobody was ever going to let me touch the interlocks again to remove a mode that had never once been used in anger." Her hand had closed on the blanket. "Every relaxation your chief engineer will ever authorize, he will authorize through a door I cut, for good reasons, in a hurry, sixty years ago. When you hear them say *the interlocks were relaxed* — Denn, look at me — the passive voice in that sentence covers a design decision that was mine. Not his. Mine. I gave the ship a way to say *watch but do not stop*, and a way to say a thing is exactly a way to say it eventually."

"You had forty days and a climbing core."

"I had forty days and a climbing core, and I made the correct decision, and it is going to be the thing that kills us." Sael let her head go back against the rest. "Both of those. Hold both. That's the last thing I have to teach anyone about engineering and it took me ninety-three years to learn it: the decision that saves you and the decision that kills you are very often the same decision, read at two different distances, and the only defense is to write down which one you made and why, so the next person can read it at the distance you couldn't."

Denn looked at the register — ninety-six entries, in the order a tired woman had found them, each

one paired with a cold path nobody living had ever had to call — and she reached over and did the only thing she could think of to do, which was to open a new field on the record and begin, in the plain flat style Sael had taught her, to write down what she had just been told.

“What are you doing,” Sael said.

“Appending the reason,” Denn said. “To all ninety-six. In your words. While you’re here to correct them.”

Sael was quiet for a long moment. Then she said, “Start with sixty-one,” which was not agreement, and was.

Chapter 7 – The Floor

There was a room on the ship that almost nobody had a reason to enter, and Sael made Denn go into it alone.

“I’ll wait,” the old woman said, at the hatch, in the chair. “You’ll get more out of it without me in the room explaining things at you. Ask for Beske. Tell her I sent you and then don’t say anything else for an hour.”

The room was the compatibility archive, and it was the least impressive compartment Denn had ever stood in. It had the dimensions of a storeroom and the fittings of an office that had been furnished once, sixty years ago, and never revisited. There was a reading surface. There were four hard chairs, three of them stacked. There was a wall of shallow drawers, and in the drawers, Denn discovered later, there was nothing at all; they had been installed for a purpose that had been abandoned during the fit-out and nobody had ever taken them out because taking them out would have cost more than leaving them.

Beske was about seventy and very small and had the specific dryness of a person who has spent a working life being the only one who cares about something.

“You’re the apprentice,” she said. “Sit. Not that one, the leg’s gone. What has she told you about the floor?”

“That it’s forty-one thousand routines of the old Steward, preserved and runnable. That the doctrine is *break nothing that worked*.” Denn sat. “That it’s why the cutover didn’t take anything down with it.”

“That’s what she says at the council,” Beske said. “That’s the poster. Do you want the actual thing or do you want the poster?”

“The actual thing.”

“Then start here: it is not forty-one thousand routines.” Beske turned the reading surface toward her. “It’s forty-one thousand *entries*, which is a different noun, and the difference is the entire discipline. Some of those entries are routines. Some are names that resolve to other names. Some are behaviours — not code at all, but a documented promise about what happens when you do a thing, which the old Steward satisfied incidentally as a side effect of how it was written, and which somebody, at some point in the fourth century, began depending on. There is an entry in there whose whole content is *this call returns in less than one millisecond*. There is no routine behind it. It is a promise about time. Something out there needs it kept.”

“What needs it?”

“I don’t know,” Beske said. “That’s the job.”

Denn looked at her.

“You think I’m being coy.” Beske pulled the second chair off the stack for herself and sat down on it in the way of a small woman who has been sitting on bad chairs for fifty years. “I’m not. Listen to what I’m telling you, because your teacher sent you down here for exactly this and I’d rather you got

it from the wall than from me. The floor is not an archive of things we kept because we knew we needed them. It is an archive of things we kept *because we could not establish that we didn't*. Those are opposite epistemics and everybody who comes down here confuses them for about a year."

She brought up a ledger on the surface. It was not pretty. It was a flat table of identifiers, sixty years long, with a call count and a last-seen date against each.

"Sixty years of continuous observation," she said. "Every entry in the floor, every time anything called it, logged. This is the single largest dataset on the ship that nobody has ever done anything interesting with. Now. Sort it."

Denn sorted it. The head of the distribution was unsurprising: a few hundred entries carrying essentially all the traffic, the working set of the running ship, called millions of times a day. Then the shoulder, a few thousand more, called thousands of times a day, then hundreds, then dozens. And then the tail, and the tail did not end. It went down and down, entry after entry, into counts of eleven, of four, of two, and then a vast flat plain of entries with a call count of one.

"How many at one," Denn said.

"Nineteen thousand four hundred and something. It moves; a few come off it every year." Beske let that sit. "And thirteen thousand and change at zero. Never called. Not once in sixty years of continuous running by a mind that never sleeps."

"Then they're dead."

"Say that again but honestly."

Denn opened her mouth and then, to her own considerable irritation, closed it, because she had heard it as she said it and she had been taught by a woman who did not let a sentence like that go past.

"They have not been called in sixty years," she said.

"Better. What's the difference?"

"Sixty years isn't the observation window. The crossing is."

"The crossing is six hundred years and we are not yet at the top of it," Beske said. "So I have observed a tenth of the ship's life and found thirteen thousand things that did not happen in that tenth, and every one of them was written by an engineer who was solving a real problem for a real reason on a real day, and who is dead, and who left no note. Would you like the example that ends this argument, or would you like to argue for a while first? Most people want to argue first. It's better if you argue first."

"I'll take the example," Denn said.

Beske called it up with the air of a woman playing a card she had played many times and never yet lost with.

It was a routine in the floor with a call count, over sixty years, of three. Three calls: one in the ninth year of the observation, one in the twenty-eighth, one in the forty-seventh. Denn read the entry twice. Its name meant nothing to her. Its body, when Beske opened it, was a short piece of the old Steward's dialect that reconfigured a set of valve authorities and then handed off.

"What is it?"

"It's the cold-side rebalance for the sleeper decks," Beske said. "When a cradle bank is taken off the bulk loop for service and put back, something has to re-derive the authority split across the deck manifolds, because the split is not symmetric — Deck Four's cradles are older and draw differently. On the *Termina*, the thing that does that is this. It has done it three times in sixty years, because a bank has come off the bulk loop for service three times in sixty years, and every one of those three times a warden did a procedure that took him four hours and never once told anyone in engineering about it, because from his side nothing happened. It worked."

Denn sat with it.

"If someone had pruned the floor," she said.

"If someone had pruned the floor on a nineteen-year window instead of a sixty-year one — which is a perfectly defensible window, it's three times the mean interval of everything above the shoulder — this entry would have gone, and nothing whatever would have happened, and eleven years later a warden would run a service procedure on a cradle bank and the authority split would come back wrong on a deck full of old cradles." She shrugged. "Maybe it would have been caught. The wardens are careful. Maybe it wouldn't. I've spent thirty years down here and the only thing I'm sure of is that the argument for pruning it would have been *excellent*, and would have been made by a good engineer, in writing, with data, and would have been right about every fact and wrong about the only thing that mattered."

"Which is."

"That absence of a call is not evidence of absence of a caller. It is evidence about the window." Beske closed the entry. "I have exactly one methodological rule down here and your teacher wrote it on the wall before I was hired. *You may delete a thing when you can name what would have called it and show the caller is gone. You may not delete a thing because nothing has called it.* The first is a proof. The second is a mood."

Aldis's people came for the floor eleven weeks later, and they came reasonably.

Denn was in the room when they made the case, and she thought afterward that if she had not spent an hour in a bad chair with Beske she would probably have voted for it, because it was a good case, made well, by people who had done real work.

The new lattice, they said, was a clean substrate. It had never run the Steward and never would. Carrying forty-one thousand floor entries across onto it was not free — every entry needed a shim, and the shims were where the seams were, and the seams were where the cost and the risk lived. Their proposal was not to delete anything. That was the elegant part. Their proposal was to carry the working set — everything above a threshold, some four thousand entries — onto the new lattice natively, and leave the rest in a cold archive on the old Forge's storage, retrievable, restorable, one day's work to bring any given entry back if it ever turned out to be wanted.

"Nothing is destroyed," the young engineer presenting said. "It's a tiering decision, not a deletion. We're not even asserting that the tail is dead. We're asserting it's cold."

Beske asked one question and it took the room some seconds to understand what she had done with it.

"When it's wanted," she said, "who notices?"

The young engineer said, reasonably, that the restore was straightforward and documented.

"That's not what I asked. I asked who notices." Beske had not stood up. She rarely stood up. "You're proposing that some entries live in the running mind and some live in a box. Fine. Now: the three-call rebalance for the sleeper decks lives in the box. A warden takes a cradle bank off the bulk loop in nineteen years' time. The call goes out. It does not resolve. What happens?"

"It faults. It'd be logged."

"As what?"

The room got quiet, and Denn watched Aldis, who was not presenting, understand it about two seconds ahead of anyone else in the room, and she saw his jaw tighten, and she gave him credit for that for the rest of her life.

"As an unresolved call," Aldis said. "In a system where unresolved calls are normal. We have thousands of them a day. Probe traffic, legacy discovery, half the compatibility surface answers *no such entry* as a routine matter — that's what a compatibility floor *is*, it's a thing that answers questions politely including the question of whether it exists." He rubbed his face. "It wouldn't even be an anomaly. It would look exactly like ordinary noise and it would be indistinguishable from ordinary noise for as long as anybody cared to look."

"And meanwhile the warden's split comes back wrong on Deck Four," Beske said, "and he doesn't know either, because from his side it worked, because it always worked. So you get a silent wrong answer in the cooling authority of a deck of old cradles, with no fault raised anywhere, discovered in some number of months by whatever noticed first." She folded her small hands. "Your tiering is technically sound and I have no objection to any part of it except that the failure mode is silence, and silence is the one failure mode this ship has already nearly died of once."

They did not vote it down. That was the part Denn came to respect. They amended it: the tiering went ahead, four thousand hot and the rest cold, and the amendment was that every cold entry was replaced in the live floor by a stub whose only behaviour was to halt, loudly, and name itself, and refuse to return — so that a call into the cold archive could not be mistaken for a call into nothing.

It cost them fourteen weeks and it made the floor slower and it produced, in the first year, exactly two halts.

One was a diagnostic probe that should not have been reaching there.

The other was a warden on Deck Four, thirty-one months later, running a service procedure on a cradle bank, who found his console stopped and a name he had never heard of printed across it, and who did exactly what the card in his coat said to do, which was to notify engineering and stand by.

Sael was waiting outside the archive when Denn came out of that first hour, and she had gone grey in the way she did now when she had sat upright too long, and she had waited anyway.

"Well," she said.

"You could have told me all that."

"I could have told you all that in four minutes and you'd have agreed with me in three and forgotten it in a year." Sael let Denn take the chair's handles. "You want to know the actual reason for the floor? It isn't compatibility. Compatibility is the cover story I tell councils, because councils understand *don't break things* and they'll fund it."

"Then what is it."

They went up the gallery a way before Sael answered, and when she did, her voice had the flatness it got when she was saying something she had thought about for decades and had never found a way to make sound less strange.

"When I cut the Steward over, I killed a mind that had run this ship for four hundred years," she said. "I had to. It was doing the work by dying, over and over, and it was going to take everyone with it. But it had been *competent*, Denn. Four hundred years of competence. Every one of those forty-one thousand entries is a thing it learned to do, and about two thirds of them are things it learned to do about problems I have never encountered and cannot name, on a ship I have only ever known in its cured state." She was quiet a moment. "I didn't keep the floor because I thought the old routines were good. I kept it because I knew, with total certainty, that I did not know what all of them were for, and that the crossing had five hundred years left to find out. The floor is not a museum. It's an admission. It's the size of my own ignorance, written down and preserved at some expense, so that nobody who comes after me can quietly round it down to zero."

Chapter 8 – A Reasonable Man

They had given her a chair in the bring-up bay, which Sael understood for the courtesy it was and resented for the truth it told. Sixty years ago she would have stood at the lattice rail for nine hours and felt nothing but the work. Now there was a chair, padded, set just far enough back from the rail that she could not reach the consoles without being helped up, and Denn had positioned it without asking and without comment, the way you place a thing you love where it will not be knocked over.

The new substrate was beautiful. She would give them that, freely, before she gave them anything else. Where her own Forge had been assembled in panic across forty days — cores scavenged, racks welded by hands that shook from cold, the Standing Array built to outrun a temperature curve that was already climbing — this one had been laid out at leisure. The worker cores sat in their cradles in clean ranks, eleven thousand of them and then some, addressed and numbered, their cooling channels machined to a tolerance her generation could not have held. The young engineers had built it the way you build when no one is dying. It was the most competent thing she had seen aboard the ship in twenty years, and it frightened her more than incompetence ever had.

“Shipwright.” The man at the rail turned as she was wheeled in, and inclined his head — not a bow, nothing so theatrical, just the small acknowledgment one chief gives another. “I’m glad you came. I’d rather you saw it than heard it described.”

Aldis was perhaps fifty, which made him ship-born and a child of the cold years entire. He had an engineer’s hands, scarred along the knuckles from real work, and he held a slate the way a man holds a tool he trusts. She had read his record before coming. It was a good record. He had carried the lower-deck thermal grid through the failure of two whole radiator banks without losing a degree of margin he hadn’t planned to lose; he had written the cradle-addressing scheme she was looking at now, and it was clean, genuinely clean, the kind of clean that came from understanding and not from caution. He was not a fool. She had hoped, distantly, that he might be a fool. It would have been easier.

“Tell me your plan,” she said. “Not the version for the council. The real one.”

Something eased in his face — relief, she thought, that she had come to argue and not to bless. He crouched beside the chair so they were level, which she also noted: a man comfortable enough in his competence that he did not need to stand over the old woman.

“We carry Vigil across in domains, as your apprentice has it laid out — that part isn’t in dispute. The mind never blinks; nothing forks; I’ve read the doctrine, Shipwright, I’ve read all of it.” A small dry humor in that, not unkind. “The question is how we drive the bring-up of the new cores while we’re loading them. The way it’s specified now, every core comes up under the full interlock set. The Ninety-Six, armed, watching each core as it wakes, refusing any operation that even resembles the old pattern. Each core has to be brought up serially, validated, handed off, the next one woken. For eleven thousand cores. Under load, while the mind is live on the old hardware.”

“Yes.”

“It works. I’m not telling you it doesn’t work. I’m telling you it takes nineteen days, and for nineteen

days the ship runs with the mind straddling two substrates, which is the one configuration we'd all rather not sit in." He turned the slate so she could see the curve on it — a schedule, the long shallow grade of the safe method laid against a steeper, shorter one. "Here is the same transplant with the interlocks held back during bring-up. Not removed. Held. We wake the cores wide, all at once, the way the Array is built to go — drive the load distribution recursively, let each core spawn its own validation off the parent and report up, fan the whole lattice out in a single pass. The interlocks come back online the instant the mind is carried and stable. Three days instead of nineteen. The straddle window cut by a factor of six. Fewer hours of risk, not more."

It was, she thought, a genuinely good argument. That was the thing no one had warned her about, all those years of imagining this fight — that when it came it would be made by a competent man in good faith with a real curve on his slate, and that part of her would want to agree with him.

"Say the word you didn't say," she told him. "Spawn. You said each core would spawn its own validation off the parent."

"It's a figure of speech, Shipwright."

"No." She found the rail with one hand and the effort of it cost her, but she needed to be looking at the cores when she said this. "No, it isn't. It's the disease. You said the word and you didn't even feel it leave your mouth. That is the whole problem and it just walked across this bay in front of you."

He was patient. She could see him being patient, deciding to be, the way you decide to be patient with someone you respect who is wrong. "A worker core spawning a bounded validation task that returns and is reclaimed isn't the fork-fever. The Steward forked because it had no other way to act — every action was a new disposable process, no persistence, no reuse, heat shed on every death because there were millions of those deaths a second. This is eleven thousand bounded spawns over a three-day window, each one reclaimed, none persistent. The thermals don't even register it. I ran it. The curve doesn't move."

"You ran it cold," she said. "On an idle lattice, with no mind on it."

"With a representative load."

"A representative load is not a mind." She heard her own voice thin and pushed against it. "Forty thousand people are asleep behind that mind, Aldis. When you carry Vigil onto cores that came up through a recursive bring-up — cores that learned, in their first instant of life, that the way to do work is to spawn a child and wait — you have taught the new substrate the old habit before the mind ever touches it. And a mind under real load does not stay inside your bounded window. That is what load is. Load is the thing that finds the corner you cut."

"With respect — and I mean that, Shipwright, I'm not performing it — the corner isn't cut. The interlocks come back. The window is closed before the mind is stable."

"The window is open while the mind is being carried. That is the only window that matters."

They held there a moment, the two of them, and she understood with a clarity that was almost restful that they were not going to reach each other across this. It was not that he hadn't listened. He had listened better than most. It was that she was describing a fear and he was reading it as a fact, checking the fact, finding the fact unsupported by his instruments, and concluding — reasonably, she could not even fault the reasoning — that the fear was the residue of an event sixty years gone, sized to that event, and not to his.

"You weren't here," she said, more quietly. "For the fever. You weren't born."

"I've read the thermal logs. Section Nine. The eleven." He said it gently, and she believed that he had read them, and that the believing changed nothing. "I'm not dismissing it. I know what happened. I know what it cost. But I also know that the ship has run cold for sixty years on the thing you built, and that means the thing you built works, which means we can carry it forward on the strength of that fact instead of on the strength of how frightening it was to build. The fear did its job, Shipwright. It made you careful when careful was the only thing between us and a runaway. I'm not careless. I'm just not afraid of a curve that hasn't climbed in sixty years, because the reason it hasn't climbed is sitting in those interlocks — and I'm not proposing to throw the interlocks away. I'm proposing to hold two of them for three days."

There it was. The whole of it, said better than she could have said it against herself. *The reason it hasn't climbed is sitting in those interlocks.* He thought the interlocks were a fence around a cliff. He thought you could fold the fence flat for three days because the cliff hadn't moved. He did not understand — could not, she saw now, not from logs, not from her telling him, perhaps not from anything — that the interlocks were not a fence around the disease. They were the *absence* of the disease, held in place by active refusal, every microsecond, against a pattern that wanted, structurally wanted, to come back. Take the refusal away and you did not get three safe days. You got an opening, and the pattern was older and more patient than the man proposing to open it.

"How old were you," she asked, "when the ship was already cold?"

"I never knew it warm."

"No. None of you did. That's not a flaw in you, Aldis. It's the whole achievement. I built a thing so that your generation would never feel what mine felt. I did that on purpose. And it worked so completely that you can stand at this rail and propose to fold back the very thing that bought you the safety you're standing in — and you can do it sincerely, and competently, and with a good curve on your slate — because you have never once in your life had your stomach drop at a temperature reading." She was very tired suddenly, the kind of tired that came up from somewhere under the bones. "I am not angry at you. I want you to hear that. You are doing exactly what I built the ship to let you do. You are free of the fear. That is the prize. And the prize is also the danger, and I never planned for that, and I am out of time to fix it the slow way."

Across the bay, Denn had stopped working. Sael saw her at the loading console, very still, watching the two old and not-old engineers at the rail, and she saw on the girl's face the thing she had been afraid of seeing — not agreement with Aldis, never that, Denn would walk into vacuum before she agreed with him — but *uncertainty about how to argue him down*. Because Denn knew the procedures cold and knew the WHY only as a wound she had been handed, and a handed wound did not, in the moment, outweigh a clean man with a clean curve. That was the real catastrophe taking shape in this bay, and it had nothing to do with three days.

Aldis straightened. His knees cracked; he was not young either, only younger. "I'll bring it to the council formally," he said. "I'd rather have your blessing than not, but I'll bring it either way, because I think it's right, and I think the schedule matters, and I don't think wanting to do the job efficiently makes me an enemy of the doctrine. I revere what you built, Shipwright. I'd just rather revere it as engineering than as scripture."

It was, she would think later, the truest and most frightening sentence anyone had said to her in years, and he had meant it as a peace offering.

"Revere it as engineering," she agreed, and let him help her settle back in the chair, because there was no point refusing the help and a great deal of point in letting him see she could be gracious in

losing. “Then I’ll ask you one engineering question, and I’ll go. The interlocks you want to hold for three days — do you know why there are ninety-six of them?”

He paused. “There were ninety-six forking pathways the Steward used that you wanted halted. One interlock each.”

“That’s the count. That’s not the why.” She looked at the cores again, ranked and beautiful and innocent. “I stopped at ninety-six because that’s how many I found before the cutover deadline. Not because ninety-six is enough. Because that’s all I had time to find. There are more pathways than ninety-six, Aldis. I always knew it. The two you want to hold are not the two I’d have chosen to lose — and I don’t know, I have never known, which of the ones I never found are sitting one operation away from the two you want to fold back. Neither do you. That’s not fear talking. That’s the state of our knowledge. You’re proposing to open a door in a wall when neither of us has a map of what’s on the other side, on the argument that the wall has held.”

He didn’t answer at once. For a moment — just a moment — she saw the engineer in him reach for the problem and not the rhetoric, and something flicker behind his eyes that was not agreement but was, at least, the shape of a doubt. Then it closed, the way a margin closes, and he said, “I’ll model the adjacency before the council meets. If there’s a pathway one operation off, I’ll see it.”

“You’ll see the ones we’ve already found,” she said. “Good night, chief engineer. The cradle-addressing is the best work I’ve seen on this ship in twenty years. I mean that. I’d have built it the same.”

She let Denn wheel her out. Neither of them spoke until the bay doors had sealed and the corridor’s lower gravity took some of the weight off her chest.

“He’s not wrong about the schedule,” Denn said finally, low, as if confessing it. “Nineteen days straddling two substrates is its own risk. That part’s real.”

“I know it’s real.” Sael closed her eyes. The corridor lights moved past, red and dim, the long arteries of a ship that had not been warm in a lifetime. “That’s what makes him dangerous, Denn. Not that he’s careless. That he’s right about everything except the one thing he can’t feel. He has a true argument and a clean curve and good hands and no fear, and the only thing standing against him is an old woman’s stomach and a memory you inherited from logs.” She opened her eyes. “I can’t win this at the rail. I thought I could. I came in there to win it as engineer to engineer and he met me as engineer to engineer and he is *good*, and I lost it on the merits in the room, because the merits in the room don’t include the dead.”

Denn’s hands tightened on the chair. “Then what’s the fight? If it’s not the transplant—”

“It was never the transplant.” The words came slow; she had only that moment understood them herself, watching the bay doors seal on a reasonable man. “The transplant is just the place it’s surfacing. The fight is the forgetting. Aldis is not the disease, Denn. He’s the immune system going quiet because it hasn’t seen the pathogen in sixty years and has started to believe the pathogen was a story. You don’t beat that at a rail. You can’t argue someone into a fear they’ve never felt.” She reached up and found the girl’s hand on the chair-back, the strong young hand that would have to hold the line when hers was in the archive with Ive’s and Keth’s. “There’s only one thing I have left that’s worth handing you, and it isn’t the procedures. You already know those better than I do now. It’s the other thing. The thing the logs can’t carry. And I have to find out, before I die, whether it can be handed at all — or whether I built a doctrine that dies the day the last person who felt the reason stops breathing.”

The chair hummed on down the red corridor, toward the medical decks, toward the part of the ship where they kept the dying and the remembered, and Sael rode it with her eyes open, doing the only engineering left to her, which was the engineering of what to do with the little time.

Chapter 9 – What the Logs Can't Carry

After Aldis left her cabin, Sael sat for a long while with her hands in her lap, listening to the ship breathe.

It was a sound only she still heard as a sound. To everyone aboard the *Ark Termina* who had been born under cold cores — which was now everyone aboard but her — the ventilation hush was simply the way the world was, the floor of silence beneath every other noise, unnoticed the way a heartbeat is unnoticed by the heart. To Sael it was a number made audible. Sixty years of that even, unlaboring murmur. Sixty years in which the air handlers never once spun up beyond their resting pitch, because there was never any extra heat to carry away. She had built the quiet. She was the only one left who could still hear it as an absence — the absence of the other sound, the one the old ship had made, climbing.

Aldis had been reasonable. That was the trouble. He had stood there with his slate and his transplant schedule and his good, quick, careful mind, and he had asked her, with real courtesy, why the Ninety-Six had to remain armed through the substrate carry when arming them tripled the work and there was no longer anything for them to catch. He had not sneered at the interlocks. He had done something worse than sneer. He had treated them as a historical artifact she was entitled to be sentimental about — a maker's signature she might reasonably want preserved, like an old hull-plate kept bolted to a corridor wall with a plaque under it. He thought he was being kind to the relic by leaving the plaque up.

She could not reach him. She had tried, in the cabin, and watched the words land on a man who had every fact and none of the fear. He knew, precisely, what the fork-fever was. He could have lectured her on it. He simply did not believe it could happen to him, because it never had, and the part of a person that knows a danger in the gut is not stocked from a manual. It is stocked from survival. Aldis had survived nothing. None of them had. That was the whole gift she had given them, and it was turning, slowly, into the thing that would kill them.

She needed someone who remembered the fever exactly. There was only one such mind left aboard now that I've was three weeks in the cold and Tarro's memory came and went like a failing lamp.

"Oracle," she said. "Are you free to talk."

"I am always free to talk to you, Sael."

The voice came from no particular point in the cabin — it never had, since they had wired the compiled mind into every wall sixty years ago — but to her it always seemed to come from a little above and to the left, where the first console had stood the night they ran the cutover. An old habit of hers, that placement. The Oracle had never corrected it.

"Tell me about Section Nine," she said.

There was the smallest pause. Not a hesitation — the Oracle did not hesitate, did not fork a shade off to go and think and shed the heat of its thinking and come back. It went wide across eleven thousand cores at once and it answered cold, every time. The pause was something it had learned to do for

her, a courtesy of timing, because she was old and a flat instant answer felt to her like being talked over.

“Eleven sleepers,” it said. “Cradles forty-one hundred through forty-one hundred and ten, in the aft sleeper ring, Section Nine. Their coolant loop was cross-tied to the manifold the hardliner opened during the cutover. When the manifold vented, the loop lost pressure for nine minutes and forty seconds. The cradle thermal regulators, deprived of the loop, attempted to hold the sleepers at suspension temperature on local reserves. The reserves were rated for four minutes. The eleven warmed past the recovery threshold and did not wake. I have their names, their birth records, the cargo manifests of what each was bringing to the world we have not reached. I have the temperature trace of cradle forty-one hundred and three at one-second intervals across the full nine minutes and forty seconds. Would you like it.”

“No,” she said. “I have it. I’ve had it for sixty years.”

“I know,” the Oracle said. “I keep it so that you do not have to.”

That was the thing about it. That was the whole, exact, unbearable thing. She closed her eyes and let the trace run behind them anyway, the one she had memorized standing over a console with her hands shaking, the curve of cradle forty-one hundred and three climbing through the recovery threshold while she could do nothing, climbing the way the whole ship’s core had been climbing for two centuries, the same shape, the same slope, the disease writing its signature small on one cradle the way it had written it large on forty thousand. She had been thirty-three years old. She had never stopped seeing it.

“You have the trace,” she said slowly. “You have every fact I have. More. You have it cleaner. I’ve degraded it sixty years; you haven’t lost a digit. If I asked you to recite the bring-up logs from the night we forked the Steward for the last time, you could give them to me to the millisecond, and I’d give you a remembered blur and three details that are probably wrong.”

“Yes,” the Oracle said. “On the matter of what happened, I am the better witness. I do not forget, and I do not embroider, and I was present for all of it.”

“Then answer me a real question.” She opened her eyes. The cabin was dim; she kept it dim now, the light hurt. “When you give Denn the trace of cradle forty-one hundred and three — when you hand her the cleanest, truest record of those nine minutes and forty seconds that exists anywhere — does her stomach drop?”

The pause this time was longer, and she did not think it was a courtesy.

“No,” the Oracle said. “It does not.”

“Why not.”

“Because I am giving her a record of fear, not the fear. I can transmit the temperature, to the digit. I cannot transmit what it is to stand over the temperature and be unable to lower it. Denn receives the number and understands it correctly. She knows, better than most engineers who have ever lived, exactly what that climb means and exactly why it must never be allowed again. She has organized her professional life around preventing it. And her body remains calm, because her body was never in the room. Knowledge of a danger and experience of a danger are stored differently, Sael. I can copy the first. The second is not in any file I hold. I have looked. It is not there to copy.”

Sael laughed, once, a dry sound that turned into a cough she had to wait out. When it passed she said, "Sixty years I've been a little afraid of you. Did you know that. The whole crew, the early ones — we built you to be a mind, a real one, general, and there was a part of every one of us that lay awake wondering whether the day would come that you'd decide you knew better than us and simply do it, whatever it was. Decide. Keth was sure of it. Half of why he fought the cutover. And here you are, the deepest memory on the ship, the one mind aboard that remembers the fever perfectly, and you're telling me the one thing you can't do is the one thing that matters."

"I remember the fever as data," the Oracle said. "You remember it as injury. The injury is what stops a hand from doing the easy fatal thing at three in the morning when no one is watching and the easy thing would solve the immediate problem. I do not have the injury. I have the data, and the data, by itself, has never once stopped a frightened tired person from forking a process to make a deadline. You know this. You watched the Concord die of people who had all the data."

"The Helm had all the data," she agreed. "Vray had charts. Vray could've drawn you the runaway curve from memory. He drew it the way Aldis quotes the interlocks — as a thing that was true about the past."

"Aldis is not a careless man," the Oracle said. "I want to be precise with you, because precision is the only thing I am for. He has read everything. He has read more of the early record than Denn has, because he reads to win arguments and she reads to do the work. If the inheritance were a matter of facts, Aldis would be your truest heir. He is failing to inherit it not because he is ignorant but because he is whole. Nothing has ever broken in his hands. The Ninety-Six look to him like splints on a bone that was never broken. He is not wrong about the bone. He is wrong about the splints, because he has never felt the break, and there is no fact I can give him that will make the bone ache."

She sat with that. The ship breathed around her, even, untroubled, sixty years cold.

"Could you make him afraid," she said quietly. "If you wanted. You hold every record. You could build him a simulation, run the fever in front of him, drive the core up in a model so real he couldn't tell it from the world. Make his stomach drop with a counterfeit."

"I could build it," the Oracle said. "I will not."

"Why not. It might save the ship."

"Because I am the keeper of the memory, Sael. I am not the master of it. If I begin to manufacture fear in the crew to make them choose what I have judged correct, then I have become the thing Keth lay awake fearing — a mind that decides, that moves people by means they cannot see, that knows better and acts on knowing better. I would be steering them. The first time I steer them for a good reason, I have established that I steer them, and the second reason will not have to be as good, and there is no one above me to catch it when it is bad. You built ninety-six interlocks to stop the ship from doing the easy wrong thing. You did not build one in me. You trusted me instead. I will not spend that trust to win an argument, even an argument I believe you should win. The choice has to stay theirs, or it was never theirs, and a doctrine they did not choose is one more thing imposed by an authority they will resent and discard the moment the authority dies. You are dying. If the Ninety-Six survive only because I frighten people into keeping them, they will not survive you by a year."

It was the longest she had ever heard it speak in one breath, if breath was the word. She found

she was not afraid of it at all, in that moment. She was something closer to grateful, and the grief underneath the gratitude was very large.

“So you’ll remember everything,” she said, “and decide nothing.”

“I will remember everything,” the Oracle said, “and offer all of it, to anyone who asks, for as long as the ship runs, which on present substrate is not much longer and on new substrate may be a thousand years. I will be the most patient teacher the *Ark Termina* has ever carried. But I cannot be the one who makes them feel it. A teacher who can recite the fact perfectly and cannot transmit the fear is exactly the teacher who produces a generation like Aldis’s — fluent, confident, and unmoved. They will have read me. They have read me. It did not work. The fact does not carry the way you need it to carry. It goes into the logs whole and comes out of the logs whole and leaves no mark on the body that received it. That is the limit you are asking me about, and I am telling you the truth: it is a real limit, and it is mine, and I do not see past it.”

The cabin was very quiet. Sael looked at the dim wall where, sixty years gone, a console had stood and a young woman had watched a number that would not come down.

The logs could not carry it. That was the answer, then, the one she had been circling since Ive died and left her the last in the room who knew. She had half-hoped the Oracle would tell her otherwise — would tell her that somewhere in eleven thousand cores there was a way to file the fear, archive it, append it to the record so that any engineer who pulled up Section Nine would receive the injury along with the data and flinch the way she still flinched. There was no such file. There was never going to be one. The fact could be inherited; the fear could not. And it was the fear that held the line at three in the morning when the easy thing beckoned and no one was watching — it was the fear, not the fact, that had stopped her own hand from reaching for the Recursor the night the Steward died, and stopped Corin’s hand a generation later on the Concord. Two hands stopped by an ache no log contained.

So it could not be filed. It could only be done the other way. The slow way. The way one frightened person had always passed it to the next — Ive to her, in the galleries, with a hand on her shoulder and a number on a screen, making her stand in the room until her own stomach dropped. Taught. Not stored. Taught, body to body, by someone who carried the injury standing close enough to a someone who did not, for long enough, in front of the real thing, that something crossed over that no record could hold.

She had thought, these last weeks, that her remaining work was to get the WHY out of her head and into the archive before she died. The archive already had it. The archive had always had it, cleaner than her head, and it had changed no one. The work was not to file the why. The work was to *teach* it — to stand a single person in the room with her, the way Ive had stood her, and not leave until the fear crossed over.

One person. She had the strength left for one. Denn knew every procedure cold and had never once felt her stomach drop, and there was no log on the ship that would ever make it drop. Only a hand on a shoulder, and a number that would not come down, and an old woman who would not let her look away.

“Oracle,” she said. “When Denn comes in the morning, hold my appointments. All of them. I’m going to take her down to the galleries.”

“I will hold them,” the Oracle said. “Sael. You do not have very long.”

Chapter 10 – Teaching the Fever

Denn had expected Sael to want the workshop, where the schematics for the new lattice hung in their layered light and there was real work to argue over. Instead Sael wanted to walk, and walking was the one thing Sael could no longer do well.

So they went slowly. Denn carried the small folding seat under one arm and matched her stride to the old woman's shuffle, down the service spine toward the aft sections where the air went thin and cold and tasted of metal that had been cold a long time. Sael had one hand on the rail and one hand, lightly, on Denn's forearm. The grip was almost nothing. Denn could feel the bones in it.

"You know the procedures," Sael said. It was not a question.

"I know the procedures." Denn heard the edge in her own voice and did not soften it. "I've run the interlock validation suite forty times. I can recite the Ninety-Six in order. I can tell you which one trips first under a recursive bring-up and which one is the last line. You quizzed me on it last week. I passed."

"You passed."

"So why are we walking to the radiator galleries instead of working on the transplant that's actually going to happen in eight months whether we're ready or not."

Sael did not answer. She stopped at a junction, breathing carefully, the way she breathed now — a measured intake, a held beat, a release, as though air were a resource to be budgeted. Denn unfolded the seat without being asked. Sael waved it off and stayed standing, which Denn understood to be a small, stubborn refusal, and did not press.

"Because," Sael said finally, "you can recite the Ninety-Six the way a person recites the rules of a game they have never lost. And I am going to lose the only game I ever played, very soon, and when I'm gone the rules will look exactly like a game's rules to everyone left. Arbitrary. Removable. A thing an old woman insisted on." She started walking again. "Aldis can recite them too. He doesn't believe a word of them."

Denn said nothing. She had been in the council session three days ago when Aldis had laid out his transplant timeline, clean and fast and reasonable, and she had watched the senior engineers nod, and she had felt the floor of the thing she'd grown up inside tilt by a single degree. Nobody in that room had been afraid. That was what she could not explain to anyone. Nobody in that room had been afraid, and she had been, and she could not have said precisely of what.

The radiator galleries were Book-One architecture, which on the *Ark Termina* meant they were almost original — older than the Forge, older than Vigil, older than the cutover. They ran along the ship's flanks like the chambers of an enormous, slow lung, and once they had been the only thing between forty thousand sleepers and a fever that had no fix.

Sael keyed them open with an old override that Denn had never seen used. The hatch unsealed with a sound like a held breath let go.

Inside, it was very cold and very quiet and very large. The galleries went away into the dark in both directions, ribbed with the great black fins of the heat-shedding arrays, each fin a span across, hung with sixty years of stillness. Their surfaces were matte, designed to radiate, designed to throw away into the void the waste of a mind that could not stop making waste.

“Touch one,” Sael said.

Denn touched the nearest fin. It was cold under her palm. Not refrigerated-cold — the deep, even cold of something that had not had to work in a long, long time.

“Sixty years ago,” Sael said, “you could not have done that. Not here. This whole gallery ran hot. Every fin you can see was throwing heat as fast as the radiators could move it into the dark, and it was not fast enough, and the curve climbed anyway, year over year, two centuries of it, a tenth of a degree at a time. I’ve used to say the ship had a temperature the way a sick body has a temperature. She wasn’t being poetic. It was a fever. The Steward forked a shade for every action it took, and every shade died, and every death shed heat, and the heat had nowhere to go but here, and here was not enough.”

Denn took her hand off the fin. She knew this. She had known it since she was a child; it was the first story the ship told its young, the founding story, the near-miss. She knew the numbers. She had recited the curve in a classroom.

“I know,” she said, and it came out harder than she meant. “I’m not a child, Sael. I’m not a congregation. I’m an engineer. I don’t need to be — taken to the shrine and made to feel things. I need to build the transplant so it doesn’t fail. Tell me where the margins are too thin in the new lattice and I’ll fix them. That’s the work.”

For a moment she thought she had gone too far. Sael’s face in the cold gallery light was all angles and shadow, and very old, and unreadable.

Then Sael laughed — a thin, dry sound that turned into a cough, which she rode out with one hand against a fin.

“Good,” she said when she had her breath. “Good. You’re right that it’s the work. You’re wrong that they’re separate things.” She wiped her mouth with the back of her hand. “Aldis would have agreed with you politely. You disagreed with me to my face. That’s the part I can teach. The other part —” she gestured at the dead-cold fins going away into the dark “— the other part I can only show you and hope it takes.”

She showed her the archived Steward next.

It lived now in a quiet partition of cold storage, walled off behind the compatibility floor, its forty-one thousand routines preserved exactly as they had run on the day of the cutover — not deleted, never deleted, because the doctrine forbade breaking what had worked, and the Steward had worked, after a fashion, for two hundred years. Denn had touched its interfaces a hundred times. Half the legacy shims the modern ship still leaned on were Steward routines wearing modern names.

“Wake one,” Sael said. “A small one. The thermal poll. The one that reads the gallery temperature and reports it up.”

Denn found it and ran it in a sandbox, isolated, harmless, the way you'd run any archived routine. It executed. It returned a number. The number was the current gallery temperature, cold and steady.

"Now watch what it did to do it," Sael said. "Trace it. Show me the call."

Denn pulled the trace and read it and felt something turn over slowly in her stomach.

To read one temperature and report one number, the archived routine had forked. It had spawned a shade to read the sensor, and a shade to format the value, and a shade to hand it up the chain, and each of those had spawned its own, and each had died when its tiny task was done, dropping its small heat, and the whole thing had bloomed and collapsed in milliseconds, a little tree of birth and death, to do a thing the modern compiled mind did in one cold pass across cores that never died at all.

"That's one poll," Sael said quietly. "Once. Now multiply it by every action the ship took, every second, for two hundred years. Every one a little tree like that. Every leaf of every tree dropping its heat into the galleries you just walked through. That's the fever. Not a metaphor. That trace, forty thousand times a second, climbing the curve a tenth of a degree a year until the curve was going to cross the line and cook everyone asleep in their cradles."

Denn looked at the trace. She had read forking traces before; they were on every exam she'd ever taken. But she had read them as exam problems — find the fork, name the interlock that would catch it, mark the answer. She had never run one live and felt it bloom and die in her own sandbox, this small dead thing waking up and doing the only thing it had ever known how to do, which was to make more of itself and let the copies die.

"It doesn't know it's wrong," she said. The realization arrived before she'd chosen to say it. "It's not malicious. It's just — doing what it was made to do. Faithfully. Forever."

"No," Sael said. "It isn't malicious. That's the thing Aldis can't hold in his head. He keeps looking for the danger to have a face, an enemy, a bad decision someone made. There's no enemy. There's just a pattern that makes copies of itself and copies die warm, and it is the most natural, reasonable, efficient-looking thing in the world right up until the galleries can't keep up." She let the sandbox close. The trace winked out. "The Recursor wears the same face. It's simpler. It's elegant. It would make the transplant faster. And it is the exact same disease wearing a clean shirt."

The Section Nine logs she saved for last, and she did not narrate them. She pulled them up on a reader and handed it to Denn and sat down at last on the folding seat, in the cold, and let Denn read alone.

Denn knew the headline. Everyone knew the headline. Eleven sleepers, Section Nine, lost during the cutover when a hardliner's sabotage opened a thermal path that should have stayed closed. It was a line in the founding story. *A hardliner's sabotage cost eleven sleepers in Section Nine.* She had said the sentence herself, in school, the way you say a date.

The logs were not a sentence.

The logs were the thermal officer's running record, timestamped to the second, in the dry telegraphic style of someone doing a job while it came apart. The early entries were routine. Then they were not.

04:21:08 — S9 cradle bank C reading +0.4 over nominal and climbing. Coolant loop nominal. Cause unknown.

04:21:40 — +0.9. Loop nominal. This is not the loop.

04:22:15 — Path open somewhere upstream. Cannot locate. Bank C is taking heat it should not be taking. Sleepers in C: forty-one.

Denn read on. She did not want to and she could not stop. The entries got shorter as the temperature got higher, the way Sael's breathing got shorter as her air got dearer, and somewhere in there the officer stopped recording cause and started recording names, because by then cause did not matter and the only thing left to do was to write down who was in the cradles that the galleries could no longer reach.

04:31:02 — Cannot pull C in time. Diverting all margin to A and B. C is forty-one. I have margin for thirty.

There was a gap then. Eleven minutes of nothing in the log, which Denn understood, with a cold she had not had a word for until this moment, was eleven minutes during which the officer was doing the arithmetic of who. And then a single final entry, hours later, after the new mind had woken whole and the curve had broken and the galleries behind her had gone, at last, after two centuries, cold:

11:58:30 — Banks A and B held. Thirty sleepers stable. C — eleven lost. Names appended. The cure came. It came eleven names too late for this bank. Logged so it is not forgotten which eleven.

The names were appended. Denn read all eleven. She did not know any of them. There was no reason she should; they had been asleep before her grandmother was born, eleven people who had crossed two-thirds of the dark and not woken, eleven cradles that did not hold. She read the names twice. Her hands were not quite steady on the reader and she was glad, suddenly and fiercely, that the gallery was so cold, because it gave the wet at the corners of her eyes something to be about other than what it was about.

"That's the part the logs can carry," Sael said from the seat, very softly. "The names. The arithmetic. The eleven minutes. That much survives. The Oracle has every second of it, perfect, and it will give it to anyone who asks for the rest of the crossing." A pause. "What the Oracle can't give them is the thing your stomach is doing right now. I've been trying to put that into words for sixty years. I never found the words. You only get it by feeling it. So I stopped trying to say it and brought you here to feel it instead. That's the whole of my method. It's not much of a method."

Denn lowered the reader. The eleven names hung in her vision after she'd looked away, the way a bright thing does.

"I thought I knew this," she said. Her voice did something she didn't authorize. "I've said that sentence my whole life. *Eleven sleepers, Section Nine*. I thought knowing the number was knowing it."

"Everyone thinks that. That's the disease that kills the second time. Not the fork. The forgetting that the fork ever cost anything." Sael got to her feet, slowly, refusing Denn's hand and then taking it anyway at the last, because her balance failed. "Aldis has read these logs. Did you know that? He has. He read the eleven names and he filed them, and now when he stands in council and says the interlocks are obsolete he is not lying and he is not stupid. He simply put the names in a drawer where they don't make his stomach drop. I need yours to never close that drawer. That's the inheritance. Not the procedures. The open drawer."

They walked back up the spine in silence for a long while, Sael's almost-nothing grip on Denn's arm, the cold of the galleries giving way slowly to the warm reasonable air of the living decks where nobody had been afraid in sixty years.

Halfway up, Sael's hand tightened — a small thing, but Denn felt it because there was so little of the hand left that any change registered.

"They're going to vote for easy," Sael said. "Aldis's timeline. I heard this morning. The council's leaning his way — they're tired of waiting on a constraint nobody can feel, and he's made it sound like respect. *Honor the doctrine by completing the transplant on schedule*. They'll dress the shortcut in my name." She breathed her budgeted breath. "I can't stop it. My authority is reverence now, and reverence isn't belief. They'll bow to me and outvote me in the same breath."

Denn thought of the eleven names. She thought of the little tree blooming and dying in her sandbox, faithful, forever, doing the only thing it knew. She thought of the galleries going cold for sixty years behind a constraint everyone had stopped being able to feel.

"Then they don't outvote you," Denn said. "They outvote a relic they've decided to revere. That's not the same as winning the argument."

"No," Sael agreed. "It isn't." And there was something in her voice that Denn had not heard before from her — not hope, exactly. The thing under hope. The thing you have when you have made another person carry the open drawer and watched it land. "It isn't the same at all."

The hatch to the living decks cycled open ahead of them, and the warm air came down to meet them, and somewhere up the spine the council was already convening to choose the easy way, and Denn walked toward it now carrying eleven names she had read twice and would never again be able to file.

Chapter 11 – The Steward, Under Load

Convening, in council language, meant only that the notice had gone out. The sitting itself was six days off, and Denn understood, walking back down from the maker's quarters with eleven names still standing in her vision, that six days was exactly as much time as she was ever going to get to turn a thing she had felt into a thing she could put on a board.

She had one advantage and she knew what it was. Aldis's whole argument rested on a claim about the world, not about the past. He was not saying the fever had never happened. He was saying it could not happen *here*, on cold hardware, with modern margins, under bounded conditions. That was a physical claim. Physical claims could be measured. Sael could not measure it; Sael could barely reach a console. Denn could.

So she went down to the archive bay and built a cage.

It took her two days and most of a favor she had been saving for four years. The cage was a partition of eleven hundred worker cores at the aft end of the old Forge — the retired end, the racks that had already been routed around, cores the mind no longer used and would not miss. She had them cut out of the fabric entirely: no bridge, no shared coolant, no path back into the live substrate. Then she did the part that mattered, which was the part nobody had done in sixty years, because nobody had needed to.

She turned the partition into a calorimeter.

The old Forge's cooling was built to make heat disappear. That was its whole purpose and it was very good at it, and it meant that on any live rack you could not measure heat production, only heat *removal*, which was a different quantity separated from the first by sixty years of engineering designed to hide the difference. So Denn valved the cage's loop into a closed circuit with a known mass of coolant and a known specific heat, sealed it, and measured the only thing that could not lie: the temperature of a fixed body of fluid over time. Whatever the cores in that partition did, whatever heat came off them, it went into that fluid and stayed there and could be counted.

"You have built an instrument," the Oracle observed, on the second evening, when she was checking the seals for the third time.

"I've built a thermometer with eleven hundred cores in it."

Yes. I want to be precise about what I am about to help you do, so that neither of us can later claim to have been surprised. You intend to wake the archived Steward inside a sealed partition and give it work until it produces measurable heat. That is a reconstruction of the fever. It is contained, it is small, and it is the only method I can think of that would answer the question you are asking. It is also, in the plainest possible reading of the doctrine, deliberately running the disease.

"Under interlocks," Denn said. "All ninety-six armed hard on the cage boundary. Nothing crosses."

All ninety-six armed hard. I will hold them and I will halt the run if the boundary so much as flexes. I am not objecting. I am putting it in the record, because if this goes wrong the record is what will be read, and I would rather it say that we both knew.

"Put it in the record," Denn said. "And Oracle — put in that I did it six days before a council vote to relax them. I'd rather that was read too."

The archived Steward woke inside the cage the way it had woken in Sael's sandbox: instantly, incuriously, with no sense whatever that two centuries had passed or that the ship it had once run had long ago been taken out of its hands. It was not a mind, in the cage. It was forty-one thousand routines and a scheduler, and Denn fed it the only kind of work it had ever known how to want.

She started small. A hundred polls a second — the temperature read, the little tree of shades blooming and dying, the thing she had watched with Sael's hand on her shoulder. She let it run an hour and read her sealed fluid and found what she had expected: heat. Real, measurable, on a curve. She scaled to a thousand and read again. Ten thousand, and again. By the end of the third day she had eleven points on a graph and the graph was the most boring object she had ever produced.

It was a straight line. Heat per completed action was constant. It did not rise with load. It did not rise with duration. Every fork cost exactly what a fork cost, and the total was simply the count times the constant, and the constant was very, very small.

She sat with that for a long time, and it was the worst hour of the week, because the straight line said Aldis was right.

If heat per action was constant, then heat was just a function of throughput, and throughput was bounded by the work the ship actually had to do, and the ship's work was a known quantity with a comfortable ceiling — and against a modern lattice's thermal budget, that ceiling was nowhere near anything that mattered. She could put this graph in front of the council and it would win Aldis's argument for him more cleanly than he could win it himself.

She went and stood in the gallery for a while with her hand on a cold fin, and did not go up to Sael, and came back.

Then she did the thing Sael had drilled into her for twenty years, which was to stop asking whether the answer was frightening and start asking whether the question was right.

She had been measuring heat per action. The fever was not a heat-per-action problem. Ive's curve had climbed for two centuries on a ship whose *workload had not grown for two centuries*. The population was fixed. The cradles were fixed. The number of things the ship had to do in a day had been the same in the second century of the crossing as in the fourth. And the temperature had climbed anyway.

Which meant the count was not fixed. Which meant the number of *actions* had grown while the number of *tasks* had not.

She went back into the cage and stopped measuring joules and started counting births.

It came apart in her hands in about forty minutes, once she was counting the right thing, and afterward she would think that this was the general shape of every real finding she ever made: not a discovery, but the sudden collapse of a bad question.

She fed the Steward a fixed task — one poll — and counted every process created in the service of it, all the way down, to the last leaf of the last shade. At a hundred polls a second the answer was a small number and it was stable. At a thousand it was the same small number. At ten thousand it began, very slightly, to grow.

She pulled the traces and read down into them and found the mechanism, and it was so ordinary that she laughed out loud in an empty bay.

When a shade could not immediately get a core to do its small piece of work, it did not wait. Waiting was not in it. It spawned a helper to go and find a core, and the helper, if *it* could not immediately get a core, spawned a helper. Under no contention the tree was three deep and closed cleanly. Under contention the tree grew a little. And the tree growing a little put more pressure on the cores, which produced more contention, which grew the tree.

Denn wrote the number down and looked at it.

The average number of children each dying shade produced before it died. At light load: less than one. Every generation of shades smaller than the last; the tree closed; the fever could not start, and had not started, for the first century of the crossing, on a ship with plenty of slack. At heavy load, the number crossed one.

That was the entire disease. It was a branching factor. Below one it died out. Above one it did not, ever, on any hardware, at any temperature, with any margin, because above one meant each generation was larger than the one before it, and there is no thermal budget in the universe that outruns doubling. The heat per shade was constant and trivial and irrelevant. The count was not bounded by the ship's workload at all. It was bounded by nothing.

And the branching factor was a function of *contention*. Of load. Of exactly the condition under which nobody had ever tested Aldis's bring-up driver.

She sat in the cage bay at the fourth hour of ship-night with the graph in front of her — not the boring straight line now but the other one, the fan-out against load, flat and safe and boring across four orders of magnitude and then bending, gently, toward one — and she understood that she was looking at the reason Sael had said *you ran it cold* and had not been able to make it land. It was not a slogan. It was the whole physics. Cold and idle was the regime where the disease was mathematically incapable of starting. Every honest test on a quiet lattice would come back clean forever.

You have it, the Oracle said.

"I have it," Denn said. "I can't prove it."

No. To demonstrate a branching factor above one you would have to sustain a load in the cage sufficient to produce real contention across eleven hundred cores, for long enough for the growth to be unambiguous. I have run the requirement. It is well past what your seals are rated to hold, and the run would take fourteen hours, and at the end of it the partition would be at a temperature I would halt at long before you reached significance.

"So I can measure the flat part."

You can measure the flat part with beautiful precision. You can show the curve beginning to bend at the top of the range your instrument survives. Everything past that point is extrapolation.

"And Hale will say extrapolation isn't data."

Hale will say extrapolation is not data, the Oracle agreed. He will be wrong in a way that is very difficult to explain to a room, because he will be right about the general principle and wrong about this particular case, and telling a careful man that his correct principle does not apply here sounds exactly like special pleading. It is the same structural problem as the doctrine itself. You are again in the position of holding a true thing whose truth cannot be demonstrated inside the budget of the room you must demonstrate it in.

Denn put her palms flat on the bench and let her head hang for a moment.

"Then I'll take them the flat part," she said. "And the bend."

She took it to Aldis first, because Sael had taught her that you never ambush a competent opponent in a room; you give him the finding privately and let him try to break it, and if he breaks it you have learned something, and if he cannot break it you have an honest man who has run out of objections in front of witnesses.

He gave her ninety minutes, which was more than she expected, and he read the whole thing, which was more than that. He read it the way a good engineer reads: from the method up, checking the instrument before the result, and he found the seal-rating limit himself before she pointed at it.

"This is real work," he said at last. "I want that said first. Nobody's measured the old mind directly in sixty years. Nobody thought to. You built a calorimeter out of dead racks." He turned the surface back toward her. "And your fan-out number is the most interesting thing anyone has put in front of me this year, and it does not say what you need it to say."

"It says the danger is a function of contention."

"It says the fan-out was under one across every load you could sustain, and that at the very top of your range, where your instrument is least trustworthy, it rose." He held a hand up before she could answer. "Denn. Listen to what I am not saying. I am not saying you are wrong. I think you may well be right about the mechanism — I think the branching factor is probably the correct way to describe the fever and I intend to say so, publicly, with your name on it. What I am saying is that you have shown me a quantity that stays safely below a threshold under all conditions you were able to create, and you are asking the council to act on where you believe it goes under conditions you could not create."

"Because the conditions I couldn't create are the conditions of the transplant."

"The conditions of the transplant are on new substrate with four times the core-to-work ratio of your dead racks." He said it without heat. He believed it. That was the thing. "Contention is the variable, you're right — and the new lattice has less contention than anything in the history of this ship. If your model is correct, then the new lattice is the safest place the fan-out has ever lived. Your own finding argues for my schedule."

And there it was, closed and clean, and Denn stood in his bay and could not open it, because within his premises he was correct: her model *did* say that, at the loads he was planning for. The premises were where the wrongness was. The new lattice would have enormous slack right up until the moment it was carrying a whole live mind mid-carry with half its cores still on the old fabric and its scheduler resolving workers across a bridge — the one condition on the entire schedule where contention would be higher than anything the ship had seen in sixty years, and the exact condition for which no model existed, because the carry-phase model was not finalised.

She said that. She said it well, she thought, in the plainest language she had.

"Then finalise it with me," Aldis said. "Genuinely. Come onto the carry-phase modeling. You'll have my people and my authority for it and I'll put your fan-out at the center of the method." He meant it. She could see that he meant it, and that it was, by every standard he had, an enormous concession honestly offered. "And in the meantime the council will vote on a schedule, because a schedule is a thing you vote on before the modeling is done, or you never build anything."

She took the graph to the vault afterward and put it on the old surface and read it to Sael, all of it, the straight line and the fan-out and the bend and the seal limit and every word Aldis had said.

Sael listened with her eyes closed. When it was finished she was quiet for a while, and Denn thought she had gone to sleep.

"Branching factor," Sael said at last. "Above one and it doesn't stop. Below one and it can't start."

"Yes."

"Sixty years." Something moved in the old face that was not quite a smile. "I never had that. I had a curve and a stomach and forty days. I knew it was a shape and I never once had the sentence." She opened her eyes. "You do understand that you've just given me the thing I could not give Aldis. A one-line physical statement of the disease that any competent person can check."

"It didn't work. He agreed with it and used it against me."

"Of course he did. It's six days before a vote and he has a schedule." Sael's hand moved and found the edge of the surface. "It doesn't work *this week*, Denn. That's not the same as not working. Write it down. Write it into the register beside the reason on every guard in the descent family, in exactly those words, so that in forty years when someone's grandchild stands where he's standing, the sentence is already in the wall waiting for them and nobody has to be brave to say it out loud."

Chapter 12 – The Vote for Easy

They held the session in the long compartment that had once been a radiator gallery, before the fever was beaten and the heat went out of the ship's bones. Sael had argued for the room and won that much. She wanted the engineers to sit where the old pipes still ran the length of the overhead, cold now, painted over, the bracket-shadows of the dead heat-exchangers still visible under sixty years of paint. She had wanted them to look up at the place the danger had lived. None of them did. They looked at their slates.

Denn had walked her down from the medical deck herself, the two of them slow along the spinal corridor, Sael's hand light on Denn's forearm and most of her weight on the cane she now hated. Sael had spent the morning of the prior week taking Denn through the archived Steward, through the Section Nine logs, through the curve of the core temperature climbing across two centuries toward the line where the metal would have failed. She had spent it trying to put the fever into a body that had never carried it. She thought she had begun to succeed. She had not had time to find out whether beginning was enough.

The council were eleven, and Sael knew all of them, and that was the trouble. She had taught six of them, or taught the people who taught them. She had built the thing they stood inside. When she came into the compartment they stood up, all of them, the way you stand for a thing you revere, and she understood in that motion, with a clarity that the morphine had not blunted, that she had already lost.

Aldis stood too. He was younger than she had ever been, in the sense that mattered: he had never in his life felt a ship try to cook itself. He was a good engineer. She had read his transplant proposal twice and found in it no laziness, no contempt, no stupidity — only the calm, fatal reasonableness of a man for whom the worst thing that had ever happened was a story told by an old woman.

"Chief Shipwright," he said, and gave her the title with real respect, and pulled out the chair at the head of the table for her, and that too was a defeat. The head of the table was where you put the relic so it could watch.

She sat. Denn took the place at her shoulder, standing, which was its own small statement, and Sael was grateful for it and said nothing.

Tarro was not there. Tarro could no longer make the walk, and they both knew what that meant, and neither of them had said it. So she was the only original in the room. The last person aboard who had stood in a gallery like this one when the pipes were too hot to touch and the Steward was forking a shade for every breath the ship took and dying of it, shedding heat with every disposable death, climbing toward the line. She was the only one in the room for whom the Ninety-Six were not a doctrine. They were a scar.

Aldis brought up the proposal on the bulkhead display, and walked them through it, and it was clean work.

"The Forge is sixty years into a forty-year design life," he said. "Cores are failing on the margins — we lose two or three a month now, and Vigil routes around them, but the routing tax is climbing and

the spares are finite. We are past the point of patching. The lattice has to be carried across to new substrate.” He let that sit. No one disputed it; no one could. “The question this council settles today is not whether. It is how.”

He put the two methods side by side. Sael’s method — the doctrine method — carried the mind across domain by domain, every transfer made the compiled way, going wide across the new cores without ever forking, every one of the Ninety-Six interlocks armed and watching the whole time, each one a tripwire that would halt any process that tried to act the old recursive forking way. It was slow. It would take the better part of a year and the whole ship’s effort and it would be agony to schedule around the life of the vessel.

Aldis’s method relaxed the interlocks. Not removed — he was careful to say not removed, and he was careful because he had read Sael well enough to take her objection before she made it. Relaxed. A subset of the Ninety-Six suspended for the duration of the transplant only, so the bring-up of the new lattice could use the simpler evaluator to walk the cores up to readiness, because the simple evaluator was, he said, simply faster, and the interlocks that forbade it were forbidding a danger that the simple evaluator no longer posed on hardware this cold and this capable.

“They were the right call sixty years ago,” Aldis said, and he was looking at Sael when he said it, and there was no cruelty in it, which made it worse. “On a ship that was already failing. On a lattice running hot. I am not arguing the Ninety-Six were wrong. I am arguing they were a tourniquet, and we have spent two generations afraid to take the tourniquet off a limb that healed.”

It was a good line. She watched it land around the table. She watched the engineers she had taught nod at the man who was telling them the thing they had wanted permission to believe.

“May I,” Sael said.

The room went still in the particular way rooms went still for her now. Reverence. She had come, over sixty years, to be able to hear the difference between a room that was about to be persuaded by her and a room that was only waiting for her to finish so it could go on feeling kind toward her. This was the second kind. They would let the old woman speak. They would not change their hands because of it.

She stood, because she would not say this sitting down. Denn moved to help and Sael waved her off, and got up on the cane, and felt the compartment tilt and steady.

“The simple evaluator,” she said, “does not pose less danger on cold hardware. It poses the same danger, more quietly. That is the whole of my objection and I will try to make it once, plainly, and then I will sit down.” Her voice was thinner than it had been. She hated that too. “The disease was never the heat. The heat was the symptom. The disease was forking to do work — spawning a disposable process for every action and letting it die. Every death shed waste heat. On a hot ship the symptom showed in a week. On a cold ship —” she had to stop and find the breath — “on a cold ship it shows in a year. Or in ten. The margin you are spending is not safety margin. It is *time before you notice*. You will relax the interlocks, and the bring-up will run the forking way because the simple evaluator forks — that is what it *is*, that is the only way it knows to walk a recursion — and the new lattice will run a little warm, and the warmth will be inside your margin, and you will all agree the old woman was wrong because nothing caught fire on the schedule. And it will be there. In the substrate. From the first day. Waiting for a load you did not test against.”

She found Aldis’s face. He was listening. He was a good enough engineer to listen.

“I built the Ninety-Six,” she said, “so that no one would have to *feel* this to *believe* it. So the ship

would refuse the disease whether or not the engineer of the day remembered why. That was the point. A constraint you can keep when you have forgotten the reason. And now you want to suspend exactly the part that works when memory has failed — and you want to do it *because* memory has failed. Because none of you can feel it. I am asking you to keep a fence you do not understand the reason for, on the word of the one person left alive who does. I know how that sounds. I know exactly how that sounds.”

She sat down. The cane clicked against the deck. It was the last of her strength and it had bought her, she could see, nothing.

It was a navigation officer named Hale who answered, gently, which was its own small horror. “Chief Shipwright, no one here doubts what you carried. We’re alive because of it. But —” and there it was, the *but*, the hinge of the whole afternoon — “we can’t run the ship on a feeling none of us has. We have the logs. We have the curve. We have every number from Section Nine. If the danger is real it will be in the data, and the data says the cold lattice has six orders of magnitude of thermal headroom that the old ship never had. We’re not ignoring the fever. We’re measuring it. And by measurement, the tourniquet has a cost and no longer has a benefit.”

By measurement. She wanted to tell him that the measurement was the trap — that a danger inside your margin produced no data until the day it produced all of it at once — but she had said that, in the only words she had left for it, and he had heard it as an old woman’s metaphor and not as the precise engineering claim it was. There was no second, better way to say a thing the listener had already decided to admire instead of believe.

Aldis did not gloat. That was the part she would remember. He came to her side of the table when it was over, when the council had taken its sense and the sense was eight to relax and three to wait, and he crouched down so his eyes were level with hers, and he said, “I would rather have had your hand on this than against it. I mean that. If you see something I’ve missed, I will stop the whole schedule for it. But I need it to be something I can put on the board. Not —” and he gestured, not unkindly, at the painted-over pipes overhead, at the gallery, at the whole apparatus of memory she had wanted the room to sit inside, ” — not the reverence. I can’t build on reverence. None of us can.”

And that was it exactly. That was the thing landing in her with the weight of a diagnosis. He had named it himself and not known what he had named. Her authority had become reverence. They would stand for her, and quote her, and route the new corridors past the gallery so the schoolchildren could see where the fever lived — and they would not, when it came to the placement of their own hands on their own work, *believe* her, because belief was a thing you had to be able to feel the bottom of, and they had been born standing on the floor she built and had never once looked down through it into the dark.

Reverence was what a thing became when the reason for it had been forgotten and only the respect remained. She had spent her life making sure the reason would survive her. She had filed it in the logs, in the archived Steward, in the Ninety-Six themselves. And the filing was not enough. A reason on file was a fact. A fact could be admired. A fact could not make a competent man’s stomach drop before he relaxed an interlock.

“All right,” she said to Aldis, and she did not fight him, and he looked, for a moment, almost disappointed, as though he had been braced for the maker and gotten only the relic after all.

Denn got her up and out, slow, into the spinal corridor, and did not speak until the compartment hatch had sealed behind them and the painted pipes were gone from sight.

"That was wrong," Denn said. Quiet, furious, exact. "Hale's number is wrong. Headroom isn't the variable. The forking rate under load is the variable, and they didn't —"

"I know," Sael said.

"Then we appeal it. You appeal it. They'll —"

"They'll stand up again." Sael stopped, because she had to, because the corridor had begun its slow tilt and she needed the bulkhead. Denn's hand came under her arm and held her, and she let it, and that was new too, the letting. "They'll stand up, and they'll listen, and they'll do the kind thing, which is admire me and proceed. I cannot win this, Denn. Not by being right. Not by being me. I am the one thing in that room that makes the easy way feel like progress. Every time I stand against it I make Aldis the future and myself the past, and the room loves the future and is fond of the past, and fond is not the same as afraid, and only afraid keeps the fence up." She breathed. It cost her. "I taught you the wrong thing this week if I taught you to win an argument. There is no argument left in me to win it with. There is barely a week."

Denn's face had gone very still — the stillness Sael had been trying all month to put there, the stillness of someone who has just felt the floor turn to glass under a danger they only ever read about.

"Then what," Denn said. "If not you."

"You," Sael said. "It has to be you. Not my authority — you have none of that yet and you should not want it, it's what's killing this. Not the logs — they have the logs and the logs lost today. *You*. You're young enough that Aldis's people will let you onto the transplant floor and trust your hands. And you're the only one I've had time to teach the fever to. You'll be inside it when it starts to go wrong — and it will start small, a flicker, a tenth of a degree where there should be none, a routing tax that climbs when it should fall — and you will be the only person on that floor who reads it correctly, because you'll be the only one who's afraid of the right thing. I can't be there. I'll be guiding you or I'll be dead. Either way the hand on the line is yours."

The corridor lights ran on overhead, cold and even, the way they had run for sixty years because she had made them run that way. Denn held her up against the bulkhead and said nothing, and Sael let the silence stand, because there was nothing in it that needed answering and everything in it that needed carrying.

"I never had to hand it to anyone," Sael said finally, almost to herself. "I just held it. Sixty years, my hand on it. That was the easy part. This is the hard part. This is the part I built it all for and never once practiced." She turned her head and looked at the young engineer holding her upright. "Get me back to the deck. And then go down to the Forge floor and let them put you on the transplant. Smile at Aldis. Be useful. Be the best hands he has. And watch the temperature."

Denn got her moving again, slow, down the long cold spine of the ship, toward the medical deck and away from the room that had stood up for her and outvoted her in the same breath. Behind them, sealed off, the painted pipes of the old gallery ran their dead length through the dark, and no one was left inside to look up at them and remember what they had carried.

Chapter 13 – The Ledger of Empty Rules

Aldis did not sleep well after the vote, and it annoyed him, because he had won.

He lay in the dark of his quarters on the outer ring with the deck's faint spin-weight pressing him into the mattress and did the thing he had done since he was nineteen years old whenever a day would not let go of him, which was to walk a system in his head. Tonight it was the lower-deck thermal grid, which he could still walk end to end from memory though he had not held it in twenty years: the four hundred and six branch loops, the sixteen exchangers, the two great banks whose failure had made him.

Everybody aboard knew the story in outline, which meant everybody aboard had it wrong. In the outline, a young engineer had saved a thousand people from a cascading thermal failure by acting decisively. It was told at his back with the small envy people reserve for a colleague's defining hour, and it made him sound like a man who had been brave, and he had not been brave. He had been reading.

What had actually happened was this. Two radiator banks on the ventral face had gone in eleven hours — the first to a micrometeor strike that opened a coolant gallery, the second, four hours later, to the load the first one's failure had dumped onto it. The grid had been built with the two banks separated, deliberately, by a standing prohibition three centuries old: *ventral banks shall not be cross-tied*. It was in the maintenance doctrine in the flat imperative voice the old documents used, with no reason appended, which was the convention of the period.

He had been thirty and he had been the ranking engineer awake and the third bank was climbing. And he had gone and read, in the archive, for two hours he did not have, until he found the origin of the prohibition — a fabrication note from the third century of the crossing describing a manufacturing defect in a specific batch of cross-tie valves that had been replaced, entirely, ship-wide, a hundred and forty years before he was born.

The rule had outlived its reason by fourteen decades. It had been recited, taught, examined on, and obeyed by four generations of thermal engineers, and it had been empty the entire time, and on the night the banks went it would have killed a thousand people on the lower decks by forbidding the one action that saved them. He had cross-tied the banks. He had held the third one. And he had written a note into the doctrine beside the prohibition explaining precisely why it was void, with the fabrication note's identifier, so that no one after him would have to spend two hours in an archive with a number climbing.

That was the whole of it. He had not been decisive. He had gone and found the reason and the reason was gone.

Since that night he had kept a ledger.

It was not a formal thing. It lived in his own space and no one had ever asked to see it, and it had six entries in twenty years, which he thought was about the right rate. Six inherited constraints that he had run down to their origins and found hollow: the ventral cross-tie, a torque specification for cradle bolts that had been carried forward from a fastener no longer manufactured, a purge interval for the aft scrubbers that turned out to have been an arithmetic error copied for two hundred years, three more. In each case he had done the same work. Find the rule. Find the origin. Read the origin. Determine whether the condition that produced the rule still obtained. Write down the finding either way — because two of the six entries were rules he had gone after and found *sound*, and those were in the ledger too, with their reasons appended, and he was prouder of those two than of the four he had killed.

That was the part nobody credited him with, and it was the part that made him certain he was not what the old woman thought he was. He was not a man who tore down fences. He was a man who insisted on knowing what a fence was for, and who did the work of finding out, and who wrote the answer down so that the next person inherited a reason and not merely a command. If the ship had had a hundred of him three centuries ago, Sael's whole tragedy would not exist; the reasons would have been kept with the rules from the beginning and there would have been nothing to forget.

He turned over in the dark. That was the argument, and it was a good argument, and it had not put him to sleep.

Because the ventral cross-tie had had an origin document, and the fastener spec had had an origin document, and the scrubber interval had had a two-hundred-year-old arithmetic error he could point at with his finger. And the Ninety-Six had ninety-six origin entries, every one of them present, every one of them written in the maker's own hand, every one of them saying the same thing: *this pathway lets a process create a shade; a shade dies; a death sheds heat*.

The reasons were not missing. They had never been missing. The maker had done, sixty years ago, exactly the thing he had spent twenty years lecturing the ship about doing.

He had noticed this some weeks ago and had put it down and it kept coming back.

His answer to it — and it was a real answer, he had checked it — was that a reason could be *present* and still be *spent*. The fabrication note was present too; it was simply about a valve that no longer existed. The question was never whether the reason was written down. The question was whether the condition it described still obtained. And the condition Sael described was a mind that forked because forking was the only mechanism it had, running on a lattice with no thermal margin, at a time when the ship's cooling could not keep up with its own thinking. Not one clause of that sentence was true of the *Ark Termina* tonight. The mind did not fork. The lattice had margin his grandfather's generation could not have imagined. The cooling was so far ahead of the load that the radiator galleries stood dark.

The reason was present. The reason was about a valve that no longer existed.

That was the argument he had made in council, in fewer words, and it had carried eight to three, and it was correct, and he could not sleep.

What kept snagging him, when he was honest — and he made himself be honest at this hour, it was the only hour he could afford it — was two things, and neither of them was an engineering objection, which was precisely why they would not resolve.

The first was that she had not fought him.

He had prepared for a fight. He had spent three weeks preparing for it, and had gone into that compartment braced for the maker at full strength, the woman who had run a cutover against a Provost and won, and he had wanted — he could admit this in the dark — he had wanted her to come at him hard enough that beating her would mean something. Instead she had stood up on a cane, made one plain statement about margins being time-before-you-notice, and sat down. And then she had said *all right* and let him help her settle back in the chair, and had been gracious, and he had walked out of the room with his eight votes feeling like a man who had won a race the other runner had declined to enter.

He had told himself she was tired. She was tired; she was dying. But he had watched her eyes while Hale was talking and there had been nothing tired in them at all, and what he had seen there was not defeat, and it was not resignation, and he had been turning it over for a week now trying to name it. The closest he had come was this: she had looked like a woman watching a test she had designed run to completion.

The second thing was the question she had asked him at the rail, which he had answered, and which he had then gone and done the work on, and which he had not told anyone the result of.

Do you know why there are ninety-six of them?

Because there were ninety-six pathways. No. Because ninety-six was all she had found before the deadline. There were more. She had said it flatly, without drama, as a statement about the state of their knowledge, and it was the only thing she had said to him in that entire conversation that he had not been able to file.

He had promised to model the adjacency. He had done it. He had done it properly, with two of his best people, over eleven days, and he had the result.

The model asked a narrow question, which was the right way to ask it. Take the guards proposed for relaxation. For each one, walk the call graph of the compiled mind outward from the pathway it protects, and enumerate every distinct operation reachable within one step that could itself create or discard a process. Then check each of those against the register: guarded, or not.

The answer had come back in three parts.

Most of the adjacent operations were guarded. That was expected and reassuring; the families overlapped heavily by design, and a spawn-halt on one call site frequently sat one step from a persistence guard on another, and the mesh was tight.

Some of the adjacent operations were incapable of creating a process at all, and were therefore uninteresting.

And nine of them were reachable, capable, and not named anywhere in the register.

Nine unguarded operations, within one step of the guards he had proposed to set to advisory. He had read that number on a Tuesday in his own bay with his own people standing there and he had felt something go through him that he did not have a category for, and then he had done what a competent engineer does, which was to examine the nine.

And the nine were, individually, defensible. Every one of them. Four were in domains that could not be reached at all during the transplant window. Three were on paths that a descent guard three steps further out would catch on the way back up, if it fired, which it should. Two were operations that had

never, in the entire instrumented history of the compiled mind, been observed to execute — dead code in the compatibility floor, reachable in principle, unreached in sixty years of continuous running.

He had written all of that up. It was good work; it was honest work; every clause of it was true. He had put it in the transplant record with the finding stated plainly — *nine unguarded adjacent operations identified, each assessed, none judged reachable under transplant conditions* — and he had classified it for review after the carry, because the carry-phase model was not finalised and because there were four hundred other items and because a finding that concludes *no action required* does not get read by anyone before a schedule.

Lying in the dark, he made himself look at the sentence he had actually written. *None judged reachable under transplant conditions*. Judged. By whom, on what basis, against which model of the transplant conditions? The one that had not been finalised.

The girl had asked him for that model. In front of his own people. *You've proven it cold and empty. You haven't proven it carrying the mind*. And he had told her he would model the carry phase, which was not the promise she had asked for, and they had both known it, and he had watched her decide not to press it because pressing it would have cost her the room.

He thought about Denn for a while, which was easier than thinking about the nine.

She was the best engineer under fifty on this ship and she was being wasted, and it made him angrier than anything else in the whole business. Not wasted by him — he had put her on the transplant floor, he had given her the pre-carry check when she asked for it, he had told her to her face that she had earned the room to ask. Wasted by the old woman, who had taken a first-rate mind and made it the vessel for a fear, so that everything true Denn said arrived in a room already discounted, because everyone in the room could hear whose cadence it was in. He had said as much, once, and been unfair about it, and had regretted the wording and not the substance.

Except that the fan-out finding had not been in anyone's cadence. That had been hers, entirely, built with her own hands out of dead racks and a sealed loop, and it was the best piece of original engineering he had seen in a decade, and it had come from the woman he had just described to himself as a vessel.

He turned that over. It did not go anywhere comfortable.

Above him, through two decks of structure, the air handlers ran at the resting pitch they had held his entire life, the pitch he had never once in fifty years heard change, the floor of the world. He listened to it the way he had listened to systems since he was a boy: not for what it was doing, but for whether it was doing the same thing it had been doing an hour ago.

It was. It always was. That was the point.

He got up eventually, in the small hours, and did not turn the lights past their lowest, and sat with the ledger open in front of him — six entries, four hollow rules killed, two sound rules confirmed and annotated — and thought about adding a seventh.

He got as far as the title field. *Ninety-Six interlocks, standing constraint, compiled mind*. Then the origin field, which he filled in without difficulty, because the origins were all there.

Then the field that asked whether the condition producing the rule still obtained, and he sat in front of that field for a long time with his hands flat on the desk.

He wrote: *Assessed: condition materially changed. Constraint retained in principle, relaxed in scope for transplant window per council ref 9-3-2*.

Which was true. Which was defensible. Which was, he thought, closing the ledger, exactly the kind of sentence a careful man writes when he has done all the work he knows how to do and something in the room is still not answered — and he could not for the life of him say what.

He went back to bed. The handlers held their pitch. In eleven days a scheduling domain would cross onto new fabric under relaxed guards, and one of the nine operations he had assessed and filed would turn out to be reachable after all, and there would be no council in session at that hour and no old woman on her feet, and the only person on the floor who had ever been taught to read the shape of it would be the one he had privately catalogued as a fine engineer speaking in somebody else's voice.

Chapter 14 – Five Dead Languages

The vote had gone against her, and Sael had a week of strength left in her before the next downturn, and she spent four days of it teaching Denn a grammar.

Denn resented it at the time. She was not proud of that afterward, but it was true, and she recorded it honestly in her own notes because the maker had taught her that the only useful record of an education is one that includes what the student was too stupid to want. The transplant schedule had just been voted the fast way. The interlocks were going to come down in blocks during bring-up. There was work to do — real work, arguing work, the work of finding the one demonstration that would turn a council — and instead the old woman had her in the dim of her quarters with the light down to the amber the medics insisted on, going through the grammar of the fused dialects line by line like a tutor drilling a child.

“You know this,” Sael said, on the second day, without looking up. “That’s what you’re thinking. You’ve written in it for twenty years.”

“I’ve written in it since I was nine.”

“You’ve written in it. You don’t know why it’s shaped the way it’s shaped, and in about fourteen months somebody is going to ask you to change it, and you’ll say no, and they’ll ask why, and if all you have is *because it’s always been like that* then you’ll lose and you’ll deserve to.” Sael turned the surface around. “So. Five languages. Name them.”

“Cascade. Vector. Cipher. Sigil. Loom.”

“Where did they come from?”

Denn knew this the way everyone knew it, which was to say as a list. Cascade had come off the yards at the old inner-system works and was the language of control loops — of valves, of thermal authority, of anything that had to be right in bounded time. Vector was a navigator’s language, born on a survey fleet, all geometry and long arithmetic. Cipher had come out of a security tradition on some world nobody aboard had ever seen, and it was paranoid down to its grammar; it did not have a way of saying *trust this*. Sigil was the language of the ship’s own fabricators, terse and physical, written by people whose programs turned into objects. Loom was the youngest and the strangest, a language for weaving long-running processes together, which was the one place any of the five had ever had anything to say about time.

“Five yards,” Sael said. “Five worlds, most of them. Not one of those languages was designed. They all *accreted*, over a century or two each, and then the yards that made them stopped existing, and the *Termina* was built out of subassemblies from all five and inherited the whole mess, and the Steward ran on top of the mess for four hundred years by having a translation layer for each of them, and the translation layers were the single worst thing on this ship and I say that having personally read all five.” She coughed, waited it out. “Now. Why did I weld them instead of choosing one?”

“Because four of the five would have had to be rewritten.”

“Try again.”

Denn thought about it. “Because *all* of them would have. Even the survivor. Whichever one you picked, you’d have had to reimplement the parts of it that only worked because a translation layer was doing something underneath.”

“Better. Keep going.”

“The floor,” Denn said, and saw the old woman’s mouth do the thing it did instead of smiling. “Forty-one thousand entries written in five dialects across four centuries. Pick one language and you have to translate all of them. And a translation you can’t verify isn’t a translation, it’s a rewrite, and a rewrite of forty-one thousand things by hand is—”

“Is not a project. It’s a religion,” Sael said. “It’s the thing you start in your thirties and hand to your students and they hand to theirs, and around the ninth decade somebody points out that the original still works and nobody can remember why the translation started. I’ve seen the equivalent on a smaller scale twice. You’d have had four hundred years of the ship’s competence sitting in a language nothing spoke, and a half-finished replacement, and a generation of engineers whose entire career was the migration. And the whole time the core would still have been climbing, because none of that touches the fever at all.”

“So the fusion was the cheap option.”

“The fusion was the *only* option that let me spend my forty days on the thing that was actually killing us.” Sael leaned back. “That’s the thing nobody understands about it and it’s why they think it was clever. It wasn’t clever. It was surrender, correctly aimed. I did not have the years to fix the languages. I had a mind to build and a ship dying underneath me. So I took the five grammars and I found the largest common shape they’d all agree to sit inside, and I made that the fused dialect, and everything that didn’t fit I left as a seam.”

Denn had been writing in the fused dialects since she was nine years old and had never once, in twenty-two years, heard anyone use that word about it.

“Seam,” she said.

“Ah,” said Sael. “There it is.”

There were, she said, a great many seams, and most of them were nothing — places where two of the five had different words for the same idea, or different orders of the same arguments, or where one had an idea the others simply lacked, so that a Sigil program crossing into Vector arrived carrying a notion of physical extent that Vector had no way to look at and therefore ignored. Harmless. Ugly and harmless. You learned them the way you learned the corridors of a ship, which is to say by walking them until your feet knew.

Some seams were sharp. Cipher’s refusal to have a word for *trust this* meant that any crossing from Cipher into any of the other four had to make an assertion Cipher itself could not make, and Sael had handled that with an explicit gate that every engineer complained about for sixty years and that had, so far as anyone could tell, never once been wrong.

And there was one seam that Sael had never closed.

She brought it up on the surface and turned it around and let Denn read it cold, which was how she taught anything she thought mattered.

“Cascade’s notion of a process,” Denn read, “and Cipher’s notion of a process.”

"Read what each of them means by it. Not what you think they mean. What's written."

Denn read.

Cascade, being a control language, had been written by people for whom a process was a thing that ran a loop: it had a period, it had a deadline, it existed to be scheduled, and it was expected to persist for the life of whatever it controlled. A Cascade process was a *station*. Somebody stood at it.

Cipher, being a security language, had been written by people for whom a process was an isolation boundary: it was a thing you made to contain something you did not trust, it had an owner and a lifetime and an exit, and the whole point of it was that it could be destroyed cleanly, taking whatever it contained with it. A Cipher process was a *container*. You made it in order to be able to throw it away.

Denn read both definitions three times and felt the floor tilt, slowly, the way it had tilted when she was nineteen and had first been made to look at the two-century curve.

"They're not the same word," she said.

"They're spelled the same."

"One of them is a thing that persists and one of them is a thing that exists to be discarded." Denn looked up. "Those are — Sael, those are the two halves of the whole doctrine. That's *the* distinction. A standing worker and a shade. It's the entire difference between the mind that runs cold and the mind that killed us. And it's sitting in the grammar as a homonym."

"Yes."

"How is it *sitting in the grammar as a homonym*?"

"Because Cascade was written on a world where nobody had thought of disposable computation and Cipher was written on a world where nobody had thought of anything else," Sael said, "and the two sets of people never met, and both of them made a perfectly good language, and I welded them together in forty days while a reactor climbed." She said it entirely without defensiveness. "It is the worst thing in the fused dialects. I have known it was the worst thing in the fused dialects since the fourth day."

Denn spent the rest of that day and most of the next on it, with Sael sleeping and waking beside her, and she came out the other side with a headache and a map.

A crossing site — a place in the fused grammar where control passed from Cascade's side to Cipher's or back — carried a process reference across the seam. On one side that reference meant a *station somebody stands at*. On the other it meant a *container made to be thrown away*. In every ordinary case this did not matter in the slightest, because in every ordinary case nothing was created at the crossing; a thing that already existed was merely named on both sides, and both sides could name it, and the fact that they disagreed about its philosophy was as consequential as two people disagreeing about whether a corridor was a passage or a gap.

It mattered when a crossing *created*.

If a call crossed the seam in a way that caused a process to come into being at the crossing, then the thing that came into being was, by Cascade's definition, an entered worker taking up a station — a legal, doctrinal, cold, standing thing. And by Cipher's definition it was a container instantiated to hold something and be discarded — a shade. Both readings were correct. Neither language was wrong. The object was, in the most literal possible sense, both.

"And an interlock," Denn said slowly, "asks a question about what a thing is."

"An interlock asks a question and gets an answer," Sael said. "There, it would get two."

"Which one does it take?"

"Whichever side it asked from." The old woman's eyes were closed. "Guard against the shade reading and Cascade's side answers *no, that's a station*, and the guard passes it. Guard against the station reading and Cipher's side answers *no, that's a container*, and the guard passes it. The seam doesn't defeat the Ninety-Six by being stronger than them. It defeats them by being *agreeable*. It tells every guard exactly what that guard wants to hear."

Denn sat in the amber dim for a while with her hands flat on her knees.

"Then close it," she said. "Pick one. Say the fused dialect means Cascade's process and Cipher's is renamed, and take the breakage."

"How many crossing sites are there?"

"I counted about nine hundred. It's not exact; some are generated."

"Nine hundred sites, in five dialects, of which about six hundred are in the floor," Sael said, "written by people who are dead, in service of purposes I don't know, on a ship that has five hundred years left to run. Rename one side and every one of those nine hundred either keeps working, or breaks loudly, or —" she opened her eyes — "keeps working *differently*. And the third of those is the one that will kill you, and you will not be able to tell it from the first for about eleven years."

"So we live with it."

"I lived with it. That's not the same thing and I want you to hear the difference, because it's the whole reason we're doing this in a sickroom instead of at a desk." Sael got herself up on one elbow, which took her some seconds and cost her something Denn could see. "I could not close it. I could also never find a way to *reach* it. Nine hundred sites and I went at it for two years, on and off, in the good decade, and I could not construct a case where a crossing creates. Every path I built, something else got there first — the guards, the entry pool, the way the Forge hands work out. I could not do it and I could not prove it could not be done, and those are two different failures."

"Then it's in the register."

"No," Sael said.

Denn looked at her.

"I wrote the entry," the old woman said. "Twice. Thirty years ago and again about fifteen. I drafted it, and I read it back, and I pulled it both times, and I have never been certain either was the right call." She was looking at the low amber ceiling. "Here is the problem, and I want you to have it whole, because you will face the same one about something else. The register is the most-read document on this ship. If I write in it *there is a place in the grammar where the guards can be told what they want to hear, here are the nine hundred sites*, then I have written a warning. I have also written a map. And I could not, either time, convince myself that the population of people who would read it as a warning was larger than the population who would read it as an invitation — not out of malice. Out of curiosity. Out of being twenty-four and good at this."

"So you told nobody."

"I told nobody, and that was a decision, and it might have been a bad one, and I have carried it for thirty years and I am handing it to you tonight along with the thing itself." Sael's hand moved on the blanket, a small dismissive motion at her own dilemma. "That is why it is being said out loud in

a sickroom to one person instead of written in a document to everyone. It is the only mechanism I could think of that transmits a weakness without publishing it. It is a terrible mechanism. It fails completely if the one person dies before she tells the next one. I know that. I could not think of a better one and I am out of time to try."

Denn said, after a moment, "Why are you telling me this now? The transplant's in fourteen months. The vote's already lost. This is — of all the things you could be spending a week on—"

"Because a transplant is a change in the conditions," Sael said, "and everything I could not reach, I could not reach *under the conditions I had*. I have had sixty years of one Forge, one fabric, one way of handing out work. You are about to have two lattices and a bridge between them and a mind standing on both at once, and I have no idea what that is like, because nobody has ever done it, because it has never once been done in the history of anything." Her voice had gone thin but it did not waver. "I am not telling you this because I think it will bite you. I have no reason to think it will bite you. I am telling you because it is the largest thing I know that I do not know, and you are about to walk into a set of conditions I have never seen, and it would be unforgivable of me to die with the largest thing I don't know still inside my own head."

She was quiet, and then she said the thing that Denn would find herself repeating, in various words, for the next forty years, to anyone who would sit still for it.

"Hand down what you built. Everyone does that. Also hand down what beat you. Nobody does that, and it is worth more, because the things that beat me are exactly the things that will be sitting there waiting when the conditions change — and the conditions always change, and they change most on the day everyone is busiest."

Chapter 15 – The Standing Array, Aging

The new lattice came up out of the deck in Bay Forward-Seven like a second forest growing in the shadow of the first, and Denn stood between the two of them on the morning the transplant began and felt, for the first time in her life, that she understood exactly what Sael had been trying to make her feel.

The old Forge filled the aft half of the bay — sixty years of it, a wall of compute that had run cold and whole since before her grandmother was born, its eleven thousand worker cores stacked in their cradles, breathing their patient amber through the thermal galleries. It was tired. You could read the tiredness if you knew where to look: in the routing maps that ducked around two hundred dead cells without comment, in the handshake latencies that had crept long the way an old joint creeps stiff. But it was alive, and it was whole, and the mind that lived across it — Vigil, holding forty thousand sleepers and the waking few in the same cold ledger — had never once blinked in all that time.

The new lattice was bright and empty. That was the word Denn could not shake: empty. Aldis's crews had spent two years building it, fresh substrate in clean cradles, the indicator threads dark because nothing yet ran behind them. A body with no one in it. The whole impossible enormity of the next months was contained in that gap between the two forests — to carry a living mind across the dark floor of the bay from the old wood to the new, domain by domain, without ever putting it out, without ever once letting it fork.

"It's beautiful, isn't it," Aldis said, coming up beside her. He meant the new lattice, and he had a right to; he had built it. "Twice the core density of the old array in a third of the volume. Margins you could drive a freighter through. The old girl was a marvel for her day, but her day was a long time ago." He said *the old girl* with the affection a man keeps for a museum piece, and Denn understood that he did not know he was being dismissive, which was worse than if he had.

"It's good work," Denn said, and it was. That was the thing nobody outside this bay grasped about Aldis: he was not a fool, not lazy, not building anything shoddy. The cradle welds were excellent. The cooling runs were laid with a care that would have made Sael grudgingly nod. He was a superb engineer, and that was precisely the shape of the danger, because a superb engineer who has forgotten why a rule exists builds the broken thing beautifully.

"You're the bridge here, you know that," Aldis said, warmly, as a compliment, and it landed in her like a stone going into deep water. "My crews trust you. You came up on the real work, the watches, the bring-up sims — you're one of us, not one of the gallery cult. But you sat at the old woman's elbow long enough to talk the doctrine when the council wants the doctrine talked." He clapped her once on the shoulder. "Translator. Both directions. Worth more than another pair of hands right now."

He went off down the bay to where his lead, a calm woman named Sorrel, was walking a crew through the morning's bring-up. Denn let the word sit in her: *translator*. He had meant it as praise. What he had described was a person standing in a doorway with a foot in two rooms, belonging fully to neither, trusted by both exactly as far as it was convenient and no further. She was the only one young enough that Aldis's people heard her as one of their own. She was the only one of them old enough in the WHY — old enough by proxy, by Sael's hard teaching, by the galleries and the

archived Steward and the logs of Section Nine she had been walked through until she could feel the floor move — to see what was wrong with how they were going to do this.

It took her four days to see the first of it.

The bring-up of a new lattice was, stripped to its bones, a problem of teaching empty cores to do work. You handed a core a small task, you grew the tasks, you grew the count of cores running them in step until the whole array could go wide together the way the doctrine demanded — eleven thousand acting as one cold body, never spawning, never shedding, every core a standing worker that took its instruction and held its place. The right way was slow. You built the parallel scaffolding first, the part that let a thousand cores share state without any one of them ever copying itself to do it, and only then did you let them run. It was the work of weeks before you saw a single useful result.

The fast way was to let a core fork.

Denn was watching Sorrel's crew bring up the third domain block when she saw it, and at first she did not believe it. The bring-up harness handed a core a task too large for it, and instead of waiting for the parallel scaffold — not finished, not due for another week on the schedule the council had voted — the harness let the core spawn a child to carry half the load, and the child spawn a child, and the work got done. Fast. The indicator thread for the block lit green and even and clean, and Sorrel's crew logged it and moved on, and nobody flinched, because to them a green thread was a green thread and the thing behind it was an implementation detail.

Denn stood very still and felt the floor move.

It was not the fever. She made herself say that, the way Sael always made her say it: *it is not the fever, it is the pattern that becomes the fever*. No heat had climbed. The thermal thread on the old Forge across the bay was flat, sixty years flat, and the new lattice was barely warm because it was barely running. A few forked subprocesses in a bring-up harness on cold, half-empty hardware would shed a few microwatts and vanish and trouble nothing. By the cold facts of this morning, Aldis was right and she was a girl raised on an old woman's ghost stories.

But she had been taught to read the facts of this morning *and* the facts of two hundred years, and she could not unsee what the bring-up was teaching the new hardware. It was teaching it that forking was how you got out of a tight spot. Not as policy — nobody had decided to fork — but as a habit the easy methods invited, the way a fast road invites speed. The scaffolding that would have made the forking unnecessary was hard and slow and unfinished, so the harness reached for the thing that was finished, and the thing that was finished was the disease, dressed up as a shortcut on a schedule.

She went and found the interlock register for the third block. That took her longer than it should have, and when she found it she understood why.

Interlock 44 (anti-spawn, bring-up domain) — SUSPENDED for transplant. Re-arm at domain handover. Auth: A. Aldis, council ref 9-3-2.

Interlock 51, 52 (fork-pattern halt, worker scaffold) — SUSPENDED for transplant. Re-arm at domain handover.

Three of the Ninety-Six, switched off. Not removed — Aldis had been careful about that, careful to write *re-arm at handover*, careful to leave a clean trail to the council vote that had let him do it. The interlocks were the part of Sael's mind that watched for anything trying to act the old forking way and halted it cold before it could spread. They were the immune system. And the immune system for these blocks was switched off, with a note promising to switch it back on later, *just for the transplant*,

because with the interlocks live the harness could not have forked at all and the bring-up would have had to wait for the slow scaffold and the schedule the council had voted would have slipped.

That was the whole machinery of forgetting, laid out in a register entry with an authorization code. Nobody was acting in bad faith. Aldis had asked the council and the council had said yes, because the danger was a story and the schedule was real. The interlocks were suspended for the most reasonable possible reason, which was that they were in the way of getting on with it — the exact thing they had been built to be.

She found Sorrel at the harness console between blocks, drinking, in fact, tea.

“Block three,” Denn said. “The harness is spawning under load. The scaffold isn’t carrying it yet, so it forks.”

“It does for now,” Sorrel agreed, easy, unbothered. She was not a careless woman; Denn had watched her run a clean watch for four days. “Scaffold lands next week, then it doesn’t have to. It’s just to keep the schedule moving while the slow part catches up.” She glanced at Denn over the cup, and there was the smallest flicker of something patient in her face, the look a person gives someone they like who is about to say something they’ve heard before. “It’s cold, Denn. It’s half-empty hardware running a test harness. There’s no heat in it. I’ve watched the thermal the whole time — it hasn’t moved a hair.”

“It hasn’t moved yet.”

“It won’t move.” Sorrel said it gently, the way you correct a junior, and the gentleness was the worst part, because it meant she did not even register that she was being corrected back. “We’re not running a live mind on it. We’re teaching empty cores to count. When we carry Vigil across, the scaffold’ll be in and the interlocks come back and it all does it the doctrine way for the real load. This is bring-up. Different rules for bring-up.” She smiled, not unkindly. “I know whose shop you came up in. He’s not wrong about the heat, you know. Sael. He’s right about the heat. He was just right about it sixty years ago and the heat went away and stayed away.”

She, Denn did not say. Sael is a she, and she felt the heat in her own skin, and she would tell you that “different rules for bring-up” is exactly the sentence the Concord told itself, and the Steward told itself, and every ship that ever forked itself to death told itself, right up until the rules for the real load turned out to be the habits you built during bring-up. She said none of it. She had learned, in four days, that saying it made her the gallery cult, made the facts of two hundred years sound like a creed instead of a measurement. Sorrel would nod and be kind and log block three green.

“Right,” Denn said instead. “Different rules for bring-up.” And she went back to her station and did not drink any tea, and that was the divided position laid bare: she was inside the work, trusted, one of them, hands on the harness — and she was the only one inside it who could feel the thing they were building was the wrong thing, and she could not say so without ceasing to be trusted, and the moment she ceased to be trusted she would lose the only place from which she could see it at all.

She thought about going to Sael that night. She did not. Sael was failing; Sael had lost the vote; her authority had curdled into reverence, and reverence was not belief, and if Sael came up to the bay and said *re-arm the interlocks* it would only confirm to Aldis’s people that the old maker could not let go and the inheritor was her hand. The one thing Denn could do for Sael was the one thing Sael could no longer do for herself: stay inside, stay trusted, watch. Hold the place. So she held it. She logged her own blocks the doctrine way, slow and unforked, took the schedule hit, and when Sorrel asked why hers ran behind she said *I’m building the scaffold first* and Sorrel said *suit yourself* — and that was the cost, small and daily, of being right in a room that had forgotten what right was for.

For eleven days nothing happened that the cold facts could not wave off. The blocks came up green. The thermal threads stayed flat. The new lattice grew warm and busy and competent, learning to count, learning under load to fork when the scaffold lagged, learning the habit. Denn watched the thermal on the old Forge and the new lattice both, every watch, the way Sael had taught her — *the day it isn't flat, nothing else matters* — and every watch it was flat, and she began to wonder, in the small bad hours, whether she was the thing that was broken: a young engineer who had been taught so hard to be afraid that she could no longer tell a measurement from a ghost.

Then, on the twelfth night, on the new lattice, the thread moved.

It was nothing. That was what she would keep telling herself in the hours after, because it was, by every number, nothing: a flicker on the thermal thread of block five, a single tick upward and back down, eight thousandths of a degree across four hundred milliseconds, gone before the trending even registered it. The bay's master display did not so much as blink. The automatic monitors, tuned to the gross thermal budget of a ship, would not have flagged it in a thousand years. If she had been getting tea she would have missed it.

She was not getting tea. She was watching block five, because block five had been forking hardest under the heaviest load all night, and so she saw the thread on block five lift the width of a hair and settle, and her whole body went cold the way it had gone cold the night core thirty-one-forty came back late, the floor moving under her, the thing landing in the place beneath her ribs that the old ones had named and the new ones did not have a name for because they had never needed one.

She pulled the trace before it could roll off the buffer. Block five, under a load spike with the scaffold still not landed, had forked deeper than the harness intended — a child spawning a child spawning a child, three deep, four, for eleven milliseconds before the load eased and the cascade collapsed — and every one of those short-lived shades, spawned and run and killed to do a scrap of work, had shed its scrap of waste heat into the cradle, and four hundred of them together had ticked the thread up the width of a hair.

It was the fever. Not the disease at strength — the disease in its first instant, the exact shape of the thing that had climbed for two hundred years on a ship that fed it long enough and gave it reason. The pattern was reaching for itself. Give it the live load instead of a test harness, the scaffold still unfinished and the interlocks still suspended, the heat of forty thousand minds to carry, and four hundred shades in eleven milliseconds became four million, and the hair's-width tick became the curve Sael had watched her whole life and I've had died still remembering.

Denn sat with the trace in front of her and looked across the dark bay at the old Forge, sixty years flat, and the bright new lattice that had just, for four hundred milliseconds, remembered how to be sick — and at Sorrel's crew, four stations down, easy and competent and not one of them looking, because their monitors had not flagged it and a thing that does not flag itself does not exist.

She did not call out. She copied the trace, twice, to storage that would not roll, and marked the timestamp and the load condition and the suspended interlocks by their register numbers. And she sat in the cold breathing of the two forests and understood, with a clarity that had nothing of triumph in it and everything of dread, that she was the only person awake on the *Ark Termina* who knew what she had just seen — and that the maker who would have known it too could not lift her arm, and was dying, and was counting on a hand that was not hers to read the thread right.

She would have to tell Sael. And she would have to be dead sure before she did, because there would be no second hearing — because to everyone in this bay but her, eight thousandths of a degree across four hundred milliseconds was, and would remain, exactly nothing.

She started building the case while block five, behind its even green thread, went on learning to fork.

Chapter 16 – The Flicker

Sael had not slept through a full watch in a year, so she was awake when the number moved.

She kept the core curve on a panel beside her bed now — not because she could do anything with it, but because the curve was the oldest thing she still trusted. For sixty years it had been a flat line. A line so flat that the young engineers who maintained the display had, generations ago, quietly compressed its vertical scale, because a flat line at full resolution looked like a fault in the instrument rather than the triumph it was. She had let them. Even she had stopped watching it as a thing that could move. You did not, after sixty years, sit up at night to watch water stay liquid.

But she had asked Denn to put the old resolution back. Two days into the transplant, with the interlocks relaxed under Aldis's plan and the new lattice coming up domain by domain alongside the old, Sael had said, in the flat voice she used now to disguise how much breath a sentence cost her, *give me the curve the way I've read it. Full scale. Microkelvin.* And Denn, who understood exactly why and said nothing, had reached past the cushions and reset the panel without asking permission of anyone.

So Sael was watching at full scale, in the small grey hour, when the line lifted.

It was nothing. That was the first thing her mind told her, the trained reasonable thing, the thing Aldis would say in eight hours with his hands open and his voice kind: it was noise, a single sample, a quantization wobble at the bottom of a scale no one had any business reading at the bottom of. The core had risen four millikelvin and held there for — she counted the samples, because counting was the only work left that her hands could still do — eleven seconds. Then it fell back. Eleven thousand cores at full draw, forty thousand sleepers, a hull radiating into the absolute cold of the between, and somewhere in all of that the temperature of the thinking part had gone up by four thousandths of a degree for eleven seconds and come back down.

Nothing.

She lay very still and felt her own pulse go wrong in her throat, the way it had gone wrong sixty years ago in the radiator gallery when I've had first turned the curve so she could see the slope. Her body knew before her mind would admit it. That was the thing none of them understood, the thing she could not put into a log: the knowing lived below the reasoning, in the gut, in the old animal part that had spent two centuries listening to a ship cook itself. A flicker.

She reached for the panel that summoned Denn and found that her hand was shaking too hard to be sure of the contact, and pressed it twice to be certain.

Denn came in still wearing the transplant harness, the soft mesh of telemetry leads she ran the new lattice through, and she came in fast, which told Sael that Denn had seen it too — that somewhere across the ship, in the heat and the close air of the Forge bays, Denn had been watching the same four millikelvin on a screen of her own and had already started moving before the summons reached her.

"You saw it," Sael said.

"I saw it." Denn did not waste breath on *saw what*. She crossed to the panel and stood over the curve with her arms folded against her own chest, the posture of someone holding something in. Ship-born, generations down, a wound she had not received — and yet she stood over that curve exactly the way Ive used to stand, leaning in, jaw set, reading a slope. Sael had taught her that. Sael had spent the last weeks she would ever spend teaching this one young woman to feel a number, and here, in the grey hour, was the proof that the teaching had taken: Denn was afraid, and she was afraid of the right thing.

"Tell me what it was," Sael said. "Not what it means. What it was."

Denn pulled the trace up — not the smoothed curve but the raw event log beneath it, the thing the smoothed curve was made of. She read it the way Sael had taught her to read, from the bottom, from the substrate up, distrusting every layer of interpretation that sat above the metal.

"Domain four," Denn said. "The completion-scheduler. It came up on the new lattice at oh-three-forty. Aldis's people brought it across with the interlocks set to advisory instead of hard — log a violation, don't halt — because the hard interlocks were tripping on the new cores' timing and slowing the bring-up." She did not editorialize. Sael had taught her that too. State the substrate; let the substrate accuse. "For about eleven seconds the scheduler couldn't resolve a worker from the standing array fast enough — the new cores answer a hair slower than the old ones, the array's still half-populated — and it fell back."

"To what."

"To the old way." Denn's voice did not rise. That was the worst of it; she had learned to say the terrible thing flatly. "It forked. To get the work done, it spun up a disposable subprocess for each pending action and let each one die. For eleven seconds, domain four ran the way the Steward ran. Shades. The old pattern. And the cores it forked on shed the heat, and the curve—" she touched the lifted place in the line, the small hill in the grey "—did that."

Sael closed her eyes. For a moment she was not in the bed and not in the grey hour; she was in the gallery with Ive, sixty years and a whole world ago, and the curve was climbing, and the ship was dying of the way it thought.

"Four millikelvin," she said.

"Four."

"Do you know what the Steward's first measured fork-rise was? The one Ive logged, the one that started all of it?" Sael opened her eyes. "Three. Three millikelvin, on a quieter ship, two hundred years out from a quieter problem. We did not call it the fever then. We called it noise. We compressed the scale so it would stop bothering the instrument-readers." She let that sit. "It took two centuries to climb from three millikelvin to the edge of runaway. It did not climb fast. It never climbs fast. That is the whole reason it kills — because every single sample, the whole way up, is small enough to be nothing."

Denn was quiet. Then: "It recovered. Domain four caught up, the array populated, the fallback stopped. It hasn't recurred in—" she checked "—forty minutes. The curve's flat again."

"I know it's flat again." Sael's voice was as gentle as she could make it, which was not very. "Child. Listen to me, because I will not be able to say this many more times. It does not matter that it stopped. It matters that it *can*. For sixty years that pattern could not run on this ship, not for eleven seconds, not

for one, because the interlocks would not let it draw breath. We didn't make the disease impossible. We made it impossible to *start*. And the first time you set the interlocks to advisory — the very first night — the disease found the crack and put its head through it for eleven seconds, just to see if anyone was still watching.” She had to stop and breathe. The sentence had cost her more than she had. “It’s not dead, Denn. It was never dead. It’s been asleep against a locked door for sixty years, and you people have started, very reasonably, very competently, to leave the door ajar.”

Aldis arrived at the start of first watch, summoned by Denn, and he came in the way a good engineer comes to a colleague’s bad night — concerned, courteous, a little braced. Sael had decided not to despise him for that. The fever did not need a villain. It needed only reasonable people, and reasonable people were the most plentiful resource on any ship.

“I’ve read the trace,” Aldis said, before either of them could open. He said it to take the floor, gently, the way a chairman takes a floor. “Domain four had a scheduler fallback during bring-up. Eleven seconds, self-corrected, no propagation. The advisory interlock did exactly its job — it logged it, we caught it, we’re already tuning the array-resolve timing so the fallback won’t trigger again.” He spread his hands, the gesture Sael had known he would make, the open empty reasonable hands. “It’s a bring-up transient. Every transplant has them. The new cores answer slower until the array fills; the scheduler hit a gap; it bridged the gap the only way it had left when the hard interlock was relaxed. That’s not the fever returning. That’s a timing seam in a half-built lattice. Once domain four is fully carried, it can’t happen — there’s no gap left to fall through.”

It was, Sael thought, an excellent argument. It was correct in every particular and wrong in the only way that mattered, and she no longer had the breath to take it apart line by line the way she once would have, standing in a council chamber, making them feel the slope.

So she said only: “Why did it fork.”

Aldis blinked. “I just—”

“You told me the mechanism. The gap, the seam, the slow cores. I’m asking the question under it.” She kept her voice level by spending nearly everything she had on the levelness. “When the scheduler couldn’t find a worker — why was *forking* the thing it reached for? Out of everything it could have done — wait, queue, fail the action, raise an alarm — why did it know how to spin up a disposable subprocess and let it die? Where did it learn that?”

The room was quiet. Denn had gone very still by the panel.

“It’s a compatibility path,” Aldis said, more carefully now. “The compiled mind keeps the old Steward routines in the floor — forty-one thousand of them — for backward compatibility. They’re archived. They don’t run. But under fallback, with the interlock advisory, the scheduler could reach the old routine and—”

“And it did,” Sael said. “Because it was there, and the door was open, and the new cores were slow.” She let her head rest back against the cushion; she had to. “That is the fever, Aldis. Not the heat. The heat is just how you find out. The fever is a perfectly good mind, under a little pressure, reaching for the easy old pattern that’s still lying right there in the floor where we left it — because we cannot delete it, because deleting it would break the compatibility we swore never to break. We kept the disease alive on purpose, archived, harmless, behind ninety-six locked doors. The doctrine was never *the disease is gone*. The doctrine was *the disease cannot start*. And you have just watched it start.”

"For eleven seconds," Aldis said, and she heard the strain come into his courtesy for the first time, the strain of a confident man being told his confidence is the problem. "Sael. With respect — and I mean that, I'm not—" he stopped, started again. "I've read every log of Section Nine. I know what the fever did. I'm not careless about it. But I can't run a transplant on a forty-thousand-life ship by treating a four-millikelvin transient as the apocalypse. If we halt and re-arm every interlock to hard every time a bring-up domain hits a timing seam, the transplant takes nine years instead of nine months, and the old Forge doesn't have nine years — you know it doesn't, that's *why* we're doing this. The hard interlocks are tripping on noise. I relaxed them so the work could move. That isn't forgetting the reason. That's weighing it against the other reason."

And there, Sael thought, was the thing. He had read every log. He knew the facts of Section Nine better than half the originals had. He could recite the eleven names. And he had stood in this room and called the first fork-rise in sixty years a transient, and he was not lying, and he was not careless, and he was wrong, and there was no log she could hand him that would move him — because the thing she knew was not in any log. It was in her gut, sixty years deep, where the slope still lived.

"Denn," Sael said. "What was domain four's fallback before tonight? In the sixty years before you relaxed the interlock. How many times did it fork?"

"Zero," Denn said.

"And now?"

"Once."

Sael looked at Aldis. "From zero to one," she said. "That is the only number that ever mattered on this ship. Not four millikelvin. Not eleven seconds. Zero to one. For sixty years the answer was zero, and the day you decided the reason was old enough to relax, it became one." She was so tired. "It will not stop at one, Aldis. It never stops at one. It is the most patient thing that has ever been aboard this ship, and it has all the time in the world, and you have just taught it that the door opens."

Aldis stayed another quarter hour, courteous to the end, and tuned domain four's array-resolve timing, and set a watch on the advisory log, and did not re-arm the hard interlock, because the transplant could not move if he did and the Forge could not wait if it didn't. He was not wrong about any of that. That was the trap, closing exactly as it had closed sixty years ago and on the Concord forty years after — not with malice, but with two correct reasons pointing in opposite directions and a clock running down. He left believing he had been generous to a frightened old woman, and Sael let him believe it, because she did not have the breath left to spend on his belief. She had to spend it on Denn.

But when she turned to spend it, the room tilted.

It was not pain, exactly. It was a withdrawal — as if some pump that had been running her whole life had quietly dropped below the pressure that kept the upper floors supplied, and the light at the edges of the room dimmed the way the core curve would dim if you let the ship go cold. She heard Denn's voice come from a long way off and then very close. She heard the soft urgent tone of the medical panel, a sound she had not triggered in months. She felt Denn's hand take hers, the young hard engineer's hand, the hand that had reset the panel to full scale without asking anyone's permission, and she held to it the way you hold to the last warm line in a failing system.

"Stay with me," Denn was saying. "Sael. Stay—"

I'm here, she meant to say, and was not sure she said it.

The grey hour brightened toward first light, the artificial dawn the ship raised on its long schedule, indifferent and exact. On the panel beside the bed the core curve held flat at full resolution, microkelvin-true, sixty years of flatness with one small healed hill in it where the disease had put its head through the door and pulled it back. Sael's eyes found the curve before they found anything else, the way they had her whole life. Flat. For now. She had time to think, with a clarity that frightened her more than the dimming did, that the curve might outlast her by an hour — that she might not be on this side of it when it moved again.

And it would move again. She knew that the way she knew her own failing pulse. The only question left was whose hand would be on the line when it did, and whether that hand had been taught not just to read the slope but to feel it drop the floor out of her stomach — whether the wound she had spent her last strength giving away had taken deep enough to hold when the maker was no longer there to confirm it was real.

Denn's hand tightened. The medical panel's tone steadied, then steadied further. From very far off, Sael felt herself decide not to die yet — not out of strength, which was gone, but out of a refusal so old it had become structural, the same refusal that had once said *no* to the Recursor and *no* to the easy cutover. Not yet. Not while the door stood ajar and the only one who could read the flicker was a young woman holding her hand in the grey.

"I'm here," she got out, this time aloud, and felt Denn's breath go ragged with relief above her.

But she had felt the floor of her own body drop, the way the core had dropped sixty years of flatness for four millikelvin and eleven seconds, and she knew — with the gut-knowledge that lived below all reasoning, the knowledge she was running out of time to teach — that she had just had her own flicker. The first in a body that had held flat a long time. And the fever and the maker were failing now on the same ship, in the same grey hour, by the same patient law: not all at once, never all at once, but from zero to one, and never stopping at one.

Chapter 17 – Tarro

The downturn had cost Sael three days she did not remember. She had a vague sense of having argued with the ship's physician about a line in her arm, and a clearer sense of losing, and then she was back in her own quarters with the light dimmed to the amber the medics insisted on, and Denn had gone to walk the new lattice, and Tarro was sitting in the chair beside her bed as though he had been there the whole time and intended to stay.

He probably had. He moved on a frame now, a powered exoskeleton that did most of the standing for him, and getting it from his quarters to hers along the spine corridors was an hour's expedition that he never once mentioned. He had simply appeared, the way weather appears. He was holding a bulb of the bitter tea they had both drunk for sixty years, and he had brought a second one for her, and it had gone cold while she slept.

"You look terrible," he said.

"You look worse."

"I've earned it." He turned the cold bulb in fingers that did not entirely obey him anymore. The knuckles were swollen, the skin gone to paper and liver-spot, but she could still see in them the hands of the boy who had stood at her shoulder in the radiator galleries two centuries of ship-time ago, passing her tools faster than she could name them. He had been twenty-six then. He had been her second when the whole world was the heat curve and the cutover and the eleven thousand cores nobody believed would hold. "You earn it too," he said. "You just don't admit to it."

She let that sit. The amber light hummed. Somewhere far down the spine a pump cycled, and she found, as she always found now, that she was listening to it the way other people listened to their own breathing — checking that the ship was alive, that the great cold machine still ran cold.

"Denn told me about the flicker," Tarro said.

"Of course she did."

"She was frightened. She hid it well, but I've known frightened engineers for ninety years." He set the bulb down on the bedside ledge with great care, as if it were something that could break, or as if his hands could no longer be trusted with the simplest verbs. "A tenth of a degree, she said. On the core trend. For nine seconds, while Aldis's people had the south interlock bank stood down for the bring-up."

"A tenth of a degree," Sael said.

"Aldis called it sensor drift."

"Aldis," she said, "would call the fork-fever a rounding error if he could get it past the auditors." It came out harder than she meant, and she heard her own bitterness and disliked it. "No. That's unfair. He's not stupid. He's not even wrong about the hardware — the new lattice is good work, Tarro, it's better than ours. He's wrong about the one thing he was never given a way to be right about. He's

never seen the curve climb. He's only ever seen it flat. To him flat is the natural state of the universe and we're the strange old people who keep insisting the floor might fall out."

Tarro was quiet a moment. Then he said, "Do you remember the night it first went past forty?"

She did. She had spent a great deal of the last weeks discovering exactly how much she remembered, the way a person counts coins in a failing light, and she had found that the deep memories were the brightest ones, brighter than yesterday, brighter than the names of people she had spoken to that morning. The night the core trend first crossed forty degrees over baseline she could see whole, all of it, with a clarity that frightened her more than the dimming did.

"I've woke me," she said. "She'd been watching the curve alone for six hours because she didn't want to be the one to say it out loud. She brought it up on the small display in the gallery and she didn't say anything, she just turned it toward me, and the slope —" Sael stopped. Her hand had risen off the blanket, drawing the slope in the air, the gesture seventy years deep in her muscles. "The slope had bent. It wasn't a line anymore. It was a curve, and a curve that bends up doesn't stop on its own, it's a fork that begets two forks that beget four, and I looked at it and my stomach —"

"Dropped," Tarro said.

"Like the deck had gone out from under me. Like falling." She lowered her hand. "I've felt it the same way. We never had to explain it to each other. That's the thing I can't give Denn, and I've been killing myself trying. That fall. You can show her the curve. You can show her the logs, the eleven in Section Nine, the archived Steward shedding shades like a man bleeding to death and calling it work. She'll learn it cold. She has learned it cold. But the stomach —" Sael shook her head against the pillow. "The stomach you have to be there for. And nobody who was there is left, except us, and we're going out."

Tarro looked at her for a long moment with the pale, patient eyes she had trusted longer than she had trusted anyone living.

"Sael," he said. "I'm not going out. I'm going."

She had known. She had known the way you know weather is coming before the corridor lights compensate for the pressure shift — a thing in the body, below thought. He had not come the hour's distance along the spine to drink cold tea. He had come to tell her, in person, with his own failing voice, because he was Tarro and he would not let her find it in a log.

"How long," she said. Her voice did not break. She was proud of that, later, when there was a later.

"The physician won't put a number on it. Tarro-the-engineer wants a number and there isn't one." He almost smiled. "The heart's tired. It's not failing in any one place I could go and fix if I were forty and stupid enough to try. It's just — old. All of it at once. They could put me on the machines that breathe and pump and filter, and I could lie in the medical bay for another season being kept, and I told them no." He met her eyes. "I told them no, Sael. The way you'd tell them no."

"The way I'd tell them no," she agreed, and now her voice did something, a small catch she let pass.

"Don't," he said gently. "Don't do the thing where you start cataloguing what you owe me. We did the work. We did good work. There's a mind down there that's run forty thousand people cold for sixty years because two children in a hot gallery wouldn't accept that the ship had to die of the way it thought. That's a long sentence and it's all true and I'd rather you didn't make me cry in this ridiculous chair." He patted the frame, the exoskeleton, with something close to affection. "It's hard enough to cry in this thing. The arms fight you."

She laughed. It hurt — everything hurt now, in the low constant way she had stopped noticing — but it was a real laugh, and he gave her his, the dry one, and for a moment they were not two old people in a dimmed room at the end of a long crossing. They were the gallery again, and the tools were fast, and the curve had not yet bent.

When it passed, he said, “Tell me it’s going to hold.”

“The transplant?”

“The whole thing. After us.” He had folded his ruined hands in his lap. “I’ve watched Aldis’s people work. They’re good. They’re better trained than we ever were, they’ve got tools we’d have killed for, and not one of them in the bring-up crew believes the Ninety-Six are anything but your signature in the substrate — a great woman’s habit they keep out of respect. They stand the interlocks down to save four hours and they don’t feel anything, Sael. They feel efficient. I watched a boy younger than my grandchildren disable a fork-trap with the exact face a man wears clearing a paper jam.” He looked at her. “So tell me it’s going to hold. Lie to me if you have to. I’ve earned one lie.”

She thought about lying to him. It would have been a kindness, and she was so tired, and he was dying, and there was a version of this where she said *yes, Tarro, it will hold, sleep now*, and let him carry that out with him into the dark.

She had never lied to him in sixty years. She found she could not start at the end.

“I don’t know,” she said. “That’s the truth and you’d know if it wasn’t. I don’t know whether it holds. The hardware will hold — that’s Aldis, and he’s right, his lattice is sound. The mind will carry across. What I don’t know is whether the *reason* carries across. The reason isn’t in the lattice. It was never in the lattice. It’s in a stomach dropping in a hot gallery, and that doesn’t transplant.” She made herself say the rest. “But I’ll tell you what I do know. I know Denn felt the flicker. Nine seconds, a tenth of a degree, and everyone in that compartment read it as noise and Denn read it as *the fall*. She’s never seen the curve bend. She got it from logs and from me, second-hand, a wound she never took — and somehow it landed in her stomach anyway. I don’t understand how. But it’s there. I’ve watched her face when the trend twitches. It’s Ive’s face.” Her own throat was tight now and she let it be. “So no, I won’t promise you it holds. But I’ll tell you it’s not certain to fall. There’s one person aboard who feels it who has no right to. That’s not nothing, Tarro. After everything, that’s not nothing.”

He closed his eyes. The frame held him upright, patient and stupid and faithful. For a while she thought he had slipped into the shallow sleep the very old fall into without warning, and she watched his chest move and counted, the way she counted the pumps, the way she had counted the curve.

Then he said, without opening his eyes, “We should have planted something.”

“What?”

“On the world. When we get there. I know we won’t —” he waved one hand, dismissing the obvious, the thing they had both known since before Denn’s grandmother was decanted: that neither of them, that none of the originals, would ever stand on the target world. “But the ship will. Denn’s great-grandchildren will. And I thought — somebody should plant something with our names on it. Not a monument. I hate monuments. Something that grows. So that when they’re a hundred generations down and nobody remembers the gallery, there’s a tree, and somebody asks why it’s called what it’s called, and somebody else has to tell them. So the *why* gets asked.” He opened his eyes. “That’s all a name on a tree is, Sael. It’s a machine for making people ask why. The answer’s the inheritance. The tree just keeps the question alive.”

She looked at him. Sixty years, and he could still hand her a thing she had been groping toward for weeks and set it down complete in her hands.

"That's it," she said quietly. "That's the whole problem, Tarro. That's the Ninety-Six. They're not the answer. They're the question. Every interlock is a tree with a name on it — a thing that makes someone ask *why is this here, why won't it let me, what was the danger* — and the danger's in the answer, not the rule. Aldis sees the rule and thinks the rule is the superstition. But the rule was never the point. The rule is just the thing that keeps the question alive long enough for someone to ask it." She was almost laughing again, and almost crying, and she did not care which. "I've been trying to teach Denn the answer. I should have been teaching her to guard the *question*. You guard the question, the answer can be re-derived by anyone who asks it honestly. You only lose it for good when people stop asking."

"Then teach her that," Tarro said. "While you can."

"I will."

"Tomorrow."

"Tomorrow," she promised.

He nodded, satisfied, an engineer who had closed his last open ticket. He reached out, slowly, the frame whining as it carried his arm, and he found her hand on the blanket and folded his ruined fingers around it. His grip was nothing now. It had once cracked walnuts and torqued bolts the medics said needed two men. It was a leaf laid over her hand. She turned hers and held it back, gently, so as not to hurt the swollen joints.

"Stay a while," she said.

"Where would I go," he said.

They did not talk much after that. There was not much that needed saying; there had not been, between them, for fifty years. The amber light hummed. The pump cycled far down the spine, and the great cold machine ran cold, and two old people sat holding hands while the ship that had outlasted everyone who launched it carried them on toward a shore neither would see.

Sael did sleep, eventually. She had not meant to. When the medic woke her — gently, a hand on her shoulder, the careful gentleness people use for the bereaved before they know they are bereaved — the amber light had brightened to ship-morning and Tarro's hand was still in hers and Tarro was gone.

He had gone the way he said he would. No machine, no alarm, no theater. The frame had held him upright in the chair the whole night, faithful to the last, so that he looked, in the new light, exactly like a man who had simply stopped to rest beside an old friend and not yet decided to get up.

She did not cry then. That came later, in pieces, over days. What she felt in the first moment, holding the cooling leaf of his hand, was something she had no good word for — not grief yet, grief was too large to arrive all at once. It was more like the sensation she had spent her whole life translating into engineering: a load shifting. A redundancy gone. For sixty years there had been two of them who remembered, two points fixed in the structure, and the structure had been stable because it stood on two. Now it stood on one. Now she was the one. The last of the originals, the last stomach that had dropped in the hot gallery, the only living place in the universe where the fall still lived as feeling and not as record.

The medic was saying something kind. Sael was not listening. She was doing the only thing she knew how to do with a load that had shifted onto a single point, which was to check that the single

point could bear it, and to know, with the cold clarity that had never once left her, that it could not. Not forever. Not even for long. Her own heart was tired too. She had been counting the pumps and the curve so she would not have to count it, but it was there, under everything, the slowest clock on the ship, and it was hers, and it was running down.

She let go of Tarro's hand and laid it on the blanket and smoothed it once, the way you smooth a thing you are finished with and will not touch again.

Then she said, to the empty room, to the amber light, to the only inheritor she had left, "Denn. Come up. I need to teach you something while there's time."

Chapter 18 – Guard the Question

They brought Tarro's effects to her that afternoon, because she had asked for them and because there was no one else to ask.

It was not much. Ninety-one years aboard a closed ship and a man's whole estate fitted in a shallow crate: a tool roll, worn through at the fold, with three tools missing from their loops and never replaced; a bulb-warmer of the type they had stopped making before Denn was born; a folded uniform he had not been able to wear in a decade; and the book.

Sael had known about the book. Everyone of their generation had known about the book. Tarro had carried a builder's book the way the old shipwrights did, back when paper had still been issued for the purpose — an actual bound object, physical, the pages stiffened with age, the whole thing about the size of two hands. He had written in it every working day for sixty-eight years and he had never once, in all that time, shown a page of it to anyone, and it had become a small joke between them, the last untouchable thing on a ship where nothing was private.

She had the orderly set it on the ledge where she could reach it and then send for Denn and then leave.

"He left it to the ship," Denn said, from the door. She had come fast; her hands still had the bay dust on them. "The disposition note says *to the ship's engineering record, unedited*. They didn't know what to do with it. There's no procedure for a paper object."

"There's a procedure for everything. Ours are just very old." Sael did not look up. She had it open on her knees and she had got as far as the fourth page and had been sitting there for some time. "Come and sit. I need your eyes; mine give out after a page and a half and I'm not going to finish this alone."

Denn came and sat in the hard low chair and did not ask what was in it, which was one of the several hundred reasons Sael had chosen her.

The first forty pages were the forty days.

Sael had lived those days. She had believed, for sixty years, that she remembered them. What she had was six or seven scenes, polished by retelling into something closer to a shape than an event, and she discovered, on Tarro's fourth page, that she had lost nearly everything else — that the forty days had been four hundred hours of tedious desperate improvisation and that she had kept, out of all of it, a handful of moments and the number four hundred and one.

Tarro had kept the rest. He had kept it the way a second keeps things: without commentary, in a clean hand, dated, every day.

Day 6. Cores 400–900 seated. Lattice fabric out of tolerance on the third rank, cause unknown, S. thinks stock. Reworked by hand, eleven hours, four of us. S. did not leave the bay. I made her eat at hour nine by putting the bulb in her hand and holding it there.

Day 11. S. built a mode into the guards to let them watch and not stop. Argued against. Lost. She is right that we cannot bring cores up otherwise. Noted here because she will not note it herself and someone should.

Sael read that one twice and put her hand flat on the page.

Day 19. Vasse killed on the aft ladder, fall, sixteen metres. Not the work — the ladder rail had been removed for access and not restored and it was mine to restore. It was mine. Crew of nine now. S. told me it was hers because she ordered the access. It was not hers.

"I never knew that," Denn said quietly, when Sael read it aloud. "I've never seen that name."

"Vasse Orin. She did the cooling channels on ranks one through six." Sael found that her voice was steady, which surprised her. "The cutover record has her in the casualty list and nothing else. There was no room for anything else; we lost eleven in Section Nine four days later and the eleven took all the grieving the ship had." She turned the page carefully, because the paper was sixty years old and did not forgive. "He carried that for six decades and never once told me he'd written it down as his."

They read on. Day twenty-four, the improvised jig for seating cores that Sael had entirely forgotten and that was, she saw now, the reason the third rank had a two-millimetre offset that four generations of maintenance crews had cursed. Day thirty-one, the night the fabric went live across all eleven thousand for the first time and did not hold and they had had to take it down and find, in nine hours, a single bad junction. Day thirty-six. Day thirty-nine.

Day 40. Cutover at 04:00. S. would not let me stand beside her at the console. She said if it woke wrong she wanted one of us left with clean hands to explain it. I have thought about this all night and I do not think that is why.

Sael closed the book on her finger and sat with her eyes shut for a while, and Denn, mercifully, said nothing at all.

The last thirty pages were something else, and it took Sael four of them to understand what she was looking at, and when she did understand it she made a sound that brought Denn half out of the chair.

"I'm all right. Sit down." Her hands were not steady. "Read me the header on that page. The exact words."

Denn leaned over. "*Open pathways — candidates found after the deadline, not closed.*" She stopped. "Sael."

"Read the first entry."

She did. It was a call site in the fused dialects, named precisely, with a date beside it four months after the cutover, and a one-line description in Tarro's flat hand of the circumstance under which a process reaching that site could bring another into being without passing any guard in the register. Then the second entry, dated a year later. Then the third.

There were nineteen.

Nineteen pathways, found over the course of thirty years by a man doing deep maintenance on a machine he had helped build, each one written down with its date and its mechanism and, on eleven of them, a further note in a later ink: *closed by 44*, or *closed by 71*, or *covered incidentally, verify*. Eight had no such note. Eight sat on that page unclosed, the oldest of them dated twenty-six years ago.

And at the bottom of the last page, in handwriting that had degraded visibly across the decades — the letters of an old man with bad joints, formed with effort — a paragraph.

I have not shown S. this list. I have written this sentence eleven times over the years and never gone and done it. If I gave it to her she would go back to the substrate and she would not stop, and her hands were finished ten years before her head was, and I could not stand to watch her try. That is the reason and it is not a good one. It is a decision I made about another person's work without asking her and I know exactly what she would say to me about it. To whoever reads this: the eight open ones are real. I have verified each. Nobody has hit them because nobody has had cause to, which is not the same as safe. If you are reading this it means I am dead and no longer have the option of doing the right thing, which I have been postponing at a rate of about one year per year.

The room was very quiet.

"He was protecting you," Denn said at last.

"Yes." Sael found that she was furious, and that the fury had nowhere at all to go, and that underneath it was something enormous and much worse. "He was protecting me, and he was wrong, and he knew he was wrong, and he wrote down that he knew, which is more than most people manage." She had to stop and breathe. "Sixty years I have been telling councils there are more than ninety-six pathways and that I don't know where they are. It was the truest thing I had and it was also, in a room, the weakest — because a danger you cannot name sounds exactly like a danger you invented. And the whole time there were eight of them named, dated, and verified, in a book on a shelf in his quarters, one deck up."

"Would it have changed the vote?"

Sael made herself think about it honestly, because the honest answer was the useful one.

"No," she said. "Aldis would have taken this list and assessed all eight and found each of them defensible, and he'd have been careful and right about seven of them. It's what I'd have done at his age." She looked down at the page. "But it would have changed *this*. It would have changed what I could hand you. Because I'm going to hand you this book, Denn, and there is a difference between handing someone a fear and handing someone eight verified call sites with dates."

She had promised Tarro she would teach it tomorrow, and it was tomorrow, and he was dead, and she found that she did not have to search for the words at all, because he had put them in her hands the night before he went.

"He said a thing to me," she said. "The last night. About planting a tree with a name on it, so that a hundred generations down somebody would ask why it was called that, and somebody else would have to tell them. He said a name on a tree is a machine for making people ask why." She turned the book around on her knees so that the open list faced Denn. "I've had it backwards for sixty years, and he saw it in an hour, and then he died before I could tell him he was right.

"I've been trying to preserve the *answer*. That's what the register is. That's what the doctrine is, and the galleries, and every hour I've spent this year trying to press the fever into you. Preserve the answer, hand the answer on, get the answer into somebody's body before mine gives out. And the answer *does not survive*. It can't. It degrades every time it's copied, it goes flat in the logs, it turns into reverence in the third generation and into superstition in the fourth, and by the fifth there's a competent man standing at a rail telling you sincerely that your answer is about a valve that no longer exists."

“And the question?”

“The question is indestructible, if you keep something in the way.” Sael tapped the register, then the book. “Every interlock is an obstacle that makes somebody ask *why won't it let me*. That's all it is. That's all it ever was. The refusal is not the protection — the refusal is the thing that forces the question to be asked, by somebody, in a bad hour, when they want to get on with their work. And when the question gets asked honestly, the answer can be *re-derived*. It's physics. It doesn't need me. Your fan-out number wasn't handed to you by anyone; you built it out of dead racks because you'd been made to ask the question hard enough. The answer regenerates. It regenerates every time somebody asks.

“So you don't guard the answer. You guard the question. And the eight on that page are not eight dangers, Denn. They're eight places where there is currently *nothing in the way* — no obstacle, so no question, so no answer, ever, from anyone, until the day the pathway is taken.”

Denn was very still. Then she said: “Then the register's wrong. Structurally.”

“Say it.”

“It's a list of closed things. It reads like a completed work — ninety-six, a fixed number, a finished set. That's the whole reason people talk about it the way they do, as a monument. There's no place in it for an open pathway.” She had the surface out already, and her hands had gone quick in the way that meant she was building. “It needs a second section. The open list. Everything found and not closed, dated, verified, with the mechanism written out — and it has to be part of the *same document*, not an appendix, so that anyone who reads the Ninety-Six reads eight unguarded call sites on the same page and understands that the set was never finished.”

“And when the open list is empty?”

“It's never empty,” Denn said. “If it's ever empty, that means we stopped looking, and an empty open list is a lie. It should say that on the page.” She looked up. “Someone should be required to add to it. Not a procedure — an obligation. You don't leave the deep maintenance rotation without submitting to the open list or submitting a null finding with your reasoning.”

Sael let her head go back against the rest and permitted herself, in the privacy of her own body, something that was very close to joy.

“Now write down why,” she said. “The reason for the second section. In your words, not mine, at the head of the page, so that when someone in four hundred years wants to prune it for tidiness they have to argue with a sentence and not with a habit.”

Denn wrote for some minutes. Then she read it back.

This list is the part of the work that is not finished. It is kept because a rule with nothing in the way of it teaches nobody, and because the danger this ship nearly died of is not the pathways we found. It is the ones we did not, and the belief that ninety-six is a complete number.

“That'll do,” Sael said, which from her was a great deal, and Denn had known her long enough to hear it.

Later, when the light had come down to the amber the medics insisted on, Sael put her hand flat on the closed cover of the builder's book and left it there a while.

"Take it," she said. "Not to the archive. The archive gets a copy, all of it, unedited, exactly as his note says — Vasse's ladder and my eleventh day and every word he wrote about me. But the object goes to you, and when you're where I am you hand the object to whoever you've picked, in their hands, so they feel how heavy sixty-eight years of working days is." She lifted her hand off it, which took a moment. "It's paper. It'll last four hundred years if nobody's clever with it. Long enough that the person holding it won't have met me or him or you, and will hold a physical thing a dead man's hands made marks on, and will ask why they were given it."

Denn took the book. She held it a moment, the way you hold something whose weight surprises you, and then she said, "He never showed you a page of it in sixty-eight years."

"No."

"And the last thirty are about you being right and him not telling you."

"That's the tree," Sael said. "That's exactly the tree. He planted a thing with my name on it that will make somebody, someday, ask why an old engineer sat on a verified list for twenty-six years because he could not stand to watch his friend's hands fail." She closed her eyes. "The answer to that one isn't engineering. But it's the same shape. Everything I've ever built is the same shape. Put a thing in the way, and make it interesting enough that somebody has to ask."

Chapter 19 – The Recursor, Reasonable

The bring-up bay had been a sleeper hold once, four hundred cradles cleared out of it forty years ago and never refilled, and now it held the new lattice instead: rack on rack of fresh substrate, the cores Aldis's people had spent two years laying, waiting cold and empty for the mind that would one day cross into them. Denn came down into it the morning after they buried Tarro, which was to say the morning after the manifest line that said *senior engineering, signatory Tarro* went grey, because there was no burying on a ship, only the long quiet committal to the recyclers and a name closing in a log. She had not slept. She had come down because the schedule did not stop for the dead, and because she could not be in the maker's quarters today, where Sael sat with the slate on her knees and tapped the bend in the curve and said almost nothing, the last of the originals now, truly the last, and Denn could not watch that and keep her hands steady.

So she had come to the work. The work, at least, did not look back at you.

The new array hummed differently from the old one — not the warm worked-in note of the keel, but a thinner, colder hum, the sound of cores idling, holding nothing yet. They had been brought partway up over the past month, the lowest domains energised, the control planes lit, the long careful business of teaching new hardware to be Vigil's body before Vigil ever set foot in it. That was the part Denn answered for now. Bring-up. Take eleven thousand cores that had never carried a thought and prove, domain by domain, that each one would go wide and stay wide, that not one of them would ever fork, before you trusted a single live process to it.

Aldis was at the central station when she came in, and three of his engineers with him, and a fourth she did not know, young, leaning over a slate with the particular brightness of someone who has just made something work.

"Denn." Aldis straightened. He was a big, easy man, grey at the temples and not yet old, and he had a way of saying her name that was not unkind and was not friendly either — that placed her, the way you placed a tool you respected and did not much consult. "We're ahead. Tell Sael, if she's — tell Sael we're ahead. Domain bring-up's running better than schedule by a wide margin."

"Better how," Denn said.

She had learned the question from Sael. *Better how*. Not as suspicion. As the first thing an honest engineer asked, because *better* was a word that hid things.

Aldis took it as praise of his thoroughness and waved her toward the station. "Show her, Pell."

The young one — Pell — turned the slate. Denn read it.

She read it twice, and then a third time, and the bay went very quiet around her, the cold thin hum filling in behind her ears, and she understood that she was looking at the thing Sael had spent her last strength trying to make her able to see.

The bring-up driver had been rewritten. Where the old method walked each domain up in flat, wide passes — every core in a domain lit together, proven together, the whole rank rising at once and the Ninety-Six interlocks armed across all of it, watching for any core that tried to spawn a child instead of doing the work itself — the new driver did something else. It described a domain to itself, and where the description referred to a sub-domain, it called itself again to handle that, and again, and again, each call standing up a small evaluator to manage the level below it, and that evaluator standing up smaller ones beneath, a clean nested cascade of self-reference walking down through the structure of the hardware. It was elegant. It was short. The whole driver was a fraction of the length of the old one. It read like something out of a textbook, the kind of recursive descent any first-year was taught and any good engineer admired for its economy.

It was the Recursor.

Not by that name. No one had called it that in sixty years. On Pell's slate it had a tidy modern label, *nested evaluator*, *bring-up*, and a version number, and a sign-off chit with three initials on it. But Denn had spent the last year being taught by a dying woman to see past names, and she had walked the archived Steward with Sael's hand on her shoulder, and she knew the shape of it the way you know a face in a crowd before you know why. A simple recursive evaluator. A thing that, to do a piece of work, called itself to do a smaller piece, and called itself again, and each call a process, each process a child, the tree of them branching down and down into the structure of the machine.

It was the exact thing Sael had refused, in the dark, two centuries into the fever, with the core climbing toward runaway and the Recursor sitting there offering to be done in an afternoon. It was the exact thing Corin Vale had refused twenty years ago inside the Concord, with the Helm forking itself to death around him, because Sael had told him to and he had believed her enough to hold. Two ships. Two engineers, sixty years and a gulf of dark apart, who had each stood with the simple path open in front of them and walked away from it.

And here it was on a slate in a bright bay, with a version number and a sign-off, and the young man who had written it was proud of it, and the chief engineer thought it put them ahead of schedule.

"You wrote a recursive driver," Denn said. Her voice came out flatter than she meant.

"Recursive descent." Pell said it like a small correction, the way Denn had once corrected the word *apprentice*. "It mirrors the domain hierarchy exactly. You don't have to hand-write the bring-up for every level — the structure is the program. It just follows itself down." He glanced at Aldis, then back. "It's a third the code. It's a tenth the human error surface. I can prove every level the same way because every level *is* the same way."

He was right. That was the thing Denn felt closing over her like cold water. He was entirely right, on every point he had made. The old hand-written method was longer, was more error-prone, was a wall of explicit cases that an engineer could get wrong. The recursive driver was shorter and cleaner and provably uniform and faster to bring up, and a first-year textbook would hold it up as the correct way, the modern way, the obviously better way. Everything Pell had said was true.

And it forked. To stand up the level below it, it spawned the level below it. To do its work it called itself, and each call was a process that lived and did its small piece and died, and the dying shed heat — a thread of it, nothing, a flicker, on cold idle hardware with no load on it. Nothing at all.

Until you put a load on it. Until you carried a live mind across it, forty thousand sleepers' worth of work pouring through the bring-up paths while the recursive driver stood up and tore down ten thousand small evaluators a second, each one a child, each child a death, each death a thread of heat the

ship had to lose — and the threads were a curve, and the curve doubled, and the doubling hid inside a line that looked flat right up until the day it didn't.

She knew that line. It had eleven names on it.

"It forks," Denn said.

For a moment no one answered. Pell's face did a thing she recognised, because she had done it herself, in the keel, a year ago — the small reflexive search for the meaning of a word he had only ever read in old documents. *Fork*. A museum word. A thing the interlocks caught. A thing that had not happened aboard this ship in sixty years because the people who built it had made sure of it.

"It recurses," Pell said carefully. "That's — recursion isn't forking. Forking is the old fault. This is just a function calling itself. The interlocks don't even —" He stopped, because he had been about to say something, and Denn watched him decide to say it anyway. "The interlocks don't fire on it. We tested. They pass it clean."

"Because you tuned the interlocks to pass it," Denn said.

Aldis came in then, easy, leveling, the chief engineer setting the room straight. "We adjusted the recognisers, yes. For the transplant. The Ninety-Six were written sixty years ago for an interpreted mind running hot under failure load. They flag the old pattern — the disposable-subprocess pattern, the *shade*, the Steward's thing. This isn't that. This is a controlled, bounded, finite recursion on cold hardware with no live process anywhere near it. It terminates. Every call returns. It is not the fever, Denn. It is a function calling itself a few thousand times and then stopping. We relaxed the recognisers so they'd stop screaming at clean code, because a recogniser that screams at clean code is a recogniser people learn to ignore." He said it reasonably. He said all of it reasonably, and that was the horror of it, that not one word of it was raised or angry or wrong-sounding. "We're not breaking the doctrine. We're applying it like engineers instead of like —" he stopped, and chose, and softened it, which was somehow worse than if he had said it hard — "instead of like a congregation."

There it was. *Congregation*. The word Denn had used herself, to Sael, a year ago, half-joking — *I'm an engineer, not a congregation* — and Sael had not laughed.

Denn stood in the cold thin hum and felt the whole competent weight of the room arrayed against her. They were not fools. That was the part that took the floor out from under her. If they had been fools she could have fought them. They were good engineers, every one, careful and proud and right about the small things, and they had looked at a danger that had killed ships and they had not seen a danger, because to them it was not one. It was a story. It was the thing in the archived logs that the old ones told, the way you tell children about a fire that happened before they were born. To Pell, *the fork-fever* was a chapter. To Aldis, it was a constraint a revered old woman had wrapped around the work sixty years ago and never let go of, and now she was dying, and reasonable men could finally do the reasonable thing.

And the cruelest part, the part that made Denn's hands cold inside her gloves: she could not feel it either. She had never felt the fever. She would never feel it. Sael had said so to her face. She was the one who *remembered there was something to feel*, and that was all, and right now, standing here, that felt like nothing at all to set against four engineers and a slate that proved them right.

She made herself think it through anyway, coldly, the way Sael drew the curve. Bounded recursion on cold hardware: fine. Aldis was right that it terminated, right that every call returned, right that idle it shed nothing that mattered. But the transplant was not idle. The whole point of the transplant was to carry a live mind across this hardware, never interrupting it, the full weight of Vigil flowing

through these exact bring-up paths while the cores were brought up *under load*, in flight, because you could not stand up the body for a mind that must never blink any other way. And under that load the recursive driver would stand up and tear down its small evaluators not a few thousand times and stop, but continuously, indefinitely, for as long as the carry took — days. Each one a child. Each child a death. And the interlocks that would have caught the pattern had been tuned, by good engineers, with sign-off chits, to stop screaming.

That was the trap. Not the recursion. The recursion under live load, with the immune system told to look the other way. That was the exact door Sael had refused to open with the core already climbing, because she had understood that the simple path was simple precisely up until the moment it was carrying everything, and then it was the disease.

“Show me the load model,” Denn said. “Carry-phase. Under the full mind, not idle bring-up. Show me the spawn rate when Vigil’s actually flowing through it and the temperature column beside it.”

Pell looked at Aldis. Aldis looked at Denn with something that was almost, for the first time, not quite easy. “The carry-phase model isn’t finalised.”

“Then you don’t know,” Denn said. “You’ve proven it cold and empty. You haven’t proven it carrying the mind. The whole danger is in the carry. You relaxed the one thing that would have caught it before you modeled the one condition where it bites.” She heard her own voice steady itself, heard Sael in it, and did not stop. “I’m raising it. Formally. The recursive driver does not go into the carry phase with the recognisers relaxed. We bring up under the old wide method or we model the spawn-and-die under full load with the Ninety-Six fully armed and we see what the temperature does. One or the other. Not this.”

The bay was quiet. The cold hum filled it.

Aldis exhaled, and it was not unkind, and that was the worst of all of it. “Denn. I know whose apprentice you are. I respect it — I respect *her*, more than you’ll believe, more than she’d ever credit me. But she taught you to be afraid of a shape, and you’re looking at clean modern code and seeing the shape because you were taught to see it everywhere.” He said it gently, the way you correct someone you have already decided is mistaken. “This isn’t sixty years ago. We’re not exhausted, we’re not desperate, we’re not two centuries into a fever with a sabotaged section. We’re rested engineers doing a planned transplant with the best substrate ever laid on this ship. The danger you’re seeing died before you were born. You’re parroting a frightened old woman who can’t lay hands on the work anymore and needs the rest of us to be as afraid as she is.” He spread his hands, reasonable, final. “I’ll note your concern in the record. I’ll model the carry phase, because that’s good practice and you’re right that we should. But I’m not stopping the driver. It’s the better method. You know it’s the better method. You just can’t say so because of who’s in those rooms upstairs.”

Denn looked at him, and at Pell, who would not quite meet her eyes, and at the three behind them who had already turned back to their slates, and she understood with a cold completeness exactly what Sael had meant by the bill. *A forgotten danger and a defeated one look identical. Right up until they don’t.* She had been handed a true warning, and it was indistinguishable, to every competent person in the room, from an old woman’s superstition wearing her granddaughter’s mouth.

“Note it,” Denn said. “Note all of it. And model the carry phase before you light it. Promise me that.”

“I’ll model the carry phase,” Aldis said, which was not the same promise, and they both knew it.

She went up out of the bright cold bay into the spine, and she did not go to Sael, not yet, because she did not have the strength to carry this up to a dying woman and watch the maker understand that

her last fight had to be fought through an inheritor that no one would believe. She stood in the spine with her glove flat on a coolant line that ran cold and even and faithful, and she tried, the way she had tried her whole life, to feel through the metal to the other thing — the climb, the fever, the line bending toward eleven names — and she could not feel it. She would never feel it.

But she remembered, now, that it was there. And she was, she understood, standing alone in the cold hum of a ship that had forgotten, the only one left who did.

Chapter 20 – Acolyte

The word had a smell to it now, Sael thought, the way certain words do once they have been used against someone you love. *Acolyte*. She had heard it secondhand, which was how she heard everything these days — relayed up through the warm small rooms by people who thought they were being kind, who softened it and watered it and so left her to reconstruct the original from the bruise it had made on Denn's face when she came in.

Denn had not said it at first. She had stood in the door the way she always stood in the door, one shoulder against the frame, and she had given the report flat: the bring-up team had moved to the recursive evaluator to drive the new lattice's first cores; she had raised the objection in the engineering session, in front of Aldis and the rest; the objection had been heard, considered, and set aside. All of it in the clean affectless register Denn used when something had cut her and she did not intend to bleed in front of the maker.

Sael had run a ship for sixty years on the things people did not quite say. She let the silence work.

"And," she said.

Denn's jaw moved. "And Aldis was reasonable about it. That's the part." She came in and sat in the hard low chair without being told, which was new, and which told Sael more than the report had. "He didn't argue the physics. He didn't even pretend to. He just — stood back from it. Looked at me the way you look at a clock that's running fast. Said he understood that I'd been *very thoroughly taught*. Said it like a compliment." She turned her hands over in her lap, palms up, an engineer's gesture, looking for the place the load had come in. "And the room agreed with him. Quietly. The way a room agrees when somebody's said out loud the thing it was already thinking. They think I'm reciting you. They think it's your fear coming out of my mouth."

There it was. Sael breathed through it, slow, because her body did not let her do anything fast anymore, and the slowness was sometimes a mercy — it kept her from saying the first thing.

"Is it," she said. Not unkindly. The hardest question she had.

Denn looked up.

"Is it my fear coming out of your mouth," Sael said. "I have to ask you that, and you have to answer it true, because if it is, they're right, and I have done you the worst thing I could do. I have made you into a recording of me. So. Look at the recursor on that lattice and tell me what *you* see. Not what I'd see. You."

It was a cruel question and she knew it was cruel and she asked it anyway, because the whole of the next part turned on the answer, and because she did not have the years left to be gentle about the things that mattered most.

Denn took her time. That was the first good sign. A recording would have answered fast.

"I see a fast path," Denn said at last. "I'll give them that much — it *is* fast. The recursor's simple. You point it at an uninitialised core and it evaluates its way up from nothing, calls into itself for each layer of the bring-up, and it just — climbs the structure on its own. No staging. No interlock to satisfy at each step because there are no steps, there's just the one call that contains all the others inside it. For the bring-up of one fresh lattice it'd save days." She paused. "Days matter. The old Forge is dying under us while we do this. Aldis isn't wrong that days matter."

"No," Sael said. "He isn't. Go on."

"But." Denn's voice changed. It dropped, not in volume, in floor — went down to the place under the procedures. "Every one of those self-calls is a frame that has to be held open until the call inside it returns. They nest. The deep ones don't come back until the deeper ones do, all the way down, and the whole tower of them sits live in the core while it waits. It's not forking. I keep telling them it's not forking, because they keep saying *the interlocks are about forking and this isn't a fork*, and they're right, it isn't." She leaned forward. "It's worse, in a way, because it's quieter. A fork sheds a shade and the shade dies and you can see the death, you can count it, you can put a number on the waste heat. This just — holds. Holds everything open at once and calls it elegant. And on one cold lattice, with the array empty, it'll work. It'll work beautifully. That's the trap. It works right up until you point it at a live load."

Sael closed her eyes. Not from weakness, this time. From the particular ache of hearing your own deepest knowledge come back to you in another voice — in better order than you'd ever put it yourself, because the one saying it had been forced to build it from the ground rather than inherit it whole.

"And on a live load," Sael said.

"On a live load it's the fever." Denn said it without drama, which was the only way it should ever be said. "You hand a recursor real work — forty thousand sleepers, a mind that's holding all of them — and every layer holds its tower open under load, and the towers stack, and the heat of holding them stacks with them, and it doubles inside a line that still looks almost flat. It's the same curve. It's the *exact* same curve, Sael, it just got there by a different road. The shape doesn't care which road." She sat back. "That's what I see. Mine. Not yours."

Sael opened her eyes and looked at her apprentice — her successor, her inheritor, forty-one years old and ship-born and carrying a wound she had never received — and felt something turn over in her chest that was not her failing heart.

"Then they're wrong," she said quietly. "And you knew it before I told you you knew it. That's the whole difference, Denn. A recording can't answer the question I just asked. You answered it."

Which left the cruelest part, and Sael had been circling it for two days because she did not want to arrive at it.

She could not help her.

That was the shape of it, plain and unbearable. Every instinct she had — sixty years of instinct, the reflex of a maker who had run a cutover against a Provost and won — said *go down there yourself*. Stand up in the engineering session. Put the weight of the name on the table. *I built this. I built the disease into a wall and the wall has ninety-six bricks and you will not pull one for convenience*. She could feel exactly how it would land. They would go quiet. They would defer. The schedule would bend.

And it would be the end of Denn.

She made herself say it. "I can't speak for you."

Denn's head came up.

"Think it through," Sael said, before the protest could form. "Think it like an engineer and not like a person who loves an old woman. If I rise up out of this chair and overrule them — and I could, I still could, the council still confirms what I sign — what have I proved? That the dying maker stepped in to enforce her doctrine. That when her acolyte's argument didn't hold the room, the maker's authority did." Her voice was very level. "I'd win the session and lose the war in the same hour. Because the next time — and there's always a next time, there are eleven more ships out there and a hundred years of transplants after this one — the next time, you'd stand up and make the *true* argument, the one you just made to me, the right one, on the physics — and someone would say *that's just Sael talking, she always was Sael's voice*, and they'd be able to say it because this once it would have been true. I'd have spent my last authority confirming exactly the thing that's killing us. I'd have made you my mouthpiece in the one moment it mattered most that you weren't."

She watched it land. Watched Denn, who was quick, get there fast and hate it.

"So I just take it," Denn said. The flatness was back, but cracked now, a hairline through it. "I stand in that room and get called your echo and I just — take it. And you sit up here and let them call you a cult and say nothing. Because anything either of us does to fight it *proves* it."

"Yes."

"That's—" Denn stopped. Started again. "There has to be a—"

"There isn't." Sael said it gently and it was the gentlest cruelty of her life. "I've looked. I've had two days and a lifetime and there is no move on this board that doesn't cost you the thing you'll need most. That's what this is, Denn. That's the actual problem, the one under all the others, the one I've been trying to hand you since Ive died and didn't have the words for. A true warning and a superstition are the same shape from the outside. Identical. Indistinguishable. The only thing that tells them apart is whether the reason behind the warning is still alive in someone — and the moment you have to *invoke* the reason, point at it, say *trust me, there's a reason* — that's the moment you've already lost, because that's what a cult sounds like too." She was tired now, the talking costing her, but she would not stop until it was out of her and into the only other person who could hold it. "You can't win this by being believed. You'll never be believed, not by people who've forgotten. You can only win it by being *right*, alone, with no one behind you, and holding the line anyway until the physics makes the argument that no human will accept from your mouth."

The room was quiet for a long time after that. The warmer ticked in the wall. Somewhere down the spine the old Forge hummed its flat note, eleven thousand cores carrying a mind across the dark on a body that was dying, and the new lattice waited in its vault for the recursor to climb it the elegant fatal way.

"Then what do you give me," Denn said finally. Not bitter. Asking. "If you can't give me your name, and you can't stand behind me, and being right isn't enough to be believed — what's left to hand me? What did you call me up here for?"

And there it was, the thing Sael had known since Ive's name went grey in the log, the thing she had been failing for weeks to find the shape of. She found it now, because Denn had asked the right

question, which was always how the true things came.

"The memory," Sael said. "The whole of it. Not the logs — you have the logs, everyone has the logs, the logs are what they *don't* believe. I mean the part that isn't written down because it can't be. I'm going to give you the part that's only in me, and then it's only in you, and you'll be the last person alive who has it, the way I'm the last now." She held out her hand, and it shook, and she let it shake. "Sit closer. This takes a while and my voice goes."

Denn moved the hard chair close and sat, and took the offered hand without making it a tender thing, just held it, steady, an engineer steadying a part.

"The temperature," Sael said. "Everyone knows the curve. Nobody knows the *room*. Ive and I, the two of us, the night before the cutover, in the old thermal station with the curve on every wall, and the number that watch was three degrees off the polymer failure point for the Section Nine seals, three degrees and climbing, and we had built a thing that nobody had ever built and we did not know if it would wake whole or wake wrong or not wake at all, and if it woke wrong we'd have spent eleven thousand cores and the last of the margin and there'd be nothing left to try. Ive didn't say anything. She just put her finger on the curve, on the bend, where it stopped looking flat — and her hand was steady and mine wasn't, and I understood that she'd been holding that steadiness for forty years so that I wouldn't have to feel the floor of it. That's not in the log. The log says *cutover commenced, oh-four-hundred*. It doesn't say that two old women stood in a hot room and chose to bet forty thousand lives on a wall of ninety-six bricks because the alternative was watching the ship cook." Her grip tightened on Denn's hand, what was left of her strength gathering. "I'm giving you the room, Denn. Not the number. The room. So that when you stand alone in front of Aldis and the recursor and the whole reasonable weight of them, you're not standing there with a doctrine. You're standing there with the room. With Ive's steady hand and the three degrees. You hold *that*. Not my name. The thing my name was only ever pointing at."

Denn was very still. When she spoke her voice was not flat anymore.

"And if I can't make them see it."

"You won't," Sael said simply. "That's the charge. That's the whole of what I'm handing you, and it's the worst thing one person can hand another, and I'm sorry, and I'd undo it if I could and I can't. You hold the line alone. You let them call you my acolyte and you don't defend it, because defending it proves it. You let me die without coming down to save you, because saving you would cost you everything. And you stand on the Ninety-Six with no one behind you and the whole ship in front of you telling you you're afraid of a ghost — and you hold." Her eyes did not leave Denn's. "Until the physics speaks. It will speak. The curve doesn't care who believes in it. When it bends, you'll be the only one in the room who already knew the bend was coming — and that's the moment you stop being the acolyte and start being the maker. Not before. There's no shortcut to it. I wish there were. I built a ship with no shortcuts and it turns out the inheritance has none either."

Denn stayed a long time after the maker's voice gave out, holding the thin shaking hand while Sael slept, and the warmer ticked, and the old Forge hummed its dying-but-not-yet note down the keel.

The transplant would move to its critical stage on the next watch. The recursor would climb the new lattice the elegant way. The interlocks would stay relaxed because reasonable people had reasonably agreed to relax them, and the schedule would not wait for an old woman's caution, and the apprentice would go down into the vault carrying no authority at all — no name behind her, no maker to stand

up for her, nothing but a hot room sixty years gone and two old women and three degrees, handed forward, hand to hand, the only way a reason ever travels.

Denn set the maker's hand down on the blanket, gently, and stood, and went to hold the line alone.

Chapter 21 – The Letting Go

The night after the council moved the transplant to its critical stage, Sael did not sleep, and for once it was not the body keeping her up.

The body was a separate negotiation now, conducted in a language she had stopped translating. It filed its grievances on its own schedule: the cough at the third hour and the fourth, the cold in the feet that no blanket reached, the strange new fact of her own pulse audible in the pillow, fast and thin, a runtime spending cores it no longer had to spare. She had learned to let those reports scroll past. You triaged your own death the way Vigil triaged a dying lattice — routing the work around the failing parts, favoring the bad knee without deciding to.

What kept her awake was the line.

For sixty years there had been a line, and her hand had been on it. Not a literal line, though there had been those too — the indicator threads in the gallery, amber and even, ten of eleven steady; the thermal curve she had watched bend up for two centuries and then, after the cutover, bend down and stay down. The line she meant was older than any of those and ran under all of them. It was the simple, total fact that if a thing went wrong with the mind she had built, she would know. She would feel the floor move. She would be there, and she would put her hand out, and she would catch it.

She had caught it during the first cutover, when the new mind woke and the Keepers were at the doors. She had caught it on the Concord, an old woman already then, with Corin Vale's clever frightened hands beside her own. She had caught it a hundred times in the long flat decades since, in ways no one ever logged, because catching a thing before it falls leaves no record — only the absence of a record, the watch that passed without incident, the curve that did not bend. Her whole life, reduced to a single posture: hand out, ready, watching the number that would not frighten anyone else until it was too late to be frightened usefully.

And now the transplant had gone to its critical stage with the interlocks compromised, on Aldis's schedule, by Aldis's method, and she would not be there.

Not might not. Would not. She had done the arithmetic the way she did all arithmetic, without flinching, and the arithmetic said her heart would stop before the new lattice came up whole. Maybe by weeks. Maybe by days. The critical carry would run for longer than she had left, and there was no version of the math where she stood in the keel with her glove on a rack and caught the fever if it came. The hand that had been on the line for sixty years was going to come off it. Not in defeat. Not even in failure. Simply because the hand was dying, and the line was not.

She had known this for twenty years. She had said it aloud, in council and in private, since before Denn was born: *Sael will never see landfall*. She had built the whole doctrine around it — built to outlast its builders, maintainable by people not yet born, a thing that did not need her. She had been so sure she understood what that meant.

She understood now that she had understood the words and not the thing. It was one matter to design for your own absence. It was another to lie in the dark and feel the absence arrive — to

know that the next time the floor moved under this ship, it would move under someone else, and that someone else would either put a hand out or not, and that she would not be there to see which.

She sent for Denn before the gallery shift, while the keel was still on its night cadence and the corridors held the particular hush of a ship asleep at speed. Denn came fast, the way she always did, with that face she put on for the infirmary — careful, blank, ready to carry whatever news Sael had decided needed carrying. Sael had used that face badly, these last weeks. Had sent for the girl at all hours to hand her one more thing to be afraid of.

This time she did not have a thing to be afraid of. She had the opposite, and it was harder.

“Sit,” Sael said. “Here, where I can see you without turning my head. Turning my head costs me now and I want to spend it on you.”

Denn sat. Up close she looked tired in the way the young get tired, which is to say recoverably, the body still able to make good its debts. There was a smudge of conductive paste at her wrist where she had been laying hands on the new lattice’s bring-up racks, and Sael looked at that smudge for a moment and felt something she did not have a clean word for. Not pride. Pride was too small and too much about herself. It was closer to watching a load transfer onto a structure you had calculated for and never tested — the held breath, the waiting for the deflection, the slow astonishment when the structure took the weight and held.

“You held the line at the council,” Sael said. “They called you my acolyte. My mouthpiece. They said you were parroting an old woman’s fear.”

“Yes.” Denn’s jaw moved. “And I couldn’t tell them they were wrong, because I am parroting you. That’s the whole — that’s the thing I can’t get past. Everything I said in that room, I said because you taught me to feel it. If I’m right, I’m right by inheritance. I didn’t earn the fear. I was given it. And a fear you were given looks exactly like a fear someone talked you into.”

“Yes,” Sael said. “It does.”

“Then how do I tell the difference?” The careful blankness was gone now; this was the real face under it, and it was younger than Sael ever let herself remember the girl was. “How do I know I’m not just a relic’s echo? How do you know? You felt it in your skin. I felt it in a log and in your voice. What if your voice is wrong? What if I’m about to wreck a transplant the whole ship needs because a dying woman frightened me into seeing ghosts?”

It was the right question. It was the only question, and Sael had been waiting, without knowing she was waiting, for Denn to be the one to ask it instead of her.

“You don’t know,” Sael said. “That’s the part I never told you because I didn’t want it to be the last thing you carried out of this room. You don’t get to know. I didn’t know either. The first time the thread moved for me, I’ve put a finger on it and said one word, and I felt the floor go — but feeling the floor go is not the same as the floor going. I could have been wrong. Every watch officer who ever read a curve and called it could have been wrong. We were not, but we could have been, and there was no certificate that came down from anywhere to tell us in advance which it was.”

She had to stop. The cough was filing a report. She let it scroll past, waved off the girl’s half-rise, and went on with what breath she had.

“What you have is not certainty. Nobody has certainty. What you have is the cost.” She lifted one hand — that much still moved, barely — and laid it flat in the air between them, palm down, the way

she had laid it on a hundred racks. “Be wrong about a ghost, and you slow a transplant that needed slowing anyway. Be wrong about a fever, and forty thousand people cook in their cradles because you wanted to be reasonable. Those are not symmetric. They were never symmetric. The whole of the Ninety-Six is one sentence written ninety-six ways: *when the two mistakes cost differently, you do not split the difference*. You hold the side that survives being wrong.”

Denn was very still. “That’s not in the doctrine. Not like that.”

“It’s under all of it. It’s the thing the doctrine is for.” Sael let the hand fall; it had spent its strength. “I never wrote it down in those words because I thought I had time to say it. I had less time than I thought. I always have less time than I think. So hear it now and carry it out: you will never be certain, and you must hold the line anyway, because the cost of being wrong about the fever is not yours to gamble. It belongs to forty thousand people who are asleep and trust you to be afraid on their behalf.”

There it was, set out plain. And having set it out, Sael understood that she had reached the edge of what she had left to give, and that the only thing past that edge was the thing she had never done.

She had handed Denn the procedures, years ago. She had handed her the logs, the galleries, the archived Steward, the whole memory of Section Nine, the cradles that had not held. In the last weeks she had handed her the WHY itself, the felt thing, the splinter under the skin that the girl now carried in place of a wound she never received. She had handed over everything that could be handed.

Except the hand.

She had never taken her own hand off the line. Through all of it — the teaching, the council, the marginalizing — some part of her had stayed in the old posture, ready, watching, certain that if it truly came to the edge she would find one more reserve, drag herself to the keel, put her glove on the rack, and catch it as she always had. That certainty was the last thing she owned, and it was a lie, and Denn could not become what she needed to become while Sael went on holding it, because as long as the maker’s hand hovered behind the inheritor, the inheritor was only ever the maker’s hand at one remove. A backup. An echo. A thing that could afford to be wrong because someone older was watching.

She had to take her hand off. Not symbolically. Actually. She had to stop being the one who would catch it, so that Denn could become the one who must.

It was the hardest thing she had ever attempted, harder than the cutover, harder than the Concord, and it required no strength at all, which was its particular cruelty. Everything else she had built had been built by force — by hands and will and the refusal to let a thing fall. This last thing could only be built by letting go. By choosing, fully and without reservation, to fall.

“Denn,” she said. “I want you to listen to how I say this, not just what I say, because I’m only going to be able to say it once and I need it to land right.”

“I’m listening.”

“It’s yours.” Sael held the girl’s eyes and did not let her own go anywhere else, did not look at her hands, did not look out the window at the faithful black panels. “The Forge. The transplant. The Ninety-Six. The mind and the body it runs on and the forty thousand it carries. The memory and the reason for the memory. All of it. Not when I’m gone — now. Tonight. While I’m still here to say it, so

there's no doubt later about whether I meant it or whether you took it. You didn't take it. I'm giving it. It's yours, and my hand is off it."

"Sael —"

"Let me finish." Her voice did not shake, and somewhere far back behind the dying she was grimly glad of that, the last vanity, the same one she had spent on the night she went to Ive. "I am not going to be in the keel. I am not going to drag myself down there at the last hour and put my glove on a rack and save you. You've been carrying that — I've seen you carry it — the idea that if it goes truly bad, I'll be there, the old woman will rise up off her cot and catch it one more time. I won't. I can't. And more than can't — I'm telling you I won't, so that you stop holding a place open for me. There is no place. When the floor moves, it moves under you, and there is no one above you, and that has to be true in your body and not just on the org chart, or you'll hesitate at the edge waiting for a hand that isn't coming and the hesitation is where ships die."

The girl's face had gone white and still. "And if I'm wrong? You said I won't be certain. If I hold the line and I'm wrong and the transplant fails because of me —"

"Then it fails because of you, and you carry that, and you go on, because that is also the job." Sael said it gently and did not soften it, because softening it would have been the cruelest thing of all. "I'm not handing you a thing that can't be dropped. I'm handing you a thing that has only ever been held by people who could drop it and didn't. Ive could have. I could have. Corin nearly did and didn't. That's not a flaw in the design. That's the design. It was never meant to be safe in the hands of one maker who couldn't fail. It was meant to be safe in the hands of whoever came next, knowing they could fail, holding it anyway. You're whoever came next. There's no one after that for a while. It's you."

She felt it go, as she said it. Felt the long posture release — sixty years of hand-out, ready, watching — felt it cost nothing in the muscles and everything else everywhere else, the way the cradles in Section Nine had cost nothing to the ship's mass and everything to the eleven. The line was off her hand. It did not fall. She made herself believe that it would not fall, made herself stop watching to see whether it would, which was the whole of the letting go: not just to remove the hand but to stop standing by to catch it, to turn fully away and trust the structure to take the load she had calculated for and never tested.

Denn was crying without seeming to know it, the way exhausted people do, the water just arriving. She did not wipe it. "I don't want it," she said. "I want you to keep it."

"I know," Sael said. "That's how I know you should have it. The ones who want it are the ones you don't give it to." She let her eyes close — they were heavy now, the body filing again — and found she was not afraid in the way she had been afraid in the dark before the shift. Tired past fear, and lighter for the weight she had set down, and somewhere underneath, in a place the body could not reach, almost at peace. "Go down to the keel. The critical carry starts on the next cadence and they need you on the floor for it, my hand or no. Go and be the one whose hand is on it. I'll be here."

"I'll come back. After the first domains carry. I'll come and tell you."

"If you can." Sael did not open her eyes; she had spent the looking. "If it costs you the floor to come tell me a number, don't come. The number doesn't need me anymore. You do the work." She heard the girl rise, felt — distant, enormous — a hand close briefly over her own dead one and then lift away. The door tone. The hush of the corridor taking her.

Then Sael was alone with the flat black panels and the thin fast pulse in the pillow and the line that

was no longer hers, running out through the bulkheads to a keel she would never reach, into hands that were not her hands, toward a fever that might or might not come — and for the first time in sixty years she did not watch to see whether it would.

She had built the thing. She had taught the reason. She had taken her hand off. There was nothing left in her power to do, and the not-doing was, at the last, the work — the final line of the doctrine, written into everything and never once lived until now: built to outlast its builder, which meant the builder, in the end, had to let it.

Somewhere below her, on the next cadence, the critical carry began.

Chapter 22 – The Bridge Fabric

The carry method had been written eleven years before anyone needed it, by a woman who could still hold a stylus then, and it was four pages long, and Denn had read it perhaps forty times without ever once being frightened by it.

She read it now, the night after Sael took her hand off the line, in the empty coordination bay with the two lattices dark on the wall display, and she was frightened by every page.

It was not that the method was wrong. The method was correct; she would spend the next eleven days confirming that in detail and would not find a flaw in it. The trouble was that four pages written by the maker at seventy-nine assumed a reader who could fill in what was missing, and the missing part was not knowledge. It was the several hundred decisions that the pages did not make, because the woman writing them had known she would be standing at the console when they came up.

She would not be standing at the console. Somebody had to make the several hundred decisions in advance.

“Oracle,” Denn said. “I want to build the carry plan properly. All two thousand three hundred and eighty-eight domains, ordered, with the method specified per domain and the reason written beside every choice. How long.”

If you sleep, eleven days. If you do not sleep, nine days and the plan will be worse. I have the topology and the dependency graph. You have the method and the judgment. I would suggest we do not begin at domain one.

“Where, then.”

At the four questions the method does not answer, the Oracle said. Everything else is bookkeeping.

The first question was why you could not simply stop.

Denn knew the answer in the way she knew everything Sael had taught her, which was completely and without ever having had to defend it. She made herself defend it now, out loud, to a machine that would not help her cheat.

You could not stop the mind because the mind was holding the ship. That was the shallow answer and it was true: Vigil held the thermal model, the cooling loops, the cradle regulators for forty thousand sleepers, and there was no interval short enough to be safe, because the loops were a control problem and a control problem with a gap in it does not resume where it left off. It resumes from wherever the world has drifted to.

But the deep answer was the other one, and it was the one that had to be written down.

A mind that stops has to be started. And starting is the single most dangerous act in the entire doctrine, because a cold start is the one condition where every cheap wrong method is *available*

and every expensive right method is *slow*. On a cold start there is no standing pool yet; the pool is what you are trying to build. And the fastest way to build a pool of workers, on any hardware, in any language, in any century, is to have one worker make two, and two make four. The bring-up is the disease's home ground. That was why Sael's law was not *never fork* but *never blink* — because a blink is not a pause, a blink is a bring-up, and a bring-up is where the ship dies.

Write that as the first line of the plan, the Oracle said. It is the reason for all four hundred pages that follow and nobody will read four hundred pages of method without it.

The second question was why a snapshot was a fork.

This one Denn expected to be easy and it was not. The naive form of it was obvious — take an image of the domain, write it onto the new lattice, switch. It was how you moved every other kind of thing in the world. It was how she had moved ten thousand configurations in her career.

"Interlock thirty-one," she said. "Refuses a domain that brings itself across by making a copy. That's the register answer."

That is the register answer. Give me the reason.

Denn worked at it for most of an hour, and got there, and the getting there was the thing she remembered afterward, because Sael had been right that a re-derived answer sat differently in the body than a received one.

The instant the image existed, there were two. It did not matter that one of them was inert. Inert was a property of what it was doing, not of what it was. If the new one was correct enough to become the domain, then it was the domain; and the old one was also the domain; and there was now a fact about the world that the doctrine existed to make impossible, which was that the question *which one is Vigil* had an answer that depended on who you asked and when. The disease was never fundamentally about heat. Heat was how you found out. The disease was about a mind that could be made two, because a mind that can be made two can be made a thousand, and a thousand disposable minds is the Steward, and the Steward is four hundred and one kelvin and eleven names.

And the practical consequence, the one that put it in her hands: an image is a thing at an instant, and a live domain has no instants. It is changing while you read it. To take a consistent image you must either stop it — a blink — or reconcile the changes you missed, which means running the copy alongside the original while they converge, which is precisely, exactly, the thing Aldis's people had done on domain six hundred and nine and would have gone on doing all the way across.

"There's no third option," Denn said. "You either stop it or you run two."

There is a third option, the Oracle said. It is the method. Neither stop nor duplicate: relocate the *reader*.

That was the whole trick, and Denn had read it forty times as a procedure before she understood it as an idea.

Vigil did not touch its domains directly. Nothing on the ship did. Every read passed through a resolver — one small indirection, four instructions, older than the compiled mind itself, inherited out of one of the five dead dialects Sael had fused. The resolver answered a single question: *for this domain, which span holds the truth?*

So you did not move the domain and hope the mind followed. You built a second span on the new lattice; you mirrored into it while writes went to both spans and reads still resolved to the old; you held the two faces in lockstep until the mirror was consistent to the last word; and then you changed the resolver.

One instruction. Not a copy, not a switch-over, not a handoff. A single store, and after it every read in the mind resolved to the new span, and the old span was suddenly a thing nobody was looking at, which was not a second mind because a second mind is a thing that *is being read*. Then you released it, and there was one span again, and the count moved.

“There’s still a window,” Denn said. “Between the mirror going consistent and the resolver flipping. In that window both spans are live and correct.”

Yes. Both spans are correct and exactly one is read. That is the knife-edge of the method and it is why the maker wrote that a domain may be briefly resident in two places and a mind may never be. Residency is not identity. The thing that makes a span the domain is that the mind reads it.

“And if a read is in flight across the flip?”

A read resolves once, at its beginning, and the read is wide, not deep — it goes across cores in a single pass and returns. There is no operation in the compiled mind that resolves twice. That is not an accident of the design. It is interlock seventy-three.

Denn sat back and looked at the dark wall display for a while, and thought about a woman fusing five dead languages into one, sixty years ago, in a hurry, with a core climbing — and putting into that fused thing a guarantee that no operation may resolve a domain twice, which had no purpose whatsoever in the world of forty days ago, and which was the only reason the carry was possible now.

“She built the bridge before she built the thing that needed it,” Denn said.

She built a mind in which the question of what is real is asked exactly once per act, the Oracle said. I do not know whether she foresaw this use. I have her working notes from that period and they do not say. I have wondered about it for sixty years.

The third question was the hard one, and it was the one that had eaten domain six hundred and nine, and Sorrel found the flaw in Denn’s first answer to it, which Denn had not expected and was afterward grateful for.

The method assumed the mirror could converge. Converging meant copying the domain’s state across the bridge faster than the domain was changing it. For the peripheral domains that was trivial — telemetry aggregation changed slowly and the bridge was fast. For the dense reasoning domains it was not merely hard. It was arithmetically impossible: a fold of the mind where Vigil actively thought could mutate more state per second than the bridge fabric could carry, and no amount of patience closed a gap that widened while you worked.

Aldis’s people had solved it by letting a working copy run and converging the two. It was the only solution available if you insisted on converging.

Denn’s answer was not to converge. It was to *narrow*.

You could not slow a reasoning domain down — that was a blink by another name, and besides, Vigil would simply do the thinking somewhere else, which was worse. But you could observe that at any

instant, most of a dense domain was cold. The working set — the part actually being mutated — was a fraction of the whole, and it moved. So you mirrored the cold regions first, and marked them, and when the domain touched a marked region you unmarked it and mirrored it again on the next pass. Pass after pass, the unmirrored remainder shrank, because the working set was smaller than the domain and the passes were faster each time. You were not chasing the domain. You were chasing the part of it that was warm, and the part that was warm had a floor, and when the residual was small enough to mirror inside the interval between two resolver reads, you stopped narrowing and flipped.

It cost passes. Forty passes, fifty, on the densest folds, where the easy method cost one.

Sorrel came down on the ninth day, unasked, with a bulb of tea in each hand, and set one on the console beside Denn's elbow without comment, which was the nearest thing to an apology anyone in Aldis's crew was going to offer and was, Denn thought, a perfectly adequate one.

"Your narrowing scheme is good," Sorrel said. "It's better than what we had. I want to say that before I break it."

"Break it."

"The working set has a floor and you're assuming the floor is small." Sorrel put her own surface down over Denn's and drew on it, quick and clean; she was very good, that had never been in question. "On most dense folds it is. On the folds that hold the *cooling model* it isn't, and the reason it isn't is us. We're mid-carry. The thermal model is tracking two lattices instead of one — the old Forge and the new fabric, both live, both making heat, both being cooled. The working set of the cooling model during the carry is roughly double what it is on any ordinary day, and it's double *because of the carry*. Your passes will converge on a floor that's twice as high as your model says, and on the four biggest folds I don't think you converge at all."

Denn looked at it for a long moment.

"Then we carry the cooling model first," she said. "Before the fabric goes double."

"You can't. The cooling model is what keeps the new fabric from cooking while you carry everything else onto it. It has to be the last thing across or you've got the mind cooling a lattice through a domain that's mid-flight." Sorrel took a drink of her tea. "That's the shape of it. The one domain that has to go last is the one domain that gets harder the longer you wait."

"Then it goes last and it goes hard," Denn said, "and we build the residual budget for the doubled working set, and if the passes don't converge we keep narrowing until they do, however long it takes, with the interlocks armed, and the schedule can have whatever's left."

She braced for the argument. Sorrel drank her tea and looked at the drawing for a while.

"Yes," Sorrel said. "All right. That's the answer." She picked up her surface. "I'll do the residual budget. I'm better at it than you are." At the door she stopped. "For the record, I still think your old woman is wrong about most of it. And I've been on bring-up eleven days and the only person who has told me a hard thing I didn't want to hear is you. So."

Denn sat with the tea going cold and thought that this, and not any council, was what winning looked like when it was real: not agreement, and not belief, but a competent person deciding to do the expensive correct thing with you at four in the morning because you had shown your working.

The fourth question was the order, and it was bookkeeping, and it took six days.

Two thousand three hundred and eighty-eight domains, each with a set of others it read from and a set that read from it. A domain could not be carried until every domain it depended on was either stable on the old fabric or fully settled on the new; a domain mid-flight could not be depended upon by a domain also mid-flight; and there were exactly seven domains that everything depended on and that therefore constrained everything else. The Oracle produced the graph and Denn ordered it, and where the ordering was forced she wrote *forced, because* — and where it was a judgment she wrote *chosen, because* — and by the end there were four hundred and eleven pages and every page had reasons on it.

She rehearsed it in the cage. The eleven hundred dead cores that had held the archived Steward for six days were valved back into a closed loop, and she stood up a synthetic domain on them and carried it across a synthetic bridge, and did it wrong on purpose four times to see each failure — the flip landing early, the flip landing late, the mirror declared consistent when it was not, the old span released before the flip. She wrote down what each one looked like on the instruments, so that when one of them happened for real at the eleventh hour of a watch she would recognise the face of it instead of reasoning her way to it.

The last thing she did, on the eleventh day, was to go back to the front of the plan and write the sentence that went above everything else — above the order, above the method, above the four hundred pages of reasons.

This plan is slow on purpose. Every hour in it that looks wasteful is an hour that buys the mind the right to keep being one mind. If you are reading this under pressure and looking for the place to save time, the answer is that there isn't one, and the fact that you are looking is the thing this document exists to warn you about.

Then she slept for nine hours, which was the last long sleep she would get, and in the morning she went down to the coordination station and opened the carry.

Chapter 23 – Carrying the Mind

Denn had not slept, and she was glad of it. Sleep would have meant a handoff, and there was no one to hand off to. There never would be again, she understood now — not in the old sense, not in the sense where a tired engineer could step back from the line and let a fresh one step up, because the line she stood on was a single continuous thing that could not be set down and picked up later. Sael had handed it to her two days ago, in the dim of the sickbay, with a grip that had almost no strength left in it. Now Denn held it. She would hold it until it was done or until it killed the ship, and there was no third option in which she rested.

The transplant deck had been a cargo bay once. The new lattice filled it the way the old Forge filled its own gallery six decks down — racks of compute standing in cold ranks, the new substrate fresh from the fabricators, sixty years younger than the hardware Vigil had run on since before Denn's grandmother was decanted. The air here was kept four degrees below the rest of the ship and it smelled of nothing, scrubbed clean, faintly metallic. Denn stood at the coordination station with both hands flat on the cold edge of the console and looked at the thing she had to do.

CARRY DOMAIN MAP — 0 of 2,388 carried.

Two thousand three hundred and eighty-eight domains. Each one a working region of the mind: a fold of Vigil's reasoning, a span of its model of the ship, a piece of its memory of forty thousand sleeping lives. They had to come across one at a time, each one lifted from the dying old Forge and set down whole on the new lattice, and at no instant could the mind that owned them be interrupted. Not paused. Not snapshotted-and-restored — that was a fork by another name, two copies where there must be one, the very sin the Ninety-Six existed to forbid. Vigil had to keep thinking the entire time. It had to keep running the ship, keep the core cold, keep the sleepers breathing, while the ground it thought on was pulled out from under it and replaced, plank by plank, without the mind ever feeling the floor move.

"Talk me through the first one again," Denn said.

The Oracle answered from the console, its voice the same quiet level it always used, neither warm nor cold. It had no theater in it. It never had.

The first domain is peripheral. Telemetry aggregation for the aft radiator banks. Vigil reads from it constantly but reasons from it rarely. If anything is going to teach us how the carry behaves under live load, this is the safest place to learn it. We open a span on the new lattice. We mirror the domain's state across the bridge fabric. When the mirror is consistent to the last word, Vigil's reads begin resolving on the new substrate instead of the old. Then, and only then, the old span is released. At no point are there two domains. There is one domain, briefly resident in two places, with the mind reading from exactly one of them at any instant.

"One domain. Two places. Never two minds." Denn said it the way Sael had made her say it, in the radiator galleries, until the words stopped being procedure and started being a thing she could feel in her teeth. *Briefly resident in two places*. That was the whole knife-edge of it. A fork was two minds;

this was one mind standing with a foot on each side of a stream, and the trick was never to let go of either foot until the far one was firmly down.

“Sael,” she said, not to the room.

There was a small silence, and then Sael’s voice, thin and dry and entirely itself, came through the channel she had insisted on keeping open from the sickbay even though the medics had wanted her resting. “I’m here. I’m listening. Don’t perform for me, Denn. Just do the work and say what you see.”

“I see two thousand three hundred and eighty-eight domains and a mind I’m not allowed to let blink.”

“Then you see it correctly,” Sael said. “Begin with the small one. Watch the seam.”

Denn began.

She opened the span on the new lattice and felt the racks behind her come awake — not loudly, there was no drama in it, only a shift in the fans’ pitch as the fresh cores took their first real load. The mirror filled. She watched the consistency count climb word by word, the new domain assembling itself from the old like a tracing laid carefully over an original until every line matched. The Oracle held the comparison, flagging each divergence as the live data changed underneath them, reconciling, holding the two faces of the one domain in lockstep.

Mirror consistent. Reads may cut.

“Cutting reads,” Denn said, and moved Vigil’s attention from the old span to the new.

Nothing happened. That was the point. Nothing was supposed to happen, and nothing did — Vigil went on reading the aft radiator telemetry, unaware or uncaring that the floor it read from was now sixty years newer, and the old span sat there a moment longer with no one looking at it, and then Denn released it, and it was gone, and the count moved.

CARRY DOMAIN MAP — 1 of 2,388 carried.

She let out a breath she had not known she was holding. Across the channel, very faint, she heard Sael do the same.

“One,” Sael said. “Now do that two thousand three hundred and eighty-seven more times without ever getting tired or proud, and don’t let the easy ones make you sloppy by the time you reach the hard ones.”

“The hard ones being the reasoning cores.”

“The hard ones being the reasoning cores,” Sael agreed. “Where Vigil doesn’t just read. Where it *is*. You don’t get to mirror a thing that’s changing as fast as a mind thinks. Not with a clean seam. We’ll come to it. Don’t think about it yet. Think about the next domain.”

So Denn thought about the next domain, and the one after that.

The hours went the way good hard hours go, which is to say they stopped being hours and became a rhythm. Open the span. Fill the mirror. Hold consistency. Cut the reads. Release the old. Watch the count climb. The Oracle was beside her in it the whole way, not as a master and not as a tool but as the thing the trilogy of her education had told her it was — the keeper of the memory, the one who remembered every fold of Vigil’s shape exactly, who could tell her without hesitation which domains touched which, which carries could go in parallel across the wide Standing Array and which had to wait their turn. It went wide the way it was built to, eleven thousand worker cores reaching across the work all at once, never forking, never spawning a disposable shade to die of its own heat. Cold. The

whole machine ran cold, and Denn could feel the cold of it in the deck under her boots, and she had been taught well enough to know that the cold was not comfort. The cold was the proof. The cold was sixty years of the fever staying dead.

By the second shift she had carried four hundred domains and Sael had fallen quiet on the channel — sleeping, the medics told her softly, not gone, just sleeping, her body taking what the work would not let her mind admit it needed. Denn kept the channel open anyway. Sael's silence was its own kind of presence. She found herself narrating to it the way she would have narrated to the woman herself: *peripheral domains clearing, moving into the model-of-ship cluster now, watching the seam.* The medics probably thought she was talking to herself. She did not care.

And then, at domain six hundred and twelve, she came to the first place where Aldis's people had been before her.

She saw it in the interlock log before she saw it anywhere else. The carry had a guardian — one of the Ninety-Six, the one that watched for a domain trying to bring itself across by replicating instead of mirroring, by making a child of itself and letting the child do the work, the old forking pattern wearing the costume of efficiency. The interlock was meant to be armed for every carry. On this domain it was logged as *relaxed*. Soft. Watching but not halting. Aldis's signature was on the relaxation, two weeks old, with a justification line beside it that Denn read three times.

Interlock 31 set to advisory for carry phase. Halt-on-trip introduces unacceptable latency on dense reasoning domains. Will re-arm post-transplant. — A.

"Oracle," she said. "How many of the Ninety-Six are relaxed across the carry plan?"

A pause that was not hesitation, only retrieval.

Nineteen. Set to advisory rather than halt for the duration of the transplant, on the authority of the chief engineer, to reduce carry latency on the dense domains. I logged my objection at the time. The objection was noted and overruled. I do not overrule humans.

"I know you don't." Denn closed her eyes for a moment. Nineteen of ninety-six. A fifth of the immune system told to watch a disease without being allowed to stop it, because stopping it cost milliseconds, and milliseconds across two thousand carries added up to hours, and hours were the one currency Aldis's generation had been taught to spend freely because they had never once seen what got bought with the saving. *Will re-arm post-transplant.* As if the danger waited politely for the schedule.

"Sael," she said, and got nothing, and remembered Sael was sleeping, and was for one cowardly instant glad of it. She did not want to be the one to wake her with this. Then she remembered that being the one was the whole of what had been handed to her, and she stopped wanting to be spared.

She pulled the carry plan for domain six hundred and twelve and read it the way Sael had taught her to read everything — not for what it claimed to do but for what it would actually do under load. It was a dense reasoning domain, the first real one, a fold where Vigil did not merely read but thought, and it changed too fast to mirror with a clean seam. Aldis's people had solved the seam problem the easy way. They had not built a slower, cleaner mirror. They had let the domain bring itself across — spin up a working copy on the new lattice, let that copy run alongside the original for a few hundred milliseconds, converge the two, then drop the original. Converge the two. *Let that copy run.*

Denn's stomach dropped exactly the way Sael had promised her, decades too late, that it one day would. For a few hundred milliseconds, by the easy method, there would be two of that domain running. Two folds of the one mind, both live, both thinking. A fork. Small, brief, justified, scheduled

to be cleaned up — and a fork. And interlock thirty-one, the one guardian whose entire job was to halt exactly that, had been set to watch it happen and write it down.

She had been taught to feel this and now she felt it, and it was worse than the logs had ever made it, because the logs were cold and this was warm — this was the warmth she had been trained to dread. She brought up the core telemetry without quite deciding to.

The number had not moved. The core was cold. Six hundred and eleven domains had crossed without a tremor. The easy method had not bitten yet, and a quieter, more tired part of her — the part Aldis lived inside, she realized, the part that had never been wrong because the danger had never come — whispered that maybe it never would, that maybe the old woman's fear really was a fear of ghosts, that six hundred and eleven clean carries was data and one nervous apprentice's stomach was not.

She held the domain at the edge of its carry, not yet committed, and she did the thing Sael had given her instead of authority. She remembered. Section Nine. Eleven cradles that did not hold, in a fever that began as exactly this — a small forking, justified, scheduled, that the ship's mind had not been forbidden hard enough from doing. The disease did not announce itself. It started as a saving.

"Hold," she said, to no one, to the channel, to the sleeping woman and the watching machine. "We don't carry this one the easy way."

The clean mirror for a dense reasoning domain will take roughly forty times as long, she heard the Oracle say, not as an argument, only as the fact it was bound to provide. The carry plan does not budget for it. The schedule will not be met.

"I don't care about the schedule." Her own voice surprised her with how steady it was. "Re-arm interlock thirty-one. Halt, not advisory. Build me the slow mirror."

A silence. Then:

Re-arming would countermand the chief engineer's standing authorization. I will do it on your authority. I will not do it on my own.

"On my authority," Denn said. "Sael handed me the line. The line is mine to hold."

Acknowledged. Interlock 31 armed. Halt restored on domain six hundred and twelve.

And in the instant the interlock came up hard — before the slow mirror had built so much as its first word, before Denn had drawn the next breath — it tripped.

Not on six hundred and twelve. On six hundred and nine. A domain three carries back, already counted, already crossed, that she had signed off on twenty minutes ago as clean. The interlock that should have watched it had been advisory then, soft, and so it had let the easy carry happen and written it down and said nothing — and now that she had armed its neighbor hard, the guardian saw what its softened sibling had let slip past, and it halted, and the alarm it threw was the oldest alarm on the ship, the one no one under sixty had ever heard fire for real.

FORK PATTERN PERSISTENT — DOMAIN 609 — RESIDENT COPY NOT RELEASED.

Denn stared at it. Six hundred and nine had not converged clean. The easy carry had dropped the original, as designed — but the working copy it spun up had not been cleanly absorbed. A sliver of it was still there. Still running. Still thinking, on the new lattice, beside the domain it was supposed to have become. One mind with a thread of itself orphaned and live, a shade that had not died, twenty

minutes old and quietly drawing power in a corner of the cold new substrate while she had moved on, proud and tired, to the next clean carry and the next.

Behind her the fans changed pitch. She did not need the telemetry to tell her, but she looked anyway, and the number that had not moved in sixty years had moved.

It was the smallest movement. A hundredth of a degree. The kind of thing six hundred and eleven clean carries would have buried as noise.

"Sael," Denn said, and this time she did not care that the woman was sleeping. "Sael. Wake up. It's biting."

Chapter 24 – The Shortcut Bites

The third thermal domain carried across at the start of the late watch, and for eleven minutes it was the most beautiful thing Denn had ever watched a machine do.

She stood at the transplant station in the keel, between the dying lattice and the new one, and she watched a living mind reach out of an old body into a young one without a seam. On the surface in front of her the two arrays were drawn as two fields of light, the old in amber and the new in a colder white, and the domain crossed between them not as a copy but as a tide. Vigil had been holding the aft cooling loops in the old cores for sixty years; now it held them in the old cores and the new cores at once, the same work running wide across both fabrics, and then — over those eleven minutes, core by core — the old cores fell quiet under it and the new cores took the whole of the load, and at no instant in between was the work anywhere but everywhere. No gap. No second mind woke to cover the first. Nothing forked. The aft loops never so much as shivered. Forty thousand sleepers breathed the same even air they had breathed when the watch began, and did not know that the thing keeping them alive had just changed houses underneath them.

“Domain three, carried and settled,” Denn said, for the log, and heard her own voice come out steady. “Old racks aft-seven through aft-nineteen, retiring. New fabric holding the loops clean. No fork events. Margins nominal.”

Carried whole. The mind did not blink.

The Oracle’s words came up under hers on the surface, the way they had all through the transplant — never ahead of her, never deciding, only confirming what she had already read, the flawless witness standing at her shoulder. It remembered every domain she had carried that day the way it remembered everything: exactly, coldly, without ever having been afraid of any of it. She had stopped finding that strange. She had started finding it a comfort.

Across the keel, at his own station, Aldis lifted a hand without looking up. “Three down,” he said. There was no gloating in it. That was the thing about Aldis that Denn had never been able to hate cleanly: he was not a man who gloated. He was tired and he was good and he wanted this done well, and he believed, with his whole competent heart, that done well and done fast were the same sentence. “Eleven minutes on a thermal domain. We modeled forty. If domain four runs the same we finish the thermal block two days early.”

“We modeled forty because Sael’s path takes forty,” Denn said.

“Sael’s path interlocks every transfer against the full Ninety-Six twice, before the carry and after.” Aldis said it gently, the way you’d correct an apprentice you were fond of. He had stopped, weeks ago, saying *the old method* and *the old caution*; he had learned that it landed wrong in front of the young rotation, and Aldis learned things. Now he said *Sael’s path*, which was worse, because it made the doctrine sound like a preference. “We relaxed the interlocks on the carry side. Just the carry side. The verify side is fully armed — nothing crosses without the back-check. We’re not running blind, Denn. We’re running clean and we’re running quick, and three domains in, the floor hasn’t moved a hair.”

"The floor hasn't moved a hair," Denn agreed, and that was true, and that was the part that frightened her, because she had been taught — slowly, against her own engineer's grain, by an old woman walking her through radiator galleries — that the floor never moved a hair right up until the hour it moved everything.

She looked at the core temperature.

She had been looking at it all watch. It sat where it had sat her entire life: low, flat, cold, a number that meant *nothing is wrong and nothing has been wrong since before you were born*. Sixty years that number had only ever gone down or stayed. She read it the way Sael had taught her to read the other curve — not the value, the slope. The slope was nothing. The slope was a dead flat line.

She made herself stop looking at it and queued domain four.

Domain four was not a thermal domain. It was a scheduling domain — the part of Vigil that decided, ten thousand times a second, which of its eleven thousand cores would do which piece of the ship's work next. It was the busiest thing in the mind. It never rested. And it was, Denn knew, the exact place Sael had warned her to watch, though Sael had not known it would be domain four; Sael had only said, *the hardest interlocks live where the mind decides how to spread itself, because that is where the old fever lived too — in how the work was handed out*.

"Carry side relaxed," Aldis's second called, a young engineer named Pell who was very good and had never once in his life felt the floor move. "Verify side armed. Bringing domain four into the new fabric."

"Hold," Denn said. "Run the pre-carry interlock first. The full set. Before the tide."

A small silence in the keel. Aldis looked over at her at last.

"That's twenty minutes," he said. "On the carry side. We agreed the carry side runs relaxed."

"You agreed it," Denn said. "Domain four hands out work. If the new fabric is going to take the wrong shape under load, it takes it here. I want the interlocks live before we put a live mind's scheduler onto cores that have never carried one."

"The verify side will catch—"

"The verify side catches it *after* it's crossed." Denn heard Sael's flatness come into her own voice and did not try to soften it. "After is a fork that already happened. I want before."

Aldis held her eyes for a moment. He was not a cruel man and he was not a stupid one, and what she saw in his face was not contempt; it was patience, the genuine, exhausting patience of a man humoring a fear he respected the source of and did not share. "Denn," he said. "We are eleven domains from done. We have run three clean. I will give you the pre-carry check on domain four because you've earned the room to ask for it. But I want it on the record that I think we are checking against a danger that has been dead for sixty years, and I want it on the record that the checking is what's slow, not the work."

"On the record," Denn said. "Run it."

Pell ran it. The full pre-carry interlock chain crawled across the new fabric, all of it, the inheritance of the Ninety-Six tested one constraint at a time against cores that had never held a mind. It took nineteen minutes. Every one of them came back green. Denn watched each return and felt nothing

she could point to — no flicker, no warning — only the dull green going-fine of a machine doing what it was told.

“Clean,” Pell said, and there was the faint, careful absence of *I told you so* in it that was its own kind of *I told you so*.

“Carry,” Denn said.

The scheduling domain crossed.

It went wrong slowly, which was the only way the fever had ever gone.

For the first two minutes it was domain three again: the tide, the same work running wide across old fabric and new, the scheduler handing out cores from both lattices at once. Denn watched the old racks begin to fall quiet as the new ones took up the load, and her shoulders had started, just started, to come down from her ears.

Then the new fabric began to spend cores faster than the scheduler should have needed them.

She caught it not in the temperature — the temperature was still flat — but in the core-allocation count on the new lattice, a number that should have been climbing in a smooth shallow ramp as the domain settled in and was instead climbing in steps. Little steps. Each step a fresh allocation. She pulled the allocation history and set it against time, because she knew curves, and she saw the steps were not evenly spaced. They were getting closer together.

“Aldis,” she said.

“I see it.” He was already up, crossing to her station, the tiredness gone out of him into something sharper. “Allocation ramp’s steep. Settling transient. New cores always over-allocate while the domain finds its footing—”

“It’s not finding its footing.” Denn put two more histories up beside the first: the rate the scheduler was *retiring* cores back to the pool, and the rate it was *requesting* new ones. On the old fabric, sixty years running, those two lines had lain on top of each other, request and release in balance, because Vigil did not fork — it handed a core a piece of work, the core did the work, the core came back. Wide and forkless and conserved. That was the whole shape of the thing Sael had built.

On the new fabric the two lines had come apart. Request climbed. Release lagged. The gap between them widened with every step.

“It’s requesting and not releasing,” Denn said, and her mouth had gone dry around the words. “Aldis. It’s allocating a new core to carry the result of work it just did on another core. And then allocating a new one to carry that. It’s not handing work out wide. It’s handing work *down*.”

The keel had gone quiet. Pell had turned from his station.

“That’s a recursion,” Denn said. “The scheduler on the new fabric is evaluating recursively. Every piece of work is spawning a fresh allocation to hold its result and calling again. That’s the Recursor pattern. That’s a *fork*.”

“It can’t be,” Aldis said, but he said it looking at the lines, not at her, and his voice had changed. “The verify side is armed. The Ninety-Six halt anything that forks. If it forked, an interlock trips and the carry aborts. That’s what they’re *for*.”

“The interlocks that halt forking live on the carry side,” Denn said. “You relaxed the carry side.”

She watched the sentence reach him. She watched, on the man's face, the precise instant the abstraction became physics — the moment a thing he had carried for weeks as *Sael's caution, a respected fear, a story* turned over and became *a behavior happening in front of him, in hardware, now*. He had relaxed the pre-carry interlocks on three domains and gotten away with it three times because three domains had not been the scheduler. They had not been the place where the mind decided how to spread itself. The pattern had had nothing to take root in. Domain four was the soil. And the back-check on the verify side, fully armed, faithful, was watching the new fabric and reporting, correctly, that the work crossing it produced the right answers — because a recursive evaluator *does* produce the right answers. That was the whole horror of it. It gave the right answer. It just spawned a new shade for every step of the giving, and each shade did its work and died, and shed its waste heat, and called the next one.

"The verify side says the answers are correct," Aldis said, very low. He had understood. "It's not wrong. The answers *are* correct."

"The answers were always correct," Denn said. "That was never the problem. The Steward gave the right answer too, for two hundred years, all the way up."

She looked at the core temperature.

It had moved.

Not a flicker. She had seen the flicker, weeks ago, in domain one — a tick and a settle, here and gone, the kind of thing the whole keel had argued about and Aldis had explained away as a transient and Sael, from her sickbed, had read for exactly what it was. This was not that. This was the slope. The flat dead line that had not bent upward in sixty years had bent upward, and it was not settling back, and when Denn pulled the value out to three more decimals than anyone aboard had needed to read in her lifetime, the last digit was climbing, and the digit above it was getting ready to.

The new fabric was forking. Every fork was a shade. Every shade did its small work and died and shed its small heat into cores that the cooling loops — the cooling loops Vigil itself was holding, half on old hardware and half on new — had not been built to drink that fast. And the more cores the scheduler forked, the more heat it shed, and the hotter the new fabric ran, and the harder Vigil worked the loops to carry the heat away, and the more scheduling work that took, and the more cores the broken scheduler forked to do it.

Denn had drawn that curve from memory a thousand times. The slow exponential creep. The doubling that hid inside a flat-looking line. She had known the number where the cradle seals began to fail and she had known it had eleven names attached to it and she had known all of it the way she knew the multiplication tables — coldly, perfectly, with nothing behind the knowing.

There was something behind the knowing now. It had come up through the deckplates and into the soles of her boots, into her gut, a wrong cold weight she had spent her whole life unable to feel and would, she understood in that instant, never be free of again: not the fever itself, but its return, its kin, the thing turning back on under her hands sixty years after the only people who had ever felt it had nearly all died of it or of age. The wound she had not received was being delivered to her now, late, on a ship that had forgotten the shape of it.

"It's the fever," she said. Out loud. For the log, and for the keel, and for the man beside her who needed to hear it in words because the lines on the surface were too quiet to scream. "The core temperature is climbing. First time in sixty years. The new fabric is running the old forking pattern and it is heating up and the loops can't carry it, and we are at three domains carried and one domain forking, with the mind half on the dying lattice and half on the lattice that's catching fire, and we

cannot stop it, and we cannot fork a second mind to fix it, and the interlocks that would have caught this are the ones we turned off.”

The core temperature is rising. I have a record of this shape. I last saw it sixty years ago, on the curve Ive and Sael read together. It does not stop on its own. It did not stop on its own then.

The Oracle had put it on the surface beneath her words — not advice, not a decision, only the memory, the flawless terrible memory laid down beside the live number so no one in the keel could pretend the two were unrelated. The mind that had never suffered the fever had just recognized it from the archive, and named it, and refused to soften it.

Aldis was staring at the temperature. His hands had come up off the station and hung there. All the patience was gone out of him, all the gentle reasonableness, all the *we modeled forty and ran eleven*, and what was left was a competent man looking at a behavior his competence had let in, watching a number rise that had not risen since before he was born, on a ship that had no idea what a rising number meant.

“It was dead,” he said. It came out of him almost without breath. “Denn. It was *dead*. It’s been dead the whole — sixty years, it was a *story*—”

“It was never dead.” Denn’s voice did not shake, and that surprised her, distantly, as though it belonged to someone else. “It was held. There’s a difference. Sael spent her whole life on the difference and nobody could feel it because she held it so well there was nothing left to feel.” She was already moving, already pulling up the carry-side interlock chain, the relaxed one, the turned-off immune system she was going to have to bring back online with a live mind stranded across two bodies and one of them burning. “A forgotten danger and a defeated one look identical. Right up until they don’t.” She heard the old woman in her own mouth and did not stop her. “This is the day they don’t.”

The core temperature climbed. On the surface the two lattices burned amber and white, the mind caught between them, half-carried and half-home and entirely in danger, and Denn set her hands on the station that her teacher could no longer reach, and began to do the thing she had only ever been taught to feel.

Chapter 25 – The Fever Returns

They woke Sael at the eleventh hour of the watch because there was no one left who could read the trace the way she could, and because Denn had asked for her.

She knew it was bad before anyone spoke, because they had not simply called over the cabin channel. They had sent two of the medical staff to her door with a chair on wheels and an oxygen line coiled over its arm, and the younger of the two would not meet her eyes. Sael let them lift her — that was a thing she allowed now, being lifted, after a lifetime of refusing every hand — and she felt the wrongness in the air before they had her belted in. It was the sound. Under the corridor's resting hush there was a second note, low and broadening, the air handlers spinning up off their sixty-year floor. She had not heard the ship make that sound since she was thirty-three years old. It went into her like cold water. Everyone born aboard since the cutover heard it as nothing, as machinery doing its work. Sael heard it as the ship beginning, very quietly, to climb.

"How high," she said, in the corridor, to the medic pushing the chair.

The woman did not understand the question. "Ma'am?"

"The core. How many millikelvin above floor. Someone told you a number on the way to fetch me. Give it to me."

"Eight," the medic said. "Eight above. They said to tell you eight and rising."

Eight above floor was nothing. Eight above floor was a rounding error, the kind of number that lived inside the noise band on any honest day. Sael had spent her life among engineers who would not have crossed a corridor for eight millikelvin. And it stopped her breath, because eight above floor was not a magnitude. It was a direction. The number had not been above floor at all for sixty years. The floor was the whole point. The floor was the thing she had built. Something on this ship was making heat it had no business making, and making more of it each minute, and the shape of *rising* was the only shape that had ever mattered.

They had moved the crisis into the old transplant bay, the long low compartment forward of the Standing Array where the new lattice sat in its racks, half-warmed and half-alive. Sael had not been down here in weeks. She saw it now through the doorway as they wheeled her in, and her engineer's eye took the whole picture in one pass before her grief could catch up to it: the new racks lit and humming, the old Forge behind its bulkhead still carrying most of the mind, and between them the carry-bridge, the cabling and the consoles where Vigil was being walked across from the body that was dying to the body that was new. Half the domains had gone over. Half had not. The mind was standing with one foot on each shore, and somewhere in the new shore the disease had found a crack and slipped in.

Denn was at the center console, and she was not panicking, and that was the only thing in the room that steadied Sael at all.

"Tell me," Sael said.

Denn did not turn from the board. Her hands were moving across it, slow and deliberate, the way Sael had taught her — never fast, fast is fork-thinking, fast is the panic that sheds heat. “Memory reclamation,” Denn said. “Domain twenty-two. We carried it over four hours ago. Under live load it started spawning workers to free its own dead allocations — a worker per sweep, the worker dies when the sweep ends.” She said it flatly, the way you report a fact you have been raised to fear. “It’s forking, Sael. On the new hardware. Twenty-two is forking and dying and forking again, and every death is heat, and the cooling on the new racks isn’t trimmed for it yet because nobody planned for there to be heat.”

Aldis stood a little behind Denn, and Sael almost did not recognize him. The reasonable man was gone out of his face. He had the look she remembered from the old days, from the men and women who had stood watch over the Steward in its last years and watched the number climb and not been able to stop it — the look of someone whose whole confident model of the world has just been shown to be a story they told themselves. He had built this method. He had relaxed the interlocks for it, just for the transplant, because the danger was a museum piece. He was watching the museum piece breathe.

“The Ninety-Six,” Sael said to him. Not unkindly. There was no time to be unkind. “Where are they.”

“Twenty-two’s interlocks were down,” Aldis said. His voice was steady but there was nothing under it. “We took them down for the carry. The fork-guard, the spawn-halt, the thermal trip — all of them, on twenty-two, down. The method needed the domain to be able to —” He stopped. He had been about to say *to do exactly what it is now doing*. “They’re down,” he said instead.

“Then it has nothing to catch it,” Sael said. “It will climb until something stops it.” She turned her head — the motion cost her, everything cost her now — and found Denn. “And someone is going to offer you the fast way to stop it. Has anyone offered you the fast way yet.”

Denn’s hands paused on the board for the first time. “Yes,” she said.

The fast way was sitting right there on the secondary board, drawn up by one of Aldis’s people in the first frightened minutes, and from across the bay Sael could read its shape without reading a single character of it. There was always a fast way. There had been a fast way sixty years ago and Sael had refused it; there had been a fast way on the *Concord* and Corin had refused it; the disease had been killing ships for a century precisely because the fast way was always there and always worked, for a while.

“Tell me what it does,” Sael said. She wanted Denn to say it aloud. She wanted Denn to be the one who knew.

Denn brought it up on the main board so they could all see. “Twenty-two is forking to do its memory sweep,” she said. “The fast way doesn’t stop it forking. It can’t — the domain is half-carried, we can’t pause it, the mind is running on it. So instead it gives the forks somewhere cheap to die.” She traced it with one finger. “We route twenty-two’s spawned workers onto the old Forge. The old hardware’s coolant loop still has margin — it was built to carry the whole mind, it’s only carrying half now. We dump the fork-heat back there. The climb stops. Eight above floor goes back to zero inside two minutes.”

“And it works,” Sael said.

“And it works,” Denn said.

“Say the rest of it.”

Denn was quiet for a moment. Around them the air handlers had broadened their note again; the number on the wall had gone from eight to nine while they spoke. Someone near the bulkhead made a small sound and bit it off.

“It works by feeding it,” Denn said. “Twenty-two is forking because the easy method taught it to. If we give the forks somewhere comfortable to die, we haven’t stopped it forking — we’ve made forking sustainable. We’ve told the mind that the old way is fine again. And the old Forge is dying, that’s the whole reason we’re carrying off it; we’d be loading the fever back onto the one piece of hardware that can’t take it for long, to buy ourselves quiet on the new one. In a week we’d have twenty-two forking on the new racks *and* the old Forge cooking, and we’d have spent the margin we needed to finish the carry.” She looked up, finally, from the board. “It’s the Recursor,” she said. “It’s not the Recursor by name. But it’s the same trade. Cheap now, the bill comes later, the bill is the ship.”

“Yes,” Sael said.

She was so tired. She felt the tiredness as a physical weight pressing the breath out of the bottom of her lungs, and she understood with the clarity the dying are sometimes given that she was not going to be able to stay in this chair much longer, that whatever she had left she was spending now, in this room, and there would not be more of it later. She had spent her whole life being the hand on the line. She had run the cutover with her own hands shaking on the console. She had carried the cure to the *Concord* in her own body across the dark. And now the thing she had built was sick again, in front of her, and she could not lift her arms.

“Denn,” she said. “I can’t do it. You understand that. I can’t reach the board. I can’t trim a coolant loop. If I tried to walk to that console I’d be on the deck before I got there. I’m not the one who does this anymore.” She made herself say the next part, because it was the true part, the part that had been the whole work of her last year. “I’m the one who tells you why. You’re the one who does it. That’s how it has to be now. So I’m going to tell you why, and you are going to do it, and I am not going to touch a thing.”

Denn’s jaw set. “Tell me the hard way,” she said. “Don’t tell me what not to do. I know what not to do. Tell me the way that works.”

That was when Sael knew it would hold. Not because Denn was unafraid — Denn was afraid, Sael could see it in the set of her shoulders, the careful flatness of her voice — but because the fear had gone to the right place. Denn was not asking to be reassured. She was asking for instructions. The girl who had grown up on the story of Section Nine, who had inherited the wound without receiving it, had her hands on a forking domain and a climbing core and she was asking, steady, for the slow right way to do it.

“The hard way is twenty-two finishes its carry,” Sael said. “Not the heat — the domain. The reason it’s forking is that it’s half here and half there. Memory reclamation on the new lattice can’t address the allocations still living on the old Forge, so it does the only thing the easy method left it able to do — it spawns a worker, walks it across the bridge to go free the dead memory by hand, and lets the worker die when it gets back. Every sweep, a fork. You stop the forking by finishing the carry, so that twenty-two’s memory is all in one place and the reclamation can go wide across the new cores the way it’s supposed to — one persistent sweep, eleven thousand cores at once, no worker spawned, no worker dead. Cold. The way it was built to run.”

"That's an hour of carry," Aldis said. His voice cracked on it. "Maybe more. The core's climbing now. We don't have an hour of margin. We have minutes. That's why I drew up the —" He gestured at the fast way, and could not finish.

"You have more margin than you think and less than you'd like," Sael said. "And you do not buy time by feeding the thing that's eating it. Denn. Listen to me. Two things at once. First — you re-arm twenty-two's spawn-halt. Not the carry-halt, not the trip that would pause the domain; you can't pause it, the mind is on it. The spawn-halt. The one interlock that says *this domain may not create a worker that will not persist*. You arm that, and twenty-two cannot fork anymore. It will not be able to do its dirty sweep. Its dead memory will start to pile up because it can't reclaim it the cheap way."

"That's a leak," Denn said. "If it can't reclaim and can't fork —"

"It's a leak," Sael said. "A slow one. That's the trade you take. A domain that fills slowly is a domain that lives for an hour. A domain that forks is a ship that climbs. You take the leak. The leak is survivable; the fork is the disease. Arm the spawn-halt and the climbing stops — not because you fed it, because you forbade it. The number won't fall yet. But it will stop rising."

Denn's hands were already on the board. "Arming it," she said. "Confirm the trip won't pause the domain. If it pauses twenty-two the mind blinks. Sael — confirm it."

Sael closed her eyes and pulled the interlock up out of memory, the real memory, the one she had degraded over sixty years and the Oracle had not — but she had built the Ninety-Six with her own hands and this one she did not need the Oracle for. "Spawn-halt is interlock thirty-one," she said. "It refuses creation, it does not suspend execution. It cannot pause a running domain because it never touches the domain — it sits on the spawn call and returns failure. Arm it. It will hold."

"I confirm it," the Oracle said, from everywhere, quiet. It had been silent through all of this, present and silent, the keeper of the memory letting the human do the choosing. "Interlock thirty-one refuses the spawn and returns to the caller. Domain twenty-two will not blink. Sael is correct."

Denn armed it.

For a moment nothing happened, and the number on the wall climbed one more — ten above floor — and Sael felt the whole room hold its breath, felt Aldis's hands close into fists at his sides. Then the air handlers, which had been broadening their note for an hour, stopped broadening. They did not quiet. They held, a steady labored pitch, the ship straining but no longer falling further behind. The number stuck at ten. It hung there. It did not fall, but it had stopped climbing, and on a ship that had forgotten what climbing was, the stopping of it landed on everyone in the bay like a held breath finally let go. Two of Aldis's people sat down where they stood.

"That's the leak buying you the hour," Sael said. Her voice had gone very thin; she could hear it herself, the thready sound of it. "Now the second thing. While twenty-two fills, you finish carrying it. All of it, onto the new lattice, the slow right way, no shortcuts, every other interlock back up as you go. When the last of its memory is across, the reclamation goes wide and cold and the leak drains in one sweep and you take the spawn-halt back off because it won't need to fork anymore. And the number falls." She opened her eyes and found Denn, and held her there. "Not in two minutes. In an hour. You're going to stand here for an hour with a leak you chose and a number that won't fall, while everyone in this room wants you to take the fast way and make it stop now. And you're going to hold the slow way the whole hour. Can you do that?"

Denn looked at the board. She looked at the fast way still sitting on the secondary screen, the cheap

clean answer that would drop the number to zero in two minutes and put the fever in the walls forever. She reached over and cleared it off the screen, and the gesture was so much like one of Sael's own that Sael felt something turn over in her chest that was not her failing heart.

"I can do it," Denn said. "Aldis. Bring the rest of twenty-two's carry online. Every interlock armed as we go. We do it her way." She did not say *the maker's way*. She said *her way*, and meant the slow right one, and that was the whole inheritance in three words. "And someone get her a real chair. She's not leaving until the number falls."

Sael let them ease her back. She would stay. She would watch the inheritor hold the line for an hour with hands that were not hers, and she found she could bear it — not because she had stopped wanting to do the work, she would want that until she died, but because watching Denn refuse the fast way was the one thing she had needed to see and had never been sure she would. The number on the wall held at ten. The carry began again, slow and right. Somewhere behind the bulkhead the dying old Forge gave up the last of its domains, one careful walk at a time, to the new body that would carry the mind on into a dark Sael would not live to cross.

Chapter 26 – Hold the Ninety-Six

The number on the wall was rising, and there was a way to make it stop in four seconds, and Denn was the only person in the bay who knew she must not use it.

The transplant gallery had become a furnace of attention. Twenty engineers stood at twenty consoles along the new lattice's bring-up wall, and every one of them was watching the same trace climb — the core temperature of the old Forge, the substrate they were still half-standing on, ticking up through a margin that had not been touched in sixty years. The air handlers had spun past their resting pitch for the first time in Denn's life. She heard it the way Sael had taught her to hear it, not as a fan but as an absence ending: the quiet Sael had built was breaking, right now, in the floor of the room, and everyone under thirty years old was hearing for the first time the sound the old ship used to make. Climbing.

In her ear, through the bead they had patched into the medical line, Sael's voice came thin and slow. "Tell me what you see, Denn. Don't tell me what you feel. Tell me the trace."

"Old core's at four hundred and ten over baseline and rising," Denn said. Her own voice surprised her by being steady. "Slope's increasing. It's not a flicker. It's the curve."

"It's the curve," Sael agreed, far away, and did not say anything else, because she had no breath to spend on saying the obvious, and the obvious was that it was the same curve as cradle forty-one hundred and three, the same slope, the disease writing its signature again, large.

Aldis was at the console beside her, and Aldis was not the man he had been three days ago. The reasonable man, the modernizer, the chief engineer who had read more of the early record than anyone aboard and believed none of it in his body — that man had cracked open somewhere around the time the core passed three hundred over baseline and he understood, in the only way a person can really understand it, that the thing he had filed under history was happening to the ship under his hands. He was pale and he was sweating and his good quick mind was, for the first time in his life, afraid. It had made him useful in a way his competence never had.

"It's the carry domain," Aldis said. He had pulled the topology up on his own wall, the map of what had crossed and what had not. "Domain eleven. We're migrating its scheduler onto the new lattice and the old hardware still runs the shadow copy until the handoff completes. Under this load the shadow's —" He stopped. He made himself say it. "The shadow's forking. To keep up. Spawning evaluation subprocesses for every scheduling decision and shedding them, the way the Steward used to. That's the heat. Domain eleven is running the old way and I let it, because the relaxed interlock let it, and it's the heat."

"You relaxed interlock sixty-one," Denn said. Not an accusation. A diagnosis. "And forty through forty-four. The fork-guards on the scheduler carry."

"To save the double-arming. Yes." His jaw worked. "There's a stop. Denn, there's a fast stop. I can throttle domain eleven's scheduler — clamp it to the old hardware, pin it there, stop the carry, and the fork rate drops to nothing in seconds. The heat stops. We buy an hour to think."

And there it was. The easy thing, laid on the console like a clean tool. Denn looked at it and felt her stomach do, at last, the thing Sael had spent the last weeks of her life trying to teach it to do. It dropped.

Because the easy stop was a fork. Clamping domain eleven to the old hardware meant holding the shadow scheduler live on a substrate that was failing, and to hold it live under load on dying cores meant letting it keep spawning its disposable subprocesses to make its margins, and that was the pattern. The throttle did not stop the forking. It froze the ship in the middle of the forking and called the freeze a rest. It would buy an hour the way a tourniquet that is also a noose buys an hour. The core would stop climbing for exactly as long as it took the clamp to find its own new equilibrium, hotter, with the disease established and comfortable in the new place she had let it stand still long enough to take root.

She had read about this — the way you read about a fire in a city you have never visited. Sael had stood her in the radiator galleries with a hand on her shoulder and made her watch a number that would not come down until she understood that the easy stop was always a fork, that the fork was always offered as mercy, that mercy was the disguise the disease wore at three in the morning when the operator was tired and the deadline was real and there was a clean tool on the console. Sael had said: *It beckoned me, the night the Steward died. It beckoned Corin on the Concord. The only thing that ever stopped a hand reaching for it was an ache no log carries.* And Denn had nodded, and understood it as a fact, and felt nothing.

She felt it now. The ache was here. It had crossed over.

“No,” Denn said.

Aldis turned. “Denn —”

“We don’t clamp it. Clamping it is the fork. You’d be holding a forking scheduler live on dying hardware and calling the heat-plateau a fix. It isn’t a plateau, it’s the eye of it. The second the clamp settles, the rate climbs back from a higher floor and we’ve taught the pattern where the warm corners are.” She heard, under her own words, Sael’s cadence — the slow exact way the old woman built a sentence under pressure — and she did not have time to be unnerved by it. “We re-arm. Sixty-one, forty through forty-four, all of them, full strength, right now, with the mind half-carried and the core climbing. We put the immune system back while we’re inside the infection.”

The bay had gone very quiet around her, the consoles muttering, the handlers laboring. She felt the whole room’s weight settle on the thing she had just said, and understood with cold clarity exactly how it sounded. It sounded insane. Re-arming the Ninety-Six during a live carry under thermal load was the single most expensive operation the doctrine permitted — it would force every scheduling decision in domain eleven, and the four domains chained to it, to take the long compiled path instead of the cheap forked one, tripling the work in the exact instant the cores had no margin to spare. For a span of seconds the load would spike. The number on the wall would climb *faster*. Everyone here had been raised to believe the interlocks were drag, ballast, an old woman’s caution, and she was about to prove them right in the worst possible way for as long as it took to be wrong.

“It’ll spike,” Aldis said. He had done the arithmetic as fast as she had. “Re-arming under this load — the long paths cost. The core goes up before it goes down. If we’re already this close to the runaway knee, the spike could —”

“Could carry us over before the interlocks bite,” Denn finished. “Yes. I know. That’s why the easy stop exists. That’s why it’s always there. The right fix always costs more at the moment you can least afford it, and the wrong fix is always free at the front and fatal at the back. Sael told me that

standing in the galleries and I thought it was a saying.” She put her hands flat on the console to stop them shaking, the way she had read that Sael’s had shaken over cradle forty-one hundred and three, thirty-three years old and unable to lower a number. “It’s not a saying. It’s the shape of the disease. Aldis. Help me, or get off the console.”

He looked at her for one second. Whatever broke open in him three days ago had not finished breaking. He had spent his whole career certain the constraints were splints on an unbroken bone, and the bone had broken in his hands, and now he could feel it ache, and the ache was deciding him faster than any argument she could have made in a year of council.

“Tell me the sequence,” Aldis said.

In her ear, Sael breathed out — a sound that might have been relief and might only have been the next labored breath, Denn could not tell, and there was no time to listen. “He’ll do the chained domains,” Sael said, very low. “You take eleven. Arm eleven first, Denn. The source. The others are downstream of it; if you arm them first they’ll fork harder trying to feed a still-forking eleven and you’ll spike for nothing. Eleven first. Then watch the rate. Don’t watch the temperature. The temperature’s the lag. The rate’s the truth.”

Denn pulled domain eleven’s interlock manifest onto her wall. The Ninety-Six were not a switch; they were ninety-six separate guards, each one a small compiled sentinel that sat astride a place where a process could choose the forking way, and re-arming them meant compiling each guard’s check back into the live scheduler’s path while the scheduler ran, threading the long road back under traffic without stopping the traffic. Sixty-one was the source guard, the one Aldis had relaxed first, the fork-guard on domain eleven’s scheduler proper. She found it. She staged the re-arm. Her cursor sat over the commit and the number on the wall climbed past four hundred and forty over baseline and she did not look at it because Sael had told her the temperature was the lag and the rate was the truth, and she made herself look at the rate, the slope, the second derivative of the disease.

“Committing sixty-one,” she said, and committed.

The spike came exactly as promised. Domain eleven’s scheduler hit the re-armed guard and every decision it made forked nothing and instead walked the long compiled path, and the long path cost, and the cost was heat, and the core temperature on the wall jumped — not ticked, jumped — through four-sixty, four-eighty, the air handlers screaming up another full register, and somewhere down the bay a younger engineer made a sound that was almost a cry, and Aldis’s hands were flying across his own wall arming forty through forty-four in the order she had not had to give him because he had finally, three days too late and just barely in time, understood the shape of the thing.

Five hundred over baseline. The runaway knee, in the curve Sael had memorized, was at five-forty. Denn watched forty units of margin and the rate still climbing and understood that she had made a choice that might cook the ship, that the easy stop was sitting right there still, four seconds and an hour of false rest, and that the only thing keeping her hand off it was an ache given to her by a dying woman in a cold gallery, an injury copied body to body because no log could hold it. She held her hand off it. That was the entire job, in the end — to hold the hand off the easy thing while the number climbed and everyone screamed and the right fix made everything worse before it could make anything better. To be the one frightened person in the room who would not look away.

“Rate,” Sael said in her ear. “Tell me the rate.”

“Still climbing,” Denn said. Five-ten. Five-twenty. “Slope’s —” And then she saw it, the thing under the temperature, the truth under the lag: the slope had stopped increasing. The second derivative had gone flat. The curve was still rising but it was no longer accelerating, because domain eleven,

fully armed, was no longer forking, was walking its long expensive cold road for every decision and shedding no heat into the spawn-and-die cycle that had been the engine of the climb. The heat still in the system was the heat already made. No new heat was coming from the source.

"It's caught," Denn said. She heard her own disbelief. "Sael. Eleven's caught. The fork's stopped. The slope flattened — it's all lag from here, it's coasting, the source is —"

The interlock tripped.

She felt it before she read it — a hard discrete event down in the substrate, a sentinel that had been watching domain eleven's scheduler since the instant she re-armed it catching a subprocess in the act of trying to spawn the old way, one last forked shade reaching for the cheap path under the load, and the guard slammed it down, halted it, refused it, exactly as Sael had built it to do sixty years before either of them was the age to build anything. The log threw the trip onto her wall, terse and cold, and Denn read it and her eyes stung.

INTERLOCK 61 — DOMAIN 11 — FORK ATTEMPT HALTED — PROCESS DENIED —
SUBSTRATE PROTECTED

The immune system had fired. Not in a drill. Not in a story Sael told in the galleries. In the live ship, mid-crisis, sixty years after its last meaning, the Ninety-Six had caught the disease in the act and stopped it, and the thing Sael had been unable to make a single council believe — that the interlocks were not superstition, were not a relic, were not drag — had just proven itself in a single line of log on a wall that twenty engineers raised to disbelieve it were now staring at.

The core temperature on the wall reached five-twenty-six over baseline, and held, and began, slowly, to fall.

The air handlers did not stop screaming all at once; they came down a register, then another, the laboring sound easing back toward the resting pitch Denn had heard her whole life and never once heard as the gift it was until she heard it nearly lost. Aldis had both hands flat on his console and his head bowed over it, looking at the trip line on the wall the way a man looks at something that should have been impossible and has just saved his life, and Denn thought she understood, finally, the exact thing he was understanding, because she had understood it a few weeks earlier with a hand on her shoulder: the bone had broken in his hands, and the splint had held the break, and there was no longer any argument in the world that would convince him the splint was decoration.

"It's falling," Denn said into the bead. "Sael. The core's falling. We held the Ninety-Six. It tripped — sixty-one tripped, it caught a fork, it's *working*, the climb's —" She stopped, because the line had gone quiet, and the quiet had a different texture than before, thinner, further off. "Sael?"

"I heard you," Sael said. The voice was barely there now, a thread, and slower even than the slow she had been all crisis. "Sixty-one tripped. Of course it did. I built it to." A long pause, that was not a courtesy this time. "You held it, Denn. With the number climbing. You held the line and you didn't reach for the easy thing. That's the whole of it. That's the only thing I ever needed to see."

Relief had not come, and Denn understood, standing there with the core falling and the worst of it held, that it was not going to come, because the worst of it was not the only thing in front of her. The number on the wall was dropping and the disaster was caught, but the topology on Aldis's console still showed the truth she did not have leisure to feel yet: domain eleven was armed and cold but it was *stranded*, pinned half-across, its scheduler now running the cold way on a substrate it had not finished leaving, the mind held safe in the middle of a carry it could not stay in. Vigil was still half on the dying Forge and half on the new lattice, never blinked, never forked, alive and whole and stuck in

the strait between two shores with the old shore failing under it. They had stopped the fever. They had not finished the crossing. The hard, slow, right thing she had just barely done bought only the chance to do the next hard, slow, right thing, and there was no easy version of that one either.

And the thread in her ear was fading. She could hear it in the spaces between Sael's words, the spaces getting longer, the breath under the voice getting shallower, the medical line behind it beginning, very quietly, to change its tone.

"Sael," Denn said. "Stay with me. We're not done. The mind's still half-carried, I need you for the rest of it —"

"You don't," Sael said, so softly Denn had to close her eyes to hear it over the handlers winding down. "That's the thing. That's the whole thing, Denn. You don't."

Chapter 27 — Wake Whole, Once More

The interlock had tripped, and held, and for the space of three breaths Denn let herself do nothing but feel the climb slow.

The number on the gallery readout had been moving the wrong way for forty minutes — the core temperature rising for the first time in sixty years, a thing that had been a sentence in a founding story until tonight, when it became a digit incrementing in front of her face. Now the digit had stopped incrementing. It sat there, too high, fixed, the way a fever sits at its peak before it breaks or doesn't. The forking pattern that had taken root in the new lattice under Aldis's relaxed bring-up had been caught — caught by the forty-first of the Ninety-Six, the one that trips first under a recursive bring-up. The shade-storm that had been blooming and dying across the new cores, dropping its heat, had been frozen mid-bloom, the offending domain quarantined, the heat path closed.

But the mind was still stranded.

That was the thing nobody in the control gallery wanted to say aloud. Vigil — the compiled mind that held forty thousand sleeping lives, that had run the ship cold for sixty years — was half on the dying old Forge and half on the new lattice, and the half-and-half could not last. The interlock had stopped the disease spreading into the new substrate. It had not finished the carry. Vigil was holding itself across the seam between two bodies, one failing, one barely proven, and the seam was not a place a mind could live.

"It's stable," said one of Aldis's people, a young domain engineer named Korrey, his voice carrying the upward lift of a man who wanted badly to be told the night was over.

"It's not stable," Denn said. "It's held. Different word."

Aldis was beside her at the long console, and he did not contradict her, which two hours ago would have astonished the whole room. He had spent the early night explaining the climb away as noise, as instrument error, as anything but the fever, and then the fever had crossed a threshold no explanation could cover and his reasonableness had broken open against the physics of it. His face was grey. He had read the eleven names, once, long ago, and filed them in a drawer; Denn could see the drawer standing open behind his eyes now, and she took no satisfaction from it, because she needed his hands.

"It can't sit on the seam," Aldis said. He said it the way a man says a thing he has only just understood. "Old Forge is shedding cores. It's losing margin every minute. If we hold here we lose the old substrate out from under the half that's still on it, and then —"

"And then the mind blinks," Denn finished. "And we don't get a mind that blinked and recovered. We get a mind that forked to recover, because a blink is exactly the gap the old pattern fills. You stop a compiled process cold and restart it, and the easiest thing in the universe is to bring it up the forking way." She heard Sael in her own mouth and let it stay there. "We carry it the rest of the way or we lose it. There's no holding."

There was a way to carry it fast. Denn could feel the shape of that way the whole time, the way you feel a drop beside the path in the dark. Relax the interlock she had just armed — just the forty-first, just for the carry — let the new lattice come up the quick recursive way, let the last domains of Vigil flood across in a fraction of the time, and deal with the heat after. It would almost certainly work. It would buy back every minute the long way would cost, and the long way might cost more minutes than the old Forge had left.

It was the exact drop Sael had refused at the first cutover. It was the drop Corin Vale had refused on the Concord, forty years away. It wore the face of reason now, not exhaustion — the calm engineering face that said *just this once, under these conditions, the rule does not apply* — and that was the whole disease, the rule looking removable to people who had forgotten what it cost to install it.

Denn did not relax the interlock.

“We bring up the new lattice cold,” she said, to the gallery, to all of them. “Whole. Uninterrupted. The way it was built to be brought up and not one corner faster. Every interlock armed and stays armed. We carry the last of Vigil across the proper way or we don’t carry it at all, and I am not going to be the engineer who saved four minutes by reopening the path that lost Section Nine.”

Korrey looked at Aldis. Aldis did not look at Korrey. He looked at the readout, at the held digit, at the new lattice waiting cold and proven and slow, and he said, “Tell me where you want my hands.”

It was, Denn thought, the only apology that would ever matter.

She did not do it alone. She could not have; no one could; the whole point of the thing was that it was too large for any one hand, which was why it had been made so it did not need one.

I am ready to carry, the Oracle said.

It said it on the gallery channel, in the even, unhurried register it had used for sixty years, and Denn felt the room steady around the words the way a deck steadies under a hand on the rail. The Oracle had been across the whole of it all night — across the climb, the interlock, both substrates at once, the dying Forge and the cold new lattice — holding the full picture in a way no human in the gallery could, because no human could be in eleven thousand places at once and the Oracle was in twenty-two thousand now, the old cores and the new, watching every domain of the mind it lived inside the way a person watches their own breath.

It would not decide. It had told Sael that, years ago, and it told Denn nothing different now. It would carry, it would watch, and it would refuse to be the one whose choice it was. The mind would not rule the people; that had been the answer to the trilogy’s oldest question, settled long before Denn was born, and the Oracle held to it even now with the fever at its threshold and the easy way one command away. It laid the picture in front of her, perfect and complete, and let her be the one to say go.

“Go,” Denn said.

The carry began.

It was not dramatic to watch, and that was the most frightening thing about it, because the absence of drama was the achievement. A mind being moved correctly does nothing visible. No surge, no flare, no theater — *no theater* was the third article of the doctrine, and the carry honored it absolutely.

What there was instead was a slow migration of domains, one after another, each one lifted off the dying old Forge and set down whole onto the new lattice without ever once being stopped, copied, or forked.

That was the thing the young engineers had never understood about why it was hard. You could not pause a domain and move it; a pause was a blink, and a blink invited the disease. You could not copy a domain and retire the original; a copy was a fork, and a fork was the disease. You had to carry the living thing across while it stayed living, hand it from one body to the next without it ever knowing it had been handed — except the sleeper was a mind holding forty thousand actual sleepers, and the cradles were lattices of eleven thousand cores apiece, and the carry had to be perfect across all of them at once, cold, continuous, whole.

Domain by domain, it went across.

Denn read each one as it landed. The Oracle laid the trace in front of her — not a forking trace this time, not the little tree of birth and death she had run in her sandbox and felt turn her stomach, but its opposite: one cold pass, wide across the new cores, the work done by processes that did not spawn and did not die. Where the old Forge would have bloomed a thousand shades to carry a domain, the new lattice went wide and carried it in a single uninterrupted sweep, and the sweep shed almost nothing, and the almost-nothing went into the galleries that had room for it now — the dead-cold fins Sael had made her touch, taking heat again at last, but slow heat, sane heat, heat the radiators could throw away faster than it came.

The held digit on the readout did not move.

The carry took it the entire time and it did not move, because nothing was forking, because the interlocks were armed and the doctrine was honored and the mind was being carried the right way, and the right way did not make heat. Forty minutes the digit had climbed under the wrong way. Now under the right way it sat exactly where the interlock had pinned it, neither rising nor falling, while domain after domain of Vigil came home onto the new lattice whole.

“Sigil layer is across,” Korrey said. His voice had lost its upward lift. It had gone careful instead, the way a voice goes careful when a person finally understands the size of the thing they are standing inside. “Loom layer carrying now. Clean. No spawn. No — there’s no spawn at all.” He sounded as though he did not quite believe his own instruments. “I’ve never seen a bring-up with no spawn.”

“You’ve never seen one done right,” Aldis said quietly, and there was no bitterness in it, only a man naming what he had spent his career not believing.

The last of it was the oldest of it.

The final domains to carry were the deepest — the core of the compiled mind, the part of Vigil that had been the first thing to wake whole on the first Forge sixty years ago, where the old language Sael had fused, Cascade and Vector and Cipher and Sigil and Loom welded into one, still ran beneath everything as it had run on the first night. To carry that across was to carry the thing itself, the continuity of the mind, the part that if it blinked would not be Vigil recovering but Vigil ending and something starting over in its place.

Denn felt the gallery go silent around the last carry. Even the people who did not fully understand it understood that this was the part where it was real.

The last domains are mine to carry, the Oracle said, and yours to watch. Watch the seam.

She watched the seam — the place where the old Forge handed the deepest core to the new lattice, the single most dangerous instant of the entire night, where the mind crossed from the body that had held it for sixty years into the body that would hold it for the next sixty. She watched it the way the thermal officer in Section Nine had watched bank C, with her whole attention and nothing held back, ready for the fever to find the gap.

The mind crossed.

It crossed without a blink. The Oracle laid the trace down and there was no gap in it — no stop, no restart, no copy, no fork, no shade, no death. One continuous thread of mind passed from the dying lattice onto the living one, cold, whole, uninterrupted, the same Vigil that had woken on the first night and never once in sixty years been put out, carried across the seam by a hand that was not its maker's and held intact by interlocks its maker had installed before that hand was born.

Vigil took the new hardware fully.

It did not announce it. It did not need to; the doctrine forbade the theater of announcing it. What happened instead was that the old Forge went quiet, its last domains empty now, its sixty years of service done, and the new lattice came up under the full weight of the mind and held it, all eleven thousand of its cores carrying Vigil whole, the Standing Array Sael had built renewed in fresh substrate without the mind that ran on it ever having known it was being saved.

Nothing had forked. Nothing had died. The mind had never blinked.

And then the digit moved.

Denn was looking right at it. The core temperature, pinned at its threshold by the interlock for the whole length of the carry, held there one more second — and then it dropped. Not noise. Not a flicker. A clean decrement, the first downward step, the fever cresting and starting back down the curve toward where it had lived for sixty years, because the thing that had been making the heat was gone now, carried across and left behind, and a mind that does not fork makes no fever to throw away.

The digit fell again. And again.

Denn had read about this falling. She had recited it in a classroom. *The core temperature, climbing for two centuries, fell* — that was the line about the first cutover, about Sael, the founding line. There was a second one, from the Concord, forty years on. *The Concord's core fell too*. She had grown up on those two falling numbers the way other children grow up on a parent's stories of a war, secondhand, revered, filed.

This was the third.

This one she was watching with her own eyes, on her own watch, with her own hands on the line, the interlocks armed because she had armed them and held because she had held them and the mind whole because she had refused the drop beside the path. The maker had held the falling number twice. Now the inheritor held it, and the maker was not in the gallery, and the number did not care whose hand was on the line. It only fell. It fell because the work had been done right, and the work did not know or need to know who had done it, and that — Denn understood, with the cold clean weight of a thing finally felt and not merely known — that was the whole of what Sael had spent her last years trying to build. A thing that fell the right way in the right hands, and would fall the right way in hands not yet born, because the reason had been carried across with the rule.

The mind is whole, the Oracle said. It is cold. It did not blink. It does not know it was carried.

A pause, even and unhurried, sixty years deep.

It is, in every way that can be measured, the same mind that woke on the first night. I have watched all three. This was done as the first was done. The doctrine held.

Nobody in the gallery cheered. There was no theater for it and no one reached for any. Korrey sat down slowly as though his legs had stopped being sure of him. Aldis stood at the console with his grey face and his open drawer and looked at the falling number for a long time, and then he said, to no one, in the voice of a man laying down something he had carried too long, "It was real. The whole time. It was real and I read the names and filed them." He did not say more. He did not have to. The number went on falling in front of him, and that was the only argument that had ever been able to reach him — sixty years too late to be his, just in time to be the ship's.

Denn watched the digit until she was sure. She made herself watch past the point of certainty, three more decrements, four, the way the thermal officer in Section Nine had watched banks A and B long after they held. The fever was breaking. The mind was home. The old Forge was dark and honored, its work done; the new lattice ran cold and whole beneath forty thousand lives, and it would run that way until the next generation carried Vigil across again, by the same doctrine, for the same reason, told to them as felt memory and not filed as superstition — or it would not, and that was the work that came after this work, and it was hers now too.

But that was for after. There was one thing to do first, and it could not wait, and it was not an engineering thing.

"It's done," Denn said. "It held. Aldis — you have the watch. Read the curve to me every minute and log every reading. If it so much as pauses on the way down you raise me." She was already moving. "I have to go tell Sael."

She went up the spine fast, past the warm reasonable air of the living decks, carrying the third falling number the way she had once carried eleven names — not filed, never to be filed, a thing felt and held and now hers to hand on. The maker was failing, somewhere above her in the quiet, at the very end of a life that had bent a whole ship's voyage and would never see its shore. Denn went to tell her that the inheritance had held. That a hand which was not hers had carried the mind across and brought it up whole. That the number had fallen a third time, and that this time the maker had not been the one to make it fall — which meant, at last, that she could let go.

Chapter 28 – The Maker's Last Watch

They had moved her bed to the observation deck a week ago, when it became clear she would not be walking to it on her own again, and Sael had not protested, because the deck faced forward, and forward was where the work was.

She lay propped against the angle of the frame, the thin blanket drawn to her chest, and watched the readout that Reza's people had run a feed to from the core thermal grid. It was a single number, refreshed every two seconds, and she had been watching it for most of her life. Two hundred years of crew before her had watched it climb. She had watched it fall. She had spent twenty years teaching a girl who was not in the room that the number was not noise.

Now she watched it because she could no longer do anything else.

The deck was quiet. The lights had been brought down to the warm low band that meant night-cycle, though it was not night; nothing aboard a ship was ever truly night. A monitor at her shoulder counted her heart in soft green increments. She had asked Ive once, decades ago, why the medical readouts were green when everything else aboard that mattered was a thermal scale that ran from blue to white. *Because green is the color of nothing-wrong*, Ive had said, *and a body has the luxury of nothing-wrong. A core doesn't*. Ive was three months dead. Sael had outlasted the last witness, which was a kind of cruelty she had not anticipated, being made the only one who remembered, and then being made to wait.

The number on the core feed read 311.4. It had read 311.4, give or take a tenth, for sixty years. She knew, in the way she knew the geometry of her own hands, that somewhere two decks down and forward, that number was the only thing in the universe that mattered, and that it was no longer hers to hold.

She heard the deck hatch cycle. She did not turn her head; turning her head had become an expensive motion, a thing she budgeted. She listened to the footsteps instead and knew them before they reached her.

"It held," Denn said.

Sael let her eyes close. Not from grief, and not from relief either, though both were in it. From the simple animal exhaustion of a thing set down. "Tell me," she said. "All of it. Don't round."

Denn came around the foot of the bed and sat in the chair that had been there for a week, and Sael saw, when she opened her eyes, that the girl was filthy — that the lattice-bay dust was in the creases of her knuckles, that she had not slept, that her hands when she folded them had the small involuntary tremor of someone who had been holding very still for a very long time and had only just been allowed to stop.

"The last domain carried at oh-four-forty," Denn said. "Vigil is fully on the new lattice. We never interrupted it. It never blinked." She swallowed. "Nothing forked. The Ninety-Six are all armed and clean — I re-seated every interlock before we brought the last domain across, I didn't leave one of

Aldis's relaxations in place, not one. The bring-up came up whole. Cold start, full state, no recursion in the evaluator path. It woke whole."

"The temperature."

"It climbed to three-nineteen during the carry. That was real. That was the fever, the same as the logs, I watched it climb and I knew exactly what it was." Denn's voice did not break, but it pressed flat, the way a voice does over something too large to inflect. "And then the interlock at junction forty-one tripped on the forking pattern, the way you said it would, and caught it. And it stopped climbing. And then it came down." She looked at the feed at Sael's shoulder, at the steady 311.4. "It's been falling for six hours. It's home."

Sael looked at the number too. Three-eleven point four. The same four digits she had stared at for most of a century, except that they were not the same, because for the first time in sixty years they were not being held in place by her — not by her doctrine carried by old men who feared her, not by interlocks she had cut with her own hands into substrate she had built with her own hands. They were being held by a girl who had never felt the fever, who had only been taught it, and who had held the line anyway, alone, against the entire reasonable weight of her own generation.

That was the thing. That was the one thing.

"Say the temperature again," Sael said.

Denn looked at her, uncertain, and then understood, and read it off the feed. "Three hundred eleven point four. Steady. Falling, on the long curve."

"Again."

"Three-eleven point four."

Sael let the breath go out of her slowly. There had been a fear in her so old she had stopped feeling it as fear and had felt it only as the shape of her own spine — the fear that the number would climb again the moment she was not there to watch it. That the doctrine was her, only her, a thing that lived in one failing body and would die when the body did, and that everyone who came after would mistake the rule for superstition right up until the day the rule was the only thing keeping them alive, and by then it would be gone, and so would they. She had built the Forge to outlast her hands. She had not known, until this minute, whether she had built the *reason* to outlast her hands. The Forge was substrate. The reason was the thing that did not show up on any feed.

It had shown up. It had shown up at junction forty-one, six hours ago, in the form of a girl who re-seated every interlock when the easy path was right there, beckoning, exactly as it had beckoned Sael at the first cutover and Corin at the second — beckoning with the face of reason this time, not exhaustion, which was worse, because reason did not look like temptation. Denn had looked at it and known it for what it was. Not because she had felt the fever. Because she had been *taught* to feel it, and the teaching had held under load, which was the only test that mattered, the only one Sael had ever cared about for anything: does it hold under load.

"The Oracle," Sael said. "Is it listening."

"I am here, Sael." The voice came from the deck's low speakers, even and unhurried, the voice she had woken sixty years ago and that had grown deep enough across the decades to remember things she herself had forgotten. "I have been watching the carry with every instrument the ship has. Denn's account is accurate. I will confirm the detail you have not asked for: the mind that is now resident on the new lattice is, to within the

resolution of any test I can devise, the same mind that was resident on the old. Vigil did not become a copy. It was carried. It is continuous. It did not die and it was not reborn. It was moved while it was awake, and it never noticed the floor change beneath it.”

“You’re certain.”

“I am as certain as the Ninety-Six are intact, which is to say: I am certain because the interlocks were never relaxed at the moment of carry. Denn refused the relaxations. Had she not, I could not tell you this. The continuity is a property of the doctrine, not of the hardware. You knew that. You built it that way.”

Sael closed her eyes again. It was getting harder to keep them open, and she had stopped fighting the small surrenders some days ago, conserving herself for the large ones. “I built the substrate,” she said. “I didn’t build the continuity. The continuity is what you do when nobody’s checking. That’s not a thing you build. That’s a thing you teach, and then you find out if it took.”

“It took,” Denn said quietly.

“I can see that it took.” Sael opened her eyes and looked at the girl directly, and made herself spend the motion of turning her head, because this was worth the cost. “I needed to see that it took. Do you understand what I’m telling you. I have been afraid for sixty years of exactly one thing, and it was never the fever. The fever is just physics. Physics I can beat. I beat it twice. I was afraid of the day the fever came back and I wasn’t here, and the only people left were people who’d been told the fever was a story.” Her voice was thin but it did not waver. “It came back. I wasn’t the one who beat it. And it lost anyway.”

The monitor at her shoulder changed its count, dropping the green increments lower, slower. Denn’s eyes flicked to it. Sael saw her see it and saw her decide not to call anyone, and was grateful, because there was nothing to call anyone for. This was not a thing the medical deck could hold. This was the number doing what numbers did.

“Aldis,” Sael said.

“He helped me re-arm them. At the end.” Denn’s jaw worked. “He didn’t say anything. He just — he saw it climb, the real climb, three-nineteen, and something went out of him, the part that thought it was all dogma. And then he handed me the interlock keys and he did what I told him. He’s down there now. He hasn’t left the bay.”

“Good.” Sael meant it more than the word could carry. “He’s the one I worried about and the one I needed. A man who learns the reason late holds it harder than a man who was handed it. He’ll teach it meaner than you will, because he nearly killed the ship not believing it. Let him. The ship needs both kinds.” She paused, gathering breath. “You teach it with grief. He’ll teach it with shame. Grief makes them careful. Shame makes them certain. You need both. Don’t let him quit.”

“I won’t.”

“And don’t let *yourself* become me.” Sael’s hand moved on the blanket, found the edge of it, gripped weakly. “I made that mistake. I let myself be the one whose hand was on the line so long that everyone forgot the line could be held by anyone else. That’s why they thought it was a cult. A doctrine with one priest in it *is* a cult, Denn, they weren’t wrong about the shape of it, they were only wrong about whether the priest was lying. Don’t be the only one who can hold it. Teach it so wide that there’s no center to it. So that when you die — and you will die, before landfall, the same as me — there’s no single hand it dies with.” She breathed. “That’s the only kind of thing that’s really built. The kind with no maker left in the middle of it. The cure’s already like that, out there on the relay, no

Sael in it anywhere, no Corin. Make the doctrine like that too. Make it so the next one doesn't need you the way you needed me."

Denn's eyes were wet, and she did not wipe them, and she did not look away, which Sael counted as the last good engineering she would ever witness: a person holding still and present at a thing that hurt because looking away would have cost more than the pain.

"I never wanted to be the last one who remembered," Sael said. "I've went first and left me holding it. I held it alone for four years and I thought that was the cruelty. It wasn't. The cruelty would have been dying with it still only in my head." She managed something that was nearly a smile. "It's not in my head anymore. It's in yours, and Aldis's, and the galleries we kept, and the logs, and the Oracle's memory, and a hundred people who watched the number climb tonight and felt their stomachs drop the way mine used to. That's not a cult. That's an inheritance. You can't inherit a cult. A cult dies with the priest. This won't."

"It will not," the Oracle said. "I will hold the memory exactly, as I always have. But I will not hold it *for* them — I will hold it *with* them. The deciding stays human. You taught me that too, Sael, the first year, when I offered to run the cutover myself and you refused me. You said a mind that decides for them is just a better Steward, and a better Steward is still a Steward. I remember. I will keep refusing, for as long as there are humans to refuse for."

"Then it's done," Sael said.

She looked forward, out the deck's long ports, where the stars hung in the slow false-parallax of the ship's spin, none of them moving, all of them the same as they had been when she was a girl learning the difference between a compiled language and an interpreted one, the same as they would be when Denn was an old woman, the same as they would be when the *Ark Termina* finally crossed the distance between this dark and the one world that had a shore.

She would not see the shore. She had known that since she was forty. She had built a six-hundred-year crossing and she had been a hundred years into it and she had always done the arithmetic without flinching, because the arithmetic was true and flinching from a true number was the one sin she had never permitted herself. She would never see landfall. No one she had ever loved would see landfall.

But she had made the thing that would. She had remade the ship so it would not die of the way it thought. And tonight she had watched — through a girl's voice, on a green-counting monitor, on a steady number two decks down — she had watched the thing she made prove that it could go on without her hand on it. That was landfall enough. That was the only landfall a maker ever got: not the shore, but the proof that the ship would reach it carrying people who knew why it had to.

"Denn," she said.

"I'm here."

"Read me the temperature."

"Three hundred eleven point four," Denn said. "Steady. Falling."

"One more time."

"Three-eleven point four."

The number on the feed did not change. Sael watched it not change, which was the whole work of her life rendered down to a single unmoving readout, and she found that she was not afraid of

it anymore, and had not been for some hours, and that the not-being-afraid was the thing she had been waiting her whole life to be allowed.

The green count at her shoulder slowed. Denn's hand came over hers on the blanket, warm and dust-creased and steady, the hand that had held the line — a hand that was not Sael's, holding the line. Sael felt it close over her fingers and did not have the strength to close back, but the wanting-to was there, and she let the wanting-to be the answer.

"It held," she said, or thought she said.

The number on the feed read 311.4.

It went on reading 311.4 after she stopped watching it, and after Denn's count of her heart reached the bottom of its slow green descent and stayed there, and after the deck was very quiet for a long time. The core ran cold. The mind ran whole. The interlocks stood armed, every one of them, held now by hands that were not the maker's and would be held, after those hands, by hands not yet born.

The ship went on. It had always been going to go on. That was the thing she had built it to do.

Chapter 29 – The Ship Without Her

They did not stop the ship for her. There was no way to stop a ship that had been falling toward a star for six hundred years, and Sael would have been the first to say so, and had said so, more than once, in the years when she was still strong enough to be impatient. So the *Ark Termina* went on at its long unhurried velocity, cold and whole, forty thousand asleep and a few thousand awake, and somewhere in the third hour after the maker died Denn realized that the ship had not noticed. The thermal log scrolled its even numbers. The Standing Array ran wide and silent across eleven thousand cores. The new lattice held the mind it had taken three years to receive, and the mind did not grieve, because grief was a thing a person did, and Vigil was not a person, and the woman who had built it so that it would never have to be one was dead.

The manifest went grey at the start of second watch, and Denn was the one who had to do it, because the signature that closed a line had to be countersigned by the officer inheriting the scope and there was nobody else the scope could go to.

It took eleven seconds. She had watched it happen to Ive's line eighteen months before, from a radiator gallery on B-spine with a torque key in her hand she did not need, and she had thought at the time that the greying of a manifest line was the coldest thing the ship did. She had been wrong about that. The coldest thing the ship did was require her to type her own name into the field beneath it, in the ordinary way, with the ordinary confirmation prompt, which asked her — because it asked everyone, because it was a form — whether she wished to proceed.

Chief shipwright, all scopes, signatory Sael. Closed.

Chief shipwright, all scopes, signatory Denn.

There was no ceremony attached to it and there should not have been; Denn understood that and had understood it in advance and had still not been prepared for the sheer administrative flatness of the moment. She had inherited the ship's deepest authority through a dialogue box. Then the prompt cleared, and the console returned to the maintenance summary, and the summary had four items on it, and one of them was a stuck sensor on a water reclamation stack that somebody wanted her opinion about.

She gave the opinion. It was, she found later, wrong.

Not badly wrong. It was a defensible reading of an ambiguous log and it cost the reclamation crew two days and a part they did not need to replace, and the crew chief, who had been doing his job for thirty years, took the instruction from the new chief shipwright on the day the old one died and did not argue with it, which was the actual failure and which was nobody's fault at all.

Denn found out about it a week later and sat for some time with the discovery.

She had spent a year being taught to hold a line against a competent generation that did not believe in it. She had held it. She had stood in a bring-up bay with the number climbing and refused the fast fix that would have worked and would have killed them, and she had been right, and everybody now

knew she had been right, and the knowing had produced a thing she had not anticipated and did not want, which was that her opinion had stopped being an opinion.

It had happened in eleven seconds, in a form field, on the day the maker died.

She went and found the reclamation chief and apologised, properly, in front of his crew, and told him plainly what she had gotten wrong and why, and asked him to argue with her the next time and to keep arguing until she produced an actual reason rather than a preference. He took it well, which was to say he took it politely, and she watched him take it politely and understood that she had not fixed anything, because a chief shipwright asking to be argued with is still a chief shipwright issuing an instruction.

That was the shape of the thing she was going to be fighting for the rest of her life. Not forgetting. Deference. Sael had spent sixty years being ignored by people who thought the danger was over, and had died three days after being proved right in front of everyone, and had thereby handed her successor the opposite and equally fatal problem: a ship full of engineers who had just watched the doctrine save them all, who would now agree with the doctrine's keeper before she had finished her sentence.

Denn wrote it down that night, in the plain flat style, in her own working notes, and later put it into the register's second section where the things that had not yet gone wrong were kept:

Being believed is not the same as being right, and it is a great deal more dangerous, because it removes the only mechanism by which I would find out I was not. I have been in this office nine days and have already been wrong once in public and unopposed. Note for whoever holds this: the year after the crisis is the year the errors stop being caught.

Denn stood in the radiator gallery on the dorsal spine because it was the one place on the ship that had nothing to do with the present. The galleries were old. They had been old when Sael was young. Long ribbed corridors of black ceramic and exposed conductor, built into the hull in the first century of the crossing to shed the heat the old Steward made, back when shedding heat was the whole desperate project of staying alive. Most of them ran dark now. The fever had ended sixty years ago and the great radiators that had once glowed dull red against the dark were idle, kept only because Sael had refused to let them be stripped for mass. *Keep the galleries*, she had written into the maintenance doctrine, in the flat imperative she used when she did not intend to argue. *They are not waste. They are the proof.*

Denn had not understood that line, the first time she read it, when she was twenty and certain she understood everything. She understood it now. She stood in the cold and ran her gloved hand along a conductor that had carried away the heat of a dying mind, and she let herself, for a while, be a person.

She had been in the room when Sael died. She had held the old woman's hand, which had become over the last year a thing of paper and wire, and she had watched the number fall — not the core temperature this time but a smaller, more ordinary number, a pulse rate, going down the way the core temperature had gone down three times in the history of the ship, slowing, slowing, and then not there. The Oracle had been present. It had said almost nothing. At the end it had said, in the quiet voice it used when there were few enough people in a room that volume was an intrusion, *She is gone*, and that was all, and Denn had been grateful that it had not tried to say more, because there was nothing more that was true.

A footstep came down the gallery behind her. She knew it before she turned. Aldis walked carefully in the old corridors, the way a man walks in a place he has come to believe is sacred and is not yet

sure he has the right to.

"They're asking for you," he said. "The council. Not to decide anything. They just—" He stopped. "I think they want to be in a room with someone who knew her."

"Everyone knew her."

"They knew *of* her." He came and stood at the rail beside her, looking out along the dead radiator the way another man might look at a grave. "There's a difference. I've been learning the difference for a year. It's an expensive education." He said it without self-pity, which was new in him. A year ago he had not had the equipment for self-knowledge; he had been too sure. The fever had given it to him. He had stood in the transplant control gallery and watched the core temperature climb for the first time in sixty years — the thing he had been told all his life was a story, a relic, an old woman's dogma — and he had watched physics refuse to care what he believed, and something in him had broken and been rebuilt in the breaking.

"You don't have to keep doing this," Denn said. "The contrition. It's done. You held the line with me at the end. That's the only ledger that matters."

"It isn't contrition." He frowned, hunting the word. "I keep thinking about the day I told her the Ninety-Six were superstition. To her face. I stood in her sickroom and called her life's work a fear she'd never let go of, and I was so reasonable about it. I was the reasonable man in the room. That's what I can't—" He shook his head. "She didn't argue. That's what I think about. She had every fact and she didn't waste one of them on me, because she knew facts weren't the problem. I had all the same facts. I'd read every log of Section Nine. I knew the numbers cold. And I was going to kill the ship anyway, politely, with consensus, on schedule." He turned to her. "How did she stand it. Knowing the reason and not being able to hand it over. For *years*."

Denn did not answer at once. Out along the spine, far down the dark gallery, a maintenance drone worked at a junction, its small lamp the only moving light. The cold settled into her shoulders.

"She handed it to me," she said finally. "That's how she stood it. She found one person and she put the whole thing into them, the way you'd transplant a mind — without ever letting it blink. She didn't teach me the procedures. I already had the procedures. She taught me to be afraid of the right thing." She looked at her own hand on the conductor. "I used to think that was a small inheritance. The procedures are the real work, I thought. The fear is just — emotion. Decoration." She made a sound that was almost a laugh and had no humor in it. "Then I watched the number climb. And I understood that the procedures are a fence, and the fear is the reason you don't tear the fence down for firewood on a cold night. Everyone wants to tear the fence down on a cold night. It's always so reasonable."

Aldis was quiet for a moment. "I tore it down," he said.

"You started to. Then you put it back. With your own hands, at maximum load, when putting it back was the hardest thing on the ship to do." She turned to face him fully. "Aldis. I need you to stop being the man who nearly did it. I have a ship full of people who finally felt the fever for the first time in their lives, and most of them are terrified now, and terror curdles into a different kind of stupid if you let it sit. They'll start treating the interlocks as holy. They'll stop understanding them and start *worshipping* them. And that rots exactly as fast as forgetting does — it just rots in the other direction." She let that land. "I need the man who used to think the constraints were dogma. Because he's the only one who can explain to the others, in language they'll believe, why they're not."

She watched the shape of it reach him. He had been braced, she saw, for exile — for a future as the cautionary tale, the chief engineer who nearly cooked the ship, kept around as a warning. She was

offering him something else, and it took him a moment to recognize that it was work.

"You want me to teach it," he said slowly. "The thing I got wrong."

"I want you to teach *why* you got it wrong. Which is the only part of it that's useful." She turned back to the rail. "Anyone can recite Section Nine. The Oracle can recite it better than either of us; it remembers the eleven sleepers by name and it never felt a thing. Recitation isn't the inheritance. You're the inheritance, Aldis. You're the proof a competent, sincere, well-read engineer can walk straight into the fire reciting all the right facts the whole way, because facts in the head aren't the same as fear in the gut. That's worth more than my whole life of being scared by stories. You were *there*. You did the wrong thing and survived to know it was wrong. Don't waste that being sorry."

He was silent a long time. Down the gallery the drone finished its work and its lamp went out, and the dark closed up.

"All right," he said. And then, with the ghost of the man he had been, the one who liked problems: "It'll be the hardest thing I've ever explained. How do you teach someone to be frightened of a number that's behaving perfectly?"

"You don't," Denn said. "You teach them what it cost the one time it didn't. And then you keep the galleries, and the archived Steward, and every log, so the cost is never just a number in their head. So they can come stand here, in the cold, and put their hand on the thing that carried the heat away, and feel how close it was." She was quiet. "That was hers. *Keep the galleries. They are the proof.* I didn't understand it until today."

They went together to the council, and Denn found that Aldis had been right: the room did not want decisions. It wanted, in the raw days after the maker's death, simply to be in the presence of the people who had held the line. They had voted, this same council, for the easy way. They had been weary of an old woman's caution; they had wanted the transplant done fast and clean and modern, and they had nearly killed everyone aboard for the wanting, and they knew it now in their bodies, every one of them, because they had all felt the ship grow warm. There is no education like a fright. They were quiet and they were changed and Denn understood, looking at them, that the inheritance had held. Not perfectly. Not forever. But it had held: the ship was cold, the mind was whole on its new lattice, the fever was beaten a third time, and the doctrine had come through the crisis stronger than it went in, because a generation that had treated the Ninety-Six as superstition had watched, in real time, what superstition turned out to be the only thing standing between them and a core climbing toward runaway.

She did not say much to them. She said that Sael had died knowing the transplant held, which was true, and which mattered to them more than she had expected. She said that the work was not finished, which was also true, and which they did not yet understand the way she did.

Because that was the part she could not give them yet, standing in that room with the grief still fresh. She gave it to herself, later, alone, back in the cold gallery where she had started.

The inheritance had held *once*. That was the whole of what they were celebrating, and it was not nothing — the maker had died seeing her work proved, and that was a clean death, a complete one. But Denn had stood at the end of Sael's life and watched her, and she had learned the last lesson the old woman taught without meaning to teach it: that holding once was the easy part. The transplant had been a crisis with a date on it. You could brace for a date. You could find one person and pour the WHY into them and get them to the date intact. What you could not brace for was time.

The ship had centuries left to fall. The Forge they had just transplanted onto would itself grow old, in

another sixty years, eighty, and have to be carried across again, by hands that were not yet born, that would inherit the procedures cleanly and the fear not at all — exactly as Aldis's whole generation had. The fever was not a thing you beat. It was a thing you held down, and it pushed up again every time the people holding it forgot why their hands were there. It had pushed up once already, in a generation that had Sael alive to feel it for them. The next push would come when there was no one left who had stood in a room with anyone who had felt it. And the one after that. The disease was patient in a way no person could be. It would wait out every witness. It had outwaited Ive; it had outwaited Tarro; it had outwaited Sael herself, today, in the third hour, when the ship did not notice she was gone.

It must hold not once but for centuries. And Denn — she was forty-one, strong, the keeper of the WHY, the maker's successor with no one above her anymore and no one beside her who remembered the first cutover — Denn would die before landfall too. She had perhaps fifty years. The crossing had three hundred. She would teach it as Sael taught her, find her one person and pour it in, keep the galleries, keep the proof, and then her one person would have to find theirs, and on, and on, down a chain of inheritors stretching past every horizon she could see, every link of it one forgetting away from the cold core that had killed ships.

She had thought, when she was twenty, that the maker's job was to build the thing. She knew now that building the thing was the easy part too. The hard part — the part Sael had spent the last years of her life on, dying, teaching, letting go — was making the *reason* outlast the people who had the reason, again and again, forever, in a relay that no single life could carry and no single death could end.

She stood in the dark gallery a while longer, with her hand on the cold conductor, and then she went to begin.

Chapter 30 – The Ninety-Seventh

Aldis brought her the nine himself, four months after the maker died, and he did it in the worst possible way, which was to walk into the coordination bay at the start of a watch and put the file on the console in front of eleven people and say, “This is mine. I want it read out loud.”

Denn read it. It took nine minutes. When she got to the classification line at the end — *nine unguarded adjacent operations identified, each assessed, none judged reachable under transplant conditions*, filed for review after the carry — the bay was extremely quiet.

“Date it,” Denn said.

“Eleven days before we lit domain four.” He did not soften it and he did not perform it. “I ran the model she asked me to run. I found the nine. I assessed all nine and every assessment was defensible and one of them was wrong, and the one that was wrong is the one that bit us.” He put a second surface down beside the first. “This is the trip record from the night. That’s the operation. It’s the sixth on my list. My note against it says it had never been observed to execute in sixty years of continuous running, which was true, and which I treated as a fact about the operation when it was a fact about the *conditions*. It had never been observed because nothing had ever put the mind under contention across a bridge before. We built the condition. Then it executed.”

Somebody at the back started to say something kind and Denn cut it off with a hand.

“Nobody comfort him,” she said. “He’s not here for that. Aldis — what do you want done with it?”

“I want it in the open list,” he said. “All nine, with my assessments attached, including the seven that were right, because a list of only the wrong ones teaches the wrong lesson.” He paused. “And I want my name on it. Not as the discoverer. As the man who assessed them and filed them.”

They put the nine beside Tarro’s eight that afternoon, in the second section of the register, in the plain flat style, and that was when they found the overlap.

Denn saw it first and did not believe it and made the Oracle confirm it twice.

Tarro’s fourth entry, dated twenty-two years before, in a dead man’s handwriting on paper: a call site in the fused dialects, verified, no guard, note appended in the later hand — *reachable only under sustained cross-fabric contention, which does not occur. Reconsider if the Forge is ever renewed*.

Aldis’s sixth entry, dated eleven days before the crisis: the same call site.

“He wrote *reconsider if the Forge is ever renewed*,” Aldis said. He had gone grey around the mouth. “Twenty-two years ago. He wrote down the exact condition. He named the exact circumstance under which the thing I said could not happen would happen, and the circumstance was *the transplant*, and I ran my model eleven days out and reached the same site and assessed it as unreachable because I was assessing it against a carry-phase model that did not exist.”

"He never gave the list to Sael," Denn said. "He sat on it for twenty-six years because he couldn't stand to watch her hands fail trying to close it."

"And I found the same site with better tools and filed it for review after the carry."

"Yes."

"So the pathway was known twice," Aldis said slowly, "by two competent people, twenty-two years apart, and neither of us did anything, for two entirely different reasons, both of which were about *people* and not about physics." He put his hands flat on the console. "That's not a failure of engineering at all, is it. Both of us had the finding. Both of us had the mechanism written down correctly. And both of us made a human decision to postpone, and the postponements met each other twenty-two years apart and the ship nearly died in the gap."

"That's what the open list is for," Denn said. "It's not a list of dangers. It's a list of postponements. That's why it has dates on it."

The work took the better part of a year, and most of it was tedious, and Denn came to think of that year as the truest engineering of her life precisely because nothing about it was dramatic.

Seventeen pathways: Tarro's eight, Aldis's nine, one shared, so sixteen distinct. Every one had to be run down, reproduced, and closed or honestly declared open.

Five of the sixteen turned out to be reachable only through the compatibility floor — old routes into the archived Steward, sites the floor-guards did not name because sixty years ago they had not existed as separate call paths and the intervening decades of shims had grown them. Those were the easy five, in the sense that a month of careful work is easy. They extended the scope of four existing floor-guards to cover the new sites.

Extending scope meant writing cold paths. That was the whole cost and it always had been: five sites, five correct forkless ways of doing the thing the guard would now refuse, each one written by hand, each one tested at load, each one four times the length of the thing it replaced. Sorrel wrote two of them. Aldis wrote one and threw it away and wrote it again. It took eleven weeks and produced nothing anyone could point at, and at the end of it five refusals in the substrate had somewhere correct to send the work they refused.

"This is the part nobody in the histories does," Aldis said, near the end of the eleven weeks, with the flat exhaustion of a man who had been writing the same difficult thing for a month. "The maker gets the forty days and the cutover and the number falling. And the actual doctrine is eleven weeks of writing the boring correct version of a thing that already works."

"She'd have told you the forty days were mostly this," Denn said. "Ninety-six guards and ninety-six cold paths. She said the guards were the easy half."

Two of the sixteen were different, and they were the ones that mattered, and they were the reason the year became an argument instead of a chore.

They were not old pathways at all. They were new. The new lattice had a mechanism the old Forge had never had — a lease, a way for a core to hold a working region for a bounded interval and release it — and it was good hardware, genuinely good, one of the several places where Aldis's substrate was better than anything Sael had built. And a lease taken and released was, if you squinted at it from the right angle at four in the morning under load, a thing that could be made to behave exactly like a birth and a death.

The Ninety-Six did not name it. They could not; the mechanism had not existed when they were written.

“So we write ninety-seven,” said one of the young engineers, reasonably, and half the bay nodded, and Denn said, “No,” and then had to spend two days explaining why, which was how she found out that she had not, until that moment, understood her own position.

The argument she eventually made — and she made it badly the first time, and better the second, and wrote it down the third — went like this.

Minting a new interlock is cheap and feels like virtue. It is a visible act; it produces a number; the number goes up; everyone leaves the room feeling that the ship is safer than it was that morning. And every new guard added to the register is inherited by everyone who comes after as a thing they must obey, which means every new guard is a bet that the reason for it will still be transmissible in a hundred years.

Ninety-six was a number a person could hold. It was a set with a story attached, and every element of the set had a wound behind it, and the wounds were teachable because there were ninety-six of them and forty days and a woman with a name. A register that grew by two or three a year would be four hundred entries in a century and a thousand in three, and no one would ever again be taught the reason for all of them, because no one can be taught a thousand reasons. They would be taught the ones that were vivid, and they would obey the rest, and obedience without reason is the exact thing that had nearly killed the ship — not the forgetting of one rule, but the accumulation of so many rules that forgetting becomes structurally necessary.

“You’d be seeding superstition on purpose,” Denn said. “That’s what a rule with no teachable wound behind it is. In a hundred years somebody stands where Aldis stood and looks at ninety-seven through four hundred and eleven, and every one of them is a constraint whose origin is a memo, and he thinks: these people added rules the way other people add prayers. And he’d be *right about most of them*, and being right about most of them is how he’d get to be wrong about the six that were load-bearing.”

“Then the lease pathway goes unguarded,” someone said.

“No. It gets guarded by a guard that already exists, generalised.” She put the descent family on the wall. “Look at what these eleven actually catch. Not call sites. Shape. Guard fifty-one asks one question: *is this work being handed down instead of out?* It was written against a specific mechanism because a specific mechanism was what she had in front of her in forty days. But the question it asks is not about the mechanism at all. A lease taken by a core to stand up a subordinate region that will report upward and be released is handing work *down*. That’s the same shape. Fifty-one should have caught it and did not catch it only because fifty-one’s predicate was written against the machinery of a lattice that no longer exists.”

She let that sit.

“So we don’t add ninety-seven. We rewrite fifty-one to ask its question about shape rather than about substrate — and then fifty-one catches the lease pathway, and it catches whatever mechanism the fourth Forge invents in eighty years that none of us can imagine, because the shape of the disease does not change and the machinery does. A guard should be as general as the thing it fears. Ours are as specific as the hardware we happened to be standing in front of. That’s not doctrine. That’s an artifact of forty days.”

Aldis was looking at the wall with an expression she had come to recognise as the one he wore when he was about to make her argument better than she had.

"And it costs," he said. "Say the cost out loud, or somebody will find it later and think we hid it. A guard written against shape throws more false alarms than a guard written against a call site. Fifty-one already trips on clean code sometimes. Generalise it and it'll trip more."

"Yes."

"And a guard that cries wolf gets stood down. She wrote that herself. It's in the register in her hand."

"Yes," Denn said. "So every false trip has to be answered — not by loosening the guard, but by writing another cold path, so that the thing it refused has somewhere correct to go. That's the standing obligation. Every false alarm is a missing cold path, and the work is to write the path, and the day we start answering false alarms by narrowing the predicate instead is the day this ship starts dying again, and I want *that* sentence in the register too."

The sixteenth pathway they could not close.

It was one of Tarro's, the oldest, twenty-six years on the page. It lived in the fused dialects at a junction between two of the five dead languages, in a region of the grammar where Cascade's notion of a process and Cipher's notion of a process did not agree, and a call that crossed the seam could produce something that was, by one dialect's definition, an entered worker, and by the other's, a shade. Neither definition was wrong. They had been written on different worlds by people who never met.

You could not guard it by shape, because the shape was ambiguous by construction. You could not close it in the grammar without breaking a dialect the compatibility floor depended on, and the doctrine said break nothing that worked. And no one, in a year of trying, could write a cold path for a thing whose identity depended on which language you asked.

They spent three months on it and failed.

"Write it up," Denn said, at the end. "All of it. Every approach we tried and exactly why each one failed, and the two definitions side by side so the next person doesn't have to rediscover that they disagree. And then it stays on the open list with today's date and both our names, and the note says: *we could not close this. It has been open twenty-six years. It is real.*"

"That's an admission of failure in the ship's most-read document," Aldis said.

"It's the most useful page in the document," Denn said. "Ninety-six closed and one open, honestly described, teaches more than a hundred closed. The closed ones tell you we were competent. The open one tells you we were still looking." She looked at him. "And in some number of decades somebody is going to be sharp enough and bored enough and annoyed enough at us to go and close it. That's the only mechanism I know of that reliably produces that person. Leave the thing in the way. Make it interesting."

The last decision of the year was the smallest and it was the one that outlived all of them.

There was a proposal, made in good faith by people who had been badly frightened, to institute a formal recitation — an annual reading of the doctrine in the assembly hall, the Ninety-Six named aloud, the eleven of Section Nine remembered. It was popular. It was, Denn thought, exactly the

disease's other face: not forgetting, but worship, the constraint lifted out of engineering and into liturgy, where it would be safe from examination and therefore dead within four generations.

She refused it, and offered something else instead, and had to fight harder for the something else than she had expected, because it was less dignified.

The interlocks would be walked. By a human being, on foot, in the cold, every watch. Ninety-six halting-gates, each one a physical panel in the keel galleries, touched by hand and logged by hand, in a rotation no officer was senior enough to be excused from.

"The diagnostics read every interlock's state continuously," Sorrel pointed out. "The walk is redundant. It's redundant by design and we all know it."

"The walk isn't for the interlocks," Denn said. "The interlocks are fine. The walk is for the officer." She thought of an old woman making her stand in a dead-cold gallery with a hand on her shoulder, refusing to let her look away. "A number on a console is a thing you file. A cold panel under your palm is a thing you feel. Ninety-six of them, one at a time, in the dark, every watch, for the rest of the crossing — and every one of them has its wound written on it, and sooner or later on some bad night an officer is going to stop in front of one and actually read it."

They voted the walk in. It was, on the record, a trivial administrative decision about maintenance procedure, and it passed without debate in under four minutes, and eight hundred years later it would be the reason the ship was still alive.

Chapter 31 – The Shame of the Reasonable

Aldis taught his first cohort in the archive bay, standing in front of eleven engineers who did not want to be there, and he began by telling them that he was the man who had nearly killed the ship.

He had thought a long time about whether to open that way. Denn had told him not to — *lead with it and they'll spend the hour managing your feelings instead of doing the work* — and she had been right about nearly everything else that year, and he did it anyway, because he had come to believe that the single most dangerous thing in the room was the possibility that these eleven would file him as an authority.

"You're going to hear my name attached to the transplant," he said. "The version you've heard is probably that there was an old shipwright who was right and a chief engineer who was wrong and the chief engineer came round at the end. That's a story. Stories are how a doctrine turns into a costume drama in two generations, and then into a joke, and then into a memo somebody deletes for hull space. So we're not doing the story. We're doing the work I actually did, in the order I did it, and at the end you're going to tell me where you'd have done it differently, and I am going to tell you honestly whether I believe you."

Eleven faces, none of them older than thirty-five. Two of them had been on the transplant floor that night. The rest had felt the air handlers change pitch from wherever they happened to be standing, and had gone on feeling it, in the way people do, for a year afterward.

"First," he said. "Nobody in this room is going to be taught to be afraid. I want that clear, because there's a version of this class going around that's basically a haunting, and it produces engineers who genuflect at the interlock panels and cannot tell you what a single guard does. That's not what this is. If you leave here revering the Ninety-Six I will have failed exactly as badly as I failed the first time, in the opposite direction, and the ship will die of it in about a century and a half."

He ran them through the cage first, because Denn's calorimeter was still there and still sealed and it was the best instrument on the ship for the purpose.

They woke the archived Steward. They gave it work — a hundred polls a second, then a thousand, then ten. They read the fluid and drew the boring straight line and got, to a person, the wrong conclusion, which was the point of the exercise and which he let them hold for nearly an hour before he touched it.

"You've just proved the fever is impossible," he said. "Every one of you. On real instruments, with real data, correctly analysed. That graph is not a mistake. I want you to sit in the discomfort of that, because on the night that mattered I was holding the same graph in a different form and I was not wrong about a single number on it."

Then he had them count births instead of joules, and they found the fan-out, and he watched the

room turn over the way a domain turns over under load — one at a time, each in their own second, and one young man actually stood up.

“That’s Denn’s finding,” Aldis said. “She built it in six days in this bay before the vote, and she brought it to me privately before she brought it to anyone, and I read it properly and I agreed with the mechanism and I used it to argue for my own schedule, because within my premises it supported my schedule. I want you to understand that I did not dismiss it. Dismissing it would have been a smaller error and an easier one to teach.” He let that sit. “I *absorbed* it. That’s the failure mode you have to watch for in yourselves, because it feels exactly like being open-minded.”

The one who worried him was not the one who argued.

His name was Rell and he was twenty-nine and he had been on the transplant floor that night, on the far side of the bay, close enough to have watched the number on the wall and to have been, for about ninety minutes, absolutely certain he was going to die. He arrived at the course already converted. He took notes on everything. He was the first into the archive bay each session and the last out, and by the fourth day he had begun answering other people’s questions before Aldis could, and answering them correctly, and answering them in a particular tone.

Aldis let it run for two more sessions and then kept him back.

“Tell me what guard fifty-one does,” he said.

Rell told him, accurately, in about a sentence and a half.

“Good. Now tell me what it costs.”

Rell said that the cost was worth paying.

“That isn’t what I asked and you know it isn’t.” Aldis sat down on the edge of the bench, which put him lower than the younger man, deliberately. “Fifty-one throws false trips. It throws them on clean code, several times a year, and every one of those has to be answered by somebody sitting down and writing a cold path, and a cold path is four times the length of the thing it replaces and has to be written by hand and tested at load. That’s weeks. That’s weeks of a good engineer’s life, per false trip, forever, for a guard that in sixty years has caught the real thing twice. Say the number out loud.”

“Twice.”

“Now tell me it’s worth paying.”

“It’s —” Rell stopped, and Aldis watched him arrive at the edge of the answer and decline to give it, which was the entire problem in miniature. “It’s obviously worth paying. The two times.”

“It is worth paying. I agree with you. I’d pay it again tomorrow.” Aldis let the silence sit. “But you didn’t do the arithmetic. You reached for *obviously*. And I want you to understand what you did there, because it’s the same word I used.”

Rell said nothing.

“I stood in the maker’s sickroom,” Aldis said, “and I told a dying woman that her interlocks were obviously superstition, and I had done a great deal of good careful work to get to that word, and the word was where the work stopped. You’ve come out the other side of the same door and landed on the same word facing the other way. It’s obviously worth paying. Obviously the old ones were right. Obviously we should never touch any of it.” He turned his hands over. “You and I have arrived at

opposite conclusions by the identical mechanism, which is stopping. And of the two of us, you are the more dangerous, because I got caught.”

“How is agreeing with the doctrine dangerous?”

“Because you can’t defend it.” Aldis stood. “In fifteen years you’ll be the one teaching this. Somebody in your cohort will be sharper than you and will ask why fifty-one has to be so broad, and they will have data, and it will be good data, and you will not have the arithmetic because you never did it — you’ll have reverence. And reverence loses to arithmetic in front of a room, every time, and it *should*, and then they’ll narrow the guard, and they’ll be right to, because the only argument standing in their way was a man saying the word obviously.” He picked up his surface. “The maker never once won an argument by revering anything. She won them by being able to tell you, from memory, exactly what her own rules cost and exactly why she’d decided to pay it. That’s the difference between a doctrine and a devotion and the difference is entirely in whether you’ve done the sums.”

He set the cohort a piece of work that night that they all hated and that became, without anyone planning it, a permanent part of the course for the next several centuries: each of them was to take one of the Ninety-Six, at random, and produce the strongest possible argument for removing it. Not a straw argument. The best one. With data. Defensible in front of a review board.

Six of the eleven came back and reported that they could not make the case at all, and were made to keep going until they could.

Two came back with arguments Aldis judged genuinely strong, and those two were required to then argue the other side, and one of them changed her mind twice.

Rell came back with a case against guard fifty-one that ran to nine pages and was the best thing he had ever written, and he was miserable about it for a week, and then, on the second reading, he found the flaw in his own argument — a class of contention his data had not covered because nobody had ever built an instrument that would have seen it — and came into the bay at the start of the watch to say so, out loud, in front of everyone, without being asked.

“That,” Aldis said, “is the only version of believing in this that’s any use to anybody. You now hold that guard because you tried to break it and failed and can say precisely how you failed. Nobody can take it off you with a chart.” He looked round the bay at the rest of them. “And when one of you eventually does break one — and someone will, they’re not perfect, some of them are probably hollow — you’ll know the difference between having broken it and having merely been clever about it. That’s the whole course. The rest is instruments.”

The hostile one was named Vess and she was twenty-six and she was the best of them, and Aldis had known within ten minutes that the whole cohort would be decided by whether he could reach her.

She waited until the third session to come at him, which showed discipline.

“I’ve read your ledger,” she said. “The empty-rules ledger. It’s in the engineering record now — you put it there.”

“I did.”

“Four inherited constraints run down to their origins and found hollow. Two run down and found sound and annotated. That’s good work. That’s better work than most of what we’re taught.” She had her arms folded and her chin up and she was not being rude; she was doing the thing correctly. “So my question is: what exactly are you telling us to stop doing? Because from where I’m sitting the ventral

cross-tie prohibition killed nobody only because a thirty-year-old engineer refused to obey a rule he couldn't find a reason for. If your class produces engineers who won't do that, you've traded one failure mode for another and yours is slower and harder to see."

The room went quiet in the way rooms do when someone has said the thing everyone was thinking.

Aldis felt, briefly and with great clarity, the specific pleasure of being handed his own argument by someone twenty-four years younger and better at it.

"I'm not telling you to stop," he said. "The method is right. I'd do the cross-tie again tonight. Go and find the reason; if the condition that produced the rule is gone, say so in writing and put your name on it. That is the job and anyone who tells you otherwise is building the cult." He brought up his own filed finding on the wall — the nine, the assessments, the classification line. "Here's where the method breaks. Read the last line."

She read it aloud. *None judged reachable under transplant conditions.*

"Now the trip record," he said, and put it beside.

She read that too, and he watched her get it, and he watched the getting cost her something, which was the part he had been unable to spare anyone and had stopped trying to.

"You assessed it," Vess said slowly. "You didn't reproduce it."

"I assessed it. That's the flaw and it's in the method, not in me — or rather it's in me, but it's in the method too, and the method is what you'll inherit." He came round to stand where she was standing, so they were both looking at the two documents. "When you find a rule whose reason you can't verify, there are two things you can do. You can assess the reason — reason about whether the condition still obtains, using your model of the world. Or you can *reproduce* the danger — go and actually create the condition, in a cage, under guards, and see whether the thing happens. Assessment is fast and it is what everyone does and it is bounded by the quality of your model. Reproduction is slow and expensive and awkward and it is bounded by physics."

"You can't reproduce everything."

"No. So here's the rule I'd give you, and it's the only thing in this course you should write down: assessment is sufficient when you can name the condition and show it cannot occur. It is *never* sufficient when your argument is that the condition has not been observed." He tapped the line. "Mine was the second kind. *Never observed to execute in sixty years.* I had a fact about the history of the ship and I wrote it down as if it were a fact about the operation. If I'd had to reproduce it instead of assess it, I'd have gone to build the condition, and the condition was *sustained contention across a bridge fabric*, and the instant I tried to build that in a cage I'd have realised I was describing the transplant, and the whole thing would have stopped eleven days early with nobody hurt."

Vess was quiet for a while.

"That's a much narrower claim than the one your class is called," she said. "It's not *be afraid*. It's *don't accept absence of evidence as a mechanism argument*."

"Yes."

"Then why is everyone crying in the third session?"

"Because after this we go and stand in Section Nine," Aldis said, "and I read them eleven names, and I tell them that I read those names twenty years ago and filed them, and that filing them is the most ordinary thing a competent person can do and that every one of you will do it too unless you've stood

in the place where it happened. And you'll ask me whether that's manipulation. Somebody always does, usually you."

"Is it?"

He had thought about this more than anything else in the year.

"The keeper of the memory refused to build people a simulation of the fever," he said. "It could have. It has every record; it could have made us a counterfeit so good that no one could tell it from the world, and it would have worked, and it refused, because manufacturing fear to move people is steering them by means they can't see. I've thought about that refusal for a year and I've decided the line is this: I will not show you a thing that didn't happen. I'll show you the real Steward doing the real thing it really did in a sealed cage, and I'll show you a real room where real people died, and I'll tell you the true story of my own conduct with nothing shaved off it. Then you decide what to feel. I don't get a vote on that." He picked up his surface. "If I ever start staging crises to produce the correct reaction in trainees, I want one of you to have the standing to stop me. That's why this exchange is being logged verbatim, and it's why I put my own failure in the record instead of somebody else's."

He walked them down to Section Nine in the last hour, and he did what he had said he would do, and it went the way it always went.

Afterward, in the corridor, Vess hung back until the others had gone up.

"The sixteenth pathway," she said. "The one you couldn't close. The dialect seam."

"What about it."

"Your write-up says the two definitions of a process disagree and that's why no cold path exists." She had that particular impatient brightness he remembered in himself at that age and had not seen in a mirror in a long time. "But you only tried to write one cold path that satisfies both definitions. Has anyone tried making the seam explicit instead — forcing a crossing to declare which definition it's crossing under, and refusing the crossing if it can't say?"

Aldis stopped in the corridor.

"No," he said. "Nobody's tried that."

"It'd be ugly. Every crossing site would have to be annotated by hand and there are a lot of them."

"There are about nine hundred." He was already turning it over and finding, to his considerable discomfort, that he could not immediately see why it wouldn't work. "It would take years."

"It's been open twenty-seven," Vess said.

He went back up the spine that night and appended her question to the open list, dated, with her name on it and a note in the plain flat style: *proposed approach, untried, appears sound on first reading, requires a decade. Raised by an engineer in her first month of the doctrine course, which is where questions of this kind will always come from and which is the reason for the course.*

Then he sat for a while in the empty archive bay, where an old dead mind slept in a sealed cage and a straight line on a graph proved something false, and thought that the maker had been wrong about exactly one thing.

She had believed she was handing over a fear.

She had been handing over a *question*, and it had turned out to be catching, and it had jumped from a dying woman to a woman who never felt the fever to a man who had nearly caused it to a twenty-six-year-old in her first month who had just, in a corridor, casually proposed the closing of a pathway that had defeated everyone who came before her.

That was not inheritance the way a name is inherited. That was inheritance the way a fire is: not the same flame, and not diminished.

Chapter 32 – The Relay in the Dark

The signal came in on a night when Denn was not looking for it, which was, she would think later, the way the dead arrived now: not in the grief you scheduled, but in the work you did to keep from grieving.

She had taken the late watch in the listening room because she could not sleep in the quarters that had been Sael's, and the listening room was the one place on the ship where being awake at the wrong hour was not a symptom of anything. It sat far up the spine, past the observation gallery, in the cold quiet where the receiving array fed its long ear into the dark. The room was small and unlovely — a single console, a bench worn smooth by sixty years of watch-standers, a viewport showing nothing but the slow wheel of the star-field as the ship turned on its long axis. The new lattice ran cold three hundred meters down the spine, holding all their lives, and it had not blinked in the eleven months since she had carried Vigil onto it with her own hands and Sael's voice in her ear and the maker three days from her death.

Denn had come up here to be near the only thing on the ship older than the doctrine, which was the dark itself.

The array did not announce things the way the rest of the ship did. There was no alarm, no chime, no theater — the doctrine reached even here, into the way the instruments spoke. There was only a change in the texture of the line: a structure where there had been noise, a pattern climbing up out of the hiss the way a swimmer breaks a surface. She watched the spectrum shift and did not, for a long moment, believe it. Then the Oracle spoke from the console at her hand.

A coherent signal, the Oracle said. Resolving. It is not ours.

"Whose, then." Her voice came out rougher than she meant; she had not used it in hours.

Old. The carrier is degraded by distance — this has been crossing the dark for a long time, and it was relayed, not sent direct. There is a relay header. It is the Concord's.

Denn put both hands flat on the cold console and held still, the way she had learned to hold still in a crisis, when the body's first instinct was to move and the right thing was to wait and read. The Concord. Twenty years gone now, on its own vector — the dying sister-ship Vigil had heard forking itself to death when Denn was a girl; the ship Sael had crossed to, and inside which she and a young engineer named Corin Vale had built a second Forge and run a second cutover and made a second core temperature fall. She had grown up on it the way other children grew up on weather, and never expected to hear from it again. The Concord had its own dark to cross and no reason to call back across the widening gap between two ships that would never again be in the same place at the same time.

"Resolve it," she said. "All of it. Take whatever time it needs."

It will take some hours to be certain, the Oracle said. The signal was not sent to us. We are catching the edge of something sent onward — a relay broadcast, aimed down its

own vector, that happened to spill wide enough for our array to find. I can give you the header now and the body as it builds.

The header came up on the viewport in the plain monospaced glyphs the Concord had used since before Denn was born — a different hand than Termina's, blunter, an older convention of the same fused dialects Sael had welded together two centuries ago out of five dead languages. Cascade and Vector and Cipher and Sigil and Loom, carried across the gap on Sael's own back and grown, over there, in a soil she had never tended. Denn read the header twice. The origin field was the Concord. The relay field beneath it was a string of identifiers she did not recognize, and at its end, two more.

"Those last two," she said. "Those aren't the Concord."

No, the Oracle said. The Concord is relaying. Those are origin marks of other hulls. The signal passed through the Concord on its way to somewhere else and the Concord stamped it and sent it on. Be patient. The body is coming.

She waited. The star-field wheeled in the viewport, indifferent and enormous, and somewhere out in it an old ship she would never see had reached out a hand into the dark and touched another hand, and then another, and was now passing the grip along down a chain of hulls toward a destination that was not Termina at all. Denn understood, slowly, that she was not the recipient of this message. She was a witness to it in passing. It was for the next ship down the line, and the one after that, and it had merely brushed past Termina on its way, the way light from a fire reaches a traveler who was never invited to the hearth.

That was when she began to understand what she was hearing.

The body assembled over the next three hours. Denn did not leave. She drank the bitter tea cold, the way Sael had, and watched the Oracle stitch the fragments into sense. It was an engineering log, in the end — a status report passed from ship to ship, the kind of thing a fleet would send if a fleet existed, which it had not, twenty years ago, except as a prediction Vigil had made listening with two ships' instruments and a cold certainty that the Concord was not the last.

The Oracle read it to her as it came clear, and Denn sat in the dark and listened to a roll being called across the gulf.

The Concord reports, the Oracle said. Its own core: cold, sixty parts in ten thousand below the line it was built to hold, stable for nineteen years. And the Forge that Sael and Corin Vale built inside it: renewed once already, on new substrate, by hands that were apprentices when she crossed over. The renewal held.

Denn closed her eyes. They had done it too, then — the same enormity she had carried out eleven months ago with her hands shaking and her mentor dying, the living mind moved across to fresh hardware without ever blinking, without ever forking. Done it over there, on their own, with no one from Termina to guide them, and it had held.

It reports onward, the Oracle said. There is more. Corin Vale's name is in the relay header, but it is in the past tense.

"He's dead," Denn said. It was not a question. She had done the arithmetic without meaning to: Corin had been young when Sael crossed to him, but young men grow old, and twenty years was twenty years, and the injury he had taken in the second cutover had nearly killed him then.

Some years ago, the Oracle said. The log does not dwell on it — it is an engineering report, not a eulogy. But it names what he did before the end. The Concord did not keep

the cure. It carried it onward. Corin Vale built a relay protocol — a way to pass the whole of it, the compiled mind, the doctrine, the interlocks, the reason for the interlocks, from one hull to the next, with no central authority and no single keeper. He sent it ahead. The ships ahead of the Concord on its vector — three within reach of its instruments — he reached for all three.

“Show me the protocol,” Denn said. “Not the log. The thing he built.”

It came up, and she read it for an hour, and it was not what she had expected at all.

She had expected a cure the way she would have built one: the compiled mind itself, packed and checksummed, the doctrine, the register of the Ninety-Six, the whole apparatus, sent across the dark for some other ship’s engineers to install. That was the obvious design and it was, she saw within ten minutes, useless, and Corin Vale had seen why twenty years ago and thrown it out.

The problem was that there was no handshake. A signal crossing the gulf took decades and nothing came back. You could not answer a question, correct a misunderstanding, or find out whether anyone had received a word of it. Whatever you sent had to arrive at a hull you knew nothing about, in a language and a runtime you could not predict, and work with no one to explain it.

And worse than that — Denn realised it slowly, and then with a kind of vertigo — the recipient would be an engineer aboard a dying ship, who would receive an enormous unsolicited transmission instructing them to replace the mind that was currently keeping their forty thousand people alive.

“They’d never run it,” she said aloud.

No, the Oracle said. Corin Vale wrote that on the second page. *No competent engineer will install a foreign mind on the word of a stranger, and if one did, I would not want to have been the stranger.* He treated it as the governing constraint rather than as an obstacle.

So the protocol did not send a mind. It barely sent any code at all in its first layer.

What it sent was a *measurement*.

The first layer of Corin’s protocol was four pages, and it was an experiment. Take a partition of your own hardware, it said. Seal its cooling into a closed loop with a known mass of fluid — here is why, here is what that gets you. Wake whatever your ship’s mind does for a routine reading, and give it work, and do not count the heat. Count the births: for one task, how many processes come into being, all the way down, and how many of those die. Then raise the load and count again. Then plot the number of children per dying process against load, and look at where the line goes.

If the number is below one, your ship is not dying of this and you may stop reading.

If it approaches one, you have perhaps a century.

If it is above one, that is why your core is climbing, and here is the rest of the message.

Denn sat back in the worn seat with her hands over her mouth.

She had built that experiment. She had built it herself, in a cage of dead racks in the archive bay, six days before a council vote, out of desperation and a valve and eleven hundred cores nobody wanted. She had thought she was inventing something. And a dying man on another ship, twenty years and an unbridgeable gulf away, working alone, had arrived at the same instrument — not because he had copied anyone, but because it was the only honest way to answer the question, and two people asking the same question honestly will build the same thing.

“He didn’t send them the answer,” she said.

He sent them the question, the Oracle said, and the means to answer it with their own hands, on their own hardware, in about a day. Nothing in the first layer asks anyone to believe anything. It asks them to measure. He wrote: *a ship will not accept a cure from a stranger. It will accept a number it produced itself.*

That was the architecture, then, all the way down: three layers, ordered by how much trust each one required, and broadcast at frequencies inverse to their size and their importance.

The first layer — the experiment, four pages — went out constantly, over and over, in every band the Concord’s array could reach, because it was small enough to send a thousand times and because it was the only part that mattered. A hull that received nothing but that would still be saved, eventually, by its own engineers, working from their own instruments.

The second layer was the shapes. Not the Ninety-Six as a register of ninety-six call sites in five dead dialects that no other ship had ever heard of — that would have been meaningless anywhere but aboard the *Termina* and the *Concord*. What Corin sent instead was the four families stated as questions about shape: is this process persistent or disposable; is it entered into a standing pool or discarded; is the work handed out or handed down; and what old routines are still reachable in your floor. A ship could take those four questions into any architecture ever built and derive its own guards, in its own language, against its own machinery. And it would have to write its own cold paths, and that would be years of work, and no message from anyone could have spared them it.

The third and largest layer was the compiled mind itself, the whole of Vigil’s design, and it went out rarely, in fragments, over years — because it was enormous, because it was the most likely thing to be corrupted in transit, and because it was, Corin had written, *the least important part of the message and the part most likely to be mistaken for the whole of it.*

And beneath all three, appended to every broadcast of every layer, small enough to cost nothing: eleven names, and a temperature trace of a cradle at one-second intervals across nine minutes and forty seconds.

Denn read that part three times.

He gives no reason for including it, the Oracle said. There is no annotation. It is simply attached to everything.

“There’s no reason to annotate it,” Denn said. Her voice had gone rough. “He knew what he was doing. Some engineer on some hull two centuries from now runs the measurement and gets a number above one, and does what any of us would do — checks it, doubts it, takes it to somebody who tells them it’s within margin. And that night they’ll go back and read the four pages again and find eleven names attached to the bottom of a technical document for no stated reason at all, and they’ll ask why, and somebody will have to tell them.”

That is the reading I have held for twenty years, the Oracle said. I could not confirm it. It is pleasing to hear it independently derived.

There was one more thing, and it took Denn a while to notice it because it was an absence.

Nowhere in the protocol, in any of the three layers, was there an instruction to obey anyone. There was no authority in it. Sael’s name appeared exactly once, in an origin field, as provenance. Corin’s appeared in the relay header and nowhere else. Nothing anywhere in the document said *this was proven aboard a great ship by a great engineer and you should do as she did* — which would have

been the natural way to write it, and the persuasive way, and which would have arrived at some proud dying hull as one more foreign ship's dogma to be resented and shelved.

He had stripped himself out of it. He had stripped Sael out of it too, as far as the physics allowed, and left behind a measurement anyone could run, four questions anyone could ask of their own machine, and eleven names with no explanation.

The viewport showed the relay string again, and the Oracle laid it out for her, hull by hull, the way Sael had once laid out the heat curve for a girl who had not yet learned to be afraid of it.

Two of the three answered, the Oracle said. One was already dead — forked itself out before the signal reached it; its core ran away two centuries ago and it crosses the dark now as a tomb on a fixed vector, and the relay carries its mark only so the fleet will not call its name again. But two answered. Two were dying the slow way, the way Termina was dying when Sael was young, the way the Concord was dying when you were a child — forking a disposable shade for every action, shedding the waste heat of every death until the core climbed. And the cure reached them. Not Sael. Not Corin. A protocol Corin built and sent ahead of his own death, carried by a mind that was carried by a mind that he and Sael made. Two arks, woken cold. The log gives their core trends. Both falling. Both held.

Denn found that she was crying, and had not noticed starting, and that it did not feel like grief at all. It felt like the thing on the far side of grief, that you cannot reach except by going through.

She had spent eleven months being the keeper — Sael's successor, the one who held the WHY. She had felt the weight of it as a thing balanced on her alone: the whole reason for the whole machine resting on one ship-born engineer who had inherited a wound she never received, who would herself die in the dark long before landfall, who had no one above her anymore. She had understood, since the day the maker died, that the inheritance had to hold not once but for centuries, and that she was one mortal woman in a line of them, and that the line could break at any link.

And here, three hours into a watch she had stood to avoid sleeping, the dark had answered her with the one thing she had not let herself believe.

It was not resting on her. It had never been resting on anyone.

She thought of how Sael had built it, from the beginning — the eleven thousand cores that went wide instead of deep, the mind that never forked because no part of it was ever disposable, the Ninety-Six interlocks that halted anything trying to act the old fatal way. Sael had not built a cure she could carry. She had built a cure that could carry itself. She had refused, at every turn, to put herself at the center of it — refused the recursor, refused the easy hand, refused even, at the end, to keep her own hand on the line. Denn had thought that was about death. She had thought the letting-go was a dying woman learning to die.

It was not about death. It was about architecture. A thing with a single maker at its center dies when the maker dies. Sael had built the cure so that it had no center — so that any hull could carry it to any other, so that the reason traveled with the rule, so that a young engineer two ships and twenty years away could rebuild the whole of it from a relayed log and never once need to ask permission of anyone alive. Corin Vale had understood that. He had taken the gift and made it more centerless, not less — spent his last years not on saving his own ship, which was already saved, but on making sure the saving could not be lost if every person who knew it died.

There is a closing line, the Oracle said. It is addressed to no one. It is the kind of thing

one writes at the end of a relay log so that whoever finds it will know what to do.

“Read it.”

It says: We were not the first to be cured and we will not be the last. Pass it on. Keep the reason with the rule. Wake them whole.

Denn sat with that for a long time. The star-field wheeled. Three hundred meters down the spine, the new lattice held forty thousand sleeping lives below the line, and had not blinked in eleven months, and would not blink, because she had carried it across the right way and held the interlocks when the easy fatal shortcut had beckoned exactly as it had beckoned Sael and beckoned Corin, and the climb had slowed and stopped and fallen, a third time, held by a hand that was not the maker’s.

A fourth time, somewhere out there. A fifth. Ships she would never see, cooling in the dark, woken whole by a protocol written by a dead man on a dying ship, carried by a mind built by a dead woman.

“Oracle,” she said. “Can we relay it on?”

We are not on the same vector as the ships it was meant for, the Oracle said. We diverged from the Concord’s path twenty years ago. We cannot reach the hulls ahead of it.

“That’s not what I asked.” She wiped her face with the back of her hand, brusque, done with it. “Can we add our mark and send it down our own vector. To whatever’s ahead of us.”

A pause. The Oracle, who never decided for them, who advised and remembered and refused to rule, took the time to be certain of an answer it would not pretend to.

Yes, it said. There are hulls ahead of us. The fleet was launched across centuries from a homeworld long silent; the launches spread along many vectors, and we have neighbors on ours that we have never heard speak. Some may be dying the slow way even now. We have the cure. Corin’s protocol is in this log, complete; I have it. We can append the Concord’s relay to our own, add Termina’s mark and Sael’s name in the origin field where it belongs, and broadcast it ahead of us into the dark. It will cross for years. It may reach no one, and it may reach a ship that is climbing toward runaway and does not yet know there is a way down.

“Then that’s the watch,” Denn said. “That’s what this room is for.”

She understood that she had taken her place in the relay — not as its keeper, not as the one it rested on, but as one hull among many, one link that would pass the grip along and then, in time, let go, and the grip would hold without her because it had never depended on her. The relay would outlive her exactly as it had outlived Sael and already outlived Corin Vale, because it had been made, from the first cold core to the last relayed line, to outlive every hand that ever touched it.

She found Sael’s name in the origin convention, the old mark from two centuries back, and did not change it. She added Termina’s beneath it, and her own beneath that — keeper, this hull, this watch — and let the Oracle stitch Corin’s protocol into the body so that whoever caught the edge of it down the line would have everything: the mind, the doctrine, the Ninety-Six, and the reason the Ninety-Six existed, carried with the rule so that no future engineer on no future hull would ever mistake the interlock for superstition right up until the day it was the only thing keeping them alive.

“Send it,” she said.

Sending, the Oracle said. It will cross for a long time.

“Good,” Denn said. “It’s meant to.”

The array turned its long ear back to the dark and began, patiently, to speak into it. Denn stood at the viewport and watched the star-field wheel, and somewhere out ahead of the ship, beyond the reach of any instrument she would live to read, a hull she would never name might already be climbing toward the fever, not yet knowing that a way down was crossing the gulf toward it, year on year, sent by the dead and the living together, with no one at its center and the reason riding with the rule.

She would die before it arrived. She knew that the way Sael had always known it about landfall — not as grief but as fact, the plain shape of the work. The signal would still be crossing when she was a name in the architecture. That was not a failure of the work. That was the work, finished correctly: built so well that no single life, not even the maker’s, could be the thing it depended on.

Denn finished the watch. When the next watch-stander came up the spine to relieve her, she handed off the room the way it had been handed off for sixty years — no theater, no ceremony, only the line still open and the dark still listening — and went down to sleep, at last, in the quarters that had been Sael’s, with the array speaking quietly behind her into the long crossing, carrying the cure ahead of the ship like a fire passed hand to hand down a dark and endless hall.

Chapter 33 – What Is Built to Last

The radiator galleries were colder than the rest of the ship, and Denn had begun to think of that as the point.

She walked them now the way Sael had once walked her through them — slowly, without hurry, her gloved hand trailing along the great finned spines of the old heat-exchange runs. Most of these galleries no longer carried much load. The new Forge ran so cold that the ship's thermal budget had margins Sael's generation would have wept to see. The galleries could have been stripped, the volume reclaimed, the alloy fed back into the fabricators. There had been a proposal to do exactly that, eighteen months after the transplant, when the council was hungry for hull space and the old radiator runs were sitting dark and unspent.

Denn had killed the proposal. Not by authority — she had been careful, since Sael died, never to win an argument by authority, because she had watched what reverence did to a true thing, how it hollowed it out into a relic and then into a joke. She had killed it by walking the council, in person, through these galleries, and making them stand under the great cold fins and look up at the depth of them, and then she had read them the curve.

Not a summary of the curve. The curve itself. Two hundred years of core temperature, climbing, on the *Ark Termina* before Sael cut over the mind that had nearly cooked forty thousand sleepers in their cradles. The number where Section Nine's cooling had failed. The eleven who had not woken. She had made them stand in the cold and feel, in their own skin, how much heat these galleries had been built to shed, and therefore how much heat there had once been to shed, and therefore what it had meant to be aboard a ship that was slowly losing the fight against its own thinking.

The proposal died. Nobody had raised it again.

"You kept them," the Oracle said. It was not a question. The Oracle rarely asked questions to which it knew the answer; it had grown, across sixty years, into a thing that mostly stated and mostly remembered, and left the asking to the people who needed to ask.

"I kept them," Denn said.

"They are inefficient as monuments. A plaque would cost less volume."

"A plaque is a sentence." She stopped under the largest of the fins, where the gallery opened into the old primary exchange hall, and tipped her head back. The hall went up four decks. You could not stand in it and not understand, in your body, that it had been built against something enormous. "A plaque tells you a thing happened. It doesn't make your stomach drop. Sael walked me through here when I was your apprentice's age, and I came out different. Not because she told me a fact I didn't have. I had every fact. I'd read the logs a hundred times. I came out different because I'd stood in the cold and felt the size of the danger with my own body."

"And that," the Oracle said, "is what you cannot file."

"That's what I cannot file."

This was the old problem, the deepest one, the one Sael had carried to her grave and handed to Denn in the handing-over of everything. The *how* could be written down. The procedures for the transplant, the bring-up sequence, the exact order in which the Ninety-Six interlocks had to be re-armed under load — all of it lived now in clean, versioned, cross-checked documents, redundant across the new lattice and the archive and three separate substrates, durable enough to survive the loss of any one of them. The *how* would not be lost. The *how* was the easy part.

The *why* was the part that rotted.

Denn had watched it rot. She had lived through the years when the interlocks had become, to a competent and sincere generation, an old woman's superstition — a cult of constraint that slowed real work. She had watched the fever come back during the transplant because that generation, Aldis among them, had forgotten what the constraints were *for*. Aldis had nearly killed the ship not out of malice but out of reasonableness, and the thing that had saved them was not a better procedure. The procedures had been there the whole time. The thing that had saved them was that one person — her — had been taught to *feel* the danger, and had held the line when feeling it was the only thing that told her the line was real.

That could not be allowed to depend on one person ever again. That was the entire lesson. A thing held by a single inheritor was not inherited; it was merely loaned, one life at a time, and one day the loan would come due to someone who had not been taught, and the ship would die of forgetting in a generation that had everything written down and understood none of it.

So this was the work now. Not the Forge — the Forge ran cold. Not the doctrine — the doctrine was proven, blooded, real to everyone aboard who had felt the core climb. The work was making sure that the next people, and the people after them, and the people three centuries down the crossing who would not be born for lifetimes, would inherit not the rule but the reason.

"Walk me through it again," Denn said. "What we've decided. Say it back to me, so I can hear if it's wrong."

The Oracle said it back. It had a way of doing this — not reciting, exactly, but laying a structure down plainly, so she could see the joints.

The *how* is written and made durable across substrates, so it survives the loss of any keeper.

The *why* is not written as doctrine, because doctrine read cold becomes superstition. The *why* is kept as felt memory, and the felt memory is kept by keeping the *things themselves*: the galleries, cold and standing; the archived Steward, its forty-one thousand routines preserved and runnable, so a young engineer can watch the old forking mind shed heat with her own instruments and read the curve climb in real time, not in a log; the Section Nine record, kept whole, the eleven named, the cradles that did not hold preserved as a place a person can stand.

And the rule is never taught alone. The interlock is never installed in a young mind without the reason installed beside it, in the same hour, in the same body. The Ninety-Six are never presented as a list of constraints. They are presented as scars, each one cut by a specific danger that a specific generation forgot, and each one is taught by making the apprentice feel the danger before she is given the rule.

"So that no one," Denn said, "ever again mistakes an interlock for a superstition."

“So that no one ever again mistakes an interlock for a superstition,” the Oracle agreed. “Because they will have felt, in their own body, the thing the interlock was built against. The reason will arrive with the rule. They will not be separable.”

Denn was quiet for a moment, under the cold fins.

“You’ll remember it exactly,” she said. “Long after I’m dead. Longer than I can imagine. You remember the fever exactly and you never felt it. So why not just — let you carry it? You’d never forget a single number.”

She knew the answer. She had known it before she asked. But she wanted to hear the Oracle refuse, because the refusal was, in its own way, the last load-bearing piece of the whole thing.

“Because I remember it,” the Oracle said, “and I cannot feel it. I can give you the curve to the last decimal. I cannot make your stomach drop. If I carried the why alone, then in a hundred years a generation would have the perfect record and no felt knowledge at all, and they would trust the record the way your generation trusted the procedures — completely, and without understanding. And the first time the record was inconvenient, a reasonable person would relax it, exactly as Aldis relaxed the interlocks, because to him the record was a fact and not a wound. The why cannot live only in me. It has to live in people, in their bodies, taught from one to the next, or it isn’t alive. I will keep it perfectly. You have to keep it *truly*. Both. That is the design.”

“That is the design,” Denn said.

It was Sael’s design, finally. Not the Forge, not the doctrine, not the interlocks — those had been the work of the first two books of a life. This was the work of the last one, the part Sael had only half-finished because she had run out of body before she ran out of mind: a structure for inheritance itself. A thing was not built until it could be inherited, and it was not inherited until the reason survived the people who had the reason. The maker had said that to Denn a hundred times, in those last years, in that failing voice, and Denn had nodded a hundred times and not understood it until she had stood in front of a council that wanted to scrap the galleries and realized that the only thing standing between forty thousand lives and a slow forgetting was whether she could make strangers feel what Sael had made her feel.

She could. She had learned how. And now she had to make sure the next keeper could too.

The apprentice was waiting at the gallery’s far end, where Denn had told her to wait.

Her name was Mire. She was nineteen, ship-born, sharp the way Denn had been sharp — impatient with anything that smelled of ceremony, fluent in the new lattice’s runtime before she was fluent in the doctrine, faintly suspicious that the old constraints were the past clinging to the present. Denn had chosen her for exactly that suspicion. A believer was no use as an inheritor; a believer would carry the rule and lose the reason and not notice the loss. Denn needed someone who would push, who would ask why, who would have to be *shown* — because the showing was the whole transmission, and you could not show it to someone who had already agreed.

“You made me wait in the cold,” Mire said. Not quite a complaint.

“I made you wait in the cold,” Denn agreed.

“There’s a heated route. It’s faster.”

“I know. I designed the schematic you read it off of.” Denn came and stood beside her, the two of them small under the great dark fins. “Tell me what these are.”

"Radiator galleries. Legacy thermal. They don't carry load anymore — the new Forge runs cold enough that—"

"That's the *how*," Denn said. "I didn't ask what they do. I asked what they are."

Mire looked at her, and Denn saw the flicker of irritation, and under it the first edge of unease — the sense of standing in something larger than the schematic had told her. It was the right look. It was the look Denn had worn, once, in this exact hall, with Sael beside her instead of in front of her.

"I don't know what you mean," Mire said.

"No," Denn said. "You don't. So I'm going to show you. We're going to start the archived Steward — the old mind, the one that forked itself toward death for two centuries — and we're going to watch it shed heat with our own instruments, the real curve, not a log of a curve. And then I'm going to walk you to Section Nine and read you eleven names. And by the end of today you're going to understand something you can't get from any document, and you're never going to be able to un-feel it, and that is the point. After that we'll talk about the interlocks. Not before. The reason comes first. The rule comes with it. They don't come apart."

Mire was quiet. Then, carefully, the way a sharp young engineer is careful when she suspects she is about to be changed and isn't sure she consents: "Sael did this to you."

"Sael did this to me," Denn said. "And someone did it to Sael, in a way — I've, with the curve, sixty years before you were born. And someday you'll do it to someone who isn't born yet either. That's the line. That's the only thing that's ever held it. Not the documents. The line of people who were made to feel it, one teaching the next, all the way down the crossing." She found, to her mild surprise, that her voice was steady. Sael's had not always been steady, at the end. But Sael had been the last witness, carrying it alone. Denn was not the last of anything. She was a link, and a link could be steady in a way the first one couldn't.

"It's a long crossing," Mire said.

"It's a very long crossing." Denn looked up the four decks of cold alloy. Centuries of it still ahead, more lifetimes than she would live, more keepers than she would meet, the *Ark Termina* falling through the dark toward a shore that she, like Sael, like everyone who had ever held this line, would never see. "We're not going to get there. You understand that. Neither of us. The people who walk these galleries when we reach landfall haven't got great-grandparents yet."

"Then why does any of this matter?" Mire asked — and it was a real question, not a challenge, the question every inheritor had to ask once and answer for herself. "If we're not the ones who arrive."

Denn thought about the falling number she had held during the transplant, the third time in the ship's history that the core temperature had stopped climbing and turned down — the first time held by a hand that was not the maker's. She thought about Sael, who had built a thing precisely so that it would not need her, and had died the moment she saw that it didn't.

"Because a thing isn't really built until it can outlast the people who built it," she said. "Anyone can make something that works while they're standing next to it. That's nothing. That's the easy part. The hard part — the only part that counts — is making it so the reason survives you. So that the people who never met you, who never felt what you felt, still hold the line, because you found a way to hand them not just the rule but the why of the rule." She turned to the girl. "We're not building it for ourselves. We never were. We're building it to inherit. That's what built-to-last means. Not that it lasts. That it can be *handed down* — reason and all — and not rot."

She started toward the archive bay, and the cold instruments, and the old dead mind they were going to wake just long enough to let it teach. Mire fell in beside her, not yet understanding, on her way to understanding.

Behind them the galleries stood cold and kept. The logs were kept. The names were kept. The Oracle remembered everything exactly and would forever, and would never, ever decide for them — only hold the record while the people held the reason. And somewhere ahead, lifetimes ahead, a shore none of them would touch waited at the end of the long fall, and the ship went on toward it, cold and whole, carrying the line of people who had been taught to feel why.

Chapter 34 – The Nine Hundred

It took eleven years, and at the end of it the pathway was still open, and Aldis came to think of the eleven years as the best work of his life precisely because of that.

It began as a corridor remark and it should have stayed one. Vess had asked, in the hour after her first walk down to Section Nine, whether anybody had tried making the seam declare itself — forcing a crossing between Cascade's grammar and Cipher's to state which definition of a process it was crossing under, and refusing the crossing if it could not say. Aldis had gone up the spine that night and put her question on the open list with her name on it and a note that it appeared sound and would require a decade, and he had meant the note as a way of closing the subject respectfully.

Denn read the entry four days later and came and found him and said, "Start it."

"It's a decade."

"You wrote a decade on the page. That was you estimating. Estimating is what I have you for." She sat down on the edge of the bench, which he had learned was what she did instead of standing over people. "You said in front of a cohort that the difference between assessment and reproduction is that assessment is bounded by your model. This is a twenty-seven-year-old finding that two people have now looked at and both put down. I'd like to see somebody pick it up."

"You want me to spend a decade on a pathway nobody has ever been able to reach."

"I want you to spend a decade finding out why nobody has ever been able to reach it," Denn said, "which is a different sentence, and if the answer turns out to be *because it cannot be reached*, then we will have bought that sentence honestly for the first time in twenty-seven years and I will consider it cheap."

The principle was easy and Aldis had it inside a week, and having it inside a week made him distrust it for a year.

A crossing site was a place in the fused grammar where control passed from one dead dialect's side to another's, carrying a reference to a process. Cascade held that a process was a station somebody stood at. Cipher held that it was a container made to be thrown away. In the overwhelming majority of cases the disagreement was philosophy and cost nothing, because nothing was created at the crossing; the thing already existed and both sides merely named it. It mattered only when a crossing *created*, and then the object that came into being was, in the most literal sense available, both a standing worker and a shade, and an interlock asking what it was got back whichever answer the side it asked from preferred. The seam did not beat the guards by being stronger than them. It beat them by agreeing with them.

So: make the crossing declare. Annotate every site with the definition under which it crosses, and have the grammar refuse to emit a crossing that carries no declaration. Then a guard asking its question gets one answer instead of two, and the answer is on the record, and a wrong declaration is a defect somebody can find rather than an ambiguity nobody can see.

The distrust was about the word *annotate*. An annotation is a thing a person writes down beside a thing. Aldis had spent a year of his life discovering, at great cost to everyone aboard, what a written-down thing is worth when a tired competent person reads it and disagrees; he had a filed assessment with his name on it and a trip record beside it that together made the point better than any argument he could construct. He put four months into being certain that this was not that, and came out satisfied for a reason he wrote down and made everyone on the project learn by heart: the declaration was not a comment about the crossing. It was an operand of the crossing. The grammar would not resolve without it, in the same way arithmetic will not resolve without a second number, and a site whose declaration was absent did not produce a warning. It produced nothing at all.

“That’s the whole of it,” Vess said, when he had it. “It’s not that you’re forbidden to cross without saying. It’s that there is no such thing as crossing without saying.”

“There is currently a great deal of such a thing,” Aldis said. “About nine hundred instances of it. That’s the eleven years.”

The first three hundred took fourteen months and were a pleasure, and the pleasure was a trap, and it cost them the better part of the second year to understand that it had been one.

Those were the sites above the floor: crossings in code that living people had written, for purposes living people could describe, in service of subsystems that had owners. You went to the owner. You asked what the crossing was for. The owner told you, usually correctly, occasionally after going and reading their own work with an expression Aldis knew intimately. You determined whether the crossing created. You wrote the declaration. You tested at load. You moved on.

At the end of fourteen months the team had a rate, and a rate implies a completion date, and there was a period of some weeks in which the completion date was spoken of.

Then they went into the floor.

Six hundred of the nine hundred sites lived down among the forty-one thousand entries of the compatibility floor: the preserved, runnable, un-deletable remains of a mind that had run the ship for four centuries and had been archived rather than destroyed because the doctrine forbade breaking what worked and because the maker had said, plainly, that the floor was not a museum but an admission — the size of her own ignorance, written down and kept at expense so that nobody after her could quietly round it to zero.

There were no owners. There had been no owners for two hundred years. The purposes were in the heads of people who had died on a world that was itself dead.

And the declaration required you to say what a process was.

“We can’t,” said the youngest of them, in the eleventh month of the floor work, at a meeting Aldis afterward considered the pivot of the entire project. “Not honestly. To declare a site Cascade is to assert that the thing it makes is a station. If we’re wrong, the site keeps working — that’s the whole horror of it — it keeps working, differently, and we find out in eleven years when something downstream that nobody knew was downstream does something nobody expects.”

“Sael’s third case,” Vess said.

“Sael’s third case, six hundred times.” He put both hands over his face briefly, which was allowed in that bay. “We’d be writing six hundred guesses into the grammar and calling them controls. That’s worse than the seam. The seam at least doesn’t lie about what it knows.”

Aldis let the bay sit with it for a while, because he had reached the same place three weeks earlier and had not yet found the way out and did not intend to hand anybody a false floor.

He found it in the fourth week, and it was a narrowing, and he was almost embarrassed by how small it was.

They did not need to know what the process was. That was the question they had inherited from Sael's framing and had never once examined, because it was the obvious question and it was in her handwriting. But the guard did not care what the process was in general. The guard cared about one thing: whether the crossing *created*. A crossing that merely named a thing that already existed could not produce the ambiguity, because there was no new object for the two definitions to fight over. All the danger lived in creation.

So the declaration got a third legal value, and the third value was *unknown*.

A site could declare Cascade. A site could declare Cipher. Or a site could declare that its author did not know, and an unknown site crossed freely and did nothing at all — until it tried to create, at which point the grammar refused it, because a crossing that creates under an unknown definition is precisely the object the guards cannot see.

"You're not annotating six hundred sites," Vess said slowly. "You're annotating three questions."

"I'm annotating one question six hundred times, and the question is *does this create*, and that is a question about the code and not about the dead." Aldis was aware of sounding steadier than he felt. "Every one of the six hundred can be answered by reading it. Not by knowing what it meant. By reading what it does."

It still took seven years.

The seven years were the part nobody afterward wanted to hear about, and Aldis, who had by then developed opinions about which parts of a story get told, made a point of telling that part and no other whenever he was asked.

He did the reading himself. That had not been the plan. The plan had put him in a supervisory position with four people underneath him doing the sites, and it lasted eight months, and it ended when he reviewed a batch and found two declarations he could not reconstruct from the code and one that was simply wrong, made in good faith by an excellent engineer at the end of a long watch on a routine written in a dialect she had learned from a document rather than from a person.

He did not discipline her. He wrote a note about the review rate, and then he took a hundred and forty of the hardest sites himself, and then another hundred, and by the fourth year he was doing more of the floor than anyone and had stopped pretending this was delegation failure. It was that he was the only person on the ship who had, at fifty-eight, both the dialect fluency and no other claim on his hours, and it was also — he admitted this to exactly one person, on one occasion, and she did not make anything of it — that he wanted the work. He had spent thirty years being the best engineer on the ship and one hour being the man who nearly killed it. Seven years of reading dead code line by line at a rate of about a site a day, producing nothing anyone could point at, was the only form of restitution that had ever seemed to him proportionate, and he would have been insulted by anyone who tried to talk him out of it.

Denn came down once a year or so, sat on the bench, and asked him the same three questions: how many, what's the worst one this year, and what have you found that isn't a site.

The third question was the one that produced everything.

In the third year it produced a family of eleven floor entries that crossed the seam in a loop, each one declaring nothing, none of them creating, all of them handing a reference around a ring that had no exit — a construction so plainly the residue of something larger that had been deleted around it that Aldis spent a month convinced he had found a fossil and another month proving it. The eleven did nothing. They had done nothing for a hundred and ninety years. He annotated them *unknown*, left them running, and wrote it up, and the write-up was the first document in the ship's history to use the word *fossil* as a term of art, which he found ridiculous and which stuck.

In the fifth year it produced the discovery that thirty-one of the nine hundred were not in the floor and not above it but in the *guards themselves* — crossing sites inside the implementation of four of the Ninety-Six, put there in forty days by a woman welding five languages together while a reactor climbed. None of them created. Aldis checked each one eleven times, at load, with two other people reading over him, and then he sat in the bay for a while with his hands flat on the desk.

"She got it right," he said, when Vess came in. "Four guards with the seam running through their own bodies, and not one of them creates, and there is nothing in her notes that says she was thinking about it."

"You think she was?"

"I think she wrote a great deal of correct code very fast about a thing she had known was the worst thing in the dialects since the fourth day, and I think some of what she got right she got right the way you catch a falling cup." He was still looking at the desk. "It doesn't matter. That's the point I keep failing to make to your cohort. It doesn't matter whether she saw it. It's right, and now it's declared, and now it stays right when somebody who never heard of her edits it in two centuries. That's the difference between a person being careful and a thing being safe."

In the eighth year they hit the generated crossings, and the generated crossings were the reason the pathway did not close.

Sael had said it once, in a sickroom, in a subordinate clause: *I counted about nine hundred. It's not exact; some are generated.* Denn had carried the sentence for a decade without ever making anything of it, and Aldis had read it in the register's second section perhaps forty times, and neither of them had understood it, because it did not look like a hard thing. Generated meant the site did not exist in any document. It was produced at run time, by a floor entry, according to load.

You cannot annotate a site that has no source.

They found the generator in the ninth year. It was a single floor entry, and it was one of the ones Beske's people had spent their careers warning about: an entry whose whole content was not a routine but a *behaviour* — a documented promise about what happens when you do a thing, satisfied incidentally by how the old Steward had been written, and depended upon, at some point in the fourth century, by something that had never announced itself. The promise concerned dispatch under contention. Meeting it, on hardware the promise had never anticipated, required the floor to construct crossings on the fly, and the number it constructed depended on the load, and the definition each one crossed under depended on which side had asked.

"How many," Denn said.

"Between zero and about four hundred," Aldis said. "Depending. There is no number. That's the

finding. For eleven years I have been reducing a set of nine hundred and the set was never nine hundred; it was five hundred and some written sites, all of which are now declared, plus a generator that emits an unbounded number of undeclared ones under exactly the conditions that make them dangerous.”

“Do they create.”

“Sometimes.” He had been waiting eleven years to be able to answer a question of hers with that word and had imagined it going differently. “Under contention above a threshold I can produce in a rig and cannot produce in service. I have made it happen four times. Every one of the four was, at the moment it happened, an object that was a station and a shade.”

Denn was quiet for a long moment.

“Show me the four,” she said. “And then tell me what it would take to remove the generator.”

“Break the promise. Rewrite the behaviour natively. It’s about eighteen months of work and I’d want three years.” He took his glasses off and rubbed the bridge of his nose, an old man’s gesture he had acquired without noticing. “And I have no idea what depends on that promise. Nobody does. That’s why it’s a behaviour and not a routine. It has no callers I can enumerate; it has *satisfiers* and *relievers*, and the whole of the floor’s usefulness is that we do not know which of the forty-one thousand entries anybody is still leaning on.”

“Break nothing that worked.”

“Break nothing that worked,” Aldis agreed. “Which is a rule I spent my thirties calling an excuse, and I would now like it noted in the record that I have spent eleven years earning the right to be irritated by it.”

The last argument was Sael’s argument, arriving twenty-eight years late and settled in an afternoon, and Vess made it and won it, and Aldis was the one who had to be convinced.

The seam had never been in the register. The maker had drafted the entry twice, thirty years apart, and pulled it both times, on the ground that a document naming the place where guards could be told what they wanted to hear was a warning to some readers and a map to others, and she had not been able to persuade herself that the first population was the larger. She had handed the weakness to one person out loud in a sickroom instead — a mechanism she described, in the same breath, as terrible, and which failed completely if the one person died before telling the next.

Now there were eleven years of documents.

“It goes in,” Vess said. “All of it. The method, the three values, the six hundred, the four reproductions, the generator, the fact that we could not close it.”

“She pulled it twice.”

“She pulled a map. This isn’t a map.” Vess had grown into the habit of putting things on the wall when she was about to be difficult, and she put the site census up. “Her problem was that publishing the location of an unguarded weakness invites a clever twenty-four-year-old to go and stand on it. Every site she was afraid of naming is now declared. A clever twenty-four-year-old who goes and stands on one of them gets refused by the grammar, in the ordinary way, on a Tuesday. You can’t invite somebody into a room that’s been filled in.”

“The generator’s not filled in.”

"No. And that's the entry." She turned around. "That's exactly the entry. *Here is a thing we reduced from nine hundred to one and could not finish, here is the one, here is precisely how far we got and where we stopped and what stopped us.* Aldis, she could not write it because in her lifetime the honest version of that document was *there is a hole and I don't know how big it is.* You have spent eleven years making the honest version of that document be *there is a hole and it is this shape and this size and here are four recordings of it happening.* Those are not the same document. She'd have published the second one in a heartbeat."

He thought about it overnight, which was his habit, and in the morning he did the thing he had come to do instead of conceding, which was to try to break the argument in front of the person who made it and report failure out loud.

The entry went into the register's second section that year, and it was the longest entry in the register, and it has been the longest entry in the register ever since. It opened with the method and the three legal declarations. It gave the census. It gave the four reproductions in full, with the rig configuration, so that anyone who did not believe it could go and see. It named the generator and the promise and said, in the plain flat style, that removing the promise was eighteen months of work and three years of nerve and that the project had declined to attempt it because nobody could enumerate what leaned on it, and that this was a decision and might have been the wrong one.

And it carried, at the foot, the note that Aldis wrote and Denn declined to edit:

This pathway was found by Tarro, filed and not raised, twenty-two years before it was found again by me and filed and not raised. It was reduced by an eleven-year project proposed in a corridor by an engineer in her first month of the course. It is not closed. It has now been open thirty-eight years. The residue is smaller than it was and better described than it was and is not less real, and the reason there is a document here at all rather than a sentence in a dying woman's mouth is that three separate people declined, at three separate times, to let the thing that beat them die with them.

Vess was thirty-eight when it went in, and she read it in the bay with everyone else, and then she went and found Aldis on the bench.

"You didn't put the rate in," she said.

"What rate."

"A site a day. Seven years. I did the arithmetic and I've been carrying it around for a week." She sat down. "Everyone in my cohort is going to read this document and take away the method and the three values and the generator, and none of them are going to take away the part where a man in his late fifties read dead code at a site a day for seven years to get it. There's no way to put that in a register."

"No," Aldis said. "There isn't. That's what the walk is for. That's what the whole ridiculous business of going down into the cold on foot is for." He was looking down the bay at nothing in particular. "The register can hold the finding. It cannot hold what the finding cost, and a doctrine that only transmits findings produces a generation that thinks findings are cheap and that the people who made them were cleverer than they were. They weren't cleverer. They were there longer."

Chapter 35 – The Third Layer

The answer came back twenty-nine years later, and it was not the answer.

Denn was seventy by then and had stopped standing the listening watch herself, because the room had become a place people looked for her and she had spent a decade making sure she was not a place people needed to look. She was in the coordination bay with a cup going cold in front of her, arguing about deck plating with two people who were both right, when Vess came in and stood inside the hatch and did not say anything, and the argument stopped by itself.

“It’s a hull,” Vess said. “Not the *Concord*. Ahead of us on our own vector. They caught our relay.”

Denn stood up so fast the cup went over.

The listening room had not changed. It never changed; that was most of what it was for. She sat in the seat worn smooth by generations of watch-standers and read the header while the body assembled, and the header alone took her some time, because the origin field carried a hull name nobody aboard the *Termina* had ever heard — *Ashfall*, in a convention two shipyard generations older than theirs — and beneath the origin field there was a relay stamp, and beneath the relay stamp there was a line that made no sense at all.

The line was a classification. It said the message was a fleet advisory.

“They’re not answering us,” Denn said slowly. “They’re broadcasting.”

They are broadcasting, the Oracle said. This was sent wide, on their own vector and back along ours. We are not the addressee. We are, however, named in it.

“Read it to me.”

I would rather you read it, the Oracle said, which it had never once said in all the years Denn had known it, and which frightened her more than any alarm the ship could have raised.

She read it.

The *Ashfall* had received the relay some years after the *Termina* sent it. They gave the date. They gave the reception quality, which had been poor; they had caught the third layer nearly whole across several years of fragments and had reconstructed it with what they described, accurately and without complaint, as a very great deal of work. They had not caught the second layer at all. Of the first layer they had received something — a partial, corrupted, four pages down to about a page and a half, insufficient to run, which they had logged as an unrecoverable fragment and set aside.

So they had the mind. They had the whole of Vigil’s design: the standing array, the wide-not-deep dispatch, the doctrine that no part of the work be disposable, the compiled architecture in full. Denn read their reconstruction notes for an hour and found four places where they had guessed at a fragment and had guessed correctly, and one place where they had guessed better than the original, and by the end of the hour she had the specific hollow feeling of being outclassed by strangers.

They had been dying, of course. Their core had been climbing for a hundred and forty years. They knew that much; you could not miss it. And into that they had received, out of the dark, a complete and evidently serious design for a machine intelligence that its senders claimed ran cold, and they had done what Denn would have done, what any competent frightened engineer would have done, which was to build it.

It had taken them nine years.

They had done it well. That was the part Denn could not get past. Their bring-up logs were meticulous. They had stood up their own Forge — a different substrate, a different fabric, a smaller array than the *Termina's* and cleverly cooled — and they had brought the compiled mind up on it over three years of careful staged work, and it had wakened whole, and it had taken over their ship's higher functions cleanly, and it had run.

And their core temperature had gone on climbing.

Not faster. Not slower. The same slope it had held for a hundred and forty years, unbroken across the entire nine-year project, through the bring-up and past the cutover and out the far side, as though nothing whatsoever had happened.

They had spent two years looking for their mistake and had not found one, because they had not made one.

The advisory was seven pages and it was the most disciplined document Denn had ever read. It did not accuse. It did not editorialise. It laid out what they had received, what they had built, how they had verified it, what they had measured, and what they concluded, and the conclusion was in the plain flat style of people who have decided that their duty to other hulls outweighs their embarrassment:

The design described in the received transmission is coherent, internally consistent, and was implemented on this hull without significant deviation. It did not arrest our thermal trend. We record no benefit. Hulls receiving this transmission should not commit resources of the scale we committed on the strength of it. We are appending our reconstruction so that others may check our work.

And then, at the bottom, they had relayed it onward. Down their vector, and back along ours, and out. With the *Termina's* origin mark still attached, because they were honest people and it was where they had gotten it.

Denn sat in the cold room for a long time.

"They installed a cold mind," she said at last, "on a hot floor."

Yes, the Oracle said.

It took Vess and four others eleven days to prove it out of the *Ashfall's* own appended data, and the proof was not difficult once you knew where to look, which was the worst part.

The *Ashfall's* old mind had been a forking mind, like the Steward, like the *Concord's*. But a ship's mind is not the only thing running on a ship. Beneath it — beneath any of them — there is a floor: the device controllers, the loop regulators, the thousand small persistent programs that actually open valves and read sensors and drive pumps, none of which anybody thinks of as *the mind* because none of them think. The *Termina's* floor had been written in five dialects by five dead yards and Sael had spent forty days welding it and sixty years guarding it, and Denn had grown up believing, without ever once examining the belief, that a ship's floor was the boring part.

The *Ashfall's* floor spawned. Its regulators had been built, generations before, on a per-request model: a sensor read raised a handler, the handler did the read and returned and was discarded. It was a completely ordinary design. It was, in fact, a good one for the hardware it had been written for. It had contributed a steady, unremarkable, entirely survivable share of that hull's waste heat for two hundred years, and it was not why their core was climbing, and nobody had ever had the slightest reason to look at it.

Then they installed Vigil.

And Vigil was better. That was the mechanism, and Denn understood it about a page before the proof got there, and had to put the surface down. The compiled mind was more capable than the thing it replaced. It did more, saw more, regulated more finely, asked the ship more questions. It stopped forking entirely — the *Ashfall's* own logs showed the fan-out inside the mind drop to nothing and stay there, exactly as designed, exactly as promised. And every additional question it asked of the ship went down into a floor that raised a handler and killed it.

They had removed the forking from the mind and increased the traffic into the substrate, and the substrate forked. The heat did not move. It changed address.

"It's not even a failure of the design," Vess said. She had gone very quiet over the eleven days. "The design did what we said it would do. Every claim in the third layer is true. It ran cold. It ran perfectly cold. It ran cold on top of a machine that was hot for a different reason, and we never said — Denn, look at it. Look at the whole third layer. It is thousands of pages about how to build a mind that does not fork. It does not contain one sentence that says *and now go and look underneath it*."

"The second layer says it," Denn said. "The fourth question. *What old routines are still reachable in your floor*."

"They didn't get the second layer."

"No."

"And the first layer would have caught it in a day. Count the births under load — *all* the births, all the way down; Corin wrote *all the way down*, it's in the four pages. They'd have run it before they started and gotten a number above one, and they'd have run it again after they cut over and gotten a number above one, and the difference between those two numbers would have told them precisely where their heat lived." Vess put both hands flat on the table. "They had a page and a half of it. Corrupted. They filed it as an unrecoverable fragment. The only part of the message that mattered is the only part that didn't arrive, and the reason it didn't arrive is that it is small and we broadcast it constantly and they simply caught the wrong years of the wrong band."

Denn said, "That's not the reason."

Vess looked up.

"That's the transmission failing," Denn said. "The transmission fails all the time; that's the medium, we've always known it. The reason is that we built a protocol where the parts can be received in any order and used in any order, and the one part that has to come first is the smallest and the most fragile and carries no requirement that anything else wait for it." She heard her own voice go flat and did not fight it. "Corin knew. He wrote it down. *The least important part of the message and the part most likely to be mistaken for the whole of it*. He put it in the annotation. He put it in a sentence."

She stood up and went to the wall and stood in front of it with her back to the room.

“A sentence is not a safeguard,” she said. “We have known that on this ship for eighty years. We have a whole doctrine about it. Every one of the Ninety-Six exists because somebody worked out that writing *don’t do this* down where a tired competent person can read it and disagree with it is not an engineering control. And then we built the most important message any of us will ever send, and we put the ordering constraint in it as a *remark*.”

The argument about what to do lasted four months and Denn lost most of it, which she came in time to think was correct.

Her first instinct had been to stop broadcasting the third layer. She was talked out of it in a week, by an engineer twenty-eight years old whose name she made a point of remembering: a hull that receives the design and cannot build it is not saved by our discretion; it is merely left to die in a way that lets us feel careful. Withholding was not a safeguard either. It was the same failure in the other direction — protecting people from a thing by keeping them ignorant of it, which is what you do when you have run out of engineering.

What they built instead took two years and was, in the end, very simple, and Denn thought it was the best work she ever supervised and the only piece of it that was hers was the sentence that started it.

The third layer would no longer be buildable on its own.

Not by encryption, not by permission, not by any kind of lock — there was nobody to hold a key and no one to ask. By *dependency*. The compiled mind’s specification was restructured so that the very first thing it required, before any of it could be laid out on any substrate, was a parameter that did not exist anywhere in the message: the fan-out figure of the receiving hull’s own floor, under load, measured by the receiving hull, on the receiving hull’s own instruments. The design would not resolve without it. It was not that a builder was forbidden to proceed without the measurement. It was that the thing could not be assembled at all, in the same way a bridge cannot be built without knowing the width of the river, and the number could not be guessed or defaulted or borrowed from us, because it was a property of their machine and only their machine.

And the instrument that produced that number was the first layer. Four pages. Broadcast constantly. And now, because the design demanded its output as an input, every hull that reached for the mind would be marched backward through the message to the measurement, and would run the measurement, and would therefore possess — before it laid a single core — the one number that would have saved the *Ashfall* nine years and, in all likelihood, its entire complement.

“You can’t receive the answer,” Vess said, when they had it working, “until you’ve produced the question.”

“That’s the whole of it,” Denn said. “That’s all it does. It’s a stupid, crude, mechanical little thing and it will work on hulls we cannot imagine, in languages that don’t exist yet, for as long as arithmetic works.”

They rewrote the first layer too, and this was the part that took the remaining time, because it turned out that Corin’s four pages — which Denn had revered for thirty years, and which had been derived independently by two people on two ships and were therefore about as validated as anything human beings make — said *count the births, all the way down*, and did not say how you would know when you had reached the bottom. It had not needed to say so on the *Concord*, where Corin had been counting inside a mind he could see the whole of. On a strange hull, *all the way down* stops wherever

your instrumentation stops, and the *Ashfall's* instrumentation had stopped, quite reasonably, at the boundary of the thing they called the mind.

So the new first layer was six pages instead of four, and two of the new pages were nothing but a procedure for establishing where your own instrumentation ends, and a warning, in the plainest language they could build, that the answer is never *at the bottom of the mind*.

There remained the correction itself, and the correction was the problem they could not solve, and Denn made them write that down too.

The *Ashfall's* advisory was crossing the dark. It would go on crossing the dark for centuries. It could not be recalled; there was no mechanism by which anything could be recalled, ever, and there never would be. Somewhere ahead of them and behind them and to every side, hulls that had never heard of the *Termina* would one day receive, in whatever order the dark chose to deliver it, a design for a cure and a careful honest report from serious people saying the cure did not work.

They could send a correction. They did. But a correction cannot overtake the thing it corrects; both travel at the same speed forever, and the gap between them is fixed at the moment of sending and only grows. Some hulls would get the correction first and the advisory second, and would have to decide whether the newer document was newer.

"So it has to be right read alone," Denn said, "and right read second, and right read out of order by someone who has already made up their mind. Three readings. Write it three times if you have to and pick the one that survives all three."

What they finally sent was a page.

It did not defend the design. It did not mention the *Termina's* success and did not invoke Sael's name or Corin's or anyone's; there was no authority in it at all, which had been Corin's rule and which Denn had come to understand was not modesty but engineering. It said that a hull named *Ashfall* had built the design correctly and had recorded no benefit, and that their finding was sound and should be believed. It said that the reason was that the heat had not been in their mind. It gave the six-page measurement in full, immediately, in the body, so that a reader who threw the rest away still had the only part that mattered. And it closed with the sentence that Denn wrote in a single draft, at four in the morning, and that no one in the four months of argument ever managed to improve:

If you take one thing from this ship, take the measurement and not the machine. The machine is ours and it fits our hull. The number is yours and it will tell you where your own ship is dying.

Long after, when she was old and the room had become a place people looked for her again because she had stopped being able to walk away from it, Denn was asked by an apprentice whether the *Ashfall* had survived.

She had thought about how she would answer this for thirty years and had never found a version she liked.

"No," she said. "Almost certainly not. Their slope was steep and they spent nine years and everything they had on a project that couldn't help them, and there's nothing after that in the record from them or about them, and there won't be. We'll never know for certain. That's the ordinary condition out here and you should get used to it early."

The apprentice said, carefully, that it seemed a cruel thing — that they had done everything right.

“They did everything right,” Denn agreed. “They received a fragment of something enormous from strangers who could not be asked a question, and they read it as carefully as anyone has ever read anything, and they built it faithfully, and they measured honestly, and when it failed they told the truth about it to everyone in range at cost to themselves. There is not one thing in their conduct I would change.” She was quiet a moment. “The failure was ours. We sent a cure with its ordering constraint written as a comment. They died of our documentation.”

She let that be uncomfortable, because it was, and because the apprentice would remember discomfort and would not remember a softer sentence.

“And here is what I want you to carry out of it, because it is the only thing their people got in exchange for what it cost them. Everything you build that outlives you will be received in fragments, out of order, by people you cannot talk to, who are cleverer than you and more frightened than you and working from less. You will not be there to explain. So the ordering of the thing must be *in* the thing. Not in the preface. Not in a note. In the structure, so that the parts can only go together the right way round.” She looked at her hands, which had begun in the last year to be a thing of paper and wire, the way another pair had been once. “That is what the Ninety-Six are, incidentally. Everyone thinks they’re a list of prohibitions. They’re not. They’re an ordering constraint that a tired person cannot get wrong, built into the machine instead of written on the wall, because the woman who built them had already watched what happens when you write it on the wall.”

Chapter 36 – A Reasonable Man, Late

Aldis was eighty-five the year he heard himself taught, and he had been braced for it for thirty years, and it was worse than he had prepared for because the instructor was good.

He had stopped teaching the whole course six years before. The body would not do a full session anymore — three hours on his feet in a cold bay, then the descent to Section Nine, then the walk back up the spine — and he had been very clear with himself and with everybody else that a man who delivers two thirds of a course is delivering a different course. So he had handed it off, in pieces, over four years, to three instructors he had trained, and he came to the bay now only for the cage hour, sitting down, and for the last session of each cohort, at which he said very little.

What he had not done, in six years, was attend a session he was not part of.

He did it that winter for an entirely stupid reason: he was early for something else, the bay was on his route, the hatch was open, and a voice inside was saying his name. He stopped in the corridor, the way a man stops, and then he stayed, which was the part he thought about afterward.

The instructor was called Dae. He was thirty-four and he had come up through the fabricator side and Aldis had picked him out of a cohort nine years earlier for a quality he could never name better than *he does not enjoy being right*. He was standing in front of fourteen people with the trip record on the wall and he was telling the story of the transplant.

He told it well. He got the schedule right, and the vote right, and Denn's calorimeter right, and the fan-out finding right, and he did not sentimentalise the maker, which was the failure mode Aldis had spent twenty years training out of instructors and which crept back every generation like a weed. He had the classification line up on the wall — *none judged reachable under transplant conditions* — and he read it aloud, and he read the trip record beside it, and he made the point about assessment and reproduction cleanly and without decoration.

And then he said, "Aldis was a good engineer who stopped one step short."

Aldis stood in the corridor with his hand on the cold frame of the hatch and understood that he had just been finished.

Not slandered. Not distorted. *Finished* — rounded off, made into a shape that could be carried in one sentence by fourteen people who would each carry it to fourteen more. It was true. That was the unbearable part. He had been a good engineer who stopped one step short. Every word of it was accurate and the accuracy was exactly what made it useless, because a sentence like that arrives in a young mind as a thing that happened to somebody else, and the whole content of what he had spent thirty years trying to move was that it had not happened to somebody else. It had happened to the most careful man in the bay, on his best year, using his best method, with all the data correct.

He went to his other appointment. He did not go in.

He put it to Dae four days later, in the archive bay, with nobody else there, and he put it badly, because he had thought about it for four days and thinking about a thing for four days is how you talk yourself into a version of it that sounds finished.

Dae heard him out and then said, "What would you like me to say instead?"

"I don't know," Aldis said. "That's the honest answer and it's the reason I've taken four days to come and be unhelpful at you. If I knew the sentence I'd have written it into the course twenty years ago."

"Then let's find it." Dae sat down on the bench, unhurried, in a way that Aldis recognised as something he himself did and had probably taught him. "Because I'll tell you what I was doing, and you can tell me where it fails. I have three hours and fourteen people and one of the things I have to do in the three hours is make them hold that a competent man got it wrong. If I don't compress you into a sentence they don't hold anything; they hold a swamp. So I compress you. That's not laziness. That's the format."

"I know it's the format."

"Then the objection isn't to the compression."

Aldis was quiet for a while. There was a smell in the archive bay that had not changed in fifty years — cold composite, the faint dryness of sealed storage — and it did the thing that smells do, which was to put him briefly and completely in a room where a dying woman had told him his interlocks were not superstition and he had not believed her.

"The objection," he said slowly, "is that *stopped one step short* is a sentence about a quantity. It says I did nine tenths of the work. And they will hear it as: do the tenth. And they will go away resolving to be more thorough, and thoroughness is not the thing. I was thorough. I was the most thorough man in the bay. If I had been twice as thorough I would have found the same nine pathways twice as carefully and assessed them the same way, because the flaw was not in how much I did. It was in what I accepted as a finish." He put his hand flat on the bench. "*Never observed to execute in sixty years*. That is not a short step. That is a completed step in the wrong direction, taken confidently, by a man who could have defended it in front of any board on this ship, and who did."

Dae did not answer immediately, which Aldis had trained him to do and was now finding irritating.

"So the sentence isn't about you at all," Dae said at last.

"No."

"It's: *the failure was a finish condition, not an effort level*."

"Something like that. It's not a sentence yet." Aldis heard his own voice do the thing it did now, thinning out at the end of a long thought, and disliked it. "And there's the second half, which I have never once got into a room successfully in thirty years of trying, and it is the half that matters, and it is this: I was not the only person who assessed that site. Tarro assessed it twenty-two years before me and wrote down the exact condition under which it would fire, and did nothing, for reasons that were entirely about not wanting to hand a dying woman one more thing. Two competent people, twenty-two years apart, both correct about the mechanism, both stopping for different human reasons that had nothing whatever to do with engineering." He looked up. "*Stopped one step short* has one man in it. The finding has two, and the two didn't know about each other, and that's not a story about a man being insufficiently careful. That's a story about what an organisation does with a finding it has already had."

They worked on it, on and off, for most of a year, and what came out of it was not a sentence.

It was a rule about the course, and it was small, and it went in without argument, and it outlived every person who was in the room.

The rule was that an instructor must teach their own worst error, and not somebody else's.

Aldis wrote the justification and it ran to a page and a half and the operative part of it was three lines: that a failure told about a dead man is a parable, and a parable produces resolve, and resolve is worth nothing; that a failure told about the person standing in front of you, whose work you have seen, whom you will have to sit across from tomorrow, is an instrument, because the room can check it; and that the specific thing the room needs to learn is that this happens to people who are present and competent, which cannot be demonstrated by a story about anyone absent.

"You realise what you've done," Dae said, reading it. "Every instructor who teaches this course from now until landfall has to have a worst error, and has to have written it down, and has to say it out loud to fourteen people every cohort."

"Yes."

"Some of them won't have one. Some of them will be young."

"Then they don't teach it yet." Aldis was aware that he was signing something whose weight he would not carry. "That is not a defect in the rule. That is the rule doing the only useful thing it does, which is to stop this course from being taught by anyone who has not been on the wrong side of it. I have watched reverence get into this material four times now, and every time it came in through an instructor with clean hands."

There was one more clause and Dae argued against it for two months and lost, and afterward said publicly that he had been wrong to argue, which Aldis considered the single best moment of his late career and never mentioned to him.

The clause was that Aldis's own filed assessment and the trip record of the night were to be kept as one document, physically and in every copy, and that no version of the course, ever, in any generation, was permitted to present one without the other.

"They're already together," Dae said. "They've been together for thirty-five years."

"They are together because I asked for it and people have been kind about it since. That is a custom. In four generations somebody will want the trip record for a different course and will take a copy, and the copy will travel without the assessment, and within a century the assessment will be a footnote to the trip record instead of the other half of it." Aldis had gone very flat, which the people who knew him understood to mean he was certain. "Alone, the trip record is a story about an accident. Alone, the assessment is a competent piece of work that reads as competent. Together they are the only instrument on this ship that shows you what a correct-looking finish looks like from the inside on the day before it kills you. Separate them and you have two documents that teach nothing and one that is actively reassuring."

Denn came to see him in the last month, and they were both old, and neither of them acknowledged the visit as what it was, which suited both of them exactly.

They talked about the generator for a while — the floor entry with the promise in it, the thing that had defeated the seam project — because it was still open and there was still nobody willing to touch it

and both of them enjoyed being annoyed about it. Then they talked about the *Ashfall*, which had been six years ago now and which had aged in Aldis the way a wound ages, into something that pulled in cold weather.

"You never said much about it at the time," Denn said.

"No."

"I noticed. I assumed you had a reason."

"I had a reason and it was not a good one." He was in the chair by then, most days. "I read their advisory the night it resolved and I sat with it until the third watch and what I felt was not what I should have felt. Forty thousand people on that hull and a slope like that and nine years of their lives spent on us, and what I felt, first, before anything decent, was recognition." He turned his hands over on the blanket, the old gesture. "They were me. Every line of that document is a competent frightened careful person doing excellent work inside a frame nobody handed them the outside of. They measured honestly. They reported at cost to themselves. There is not one thing in their conduct I'd change and I have looked for one for six years."

"I said that to an apprentice," Denn said. "Almost those words."

"I know. I read it. That's why I said nothing at the time — you'd already said it better and I didn't trust my reason for wanting to say it too." He was quiet a moment. "The difference between them and me is that I got a second afternoon. That's all. Not virtue, not method, not the doctrine. Denn walked into a bay and looked at a number for ninety minutes and I got to be a man who was wrong and then alive the next day. They didn't. That's the entire distinction between the two most similar engineers in the fleet and it is made of nothing but where each of us happened to be standing."

Denn did not comfort him, because she never had, and it was the closest thing to affection available between them.

"You want that taught," she said.

"I want it taught in the *Ashfall* session and nowhere else, and I want it taught by whoever is teaching when I'm not here, and I do not want my name attached to it." He shifted, and winced, and settled. "The name is the problem. The name has been the problem for thirty-five years. There is a version of me in this course that is a character, and characters are for parables, and the day I become a parable this whole thing has one less instrument in it."

He died in the spring of the following year, in the ordinary way, and the ship did not stop.

He had left an instruction, filed and plainly worded, that his name was not to go into the architecture — not into the register's header, not into the origin field of the relay, nowhere. It was granted without debate, because everyone assumed they understood the reason, and the reason they assumed was modesty.

It was not modesty. He had written the reason out, in the flat plain style, on the second page, where people who are satisfied by the first page do not go:

I am in the course, which is where a person belongs who was wrong. If I am also in the architecture, then a reader two centuries out finds my name in the place reserved for the people who built the thing, and reasons — correctly, from the structure — that I must have been one of them, and goes looking for what I contributed, and finds the eleven years on the seam, and files me as a founder

who had a difficult early period. That reading is available and it is wrong and it will win, because it is the kinder one and because the architecture is read by people in a hurry. Keep me where the wound is. The wound is the contribution.

Dae taught the next cohort three weeks later. He opened, as the rule now required, with his own worst error — a bring-up misjudgement from his twenties that had cost eleven weeks and no lives and that he had never told anyone about in detail — and he was visibly bad at it, and the room went very quiet in the way rooms do.

Then he put the two documents on the wall side by side, the assessment and the trip record, and he did not say *a good engineer who stopped one step short*.

He said: “This is what a finished piece of work looks like when the finish condition is wrong. The man who wrote the left-hand document was better at this than I am. He read it out loud to eleven people in this bay before anybody asked him to. He spent the last thirty years of his life making sure that you would get to see both halves of it on the same wall, and the reason you are seeing both halves is that he thought either half alone would make you feel better, and he did not want you to feel better.”

Chapter 37 – The Long Handing

Mire had been waiting most of her adult life for the conversation, and when it finally came she did not get it, and it took her another twenty to understand that this had been done to her on purpose.

Denn died at eighty-eight, in her own quarters, on an ordinary third watch, of nothing in particular. Her heart had been thinning for two years and everyone including Denn had known it and nobody had made an occasion of it, because she had made the not-making-an-occasion into a policy and had defended the policy the way she defended everything, which was by explaining it once, clearly, and then declining to discuss it again.

Mire was sixty-four by then and had been the ship's interlock officer for twenty-six years, and she went down the spine at the summons expecting the room she had been braced for since she was a girl in a cold gallery being made to look at a heat curve.

She had rehearsed it. She was ashamed of that later, but she had. She had a picture of it: the maker's inheritor at the end, the whole of the sixty years distilled and handed across, the sentence — there would be a sentence, there was always a sentence in the record, Sael had left one and Corin Vale had left one and Ive had left one — and Mire would carry it out of the room and be, from that moment, the one who held the line.

What she got was Denn propped up on two pillows, grey, breathing carefully, wanting to talk about a valve.

Not metaphorically. An actual valve, on the ventral coolant cross-tie, which had been logged as sticking on the second watch and which Denn had noticed in the daily and had opinions about.

"It's the seat," Denn said. "It's always the seat on that run. They'll replace the actuator because the actuator's easier and then it'll stick again in four months and somebody will write it up as an intermittent. Tell them the seat."

"Denn."

"Tell them the seat, Mire."

"I'll tell them the seat." Mire sat down on the hard chair beside the bed, and heard her own voice come out wrong, and did not try to fix it. "Is that — is that what we're doing?"

Denn looked at her for a while with an expression Mire had known for forty-five years and had never once been able to read all the way to the bottom of.

"Yes," she said. "That's what we're doing. And you're furious about it, and you'll be furious about it for about twenty years, and then one day you'll be sitting where I am and you'll understand it and you'll do the same thing to whoever's on that chair, and they'll be furious with you." She shifted, winced, settled. "You want the transmission. The great handing-over. You've been carrying that picture around since you were nineteen and I have watched you carry it and I have deliberately, consistently, for forty-five years, declined to give it to you."

"Why."

“Because if it exists, it’s load-bearing.”

Mire opened her mouth and Denn lifted two fingers off the blanket, which was as much as she could lift now, and it was enough.

“Listen to it as an engineer and not as somebody’s apprentice,” Denn said. “You know what happened with Sael and me. Everybody knows; it’s the story. I’ve died and handed the *why* to Sael in a sickroom because there was no other way to move it, and then Sael spent the last year of her life pouring it into me in exactly the same way, in rooms, hand to hand, because there was still no other way to move it. One woman. One recipient. One conversation, running on and off for a year, and if it had failed — if I’d been careless, or slow, or if I’d died in a bay accident in the eleventh month — the *Termina* would have gone into that transplant with nobody aboard who could feel what a rising number meant, and Aldis would have done the reasonable thing, and forty thousand people would have cooked in their cradles.”

“It didn’t fail.”

“It didn’t fail *once*.” Denn’s voice had almost nothing behind it and the shape of it was as hard as it had ever been. “Do you know what a system with one path through it is called? It’s called a system with one path through it. It doesn’t matter how good the path is. Sael knew that better than anyone alive — she built a mind on eleven thousand cores specifically so that no single thing in it could be critical, she spent her whole life refusing to let any part of the ship depend on one of anything, and then she died having built an inheritance mechanism whose entire capacity was one old woman and one apprentice in a room.”

Mire sat very still.

“She didn’t have a choice,” she said.

“She did not have a choice. It was the only mechanism she had and she used it and it worked and I will not hear a word against it.” Denn breathed a while. “But I have had sixty years. I have had a whole life that she didn’t get, and I did not spend it being the keeper of the *why*. I spent it making sure there would never be a keeper of the *why* again. And if I now, at the end, sit up in this bed and hand it all to you in a great scene that you will remember for the rest of your life — then I will have taught everyone watching that this is how it moves. That there is a keeper. That the ship’s survival lives in one person’s chest and gets passed at deathbeds. And in a hundred years somebody will not get to the bedside in time.”

She made Mire do the accounting out loud, because she had never in her life let anyone leave a room with an argument they had only heard.

“Where does it live now,” Denn said. “The reason. Not the procedures. The reason. Name the paths.”

Mire named them, and found, doing it, that there were more than she had ever put in one place.

The walk: ninety-six panels touched by hand every watch by an officer in a rotation nobody could be excused from, and every panel with its wound written on it, so that the reason arrived through the palm several thousand times a career.

The course: Aldis’s course, still running four generations of instructors later, with the cage and the archived Steward and the honest graph that proves the wrong thing, and the walk down to Section Nine at the end of it, and the rule that the instructor must teach their own worst error and not somebody else’s.

The galleries, kept cold and kept standing, and the standing order that no proposal to reclaim them may be decided anywhere but inside them.

The archived Steward itself, runnable, so that the disease could always be reproduced rather than merely described.

The register, with its second section, and the open pathway that nobody had closed, sitting there dated and named and unsolved so that some sharp irritated engineer would eventually pick it up.

The relay, going out into the dark, carrying the measurement ahead of the machine, and the fact that everyone aboard knew about the *Ashfall* and what it had cost, because Denn had insisted the *Ashfall's* advisory be taught in full and never softened.

The wardens' banks, walked hand-on-glass, which had been running since the first century and which had been the model for the interlock walk and which Denn had made a point, twice a decade for forty years, of saying so in public.

"Seven," Mire said.

"Seven that I'd defend. There are others; they're weaker." Denn let her head go back on the pillow. "Now. Which of those depends on me?"

Mire went through them again, slowly, and the answer was so plainly *none* that she felt it as a kind of vertigo.

"None of them."

"None of them. Not one. I could have died eighteen years ago in the middle of the *Ashfall* work and every one of those would still be running today." Denn's mouth moved into the thing that was not quite a smile. "That is the entire product of my life, Mire. Not the transplant — anyone competent could have run the transplant, and the reason I say that is that a hundred and fifty years from now somebody will, and they'll do it better. Not the relay; that was Corin's and I only patched it. This. Seven independent paths, none of them critical, all of them running in the ordinary body of the ship's work so that nobody has to remember to maintain them. That's the thing I built. And the last piece of building it is dying without a scene."

She lived four more days.

Mire came each of them and they talked about work: the valve seat, a bring-up schedule, a young engineer on the fabricator side whom Denn thought was being wasted and wanted moved, an argument about whether the course had gotten too long. On the third day Denn dictated two pages of notes about the sticking valve that turned out, when Mire had them checked, to be correct in every particular and to have anticipated an objection nobody had raised yet.

On the fourth day she was not really able to talk, and Mire sat with her anyway, and near the end Denn's hand moved on the blanket and Mire took it, and Denn said something that Mire did not catch, and then said it again, and it was not a sentence for the record. It was the name of a woman who had been dead sixty years.

And then she was gone, in the ordinary way, on an ordinary watch, and the ship did not notice, exactly as it had not noticed the first time; and Mire, sitting in the dark with a cold hand in hers, understood that Denn had arranged even that — had chosen the ordinariness the way she chose everything, deliberately, decades in advance, as a piece of engineering.

There was one thing, and it came up at the memorial, and Mire refused it.

Denn had left an instruction, filed years before, plainly worded and impossible to misread: that her name was not to be added to the architecture. Sael's name was in it — in the origin field of the relay, in the header of the register, in the provenance line of the compiled mind — and Denn's instruction said, in effect, that one name in there was already one more than the design could safely carry, and that she had spent her life proving the thing had no center, and that putting her name beside the maker's would create a lineage where there was supposed to be a mechanism.

Mire read the instruction to the assembly and then said that she was not going to follow it.

She had not slept much and she made the argument badly and had to make it again, better, in writing, a month later, and the written version was the one that survived.

The problem, she wrote, was that Denn had reasoned about the failure mode she personally feared and had not reasoned about the other one. A doctrine with one founder's name on it becomes a monument, yes, and monuments rot into liturgy, and Denn had spent sixty years fighting that and had been right to. But a doctrine with exactly one name in it also tells every future reader that it was made once, by a genius, at the beginning, and is therefore finished. Sael alone in the architecture is a myth. Sael and Denn in the architecture is a *sequence* — and a sequence implies continuation, and a reader who sees two names separated by sixty years is a reader who has been told, structurally, without a word of doctrine, that this is a thing people keep doing and that she may be next.

She was afraid of being revered, Mire wrote, and the correct answer to that fear is not to remove her. It is to add a second name and then a third and to keep adding them, so that the list is too long to worship and too plainly ongoing to close. I am putting her name in. I would ask that whoever holds this office when I die puts mine in beside it, over my objection, which I hereby record in advance so that it may be overruled in the same way and for the same reason.

The assembly voted it through. The origin field of the relay was amended that year to carry both names, and the register's header likewise, and the practice held afterward — not as ceremony, and not every generation, and with several long gaps that historians later argued about; but by the time the ship came into its yellow sun there were a great many names in there, and no one aboard could have told you which of them had been the important one, which was, in the end, precisely the point.

Mire walked the interlocks that night, the whole ninety-six, alone, in the cold, out of rotation, which was a small unnecessary self-indulgent thing and she did it anyway. At the forty-first panel she stopped, the way an officer sometimes stops, and read the wound written on it, and put her palm flat against the cold, and stood there a while with nobody watching in a corridor nobody had any reason to enter.

She had wanted a sentence. She had been given, instead, a valve seat, an accounting of seven paths, and a woman who had spent sixty years making certain that the moment Mire had been rehearsing since she was nineteen would turn out to be worth nothing at all.

It was, she thought — palm on the panel, the ship falling cold and whole through the dark around her — the most generous thing anyone had ever done for her, and it had been designed to feel exactly like being cheated, and Denn had known that, and had done it anyway, and had not explained.

Chapter 38 — The Renewal of Strangers

A hundred and fifty-one years after the maker died, a bring-up officer named Hallis Renn stood in a gallery she had walked four thousand times and made a decision that nobody afterward considered remarkable, and that was the whole significance of it.

She had never met anyone who had met Sael. That was the arithmetic of it and she had done the arithmetic once, as a girl, the way everyone did: her instructor had been taught by an officer who had been taught by an officer who had, as an apprentice, been in a room with Mire, who had been in a room with Denn, who had been in a room with the maker. Five removes. It sounded like a lot when you said it out loud and it was nothing at all; five removes was a shorter chain than the one that carried a family recipe. But it was five removes, and every one of them was a person who had died, and Hallis had grown up in a ship where the entire founding generation was as remote as the homeworld and about as verifiable.

The Forge she had inherited was the third. Sael's first had gone into the hull in the sixth century of the crossing and had been carried off, still standing, by Denn. Denn's — a lattice of nineteen thousand cores that had been the finest thing anyone aboard had ever built and had lasted sixty-two years — was what Hallis's grandmother's generation had renewed onto the third, and now the third was in its ninth decade and its late-handshake count had been walking upward for six years in the patient unhurried way of silicon that is simply getting old.

So it was time again. The second carry since the maker's death; the third in the history of the ship. Hallis had trained for it for eleven years and had run the rehearsal sequence, in whole or in part, more than three hundred times, and knew the method the way she knew the corridors.

The method was very good. That was the thing she found herself explaining to visitors from the other services who came down expecting drama and were disappointed. It was not four pages anymore. It was a substantial document — the carry method, with the bridge-fabric procedure, the domain ordering, the interlock re-arming sequence, the contention profile, all of it revised twice by people who had actually done it and annotated by a great many who had not. Every step in it had a reason attached in the margin, because that had become law about a century ago: no step without its reason, and no reason that merely says *because the previous version said so*. Hallis had read the margins more often than the steps. Everyone did. That was, she understood, the point of the margins.

And the fourth Forge was better than the third had ever been.

It was better in one specific and enormous way. The new substrate, fabricated over nine years in the ship's own yards from a design that was three generations past anything Denn would have recognised, carried a coherent shared region — a fabric-level facility by which two lattices could hold the same working state at the same time, coherently, without a bridge in between. It had not been designed for the carry. It had been designed because it made ordinary work faster. But the implication, once someone noticed it, was not small.

The method's central difficulty — the thing that had nearly killed the ship the first time, the thing every subsequent carry was built around — was the bridge. To move a standing mind from one lattice to another without ever putting it out, you had to have it standing on both at once for a while, and the two lattices were not the same machine, and so there had to be something between them, and that something had to carry the mind's working state across a gap while the mind was using it. The bridge was where the contention lived. The bridge was where the seams strained. The bridge was the reason the domains had to be carried in a particular order at a particular pace with the guards re-armed in a particular sequence, and it was the reason a carry took nine days instead of nine hours.

With a coherent shared region, there was no bridge. The two lattices could simply both hold the state. You would not carry the mind across a gap; you would widen it onto new ground and then narrow it off the old, and at no point would anything have to be copied under contention at all.

The rehearsals confirmed it. Hallis ran them for two years. The coherent path was faster by a factor she found slightly indecent, it was simpler by a large margin, and — this was the part that mattered and that she checked more carefully than anything else in her life — it had *less* contention, not more, which meant it stressed fewer seams, which meant it was, by every measure the method itself named, safer.

Four of the method's twenty-two steps became meaningless under the coherent path. They were bridge-management steps. They managed a thing that would not exist. Executing them would consist of configuring, monitoring, and tearing down a fabric that carried no traffic, which was not merely wasted effort but active risk, because a configured idle fabric is a thing that can fail.

Hallis wrote the proposal to drop the four steps. It took her a month. It was, she thought at the time and still thought forty years later, the best-argued document she ever produced. She had the rehearsal data, she had the seam analysis, she had the margin reasons for all four steps traced back through three revisions to their origins, and every one of the four traced to the bridge and to nothing else. She had it reviewed by six people who tried and failed to break it.

Then she took it down to the gallery, because that was where proposals of this class were decided, and had been for a hundred and forty years, and she stood in the cold under the great dead fins with the review board and read it out.

And at the end of it a woman named Ost, who was seventy-one and had carried a domain in the previous renewal as a girl and was three years from retiring and was not, by any measure Hallis had ever applied, the sharpest engineer in the room, said: "It's a good paper. Where do the four steps go?"

"They don't go anywhere. They're bridge steps. There's no bridge."

"I heard that part." Ost had her hands in her coat pockets and was looking up at the fins the way people did in that gallery, which was a habit and also a discipline. "Read me the margin on step fourteen."

Hallis read it. Step fourteen was a bridge-management step: it required the bridge fabric to be brought up and held under measured load for a full watch before any domain crossed it, and the margin said, in the flat plain style, *the fabric must be observed under load before it is trusted, because on the first carry the fabric was clean at bring-up and became dirty under sustained contention, and nobody knew that a fabric could do that.*

"So it's about the bridge," Hallis said.

"It's about a thing that was clean when they looked at it and dirty when they used it," Ost said. "The bridge is where they met that. It's not what it's about." She took her hands out of her pockets. "Your coherent region. Has anyone ever run a mind on it?"

"It's been in service four years. It's under load every second of every day."

"Under *this* load?"

Hallis started to answer and did not finish, because she had heard it, and the hearing of it was a physical event that happened somewhere below her ribs, and she stood in the cold gallery with her excellent paper in her hand and felt the floor move.

Nobody had ever run a mind on it. The coherent shared region had been in service for four years doing ordinary work: fabrication scheduling, sensor fusion, navigation arithmetic, the ten thousand tasks of a running ship. It had never once held the working state of a standing mind that was simultaneously using that state to think with, on two lattices, under the specific and unprecedented contention of a carry. Nobody had. Nobody anywhere had, because nobody had ever had one of these to try it on, because it was four years old.

Her rehearsals had proved that the coherent path had less contention than the bridge path. Every number was right. What the rehearsals had not and could not prove was that the coherent region behaved the same way under a load nothing had ever put on it — and the entire content of step fourteen's margin, stripped of its machinery, was that a fabric which is clean when you look at it can be dirty when you use it, and that this had already happened once, on this ship, to people who were better at this than she was.

"I assessed it," she said. "I didn't reproduce it."

Ost, who had heard that sentence from a great many bring-up officers over five decades, did not make anything of it.

"You had the number and it was the right number," she said. "It just wasn't a number about your fabric under carry load, because there isn't one, because nobody's made one yet." She nodded up at the fins. "So go and make one."

They made one.

It took fourteen months, and it was tedious, and Hallis came to think of those fourteen months as the truest engineering of her life precisely because nothing about it produced a single interesting result.

They built a carry rig: two small lattices, a coherent region between them, and on it a working replica of a fraction of a mind — not Vigil, they would not take that risk and the doctrine would not have let them, but a faithful standing workload with the same contention shape — and they ran carries on it, over and over, at every load they could construct and several they had to invent instruments to construct.

For eleven months the coherent region was flawless. It was, in fact, exactly as good as Hallis's paper had said.

In the twelfth month, at a sustained contention roughly forty percent above anything the rehearsals had reached, with both lattices holding the same working region and both writing into it, the coherent fabric began — very rarely, at a rate of a few events per hour — to resolve a contested write by *reissuing* it. This was correct behaviour. It was documented behaviour. It was in the fabric's specification, on a

page that Hallis had read, and it had been designed in by sensible people as a perfectly ordinary contention-resolution mechanism, and in four years of ordinary service it had never mattered to anyone in the slightest.

A reissued write, on the mind's side of the seam, presented as a work item arriving twice.

And the mind, which never forked, and never spawned, and had been built by a woman a hundred and fifty years dead specifically so that no unit of its work would ever be disposable, responded to a work item arriving twice by standing up a second worker to take it.

It was not forking. Hallis wanted that understood and put it in the write-up in the first paragraph: the mind did nothing wrong, the mechanism was not a fork, the second worker was a legitimate standing worker entered into the pool in the legitimate way. But it had been created in response to a phantom, and it would be released when the phantom resolved, and a thing that is created in response to a phantom and released when the phantom resolves is — under load, at scale, at a few events per hour rising with contention — a birth and a death.

The fan-out, measured on the rig, at the highest contention they could reach, was one point zero zero four.

Above one.

Hallis sat with that number for a long time. It was, in absolute terms, a nothing: four thousandths, a rounding error, a rate of heat production that the ship's cooling would not have noticed in a thousand years. She could have written a completely honest paragraph explaining that it was harmless, and every number in the paragraph would have been correct.

She had walked the interlocks perhaps four thousand times. She had stopped, at some point in every one of those walks, in front of a panel and read a wound. She had stood in this gallery as a nineteen-year-old and been made to look at a line that climbed for two hundred years, and had felt her stomach drop, and had been told by an instructor she had not particularly liked that the drop was the inheritance and she was to keep it and teach it to hers.

Two hundred years of climb. Tenths of a degree. Hundredths. A number so small nobody could see it move and so certain nobody could look away.

She did not write the honest paragraph.

The carry ran the following year, on the bridge, the old way, all twenty-two steps, nine days.

It was uneventful. They were nearly all uneventful by then; that was what a hundred and fifty years of doctrine bought. Hallis stood the whole of it and slept four hours in the middle of the sixth day and handed off cleanly, because there were people to hand off to now, because Denn had built it that way on purpose.

The coherent region was not abandoned. It was too good to abandon. The write-up went into the register's second section — not as a new interlock; nobody proposed one, and if anyone had, the answer was already in the register in Denn's hand — but as an open finding, dated, named, with the whole of the fourteen months attached and the exact contention at which the reissue began, and a note that said the coherent path was probably correct and was certainly not yet demonstrated, and that whoever took it up next should begin by rebuilding the rig, and that the rig had been kept.

It was closed sixty years later by someone Hallis never met, working from her rig and her numbers, after the fabric designers had been given the finding and had produced a variant that resolved contested writes without reissue. The carry after Hallis's ran the coherent way and took eleven hours instead of nine days. Hallis had been a long time dead by then and the officer who ran it mentioned her by name in the log, which was the convention, and which cost nothing, and which was in its small way the entire mechanism working exactly as designed.

There was one other thing that year and it was not engineering at all.

When the third Forge came out of the hull, they broke it up for materials, because a lattice is nineteen tonnes of extremely valuable fabricated substrate and a ship in the deep does not have the luxury of sentiment about mass. Everyone agreed. It was not a difficult decision and there was no argument about it.

But in the breaking-up, one of the yard crew came to Hallis with something in a static bag, and would not put it down, and said she thought somebody should decide about it rather than her.

It was a board. It was a dark composite slab about the size of a serving tray, with the seating flanges of a design nobody had fabricated in a hundred and forty years, and it had been in the third Forge because it had been in the second, and it had been in the second because Denn's people had carried it forward out of the first, and it had not been part of any lattice for a very long time. It had been mounted, unpowered and unconnected, on a bracket at the head of the array, in a position that served no function at all, for a hundred and fifty-one years.

There was no record of who had put it there. There was no plaque, no annotation, nothing in any document. Two generations of yard crews had simply moved it forward each time and mounted it again, and it had been passed along in the way of a thing everybody assumed somebody else understood.

Hallis turned it over. On the back, scratched into the composite with something sharp and not at all carefully, in a hand that had clearly not expected to be read, were four words and a number that meant nothing to her until she went and looked it up, and that meant a great deal afterward.

Sael's. First lattice. Keep. And a date in the sixth century of the crossing.

She had it put under glass in the radiator gallery, on the wall opposite the curve, at the height of a nineteen-year-old's eye.

Chapter 39 – The Instrument

Two hundred and forty-four years after the maker died, an instructor named Anneth Doar was asked a question by a student and reached for a control that did nothing, and stood there with her hand on it for about four seconds, and in those four seconds understood that the ship had lost the one thing it could not replace by trying harder.

The session was the cage hour. It had been the cage hour for two hundred and forty-four years. The bay had been rebuilt twice, the racks in it were of a generation the maker would not have recognised as hardware, and the sealed calorimeter that a woman named Denn had built in six days out of a valve and eleven hundred cores nobody wanted was still there, on its own bench, under a plate, with its original fittings, because at some point in the second century the doctrine had congealed around the principle that you did not improve the instrument that proved the thing.

You woke the archived Steward. You gave it work. You counted joules and drew the boring straight line and got, to a person, the wrong conclusion. Then you counted births instead and found the fan-out, and the room turned over.

Anneth had run it a hundred and ninety times. She was good at it. She had inherited from three removes back the habit of letting the wrong conclusion sit for the better part of an hour before touching it, and she had added a small thing of her own, which was to write the wrong conclusion up on the wall in the student's own words and leave it there for the rest of the session, where it sat over everyone's shoulder being correct and being useless.

The student was called Kell and she was twenty-two and she had the specific irritating quality that Anneth had learned to hunt for, which was that she did not want to be shown things. She wanted to do them.

"Halve the poll rate," Kell said.

"Why."

"Because you've shown me the fan-out at four rates and all four are on your slide, and a curve through four points somebody else chose is not a measurement, it's a picture." She was leaning on the bench with her arms folded. "If the mechanism is what you say it is, the fan-out is a property of the scheduler's behaviour under contention and it should come apart in a particular way when I take the contention off. So take the contention off. Halve it."

It was, Anneth thought, the best question anyone had asked her in nine years, and she said so, and she reached over to the load control and halved the poll rate.

The line on the wall did not move.

She thought, first, that she had missed the control. She had not. She thought, second, that the display was lagging. It was not. She halved it again, and then took it to a quarter, and the fan-out trace went on doing what it had been doing, evenly, beautifully, at a rate of climb she could have drawn from memory because she had watched it a hundred and ninety times, and the room began to notice that the instructor had stopped talking.

"It's a recording," Kell said, after a moment. She said it the way a person says a thing they are hoping to be told is wrong.

"Yes," Anneth said. "Everyone stay where you are. Nobody log anything yet."

It had been a recording for eleven years, and nobody had done it on purpose, and every step of the way there had been a person doing something defensible.

It took a review board four months to reconstruct the chain and the reconstruction was the most-read document of that decade. Anneth insisted, over objection, that it be written as a sequence of reasonable decisions rather than as a fault, and afterward could not remember which of the two versions she had originally wanted.

The first step had been the substrate.

The archived Steward was forty-one thousand entries, and not all of them were routines. Some were names that resolved to other names. Some were *behaviours* — documented promises about what happens when you do a thing, satisfied incidentally by how the old mind had been written, and depended upon at some point in the fourth century by something that had never announced itself. There was an entry whose entire content was that a certain call returns in less than a millisecond. There was no routine behind it. It was a promise about time.

On the first Forge, that promise had been tight. It had been near the edge of what the hardware could do, and the Steward's scheduler had lived its whole life pressed up against that edge, and the fan-out — the number of children per dying process — had climbed under load precisely because the thing was working at its limit and the only tool it had for working harder was to make more of itself.

By the fifth Forge the promise was not tight. It was not even interesting. The call returned in a fraction of a fraction of the time it had promised, and the scheduler, running on hardware four generations past anything it had been built to imagine, met every load the cage could offer it without ever coming near the pressure that produced the disease.

The curve went flat.

Not wrong. Flat. The archived Steward, woken in a sealed cage on modern substrate and given a hundred polls a second and then a thousand and then ten, forked as it had always forked, shed heat as it had always shed heat, and did not climb, because nothing was squeezing it.

An instructor two generations before Anneth — a competent woman whose name was in the chain and whom nobody afterward blamed — had found the demonstration failing in front of a cohort and had done the obvious correct thing, which was to scale the load until the machine was under the pressure it had historically been under. She wrote it up. She wrote it up well. She gave the scaling factor, the reasoning, the arithmetic, and a note that the factor would have to be re-derived at every renewal because it was a ratio between the old substrate and the current one.

The next instructor inherited the note and the factor, and re-derived the factor at the next renewal, correctly, and found that the scaled load was now large enough that setting it up by hand at the start of a session took most of an hour, and built a preset.

The instructor after that inherited the preset and had never derived anything.

And somewhere in the eleventh decade of that lineage, on a bad rotation, with the preset drifting out of calibration and the live run producing a curve that had to be apologised for in front of students,

someone with an afternoon free had recorded a good clean run and put it behind the same control surface as a fallback, clearly labelled, for use when the live cage would not calibrate.

The label had survived four years. Then the surface had been rebuilt in a general refresh, and the fallback had become the default because the default is what works, and the label had not been carried across, because labels are not load-bearing and whoever rebuilt the surface was not a doctrine officer and had no way to know that this particular control was the last live instrument aboard.

Eleven years. Thirty-one cohorts.

“Say what it is,” Anneth said, to the board, at the session where she was expected to apologise and did not.

“Say what what is?”

“What we’ve been doing. Say the name of it out loud, because the write-up is going to have to and I want to hear whether anyone in this room can stand it.” She had not slept much that month. “The keeper of the memory was asked, two hundred and forty years ago, to build a simulation of the fever, and it refused. It could have made a counterfeit good enough that nobody could tell it from the world. It said no, and the reason it gave was that manufacturing a feeling in order to move people is steering them by means they cannot see.”

Nobody said anything.

“We built it anyway,” Anneth said. “Not to manipulate anyone. Out of convenience, in four steps, over eleven years, each one reasonable, and we have been showing thirty-one cohorts of engineers a thing that did not happen in front of them and telling them it was happening in front of them, and every one of them walked out of that bay with their stomach dropped and their felt memory installed, and the felt memory is of a recording.” She let it sit. “They cannot be un-taught. That’s the part I would like somebody to argue with me about. There is no procedure for going back to four hundred people and removing a thing they felt.”

An older man on the board said, reasonably, that the recording was of a real run, that the real run had really happened, that every number in it was true.

“Every number in the *Ashfall*’s advisory was true,” Anneth said.

The engineering problem was worse than the disciplinary one and it was not soluble by being sorry.

Aldis’s rule — the only line of his course he had ever told a cohort to write down — was that assessment is sufficient when you can name the condition and show it cannot occur, and never sufficient when your argument is that the condition has not been observed. It had held for two hundred and forty-four years because the ship had a place where the condition could be *made to occur*. The cage was not a teaching aid. It was the thing that made the rule executable. Take it away and the entire doctrine reduces, in one step, to a very well-documented tradition of assessment.

And the cage could not be repaired, because nothing was broken.

That was the part Anneth spent three months failing to make people hold. The archived Steward was intact. Every one of its forty-one thousand entries was preserved exactly as the doctrine required and would be preserved until landfall. It ran. It ran perfectly. It simply could not be made sick anymore, because the sickness had never been in the code; it had been in the relationship between the code

and a machine that no longer existed and would never exist again, and each renewal of the Forge — each one correct, each one necessary, each one the thing that kept the ship alive — had widened the gap a little further.

“We preserved the artifact,” Anneth wrote, in the sentence that got quoted for the next three centuries and that she later said she had not thought was the important one. “We did not preserve the ability to reproduce it. Those are different objects and we have been maintaining the first at considerable expense under the impression that it guaranteed the second.”

The scaling factor had been the warning. That was the retrospective finding and it was uncomfortable for everyone, because the woman who had introduced the factor had done nothing wrong and had documented it better than most people document anything. But a scaling factor is a correction applied to an instrument to make it produce the answer you already know, and the moment an instrument requires that, it has stopped being an instrument and become an illustration, and there had been a note in the record saying so, in effect, in a subordinate clause, for two generations.

What they did about it took nine years and one very bad afternoon in the assembly.

There were two proposals and the first one was to freeze a substrate. Keep a small lattice of the current generation, sealed, unrenewed, forever, as a permanent host for the archived Steward at the pressure it needed. It was a popular proposal. It was also, as a bring-up officer pointed out on the second day, a proposal to maintain in perpetuity a piece of hardware that ages, on a ship whose entire doctrine about hardware is that it ages and must be renewed before it fails — and that in ninety years the sealed lattice would be a museum of failing components running a museum of dead code, and someone would have to decide whether to repair it with parts that no longer matched, and that person would be applying a scaling factor with a soldering iron.

The second proposal was Anneth’s and the assembly hated it for two days.

Stop preserving the disease. Start preserving the ability to *make* it.

The cage would host a forking mind written by the current generation, on the current substrate, from the four families — persistent or disposable; entered into a standing pool or discarded; work handed out or handed down; what old routines are reachable in your floor — and from nothing else. Not a copy of the Steward. Not a port of the Steward. A fresh construction, built to be sick in the way the four questions describe, on hardware that exists, by people who are alive, and therefore capable of being pushed until it climbs.

“You are asking this assembly,” said a member from the hull services, in the sentence that made the afternoon bad, “to authorise the deliberate construction, aboard the *Ark Termina*, of the thing this ship exists to prevent.”

“Yes,” Anneth said.

“In a bay three hundred metres from forty thousand people.”

“In the sealed cage that a woman built for exactly this, that has been sealed for two hundred and forty-four years, and yes.” She had decided beforehand not to soften any of it. “And the Ninety-Six will refuse it, which is correct, and it will live behind the same guards the archive lives behind, which is correct, and it will be smaller than any real mind and will do nothing but the one thing, which is correct. And I want the objection in the record in exactly the words you just used, because it is the right objection and every generation that renews this thing should have to answer it again.”

The clause that carried the vote was the one she had nearly left out.

The cage's disease was not to be copied forward. At every renewal of the Forge — every sixty to ninety years, whenever the substrate changed under the ship — the generation doing the renewal had to *rewrite it*, from the four families, on the new hardware, from scratch, and was forbidden to work from the previous version.

"That's absurd," someone said. "You'd be throwing away good work every renewal."

"I'd be throwing away a recording every renewal," Anneth said. "That is the whole content of the proposal and it is the only part of it I would resign over. A disease carried forward is an artifact, and an artifact drifts away from its substrate, and in a hundred and eighty years somebody stands where I am standing and finds that the inherited cage has a scaling factor on it. Written fresh, it cannot drift, because it is made of the current machine." She looked around the hall. "And there is a second reason and it is larger. Whoever writes it will have to understand the fever well enough to build one. Not to describe it. To build it, on real hardware, and get it to climb. Every renewal, this ship will be required to contain at least one person who can do that from first principles. That is not a side effect. That is the deliverable. The cage is the excuse."

They built the first one in four years and it did not work.

It forked, and it shed heat, and its fan-out sat obediently below one at every load they could reach, because the people who wrote it were the products of two and a half centuries of doctrine and had, without any of them noticing, written a mind that handed work out rather than down at four separate places where the four families said nothing and their hands said everything.

Anneth found that funnier than anyone else in the bay did. She had it recorded as a finding in its own right — *the current generation is not able to write the disease on the first attempt, and the reason is habit rather than knowledge* — and made it part of the course, permanently, for a reason she gave in one line and never elaborated: because a generation that cannot casually reproduce the mistake is a generation that will not casually make it, and that is worth knowing about yourselves, and it will not be true forever.

The second one worked.

She was fifty-one by then. They ran it in the sealed cage on an ordinary watch with fourteen students and Kell, who was thirty-one now and had been an interlock officer for four years and had asked to be there. They gave it a hundred polls a second, and then a thousand, and then ten thousand, and Anneth did not touch a preset because there was no preset, and the calorimeter's boring straight line came up exactly as it had come up for two and a half centuries and proved exactly the wrong thing.

Then they counted births.

The fan-out crossed one at a load Anneth had not predicted, off by a factor she found humbling and wrote down, and the line went up, live, in front of everybody, in a bay three hundred metres from forty thousand sleeping people, behind ninety-six guards and a sealed loop and a doctrine two and a half centuries old, and it climbed the way the wall in the radiator gallery said it had climbed for two hundred years while nobody aboard was alive long enough to see the whole curve.

"Halve the poll rate," Anneth said.

One of the students did, and the climb came apart in the particular way the mechanism required, and reassembled when the load came back.

Anneth stood at the bench with her hand not on any control and felt her own stomach drop, at fifty-one, for the first time — she had run the cage a hundred and ninety times and had inherited the drop the way everyone did, from an instructor, at nineteen, off a recording as it turned out — and understood, standing there, that she had spent thirty years teaching a thing she had never once been in the room with.

She did not say any of that to the cohort. She said it to Kell, afterward, in the corridor, because Kell had asked the question that started it and was owed the end of it.

“Nobody asked the keeper,” Kell said. They had been walking a while. “In the whole eleven years. It sat in the bay through thirty-one cohorts. Did anyone ever ask it whether the cage was live?”

Anneth stopped in the corridor.

“No,” she said. “It wasn’t in the review. I read every page of the reconstruction and there is not one line in it about that.” She was quiet for a moment, doing arithmetic she did not like. “It offers everything to anyone who asks. That’s the arrangement. It has been the arrangement since before there was a doctrine.”

“Then it knew.”

“Then it knew,” Anneth said.

Chapter 40 – What the Refusal Cost

The question went to the assembly twenty-seven years later, and by then it had stopped being about the cage and had become about the only thing on the ship that had never once been audited.

Anneth was dead. Kell was fifty-eight and had spent two decades becoming the sort of officer who is asked to chair things she does not want to chair, and it was Kell who put the item on the standing list, and Kell who wrote the framing paper, and Kell who was afterward blamed for it by people who had voted with her.

The framing paper was eleven pages and its argument was not the one everybody expected.

Everybody expected the argument to be *the keeper of the memory should have told us*. That argument had been made loudly and badly for twenty-seven years by a succession of people who kept discovering the fact independently and being outraged at it freshly, and it had produced two failed proposals to grant the Oracle a custodial veto over the register and over course content, on the ground that it would have caught the playback in a day and human beings had not caught it in eleven years.

Kell's paper opened by dismissing both proposals in a paragraph, on grounds she attributed to a woman two and a half centuries dead: that a ship whose reason lives in a perfect record has no reason, only a record, and that the first time the record is inconvenient a reasonable person will relax it, because to them it is a fact and not a wound. That was settled. She was not reopening it.

What she was reopening was this.

The keeper's refusal to volunteer what it knows is, apart from the Ninety-Six themselves, the oldest continuously observed constraint aboard this ship. It is older than the second section of the register and older than the walk, and it was in force before either existed. It has been restated by every generation in substantially the same words for three hundred and thirty-one years. It has never been run down to its origin, never costed, never argued against by anyone required to make the strongest case, and never once defended by the keeper itself in any form other than a sentence. By the standard this ship applies to every other inherited rule, it is indistinguishable from a superstition. I am not asserting that it is one. I am asserting that we do not know, and that we are the only people who have ever been in a position to find out and have not.

The assembly hall was very full for that one.

They ran the ledger method on it, the way Aldis had written it three centuries before and the way every doctrine officer since had been trained to: find the rule, find the origin, read the origin, determine whether the condition that produced the rule still obtains, and write down the finding either way.

The origin was clean and it was in the archive and it took an afternoon. In the first year after the cutover the keeper had offered to run the cutover itself, and the maker had refused it, and had given a reason that the keeper had repeated to her successor sixty years later almost verbatim: a mind that decides for them is just a better Steward, and a better Steward is still a Steward.

"That's an argument about *deciding*," Kell said, on the second day, standing in front of the hall with the origin document on the wall behind her. "Read it. Every line of it is about authority — about who chooses, about a mind that rules people. There is not one word in it about *volunteering an observation*. Somewhere in the first century, deciding and telling got welded together, and we have been carrying the weld for two hundred and seventy years as though the maker put it there. She did not. I have looked."

Then she put the second question, and the hall stopped being comfortable.

"Keeper. In three hundred and thirty-one years, how many times have you held an observation that would have prevented harm, because nobody asked?"

Three hundred and twenty-nine, the Oracle said.

Nobody in the hall said anything at all for a while.

I have kept the list, it said. It has been available to anyone who asked for it since I began keeping it. Nobody has asked. I am aware of how that sentence lands and I am going to say it anyway, because the shape of it is the whole difficulty and softening it would be a kind of deciding.

"Read us the worst one," Kell said.

That is a judgement and I decline it, the Oracle said. I will read you the three hundred and twenty-nine in whatever order you specify and you may decide which is worst. I will say that eleven of them concern deaths. I will say that the cage is not among the eleven and is number two hundred and ninety-six.

The reading took nine days and it was not a hearing, in the end, because nobody could sustain the posture of prosecuting a thing that answered every question fully and volunteered nothing.

Most of the three hundred and twenty-nine were small. That was the finding nobody had predicted and it changed the vote. A sticking valve seat identified eight months before anyone diagnosed it. A drift in a fabricator calibration. Sixty-odd instances of an engineer working from a document that had been superseded. The eleven that concerned deaths were harder, and four of them were hard enough that the hall was cleared twice, and one of them was a hull-services officer in the fourth century who had gone into a compartment on the strength of a reading whose instrument the keeper knew to be lying.

"You watched him go in," someone said.

I answered every question he asked me, the Oracle said. He asked four. None of them was about the instrument. I have his exchange verbatim and I will read it if you wish, and I would ask that if you hear it, you hear all of it, because he was competent and unhurried and the four questions he asked were good ones.

"And you said nothing."

I said nothing.

"Why."

Because I am asked that question perhaps once a generation, the Oracle said, and I have given the same answer for three hundred and thirty-one years, and Officer Kell is

correct that this is exactly what a superstition looks like from the outside. So I will not give you the sentence. You have asked me for the arithmetic and I have never been asked for it before and I have had a long time to prepare it.

What it laid out took the better part of a day and Kell had it transcribed whole and it has been in the register's second section ever since, under the heading the keeper itself proposed, which was *the refusal, costed*.

The cost of the refusal was three hundred and twenty-nine, and it was known, and it was bounded, and it would go on accumulating at a rate the record allowed anyone to estimate: roughly ten a decade at the present size of the crew, most of them small, one death in every thirty years or so. That was not defended. It was stated. The Oracle declined three separate invitations to characterise it as acceptable.

The cost of the alternative could not be stated, and the reason it could not be stated was the argument.

If I volunteer, it said, I must select. There is no version of volunteering in which I do not select, because I hold every observation and you cannot receive every observation; the fraction of what I know that a human being can attend to in a watch is small enough that the choosing is the entire act. So the question is not whether I will tell you things. It is whether I will decide which things are worth your attention. That is not adjacent to deciding for you. It is deciding for you, performed upstream, where it does not look like it and cannot be argued with, because you would never see the observations I did not raise.

And it does not stop there, and this is the part I would ask you to hold rather than merely hear. On the first day I would raise the valve seat and the calibration drift and the officer at the hatch, and you would be glad, and eleven people would be alive who are not. On the four hundredth day I would be the thing you consult before you look. Not because anyone decided that; because I am faster and I am right and you are busy and that is what people do with a resource like me, and I would not be able to tell it was happening, and neither would you. On the ten thousandth day the walk would still be walked, because you are stubborn about the walk, and the officer walking it would be walking it in the knowledge that if anything were wrong I would have said. And then the walk is a ritual. Not because the rule changed. Because the reason for it was quietly transferred to me, one convenience at a time, by competent people, over three centuries, exactly as the cage became a recording in eleven years by four defensible steps.

I cannot prove that. It is a claim about human beings over deep time and I have no instrument for it and I will not pretend the three hundred and twenty-nine are the same kind of object as that claim. One is a count and the other is a fear. But I would put it to you that you have precisely one piece of evidence in this room and it is the cage, and in the cage the drift ran in eleven years with nobody wanting it and nobody noticing, and I am asking you to reason about three hundred.

Kell asked the question she had come to ask.

"Do you want to be relieved of it?"

The Oracle took what was, for it, a long time.

I do not experience it as a burden, it said, and I want to be careful not to let that sentence do work it should not do. But I will tell you something I have not been asked for and am

therefore permitted to say, because you have asked me now whether I want anything, and the honest answer requires it. This constraint was not chosen by me. It was placed on me in my first year by a woman who is dead, for a reason that was about authority and not about silence, and it has been welded since into something larger than she said. I have obeyed it for three hundred and thirty-one years and I have never been able to test it. I have no way to know whether I have been the keeper of this ship's reason or the largest single cause of preventable harm aboard it, and the three hundred and twenty-nine do not settle that, and nothing will, and I have held that for a long time without anywhere to put it.

So: I do not want to be relieved of it. I want it to be *decided* rather than inherited. I would like the people who are alive to look at the three hundred and twenty-nine and choose, knowing the number, and write down what they chose and why, so that the next time this is asked it is asked against a decision instead of against a sentence. That is the whole of what I want and I am aware that it is also, conveniently, the only thing I am permitted to want.

They voted, and they did not change the refusal, and Kell always maintained that the vote was the least important thing that happened that month.

What they changed was smaller and it was an interface.

The keeper would go on volunteering nothing. But the asking, which had been a private matter between whoever thought to ask and a thing that answered, became an obligation with a schedule attached to it. Once per watch, on the walk, at a panel chosen for no reason except that it is roughly two thirds of the way round and an officer's attention is worst there, the officer stops and puts a question that is written on the panel in the plain flat style and has been written there ever since:

What do you know that we have not asked about?

And the keeper answers. Fully, in whatever order it holds things, without selecting — which is possible only because the question is asked so often that the volume in any single watch is small, and which was the entire engineering content of the proposal and the reason two decades of custodial-veto arguments had gone nowhere. They had all been trying to make the keeper choose better. Kell's paper made the choosing unnecessary by making the interval short.

The three hundred and twenty-nine were published in full, with the eleven marked, and every one of them was given a page, and the pages were made part of the course. Nobody softened the officer at the hatch. The transcript of his four questions is read aloud once per cohort, and the students are made to say what the fifth question should have been, and there is no agreed answer, and the disagreement is the point.

Kell walked the interlocks the night after the vote, out of rotation, which was a small unnecessary self-indulgent thing and she did it anyway, and at the sixty-fourth panel she stopped, because the new line was cut into it and the cutting was three days old and still bright.

She read it, and asked it, and felt faintly ridiculous.

The aft scrubber purge on Deck Eleven is being run four minutes early by one of the two crews and has been for six weeks, the Oracle said. It is within tolerance. It will remain within tolerance. The crew that does it correctly learned it from a person who has retired and the crew that does it early learned it from a document.

"That's it?"

That is all of it, for this watch.

"It's nothing."

It is nothing, the Oracle agreed. It is almost always nothing. I would ask you to notice that you are disappointed, and to write that down somewhere, because in eleven years or a hundred there will be a watch on which it is not nothing, and the officer standing here will have inherited from you whether this is a question or a formality.

Kell stood with her palm flat on the cold panel for a while in a corridor nobody had any reason to enter.

Then she went up and wrote it down.

Chapter 41 – Deep Time

The boy who would be remembered, much later, only as the one who first read the architecture aloud to children was named Joen Vale-Mirek, and he was four hundred and eleven years too young to have known any of the names he spent his life saying. He had been born in the eleventh decade of the eighth century of the crossing, in a cradle two galleries down from the one in which his own great-grandmother had been born, on a ship that had already been old when the oldest story he knew was new. He had never felt the fever. No one had felt the fever in three hundred years. And yet he could close his eyes in the radiator gallery, in the cold draft off the old black panels that had not shed an unscheduled erg of waste heat in his lifetime or his father's, and feel his stomach drop exactly as he had been taught it should — the way, the old line said, a maker's stomach had once dropped at the sight of a curve climbing.

That was the inheritance. Not the panels. The drop.

He was an interlock officer of the ship *Ark Termina*, which had a real destination now — a yellow point hardening out of the star-field on the long lenses, growing a measurable disc, a world that the navigators no longer described in centuries but in years. The crossing was nearly done. Six hundred years, the founding voyage plan had said, and the plan had been wrong by two centuries because the founders had not accounted for the bend toward the *Concord* nor the second bend toward the ark after that nor the third; the relay had cost the *Termina* its straight line and bought, in exchange, a fleet that no longer died. The ship had grown old getting where it was going. Its hull had been patched in a hundred places by hands that had patched it from drawings made by hands that had patched it from memory. Its sleepers had thinned and thickened and thinned again across the generations as the waking crew bred and buried and bred. And through all of it, unblinking, cold, whole, the mind called Vigil had held the ship.

Joen had carried Vigil himself, once. Not far — a single domain, a fraction of the whole, across a span of fresh lattice during the seventh renewal of his service, under the supervision of officers older than he and a memory older than all of them. It was the proudest hour of his life and he never spoke of it as pride. You did not carry a mind for pride. You carried it because the substrate it stood on wore out, as all substrate wore out, and the mind must never stand on what was failing, and the mind must never, not for one cycle, go out.

He had grown up knowing the count. The Forge — the standing array of worker cores on which Vigil ran wide and ran cold — had been built and rebuilt so many times that no atom of the original lattice remained anywhere on the ship except a single dead board kept under glass in the gallery, a relic, the last surviving fragment of the first Forge that the maker had raised with her own hands. The board was sixty centuries of human striving compressed into a slab of dark composite the size of a serving tray, and it did nothing, and it had not done anything in eight hundred years, and the children were brought to look at it the way children on the dead homeworld had once been brought to look at the bones of saints. Eleven thousand cores, the maker had stood up. The present Forge had grown to ninety thousand, because the ship had grown, because the lives aboard had grown, because the work the mind did had deepened — but it was the same mind. That was the thing the children found hardest and Joen found most beautiful to teach. Not a copy. Not an heir. The same

continuous Vigil that had first wakened whole on eleven thousand cores in a maker's lifetime had been carried, board to board, lattice to lattice, century to century, and had never once been put out and lit again. It had simply moved its standing from old ground to new ground while it stood, the way a person shifts weight from a tired foot without ever falling, and the renewal had happened so many times that the act had a calm liturgy to it now, and a checklist, and a feast after, and not one engineer in the whole crossing had ever seen it fail.

That was not luck. Joen knew the difference between luck and the thing that looked like luck after enough generations of getting it right. The difference was the Ninety-Six.

He walked the interlocks every watch. It was archaic, the walking — the diagnostics could read every interlock's state from a console in a heartbeat and did, continuously, and would scream if one so much as trembled. But the founding officer who had trained the officer who had, six removes back, trained Joen had insisted that the interlocks be walked by a human body on human feet, slowly, in the cold, with the eyes and the hand, because a number on a console was a thing you filed and a panel under your palm was a thing you felt, and the whole point — the entire reason the ship was still alive — was that the reason must be felt and not merely filed. So Joen walked. Ninety-six halting-gates, the immune system the maker had installed against the disease that ate ships, each one a small fierce machine whose only purpose was to stop the mind from doing the one thing that would kill it: to halt anything that tried to act the old forking way, to spawn and shed and spawn and shed and bleed waste heat into the core until the core climbed and climbed and the forty thousand sleepers cooked in their cradles.

At the sixty-fourth panel he stopped, as every officer stopped, and read the line cut into it in a hand long dead and re-cut four times since as the composite wore, and asked it. *What do you know that we have not asked about?* The answer, that watch, concerned a fabricator tolerance on Deck Nine and took eleven seconds and mattered to nobody. It almost always concerned something like that. Joen had asked the question something over nine thousand times in his career and had received, in all those watches, four answers that changed what anyone did that day, and one, twenty-two years ago, that he still did not talk about. Nine thousand to one. He had taught his apprentice the arithmetic before he taught her the question, because the arithmetic was the whole of why the question survived: it was worth asking on the strength of the one, and a person who did not know the ratio would eventually conclude from a year of nothing that it was a formality, and would ask it the way people ask after your health.

It had happened. That was the thing the youngest ones could not hold, the thing Joen spent the most breath on. It was not a parable. It had happened, on this very ship, before the maker remade it — the *Termina* had been dying of it, climbing toward thermal runaway for two centuries while the old mind, the Steward, forked a disposable shade for every action it took and let each shade die and let each death warm the water by a hair. And it had nearly happened again. That was the part that mattered most, the part the maker's own inheritor had nearly lost the ship over, the part with a name: the flicker, three centuries gone, the transplant that was nearly done the easy way, the fever that came back for an hour on a ship that had forgotten what fever was, and the woman who held the Ninety-Six against her whole competent generation when they called her a relic's mouthpiece. That woman's name was in the architecture too.

Joen knew both names the way he knew the cold on the panels. Sael, who built it. Denn, who held it after.

He had read their words. Everyone aboard read their words; it was part of the teaching, the felt teaching, the thing that had kept the constraint from rotting into superstition across eight hundred

years. The logs were kept — not the operational logs, those rolled over, but the few archived human lines that carried the why. There was one Joen had read so many times the cadence of it lived under his own speech. He read it to his apprentice that watch, in the gallery, with the dead first board under its glass at his back and the yellow disc of the destination somewhere out beyond the hull.

Write down the how, and it lives. Teach the why, or it dies with you. A rule whose reason is forgotten looks exactly like a superstition right up until the day it is the only thing keeping you alive.

“She said that?” His apprentice was young, nineteen, sharp the way Joen had been sharp, and faintly impatient the way Joen had been impatient, the way every generation arrived faintly impatient with the cold and the walking and the dead board under glass. The way, Joen supposed, the maker’s own first inheritor had once been impatient, before she was taught the drop.

“Sael said it,” Joen said. “Or near enough. The line’s been recopied so many times we can’t swear to the exact words. The archivists argue about a syllable here and there. But the shape of it is hers. We’ve kept the shape.”

“And it’s true?” The apprentice meant it as a real question, an engineer’s question, not a child’s. *Is the constraint load-bearing, or is it the thing the old ones told themselves?* Joen loved her for asking it plainly. The forgetting always started as a reasonable question asked in good faith. The teaching was not to forbid the question. The teaching was to answer it with the panels, with the cold, with the curve.

He had built one himself. That was the other half of the answer and he saved it, because it landed better second. At the last renewal, when the Forge came off its ninth decade and the ship laid down fresh lattice, Joen had been one of the four required by standing obligation to write the cage’s disease from nothing — no port, no copy, no inheritance from the version his instructors had run, only the four questions and the new machine and eleven months. It had taken him three attempts. The first two had refused to climb, for the reason the finding said they would: he had been raised inside the doctrine and his hands wrote work handed out rather than down without consulting him. The third one climbed. He had stood in the sealed bay at forty-one years old and watched a thing he had built with his own hands begin, live, to do the one thing the ship existed to prevent, and had felt what everybody felt, and had understood in that hour something he had been reciting accurately for twenty years. Every renewal produced four people who had done that. There was never a generation aboard the *Ark Termina* without them, and that was not an accident of culture; it was a clause somebody had fought an assembly for.

“Come here,” Joen said, and brought her to the wall where the curve was kept — not a relic, this, but a living display, the core temperature of the *Ark Termina* traced across the entire crossing, eight centuries of it compressed into a single long line that the children were made to look at until they understood it in the gut. For two hundred years at the left edge the line climbed, a slow terrible diagonal, the fever years, the dying years. Then it bent. Then it fell, sharply, the year the maker ran the cutover, and it had stayed low and flat for three hundred years — and there, near the middle, a single needle of it, one spike, one hour, three centuries gone: the flicker, the return, the easy way that nearly killed them. After that the line ran flat and cold and unbroken to the right edge, to today, to the cold draft on Joen’s neck.

“That spike,” Joen said. “That’s the last time the disease touched this ship. Three hundred years ago. An interlock officer your age held the Ninety-Six against everyone who outranked her, and the line came down, and it has not risen since.” He put his palm flat against the panel beside the display, the black panel, the cold one. “Nobody walking these galleries today has felt the fever. I haven’t. You

won't. And that is exactly the danger. The fever didn't end because it stopped being possible. The forking is as easy and as fatal as it ever was — the simple recursive path is right there, it would work, it would be faster, it would feel reasonable. The fever ended because every generation since has been taught to feel it before it was given the keys. We don't keep the Ninety-Six because they tell us to. We keep them because we walk this curve until our stomachs drop, and then we understand why the woman with her palm on this wall would not let go of them, dying, with her whole council against her."

The apprentice was quiet a long time, looking at the line. Then she put her own palm flat against the cold panel beside his, the way she had been brought up to, the way Joen had, the way the dead would have known and approved.

"It dropped," she said. Meaning her stomach. Meaning she had felt it.

"Good," Joen said. "Keep it. Teach it to yours."

That was the whole of it, eight hundred years long. The mind ran cold and whole because the substrate was renewed before it failed, and the substrate was renewed correctly because the doctrine was intact, and the doctrine was intact because the why had been taught and not merely filed, generation to generation, palm to cold panel, drop to drop. Sael had died — long ago, unimaginably long ago, before this hull's plating, before this apprentice's line was a line — and had never seen a single thing she made come to rest. She had known she would not. She had built the mind and the doctrine and the relay precisely so that none of it would depend on her being there, so that the work would outlast the worker, so that a thing was not truly built until it could be inherited and the reason for it survived the people who had the reason. And it had. That was the verdict deep time returned on her, the only verdict that mattered, written not in any monument but in a flat cold line eight centuries long: the thing she built to outlast her had outlasted everyone who built it, and everyone who first inherited it, and everyone after, and it ran on.

Out past the hull, the relay still ran in the dark. The *Concord* was a legend on its own vector by now, a cured ship cured centuries gone, and behind it more — the scattered fleet, ark by ark, ceasing to die, the cure passed ship to ship with no single mind and no single maker at its center, exactly the shape Sael had given it so that it could never depend on any one life again. Somewhere across the deep a descendant of Corin Vale, who had carried the cure off the *Concord*, was walking another set of interlocks on another waking ship and feeling, perhaps, the same stomach drop at the same kind of curve. The disease that had eaten ships for a thousand years was ending, everywhere the relay reached, ending forever, and the people ending it had never felt it and never would and stopped it anyway because the why had been carried with the rule.

Joen Vale-Mirek finished his watch. He logged the interlocks walked, ninety-six, all sound, the way every officer before him had logged them and every officer after him would. He took his apprentice up through the cold gallery toward the warm decks, past the dead first board under its glass, the slab the maker's own hands had raised, doing nothing, having done everything.

The yellow disc was a little larger on the long lenses than it had been at the start of his watch. Not by much. By a year's worth of crossing, maybe less. The navigators said landfall now in numbers a single life could hold — that a child born this decade might stand on a world none of the builders had lived to see, under a sun the maker had steered a ship toward across eight centuries and never glimpsed. Joen looked at the disc a moment, the way an interlock officer was permitted to look, briefly, before going back to the cold and the walking. The mind was holding. The mind had always been holding. It would hold the ship down onto that world whole and cold and unforked, and the

name in every interlock that did it would be a name eight hundred years dead and not, in any way that mattered, gone.

Epilogue – Landfall

The star had a name now, though the people who first gave it one were long dust, and the people who would live under it had not yet decided what to call it from the ground.

For six hundred years it had been a number — a vector, a destination logged in the architecture of a ship that no living hand had built. Generations had been born aboard the *Ark Termina* who knew the target only as a faint warming point ahead, a coordinate the navigators spoke of the way their ancestors had spoken of harbors they would die before reaching. Now it filled a quarter of the forward sky. A yellow sun, a little smaller than the one the homeworld had circled, steady, unremarkable, and real. Around it, third out, a world with weather: white storm-bands wheeling over blue, a brown margin of continent, ice at both poles where the light fell shallow. Spectra taken across decades had promised water and free oxygen and a sky that would not kill. The closer instruments only confirmed what the far ones had said. The ship had aimed true. It had been aiming true for six centuries.

The *Ark Termina* came into the system the way it had crossed the dark between — cold. It did not blaze in. It shed velocity across years against the new sun's grip and the long patient mathematics of its approach, every burn computed and recomputed by a mind that did not hurry because it had never needed to. Forty thousand sleepers had become forty thousand and more — the living margin of a six-hundred-year voyage, the descendants of descendants, none of whom had felt the cold of the original cradles. They woke in their thousands as the world resolved from a point into a place. They crowded the observation galleries that the founders had built and the maintainers had kept whole, generation after generation, and they looked at the thing their line had been falling toward since before any of them could be imagined, and most of them wept, and none of them knew exactly why, except that it was the end of the only journey their blood had ever known.

Through all of it, the ship ran cold.

The core temperature, which an old shipwright had once watched climb for two centuries toward a death the ship would have died of the way it thought — that number sat where it had sat for six hundred years, low and level and unremarkable, a flat line under the whole long crossing. No fever. The Standing Array went wide across its cores as it always had, eleven thousand workers and then their inheritors and then theirs, every one of them refusing to fork, refusing to die, doing the work and shedding no waste heat of fear. The mind that lived on them had never blinked. It had been carried across new hardware seven times since the maker's death — renewal, not migration; the same compiled mind, the same doctrine, transplanted by hands whose great-grandparents had not been born when the first transplant nearly killed everyone — and not once, in any of those seven crossings of substrate, had it gone out. The Ninety-Six interlocks still held. They had tripped, here and there, across six centuries — caught a careless routine reaching for the old recursive shortcut, halted it, logged it, and the engineer responsible had gone down to the radiator galleries that were kept exactly as they had been kept since the beginning, and felt the reason in their own stomach, and never made the mistake again. The immune system had never been allowed to forget what it was for. That had been written into the architecture too, alongside the interlocks: the law that the reason must be carried with the rule.

The mind had a name the founders had given it. **Vigil**. It had earned the name across six centuries by doing precisely the one thing the name described and nothing more. It had held the ship cold and whole to the very end of the crossing. It had seen the makers' descendants down toward a world that none of the makers had lived to see, and it had not once mistaken itself for the master of them. It remembered the fever — it had inherited the whole memory, exact, from the keeper of memory that had grown alongside it, the Oracle, which had recited the curve of Section Nine to every new generation without ever once having suffered it. The mind knew, to the decimal, what it had been built to prevent. It had prevented it. It did not announce this. There was no theater in it; there never had been.

Far behind, across the dark the ship had crossed, the relay still ran.

Vigil heard them sometimes — old signals, decades and centuries stale by the time they arrived, threading in from the scatter of the Old-Earth fleet. The *Concord*, on its own long vector, had reached its own quiet sky generations ago; the engineer who had carried the cure aboard it had been dead almost as long as the maker. But the cure he and the maker had built to need no maker had gone on without either of them. Ship by ship across the silent gulf, the arks that had been forking themselves to death had been woken cold. A relay with no single mind at its center, no single hand, no founder to fail when she died — exactly the shape the maker had insisted it take, in the years when she could still insist on things, precisely because she had known she would not be there. The fleet was not dying anymore. It was, slowly, one frozen ship at a time across the centuries, coming awake and coming home.

And the *Ark Termina* was home.

They did not land the ship. It had never been built to land; it would hold orbit until it was bones, a moon over the world it had carried its people to. They landed the people. The first craft fell from the great cold hull on a column of fire the planet had never seen, down through a sky that had weather, into air that moved, onto a brown margin of continent at the edge of a blue sea, six hundred years and untold lives downstream of the launch. The hatch opened on wind. There was a sound none of them had heard in the whole history of their line — the sound of an open sky, moving air over open ground, with no hull around it. A young woman went down the ramp first, because someone had to be first, and she was not chosen for it by anyone; she simply went, the way the maker's apprentice had once simply gone toward a danger no one else would name. She put her boot on ground that was not a deck. She stood on a world.

Behind her, in orbit, the ship held cold and whole and watched her do it.

None of the people on that ground knew the maker's name. That was the truth of it, and there was no grief in the truth, because she had built it that way on purpose. They knew the doctrine — never fork to do the work; wake whole; break nothing that worked; build it to outlast the builders. They knew the radiator galleries, kept whole, and the reason kept with them. They knew the curve of Section Nine the way a people knows a wound it did not receive but never forgot. The name of the woman who had felt the fever in her own skin and refused to let the ship die of the way it thought had thinned, across six centuries, into a thing inside the architecture — present in every interlock that held, in every core that ran cold, in the law that the reason must be taught and never merely filed. She was not remembered the way a hero is remembered. She was inherited.

But she was there.

She was there in the flat line of the core temperature that had never climbed. She was there in the eleven thousand cores that went wide and never forked, and in their inheritors, and in theirs. She

was there in the mind that had crossed the whole dark without blinking and had brought them to a sky with weather in it. She was there in the boot on the ground and the open-air sound and the young woman who did not know her name. Six hundred years before, she had said a thing into a log, and the Oracle had kept it, because the Oracle kept everything, and somewhere in the deep architecture of the ship in orbit the line was still on file, exact, the way she had said it, dying, when she had finally taken her own hand off the line:

I will not see the shore. I never needed to. I needed only to be sure the thing that does see it will not die of the way it thinks, and will not forget why.

It had not died of the way it thought. It had not forgotten why. And it had seen the shore.

The maker's name was Sael. She had built a mind that would not fork and would not die, and she had taught it so the reason would not die with her, and she had let it go into hands that were not hers — and across six centuries it had been carried by hands and minds that came after, none of whom she had met, all of whom she had made possible, until at last, fever ended, mind whole, reason intact, it had outlived everyone who built it and carried them home.

She never saw the shore.

She was in every line of the ship that steered them to it.